

BINOS® 100 Series Gas Analyzers

BINOS® 100, BINOS® 100 M

BINOS® 100 2M, BINOS® 100 F

OXYNOS® 100, HYDROS® 100, CAT 100



ROSEMOUNT®
Analytical

ESSENTIAL INSTRUCTIONS

READ THIS PAGE BEFORE PROCEEDING!

Emerson Process Management designs, manufactures and tests its products to meet many national and international standards. Because these instruments are sophisticated technical products, you **MUST properly install, use, and maintain them** to ensure they continue to operate within their normal specifications. The following instructions **MUST be adhered to** and integrated into your safety program when installing, using and maintaining Emerson Process Management products. Failure to follow the proper instructions may cause any one of the following situations to occur: Loss of life; personal injury; property damage; damage to this instrument; and warranty invalidation.

- **Read all instructions** prior to installing, operating, and servicing the product.
- If you do not understand any of the instructions, **contact your Emerson Process Management representative** for clarification.
- **Follow all warnings, cautions, and instructions** marked on and supplied with the product.
- **Inform and educate your personnel in the proper installation, operation, and maintenance of the product.**
- **Install your equipment as specified in the Installation Instructions of the appropriate Instruction Manual and per applicable local and national codes.** Connect all products to the proper electrical and pressure sources.
- To ensure proper performance, **use qualified personnel** to install, operate, update, program, and maintain the product.
- When replacement parts are required, ensure that qualified people use replacement parts specified by Emerson Process Management. Unauthorized parts and procedures can affect the product's performance, place the safe operation of your process at risk, **and VOID YOUR WARRANTY**. Look-alike substitutions may result in fire, electrical hazards, or improper operation.
- **Ensure that all equipment doors are closed and protective covers are in place, except when maintenance is being performed by qualified persons, to prevent electrical shock and personal injury.**

The information contained in this document is subject to change without notice.

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PREAMBLE

This instruction manual provides information about BINOS® 100 series gas analyzers concerning subassemblies, functions, procedures, installation, operation and maintenance.

This instruction manual covers several BINOS® 100 series analyzer variations and therefore may describe configurations and/or options not part of your specific analyzer.

Installation and operation of instruments intended to be installed and operated in HAZARDOUS AREAS is NOT COVERED by this instruction manual, but part of the specific instruction manual shipped together with such analyzers because of the special requirements for working in hazardous environments!

DEFINITIONS

The following definitions apply to WARNINGS, CAUTIONS and NOTES found throughout this publication.

WARNING

Highlights an operation or maintenance procedure, practice, condition, statement, etc.

If not strictly observed, could result in injury, death, or long-term health hazards of personnel.

CAUTION

Highlights an operation or maintenance procedure, practice, condition, statement, etc.

If not strictly observed, could result in damage to or destruction of equipment, or loss of effectiveness.

NOTE



Highlights an essential operating procedure, condition or statement.

IMPORTANT

SAFETY INSTRUCTIONS WIRING AND INSTALLATION OF THIS APPARATUS

The following safety instructions apply specifically to all EU member states. They should be strictly adhered to in order to assure compliance with the Low Voltage Directive. Non-EU states should also comply with the following unless superseded by local or National Standards.

1. Adequate earth connections should be made to all earthing points, internal and external, where provided.
2. After installation or troubleshooting, all safety covers and safety grounds must be replaced. The integrity of all earth terminals must be maintained at all times.
3. To ensure safe operation of this equipment, connection to the mains supply should only be made through a circuit breaker which will disconnect all circuits carrying conductors during a fault situation. The circuit breaker may also include a mechanically operated isolating switch. Circuit breakers or switches must comply with a recognized standard such as IEC947. All wiring must conform with any local standards.
4. Where equipment or covers are marked with the symbol to the right, hazardous voltages are likely to be present beneath. These covers should only be removed when power is removed from the equipment — and then by trained service personnel only. 
5. Where equipment or covers are marked with the symbol to the right, there is a danger from hot surfaces beneath. These covers should only be removed by trained service personnel when power is removed from the equipment. Certain surfaces may remain hot to the touch. 
6. Where equipment or covers are marked with the symbol to the right, refer to the Instruction Manual for instructions. 
7. Further graphical symbols used in this product:



Elektrostatic discharge (ESD)



Explosion Hazard!



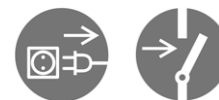
Heavy Instrument!



Harmful (to Health)!



Toxic!



Disconnect from Mains!

All graphical symbols used in this product are from one or more of the following standards: EN61010-1, IEC417, and ISO3864.

SAFETY INSTRUCTIONS

OPERATING AND MAINTAINING THIS APPARATUS

This instrument has left the factory in compliance with all applicable safety regulations.

To maintain this operating condition, the user must strictly follow the instructions and consider the warnings in this manual or provided on the instrument.

Before switching on the instrument, verify that the electrical supply voltage matches the instrument's operating voltage as set in the factory.

Any interruption in the instrument's ground line, whether inside or outside the instrument, or removal or interruption of its ground line connection, could result in hazardous operating conditions. Intentionally interrupting the instrument's protective ground is strictly prohibited.

Opening cover panels could expose voltage-carrying components. Connectors may also be under voltage. The instrument must be disconnected from all electrical supplies before attempting any calibrations, maintenance operations, repairs or component replacements requiring opening of the instrument. Any calibrations, maintenance operations, or repairs that need the instrument to be opened while connected to electrical supplies should be subject to qualified technicians familiar with the hazards involved only!

Use only fuses of the correct type and current ratings as replacements. Using repaired fuses and short circuiting of fuse holders is prohibited.

Observe all applicable regulations when operating the instrument from an auto-transformer or variac.

Substances hazardous to health may emerge from the instrument's exhaust.

Please pay attention to the safety of your operation personnel. Protective measures must be taken, if required.

SAFETY INSTRUCTIONS**INTENDED USE STATEMENT**

BINOS® 100 series gas analyzers are intended to be used as analyzers for industrial purposes. They must not be used in medical, diagnostic or life support applications, and no independent agency certifications or approvals are to be implied as covering such applications!

SAFETY SUMMARY

If this equipment is used in a manner not specified in these instructions, protective systems may be impaired.

AUTHORIZED PERSONNEL

To avoid loss of life, personal injury and damage to this equipment and on-site property, do not operate or service this instrument before reading and understanding this instruction manual and receiving appropriate training. Save these instructions.

**WARNING****ELECTRICAL SHOCK HAZARD !**

Do not operate without covers secure. Do not open while energized. Installation requires access to live parts which can cause death or serious injury.

For safety and proper performance this instrument must be connected to a properly grounded three-wire source of power.

WARNING**TOXIC GASES !**

This unit's exhaust may contain toxic gases such as sulfur dioxide. These gases can cause serious injuries.

Avoid inhalation of the exhaust gases at the exhaust fitting.



Connect exhaust outlet to a safe vent. Check vent line and connections for leakage.

Keep all fittings tight to avoid leaks. See section 9-2, page 9-3 for leak test information.

SAFETY INSTRUCTIONS

WARNING

EXPLOSION HAZARD !



Do not operate nor install these instruments in hazardous areas without additional measures!

CAUTION

BINOS® 100 F and CAT 100 -- HEAVY INSTRUMENTS !



The analyzer variations BINOS® 100 F and CAT 100 intended to be wall mounted and/or outdoor installed weigh up to 30 kg resp. 70 kg, depending on included options!

Use two person and/or suitable tools for transportation and lifting these instruments!

Take care to use anchors and bolts specified to be used for the weight of the units!

Take care the wall or stand the unit is intended to be installed at is solid and stable to hold the units!

CAUTION

HIGH TEMPERATURES !



While working at photometers and/or thermostated components inside the analyzers hot components may be accessible!

SAFETY INSTRUCTIONS

GASES AND GAS CONDITIONING

WARNING

Take care of the safety instructions applicable for the gases (sample gases and test gases) and for the gas bottles containing these gases!



Before opening gas paths they must be purged with ambient air or neutral gas (N₂) to avoid hazards caused by toxic, flammable, explosive or harmful to health sample gas components!

**WARNING****EXPLOSION HAZARD !**

Supplying flammable gases of concentrations above 25 % of lower explosion limit (LEL) we recommend to utilize one or more of the following measures:

- Purging the housing with inert gas
- Internal tubing with stainless steel
- Flame arrestors at gas input and output fittings
- Intrinsically safe paramagnetical or thermal conductivity sensors

Supplying explosive gases is not permitted !

SAFETY INSTRUCTIONS

POWER SUPPLY

CAUTION



Verify the mains voltage at site of installation corresponds to the analyzer's rated voltage as given on the nameplate label!

Verify the safety instruction given by power supply unit manufacturer !

Instruments with External Power Supply Unit

CAUTION



The mains socket has to be nearby the power supply unit and easily accessible! Disconnecting from mains requires unplugging the power plug!

To comply with the CE mark requirements use only power supply units of type VSE 2000, UPS 01 T, DP 157, SL5, SL10 (DP 157 and SL for rack installation only) or equivalent units. Equivalent units must provide SELV output voltages!

Verify proper polarity when connecting DC 24 V operated analyzers !

Instruments with Internal Power Supply Unit

WARNING

ELECTRICAL SHOCK HAZARD !



These instruments provide a protective earth terminal. To prevent electrical shock hazards the instrument must be connected to a protective earth. Therefore the instrument has to be connected to mains by using a three wire mains cable with earth conductor!

Any interruption of the earth connector inside or outside the instrument or disconnecting the earth terminal may cause potential electrical shock hazard! Intended interruption of protective earth connections is not permitted!

SAFETY INSTRUCTIONS

WARNING

ADDITIONAL HINTS FOR MODELS BINOS® 100 F AND CAT 100

Installation and connecting mains and signal cables are subject to qualified personnel only taking into account all applicable standards and legislative requirements!

Failure to follow may cause warranty invalidation, property damage and/or personal injury or death! Mains and signal cables need to be connected to internal screw terminals requiring to work at open housing near live parts!

Installation of this instrument is subject to qualified personnel only, familiar with the resulting potential risks!



These gas analyzer do not provide a mains power switch and are operable when connected to power.



These gas analyzers do not provide a mains switch! A mains switch or circuit breaker has to be provided in the building installation. This switch has to be installed near by analyzer, must be easily operator accessible and has to be assigned as disconnecter for the analyzer.

CAUTION

ADDITIONAL HINTS FOR MODEL BINOS® 100 F



Cables for external data processing must be double insulated for mains voltage when used inside the instrument!

If double insulation is not available signal cables inside the analyzer must be installed in a way that a distance of at least 5 mm is ensured permanently (e.g. by utilizing cable ties).

SAFETY INSTRUCTIONS

CAUTION

ADDITIONAL HINTS FOR MODEL BINOS® 100 2M



Supplying external components utilizing the DC 24 V power output provided by the BINOS® 100 2M internal power supply unit requires a fuse to be connected in series to the load limiting the power to 2 A max!

Verify proper DC 24 V polarity when connecting any external load!

General Operating Instructions

General operating instructions

WARNING**DANGER TO LIFE ! EXPLOSION HAZARD !**

Verify all gas lines are connected as described within this manual and tight!

Improper gas connections may cause explosion, serious injury or death!

Exhaust may contain hydrocarbons and other toxic gases, e.g. carbon monoxide. Carbon monoxide is toxic!

- Installation area has to be clean, free from moisture, excessive vibration and frost-protected. Take care to meet the permissible ambient temperatures as given in the technical data section! Instruments must not be exposed to direct sunlight, fluorescent lamps nor sources of heat. For outdoor installation it is recommended to mount the instruments into a cabinet. At least sheltering against rain is required. Do not cover venting openings and take care to mount the instrument in a distance to walls not affecting venting.
- Do not interchange gas inlet and outlet! All gases must be conditioned before supplying! When supplying corrosive gases ensure that gas path components are not affected!
- Max. permissible gas pressure: 1,500 hPa, except with integrated valve blocks (see hint at page 3-4) and/or paramagnetic Oxygen sensor (see page 2-15)!
- Exhaust lines must be installed in a descending way, need to be pressureless, frost-protected and in compliance with applicable legislative requirements!
- When it is necessary to open gas paths seal the analyzer's gas fittings by using PVC caps to avoid pollution of the internal gas path by moisture, dust, etc.
- To stay in compliance with regulations regarding electromagnetic compatibility it is recommended to use only shielded cables, as optionally available from Emerson Process Management or equivalent. Customer has to take care that the shield is connected in proper way (see section 3-5). Shield and signal connector enclosure need to be conductively connected, submin-d plugs and sockets must be screwed to the analyzer.
- Using external submin-d-to-terminal adaptor elements (option) affects electromagnetic compatibility. In this case the customer has to take measures to stay in compliance and has to declare conformity, when required by legislation (e.g. European EMC Directive).

Magnetically Operated Front Panel

Magnetically Operated Front Panel

WARNING

DANGER TO LIFE !



Persons with cardiac pacemakers should absolutely avoid magnetic fields !



Negative effects on persons beyond those described above caused by magnetic fields are not known. It is presumed that persons showing allergic reaction on contact with ceramic or metallic material show the same behavior on contact with magnetic material.

CAUTION



Permanent magnets are surrounded by magnetic fields. These magnetic fields can disturb and even destroy sensitive electronic measuring devices, but also mechanical watches, credit cards, etc.

Usually a distance of 0.5 m is enough to avoid damages. All sintered permanent magnets are hard and brittle. Hitting of sintered permanent magnets by the magnetic attraction causes splitting into fragments with many sharp edges. This especially occurs with high energy magnets, and can also cause skin bruises by high attraction.

High energy magnets made of rare-earth materials have to be stored dry, otherwise the surfaces would oxidise. Unprotected operation in a humid environment may cause corrosion. Avoid damaging the protective galvanic coating.

A storage in a hydrogen atmosphere destroys these magnets. A demagnetisation is caused when permanent magnet materials have been exposed in a radioactive radiation for a long time.

For air transportation of magnetic material the IATA instructions have to be observed:

Magnetic fields are not allowed to penetrate the package, if necessary the magnets have to be shorted using a metal plate.

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Chapter 1 Technical Description

1-1 General Description

BINOS® 100 series analyzers are characterized by

- compact design
- highly integrated main board, containing all necessary basic functions and interfaces
- microprocessor based user interface
- multi-component and multi-method analysis with the opportunity to combine several methods in one single analyzer.

BINOS® 100 series analyzers are designed to measure 1 or 2 gas components, except OXYNOS® 100 for Oxygen measurement with paramagnetic cell and HYDROS® 100 (Thermal conductivity measurement): These two analyzers are single channel units.

The following methods are available:

IR = non-dispersive infrared measurement
 PO₂ = paramagnetic Oxygen measurement
 EO₂ = electrochemical Oxygen measurement
 TC = thermal conductivity measurement

For applications with solvent and/or corrosive components in the gas stream special resistant measuring cells are available.

For measuring flammable gases special solutions are available too (e.g. intrinsically safe cells).

Standard General Purpose Applications

Several enclosure variations are available:

Tabletop and rack mount versions of 1/4 19" size (BINOS® 100 (M), OXYNOS® 100, HYDROS® 100) and 1/2 19" size (BINOS® 100 2M) are available. They are IP 20 protected (acc. to EN 60529), tabletop versions are equipped with a handle and rubber feet.

An IP 65 protected field housing (BINOS® 100 F) for outdoor installation is available, too. This enclosure is intended to be wall mounted.

Installation in Hazardous Areas

Note!

This manual does not deal with special conditions for analyzers in hazardous areas, related to installation, operation, maintenance etc. For such applications refer to the separate instruction manuals, delivered together with the analyzers.

For installation in hazardous areas the BINOS® 100 F analyzer is provided with an adapted pressurization system (ATEX type approved for Zone 1 resp. Zone 2 in Europe) and an impact tested magnetically operated front panel. Optionally intrinsically safe signal couplers are available, too.

A simplified z-purge system permits installation in North America Zone 2 environments.

The CAT 100 analyzer, designed to be installed in hazardous areas, too, comprised of a BINOS® 100 2M analyzer installed into a flameproof enclosure. This model is CSA-C/US and ATEX approved for installation in North-American and European hazardous areas.

Overview

The BINOS® 100 series analyzers offer a wide range of available configurations and options, to be combined according to the selected model:

■ Measuring principles

All offered principles may be combined, depending on analyzer model and other options:

	Variation 1 Variation 2 Variation 3 Variation 4 Variation 5 Variation 6 Variation 7 Variation 8 Variation 9 Variation 10 Variation 11 Variation 12 Variation 13												
IR photometer	1	2							1	1	1		
EO2 sensor			1	2					1			1	
PO2 sensor					1	2				1			1
TC sensor							1	2			1	1	1

Table 1-1: BINOS® 100 Series
Combinations of Measuring Principles

■ Options

Depending on which model is selected, one or more of the following options are available:

- Solenoid valve block
(BINOS® 100 2M / BINOS® 100 F / CAT 100 only; (special solution with 2 valve blocks and parallel gas paths available)
This option uses 4 (8) internal solenoid valves to control sample, zero, span gas 1 and span gas 2. These gases are fed to the analyzers to provide manual or controlled automatic calibration (initialized by keypad, digital inputs, RS 232/485 or FOUNDATION™ Fieldbus commands).
If any valve is open this is indicated by a green LED on the analyzer's front panel.
- Barometric Pressure sensor
(Measuring range 800 to 1,100 hPa)
Used to compensate atmospheric pressure variations to improve precision of results (see measurement specification, pg. 2-6).
Special sensors for e.g. corrosive gas on request.
- Sample pump
(BINOS® 100 2M / BINOS® 100 F / CAT 100 only)
Maximum flow rate 2.5 l/min (special solution with 2 pumps and parallel gas paths available)

Overview

All models may be configured to use several interfaces (in part optional), depending on model and other configuration:

- analog outputs (standard)
- digital outputs (standard; threshold contacts / valve control)

Options:

- analog signal inputs (external cross compensation, **for TC cell only!**)
- serial interface RS 232 / RS 485 *)
- status signals (relay outputs)
- 7 digital inputs **)
- Foundation™ Fieldbus **) ***)

Analyzers use several software revisions, depending on selected options (actual installed revision is shown during analyzer boot sequence in lower 7 segment display on the front panel)

BINOS® 100 (M), OXYNOS® 100, HYDROS® 100:

Version 5.x0 with optional RS 232/485 interface

BINOS® 100 2M (Variation with internal power supply):

Version 5.x0 with optional RS 232/485 interface*)

BINOS® 100 2M (Variation with external power supply):

Version 5.x0 with optional RS 232/485 interface*) and/or 7 digital inputs

Version 5.x1 with optional 7 digital inputs and FOUNDATION™ Fieldbus**)

BINOS® 100 F:

Version 5.x0 with optional RS 232/485 interface*) and/or 7 digital inputs

Version 5.x1 with optional 7 digital inputs and FOUNDATION™ Fieldbus**)

CAT 100:

Version 5.x0 with optional RS 232/485 interface*) and/or 7 digital inputs

Version 5.x1 with optional 7 digital inputs and FOUNDATION™ Fieldbus**)

*) not in combination with FOUNDATION™ Fieldbus

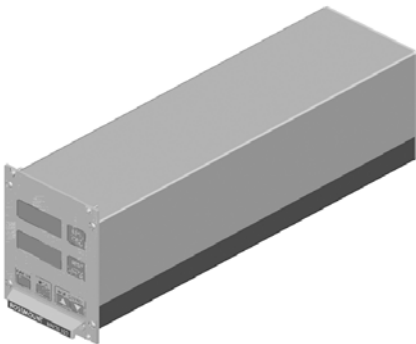
**) BINOS® 100 2M with external power supply resp. BINOS® 100 F / CAT 100 only

***) not in combination with RS 232/485 interface

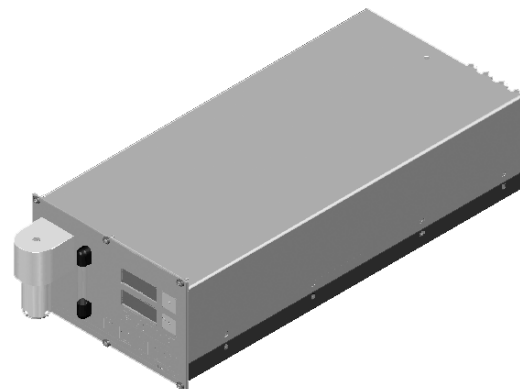
Overview

The different BINOS® 100 series models and their appearance.

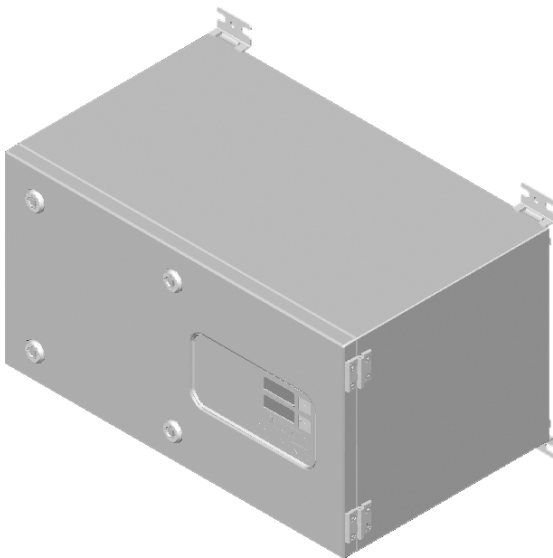
The following sections 1-2 to 1-5 give detailed descriptions for all available configurations.



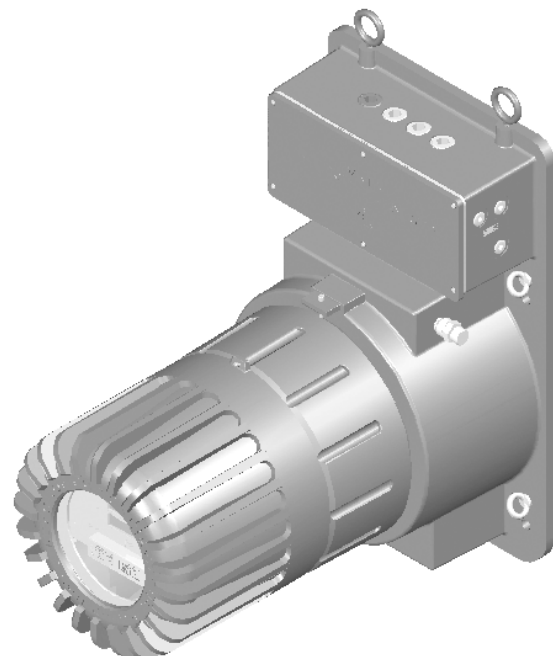
BINOS® 100 (M), OXYNOS® 100, HYDROS® 100
(see section 1-2, page 1-5)



BINOS® 100 2M
(see section 1-3, page 1-9)



BINOS® 100 F
(see section 1-4, page 1-15)



CAT 100
(see section 1-5, page 1-22)

1-2 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

1-2 BINOS® 100,
BINOS® 100 M,
OXYNOS® 100,
HYDROS® 100

These analyzers are the most compact variations within the BINOS® 100 series:
1/4 19" enclosure, external power supply unit.

The various designators are based on the measuring principles used with the analyzer:

BINOS® 100: IR measurement only

BINOS® 100 M: IR measurement combined with Oxygen measurement

OXYNOS® 100: Oxygen measurement only

HYDROS® 100: Thermal conductivity measurement

	BINOS 100		BINOS 100 M		OXYNOS 100	HYDROS 100
IR photometer	1	2	1			
EO2 sensor			1	1	2	
PO2 sensor					1	
TC sensor						1

Table 1-2: BINOS® 100 (M), OXYNOS® 100, HYDROS® 100,
Combinations of measurement principles

All variations with 2 measuring channels allow serial tubing (1 gas inlet) as well as parallel tubing (separate gas inlet for each channel).

The front panel shows 7 segment displays for both channels and pushbuttons for operating the analyzer (see fig. 1-1).

Up to 4 gas connections, DC 24 V power input and submini-d connectors for interfaces (some of

them optional) are located at the rear panel (see fig. 1-2).

Compared to the other analyzer versions the rear panel of OXYNOS® 100 with paramagnetic measuring cell is rotated by 90° (see fig. 1-3).

Chapter 8 shows several internal variations of components as a result of combinations given by table 1-2.

Note!

The analyzers may optionally provide a gas fitting for purging the enclosure (e.g. with nitrogen) when measuring

- low CO₂ concentrations: N₂ minimizes interference by ambient air
- flammable gases: N₂ gives additional safety in the event of leakage inside the instrument

- aggressive components: purging with air gives additional safety in the event of leakage inside the instrument.



Purge medium must be dry, clean and free of corrosive and solvent components!

1-2 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

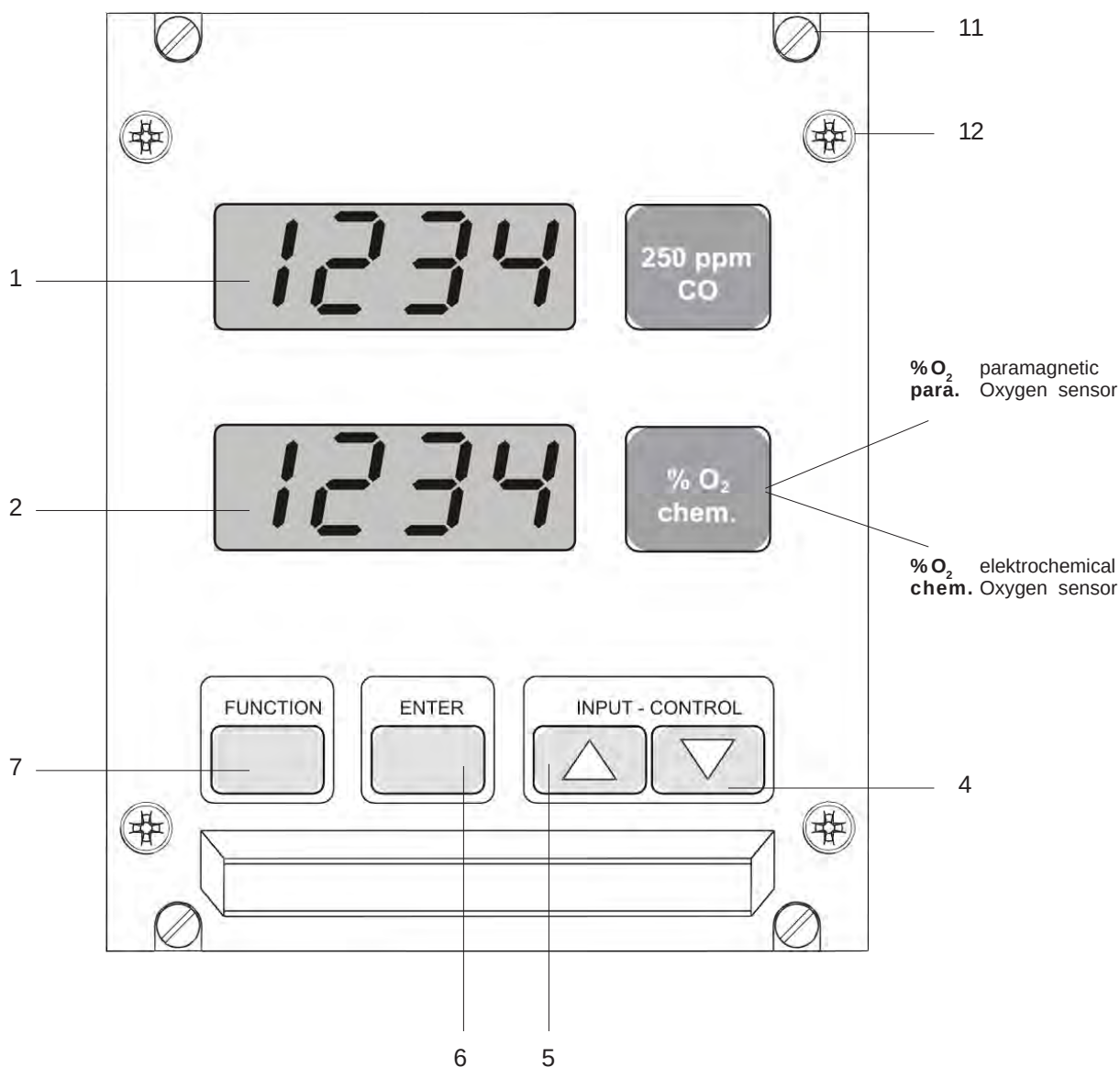


Fig. 1-1: BINOS® 100 (M), OXYNOS® 100, HYDROS® 100, front side view

- | | | | |
|---|-------------------------------|----|--|
| 1 | 7 segment display (channel 1) | 6 | Pushbutton ENTER |
| 2 | 7 segment display (channel 2) | 7 | Pushbutton FUNCTION |
| 4 | Pushbutton DOWN | 11 | Screw for fixing handle/plug-in module |
| 5 | Pushbutton UP | 12 | Screw for fixing enclosure cover |

1-2 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

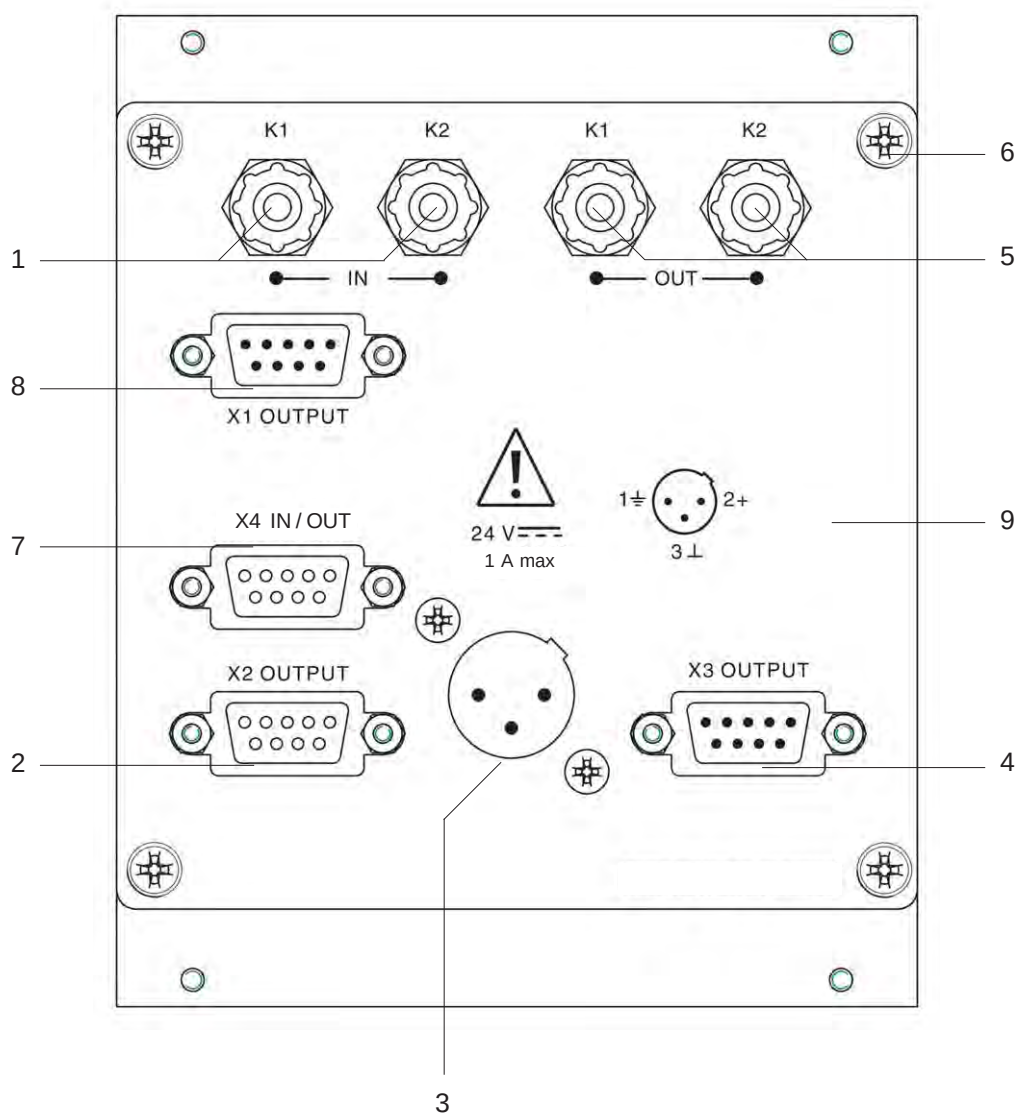


Fig. 1-2: BINOS® 100 (M), OXYNOS® 100 (EO₂), HYDROS® 100, rear side view

- | | | | |
|---|-------------------------------|---|--|
| 1 | Gas inlet fittings | 6 | Screw for fixing enclosure cover |
| 2 | Socket "analog signal output" | 7 | Socket "serial interface" [RS 232 / 485] (option) |
| 3 | 24 V DC power connection | 8 | Plug "status messages" (relay outputs, option) |
| 4 | Plug "digital outputs" | 9 | Plug "analog signal inputs" |
| 5 | Gas outlet fittings | | (external cross compensation, HYDROS® 100 only) |

1-2 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

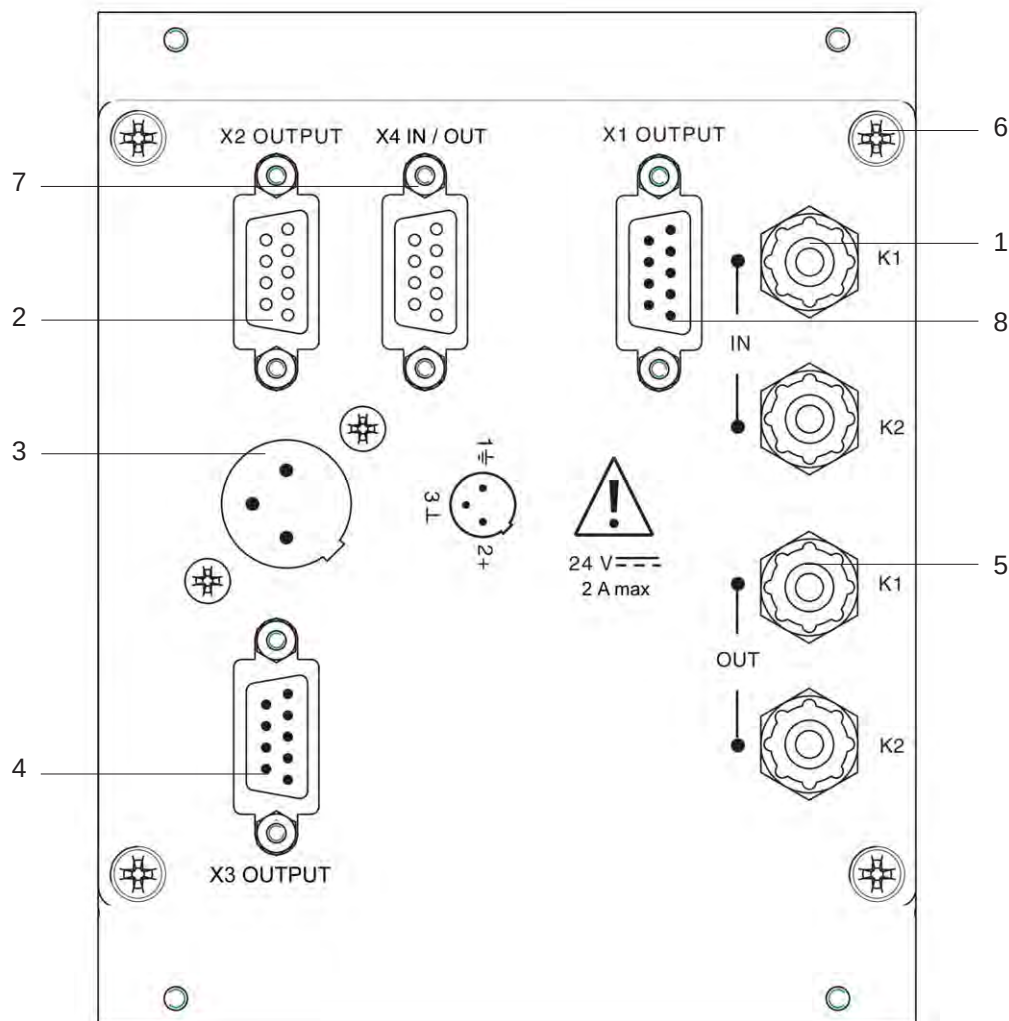


Fig. 1-3: OXYNOS® 100 (PO₂), rear side view

- | | | | |
|---|-------------------------------|---|---|
| 1 | Gas inlet fitting | 5 | Gas outlet fittings |
| 2 | Socket "analog signal output" | 6 | Screw for fixing enclosure cover |
| 3 | 24 V DC power connection | 7 | Socket "serial interface" [RS 232 / 485] (option) |
| 4 | Plug "digital outputs" | 8 | Plug "status messages" (relay outputs, option) |



*Continue with section 1-6
"Internal gas paths" , page 1-30*

1-3 BINOS® 100 2M

1-3 BINOS® 100 2M

The BINOS® 100 2M model represents the variation that may be upgraded from a tabletop/rack mountable analyzer to an analyzer system by integrating optional sample handling components (sample pump, valve blocks, flowmeters and fine dust filter).

The integrated power supply unit allows to use the BINOS® 100 2M analyzer for stationary as well as for mobile applications. In addition the internal power supply unit provides a DC output to supply external components with 24 V / 2 A max.

The 1/2 19" enclosure provides twice the volume compared to a standard BINOS® 100 enclosure to include e.g. 2 thermal conductivity cells.

The option of alternatively using an external power supply unit allocates even more internal volume, which may be used to integrate 2 paramagnetic and/or up to 2 software controlled solenoid valve blocks.

An optional "FOUNDATION™ Fieldbus" interface enables integration into process control systems.

Nevertheless not all imaginable combinations of measuring principles, measuring ranges and options are available - refer to the pricebook for restrictions!

	Variation 1 Variation 2 Variation 3 Variation 4 Variation 5 Variation 6 Variation 7 Variation 8 Variation 9 Variation 10 Variation 11 Variation 12 Variation 13												
IR photometer	1	2							1	1	1		
EO2 sensor			1	2					1			1	
PO2 sensor					1	2				1			1
TC sensor							1	2			1	1	1

Table 1-3: BINOS® 100 2M
Combinations of Measuring Principles

All variations with 2 measuring channels allow serial tubing as well as parallel tubing, also in combination with up to 2 solenoid valve blocks.

The front panel shows 7 segment displays for both channels and pushbuttons for operating the analyzer (see fig. 1-4).

In addition the BINOS® 100 2M front panel shows status LEDs for solenoid valves and sample pump (options) and a pushbutton to switch the pump.

1-3 BINOS® 100 2M

Interfaces are accessible at the analyzer's rear panel via submini-d connectors (male / female), except FOUNDATION™ Fieldbus and 7 digital inputs, which are available on screw terminals.

The multitude of internal configurations (see table 1-3) result in various rear panel designs, shown on page 1-12 et. sqq., while some representative configurations of the internal configurations are given in chapter 8.

Note!

The analyzers may optionally provide a gas fitting for purging the enclosure (e.g. with nitrogen) when measuring

- *low CO₂ concentrations: N₂ minimizes interference by ambient air*
- *flammable gases: N₂ gives additional safety in the event of leakage inside the instrument*

- *aggressive components: purging with air gives additional safety in the event of leakage inside the instrument.*



Purge medium must be dry, clean and free of corrosive and solvent components!

1-3 BINOS® 100 2M

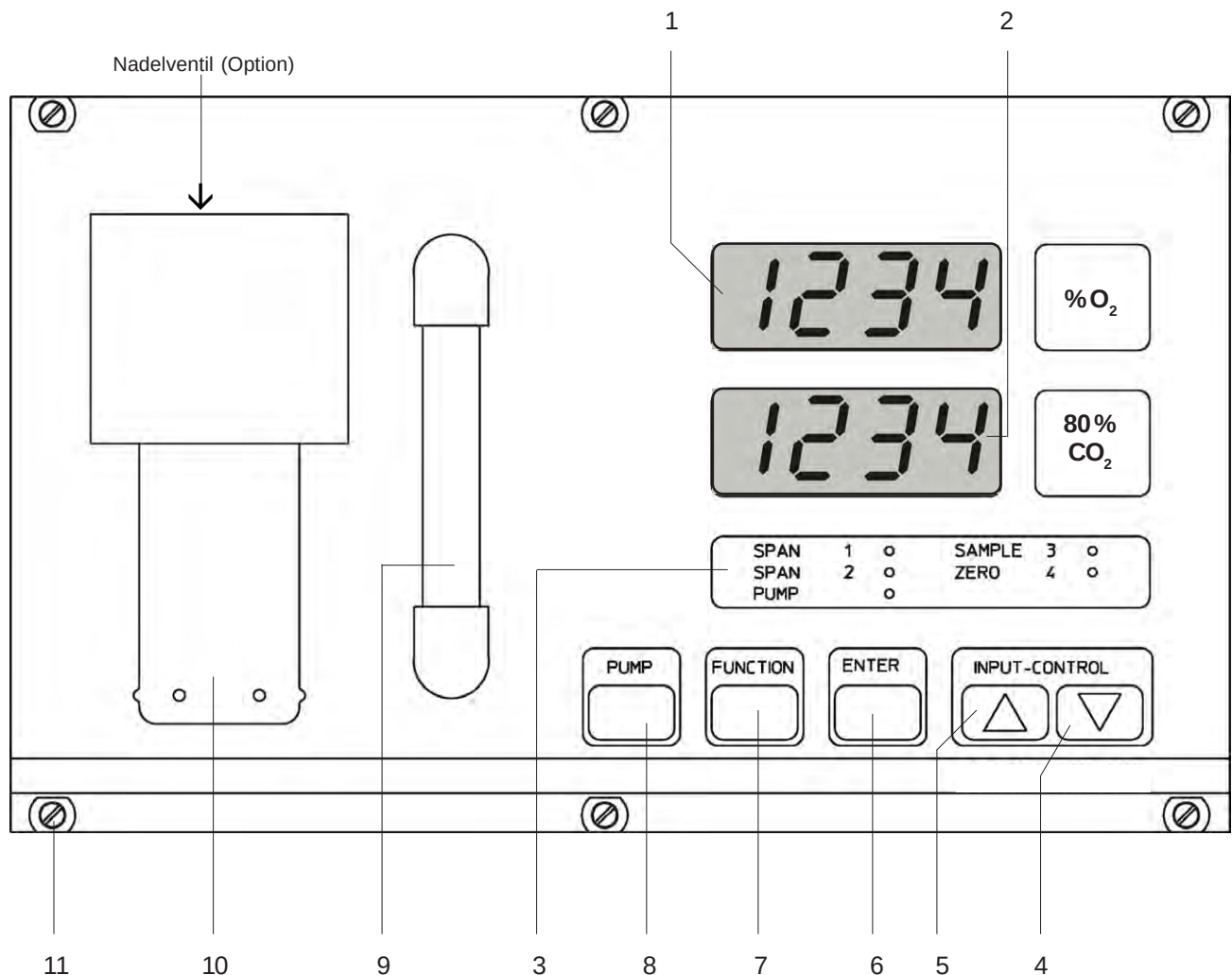


Fig. 1-4: BINOS® 100 2M (standard version)
front side view

- | | | | |
|---|-------------------------------|----|---|
| 1 | 7 segment display (channel 1) | 7 | Pushbutton FUNCTION |
| 2 | 7 segment display (channel 2) | 8 | Pushbutton for optional sample pump |
| 3 | LEDs "valves / sample pump" | 9 | Flowmeter (option) |
| 4 | Pushbutton DOWN | 10 | Fine dust filter with needle valve (option) |
| 5 | Pushbutton UP | 11 | Screw for fixing handle/plug-in module |
| 6 | Pushbutton ENTER | | |

1-3 BINOS® 100 2M

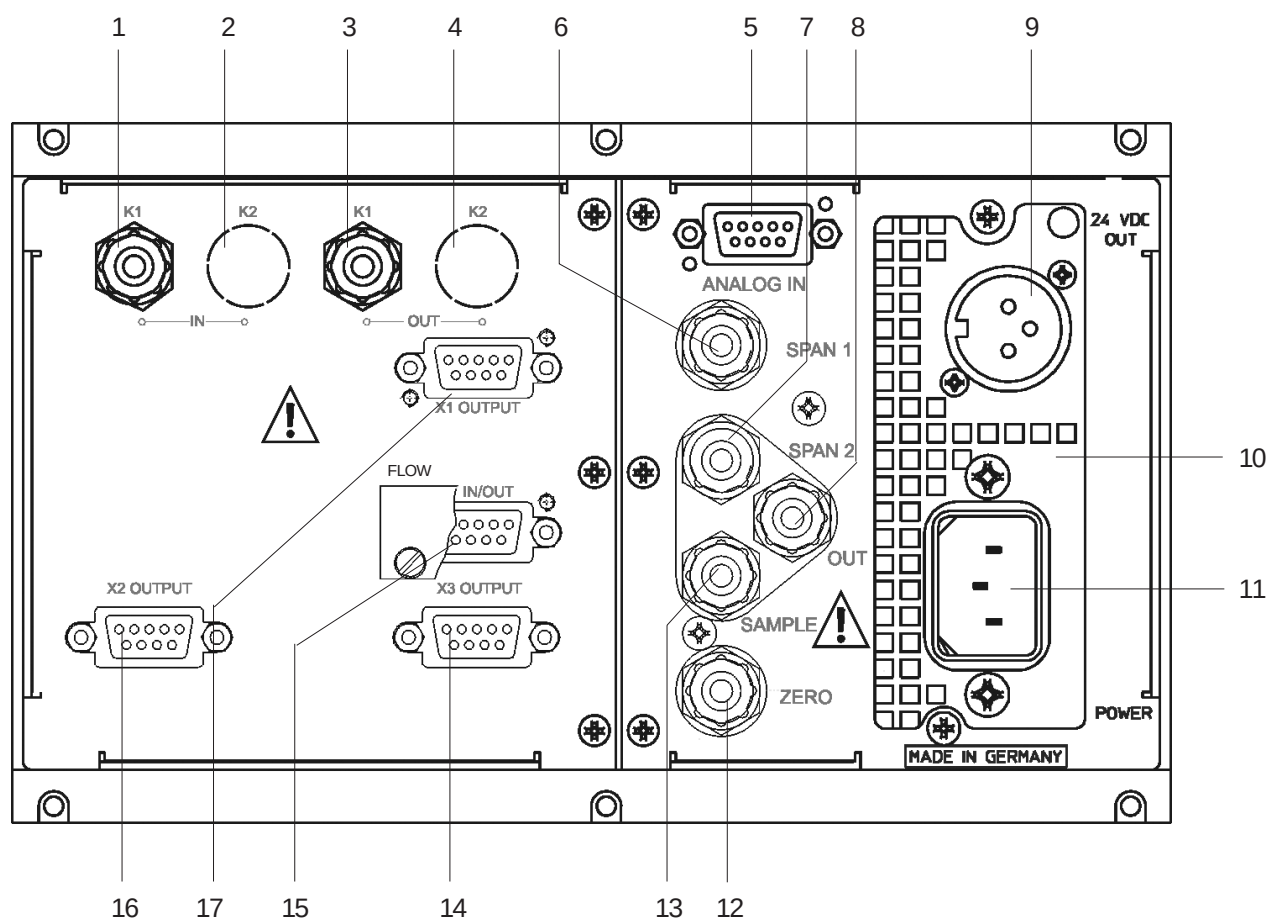
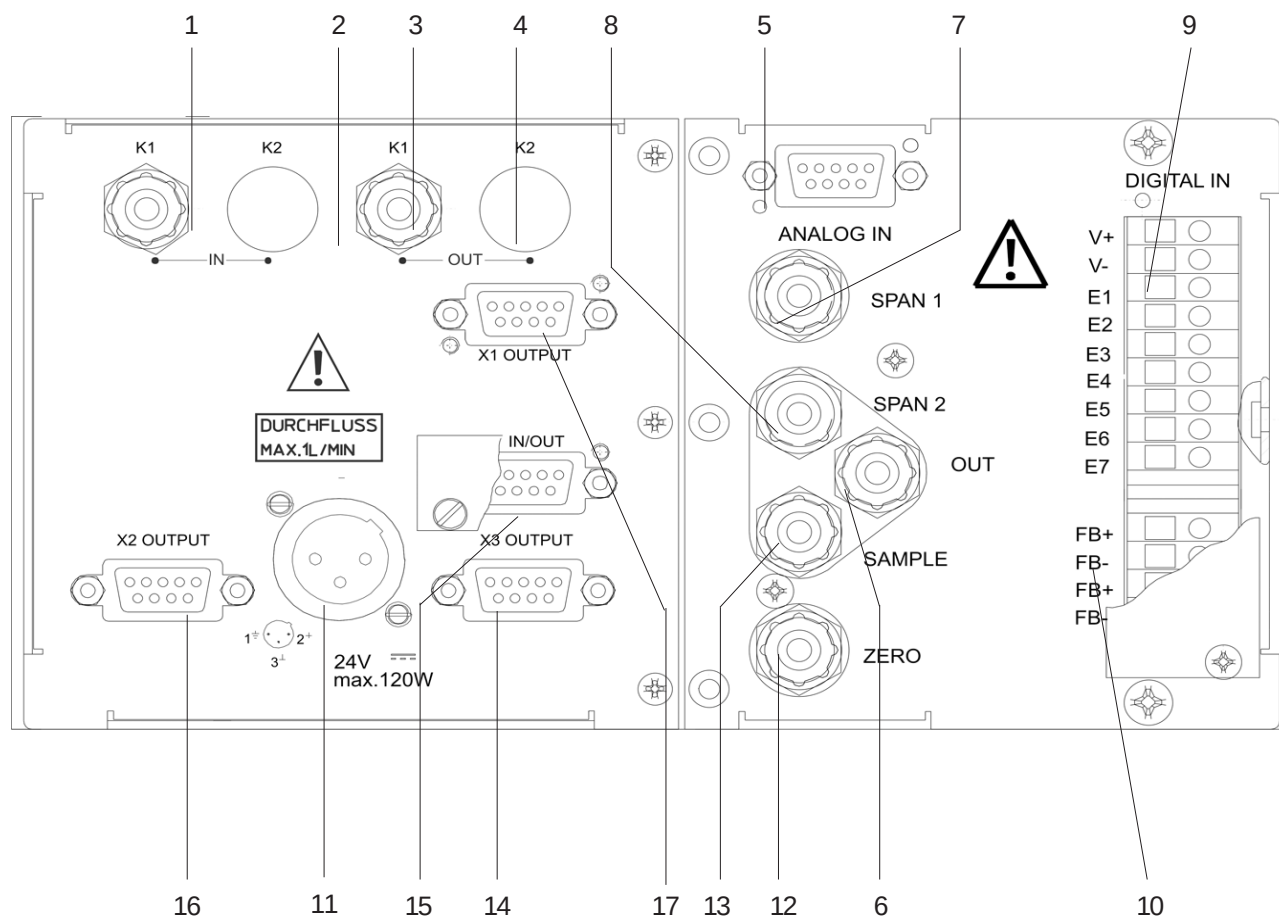


Fig. 1-5: BINOS® 100 2M, variation A (internal power supply)

Rear side view with all options

- | | | | |
|---|---|----|---|
| 1 | Gas inlet fitting channel 1 | 10 | Power supply [UPS 01 (Universal Power Supply)] |
| 2 | Gas inlet fitting channel 2 (option) | 11 | Mains connector |
| 3 | Gas outlet fitting channel 1 | 12 | "Solenoid valve block": zero gas in |
| 4 | Gas outlet fitting channel 2 (option) | 13 | "Solenoid valve block": sample gas in |
| 5 | Plug "analog signal inputs"
(external cross compensation, TC only) | 14 | Plug "digital outputs" |
| 6 | "Solenoid valve block": span gas 1 in | 15 | Socket "serial interface" [RS 232 / 485] (option) |
| 7 | "Solenoid valve block": span gas 2 in | 16 | Socket "analog signal output" |
| 8 | "Solenoid valve block": shared gas out | 17 | Plug "status messages" (relay outputs, option) |
| 9 | 24 V DC output (2 A max., see techn. data) | | |

1-3 BINOS® 100 2M



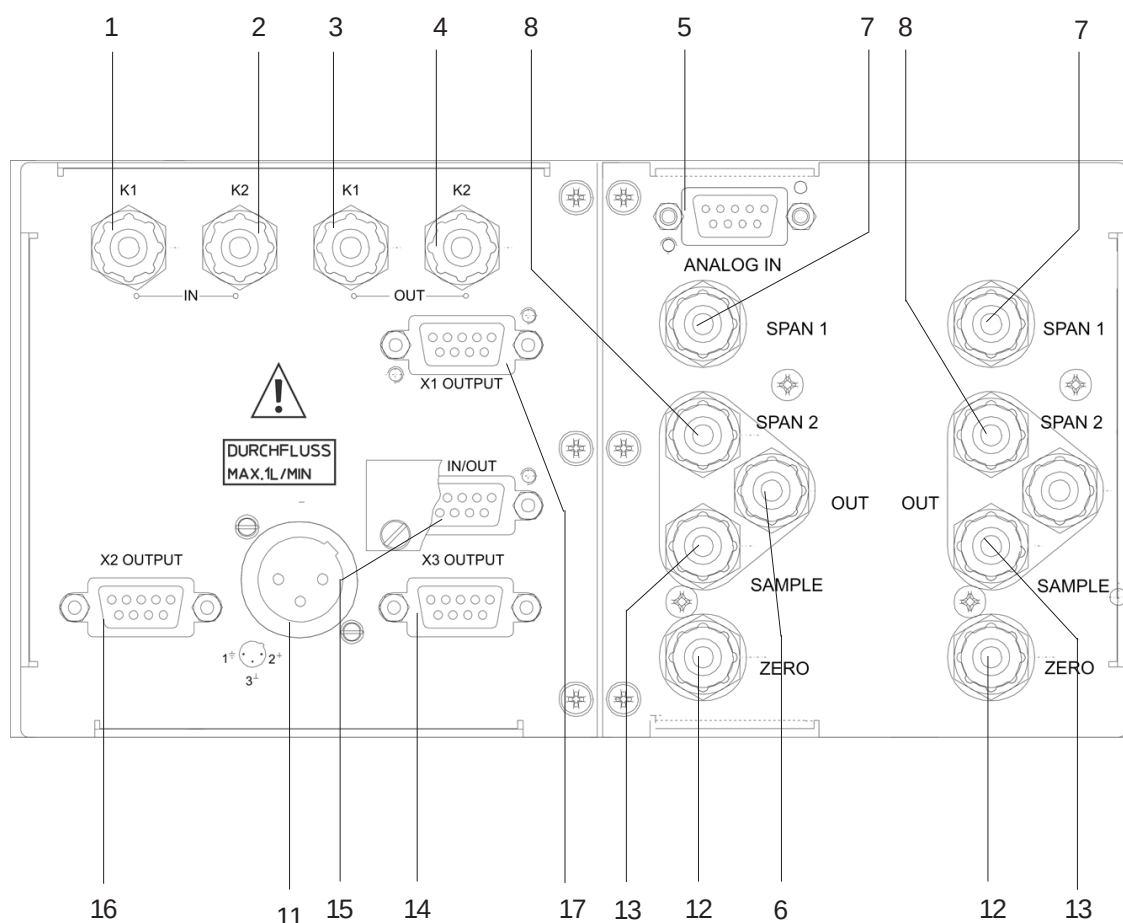
**Fig. 1-6: BINOS® 100 2M, variation B (external power supply),
Rear side view with all options**

- | | |
|---|--|
| 1 Gas inlet fitting channel 1 | 10 Terminal block for FOUNDATION™ Fieldbus (option)*) |
| 2 Gas inlet fitting channel 2 (option) | 11 24 V DC supply |
| 3 Gas outlet fitting channel 1 | 12 "Solenoid valve block": zero gas in |
| 4 Gas outlet fitting channel 2 (option) | 13 "Solenoid valve block": sample gas in |
| 5 Plug "analog signal inputs"
(external cross compensation, TC only) | 14 Plug "digital outputs" |
| 6 "Solenoid valve block": shared gas out | 15 Socket "serial interface" [RS 232 / 485] (option) **) |
| 7 "Solenoid valve block": span gas 1 in | 16 Socket "analog signal output" |
| 8 "Solenoid valve block": span gas 2 in | 17 Plug "status messages" (relay outputs, option) |
| 9 Terminal block for 7 digital inputs (option) | |

*) not in combination with serial interface RS 232/485

**) not in combination with FOUNDATION™ Fieldbus

1-3 BINOS® 100 2M



**Fig. 1-7: BINOS® 100 2M (special variation),
Rear side view with all options**

- | | | | |
|---|---|----|---|
| 1 | Gas inlet fitting channel 1 | 11 | 24 V DC supply |
| 2 | Gas inlet fitting channel 2 (option) | 12 | "Solenoid valve block": zero gas in |
| 3 | Gas outlet fitting channel 1 | 13 | "Solenoid valve block": sample gas in |
| 4 | Gas outlet fitting channel 2 (option) | 14 | Plug "digital outputs" |
| 5 | Plug "analog signal inputs"
(external cross compensation, TC only) | 15 | Socket "serial interface" [RS 232 / 485] (option) |
| 6 | "Solenoid valve block": shared gas out | 16 | Socket "analog signal output" |
| 7 | "Solenoid valve block": span gas 1 in | 17 | Plug "status messages" (relay outputs, option) |
| 8 | "Solenoid valve block": span gas 2 in | | |



*Continue with section 1-6
"Internal gas paths" , page 1-30*

1-4 BINOS® 100 F

1-4 BINOS® 100 F

With BINOS® 100 F Emerson Process Management offers a gas analyzer designed for wall mounting and outdoor installation due to its stainless steel enclosure.

Without the optional fine dust filter the analyzer is IP 65 protected against dust and water.

The BINOS® 100 F offers similar potentials as the BINOS® 100 2M, that is it can be upgraded to an analyzer system by integrating sample handling options like sample pump, solenoid valve blocks, flowmeters and fine dust filter.

The enclosure has enough volume to integrate 2 paramagnetic or thermal conductivity cells (see table 1-4), if required as well in combination with up to 2 software controlled solenoid valve blocks. Variations with 2 measuring channels allow serial or parallel tubing, again also in combination with up to 2 valve blocks, if required by application.

For examples of internal configurations see chapter 8.

	Variation 1 Variation 2 Variation 3 Variation 4 Variation 5 Variation 6 Variation 7 Variation 8 Variation 9 Variation 10 Variation 11 Variation 12 Variation 13												
IR photometer	1	2							1	1	1		
EO2 sensor			1	2					1			1	
PO2 sensor					1	2				1			1
TC sensor							1	2			1	1	1

Table 1-4: BINOS® 100 F
Combinations of measuring principles

An optional "FOUNDATION™ Fieldbus" interface enables integration into process control systems.

The front panel shows 7 segment displays for both channels and pushbuttons for operating the analyzer (see fig. 1-9).

In addition the BINOS® 100 F front panel shows status LEDs for solenoid valves and sample pump (options) and a pushbutton to switch the pump (pushbutton "Pump" not in combination with optional impact tested front panel).

1-4 BINOS® 100 F

The optional impact protected front panel enables installation at harsh locations (or even in hazardous area when combined with a pressurization system for the enclosure).

Such analyzers are contactless operated using a special magnetic tool. The pump (option) is controlled by a software menu entry, the front panel pushbutton is omitted.

All interface signals (partial options) are accessible on internal screw terminal blocks. That applies to mains power supply, too.

Take care of the special installation instructions in chapter 3 "Installation"!



Provided with an appropriate pressurization system the BINOS® 100 F is suitable for installation in hazardous areas. Installation, startup and maintenance are described in detail in a separate instruction manual, shipped together with each such analyzer, and are not subject of this current instruction manual!

WARNING



Connection of mains and interface signals are to be established via internal screw terminal blocks. This requires working inside an opened analyzer, near by potentially live components!



Installation of this analyzer is allowed by qualified personnel only, familiar with the potential risks of working near live components!



The model BINOS® 100 F (field housing) does not provide a mains switch. A mains switch or circuit breaker must be provided in the building installation. This switch has to be installed near by the analyzer, must be easily operator accessible and has to be designated as disconnecter for the analyzer.

Cables for external data processing must be double insulated for mains voltage. If not available, signal cables inside the analyzer must be installed with a distance of at least 5 mm to mains cables. Distance has to be ensured permanently, e.g. by using cable ties!

1-4 BINOS® 100 F

WARNING

Risk of electric shock!



BINOS® 100 F analyzers provide earth connection terminals. To minimize risk of electric shock the enclosure must be connected to earth! Therefore connecting the analyzer is allowed only by using a 3 pole mains cable providing a separate earthing conductor.

Any interruption of the earthing conductor inside or outside the analyzer as well as loosening the earthing connection may cause serious injury! Intended interruption of earthing connections is not permissible!

CAUTION

Heavy instrument!



BINOS® 100 F analyzers ,designed for wall mounting and/or outdoor installation may weigh up to 30 kg, depending on installed options!

Use two persons or a suitable lifting device to move or carry the instrument!

Note!

The analyzers may optionally provide a gas fitting for purging the enclosure (e.g. with nitrogen) when measuring

- *low CO₂ concentrations: N₂ minimizes interference by ambient air*
- *flammable gases: N₂ gives additional safety in the event of leakage inside the instrument*

- *aggressive components: purging with air gives additional safety in the event of leakage inside the instrument.*



Purge medium must be dry, clean and free of corrosive and solvent components!

1-4 BINOS® 100 F

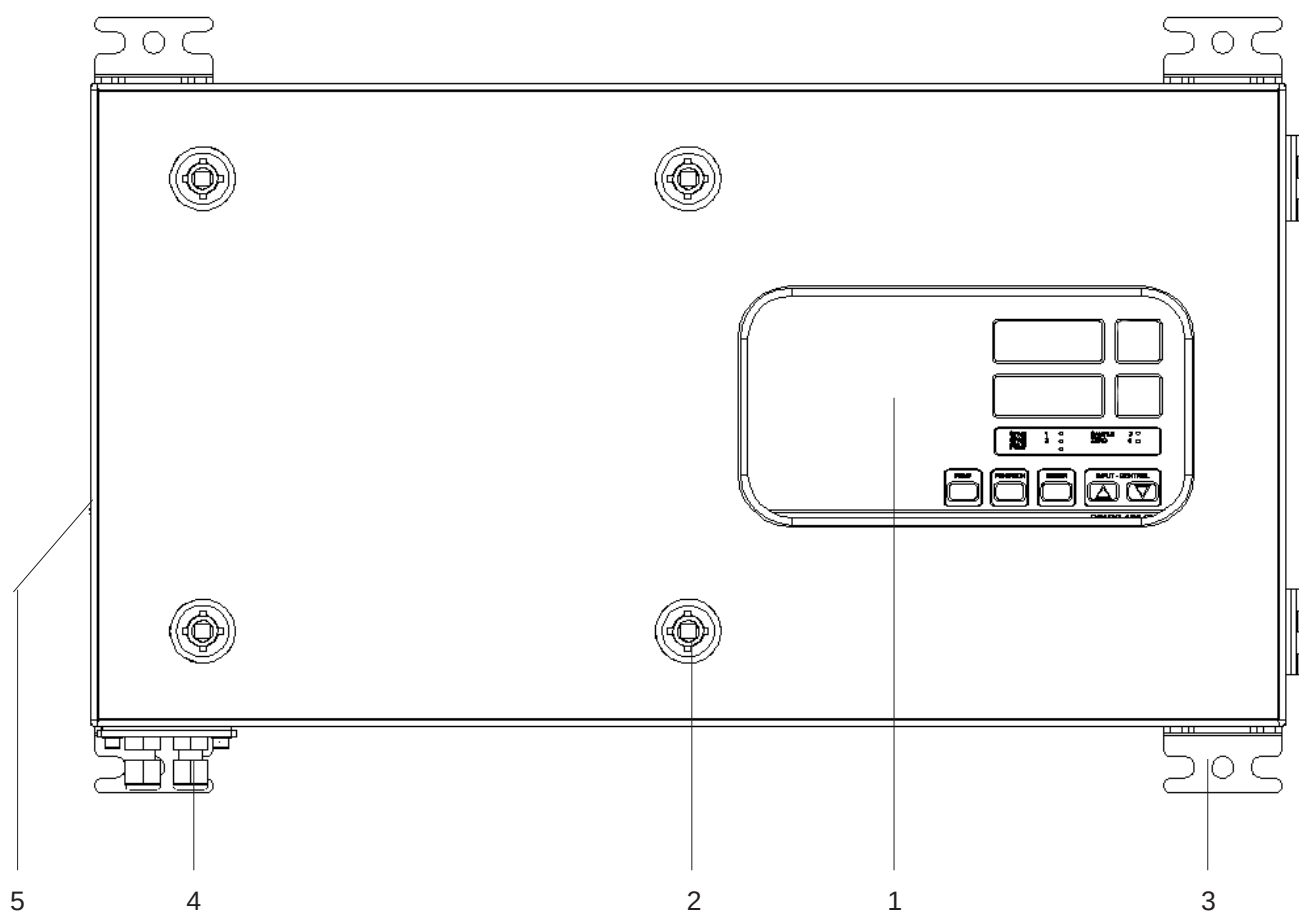


Fig. 1-8: BINOS® 100 F
Standard version, front side view

- 1 Front panel (without options; for details see fig. 1-9)
- 2 Sash fasteners (4 pcs)
- 3 Wall mount bracket (4 pcs)
- 4 Gas fittings
- 5 Cable glands for signals and mains

1-4 BINOS® 100 F

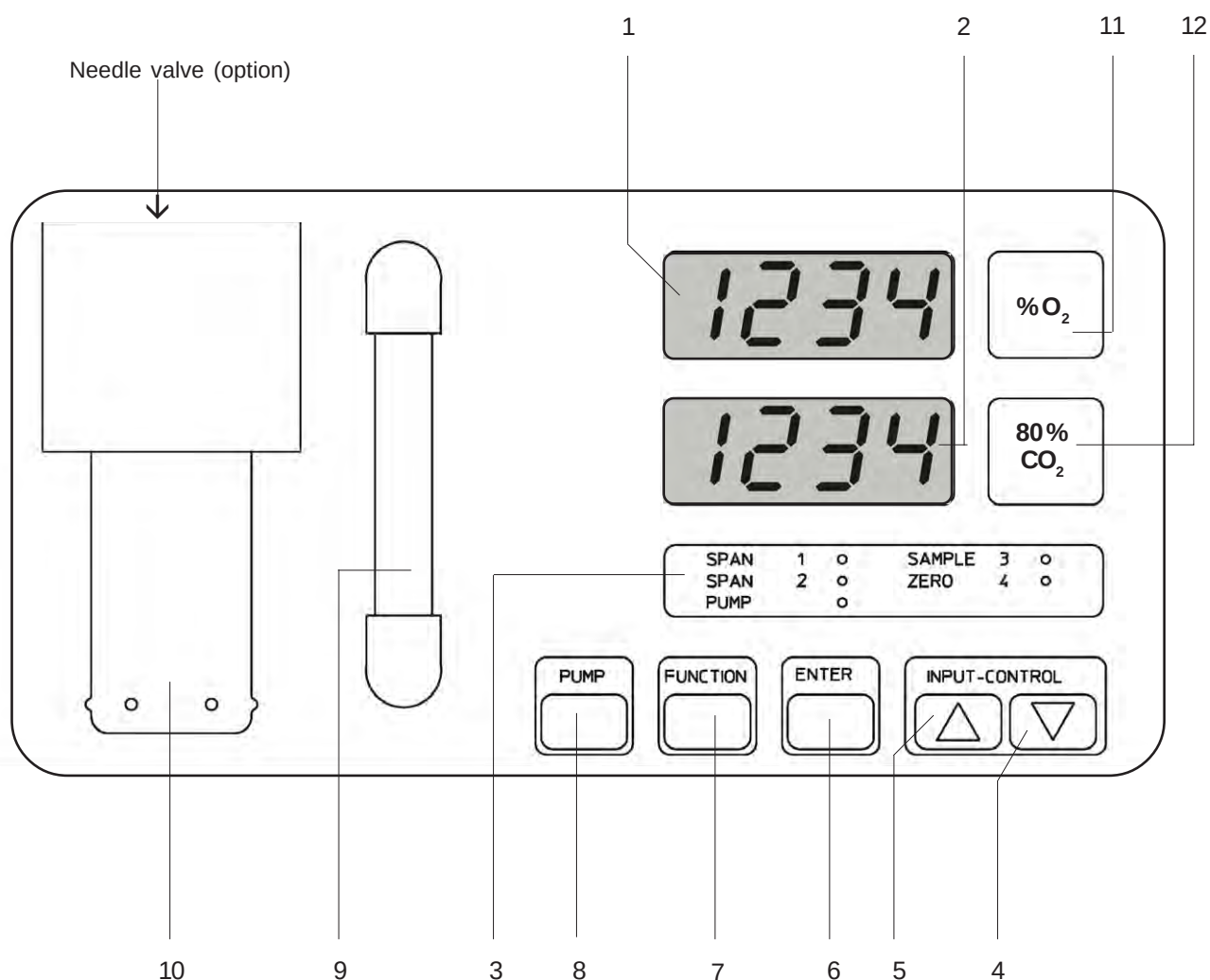


Fig. 1-9: BINOS® 100 F
Standard front panel incl. sample handling options

- | | | | |
|---|-------------------------------|----|---|
| 1 | 7 segment display (channel 1) | 7 | Pushbutton FUNCTION |
| 2 | 7 segment display (channel 2) | 8 | Pushbutton for optional sample pump |
| 3 | LEDs "valves / sample pump" | 9 | Flowmeter (option) |
| 4 | Pushbutton DOWN | 10 | Fine dust filter with needle valve (option) |
| 5 | Pushbutton UP | 11 | Sample gas designator (channel 1) |
| 6 | Pushbutton ENTER | 12 | Sample gas designator (channel 2) |

1-4 BINOS® 100 F

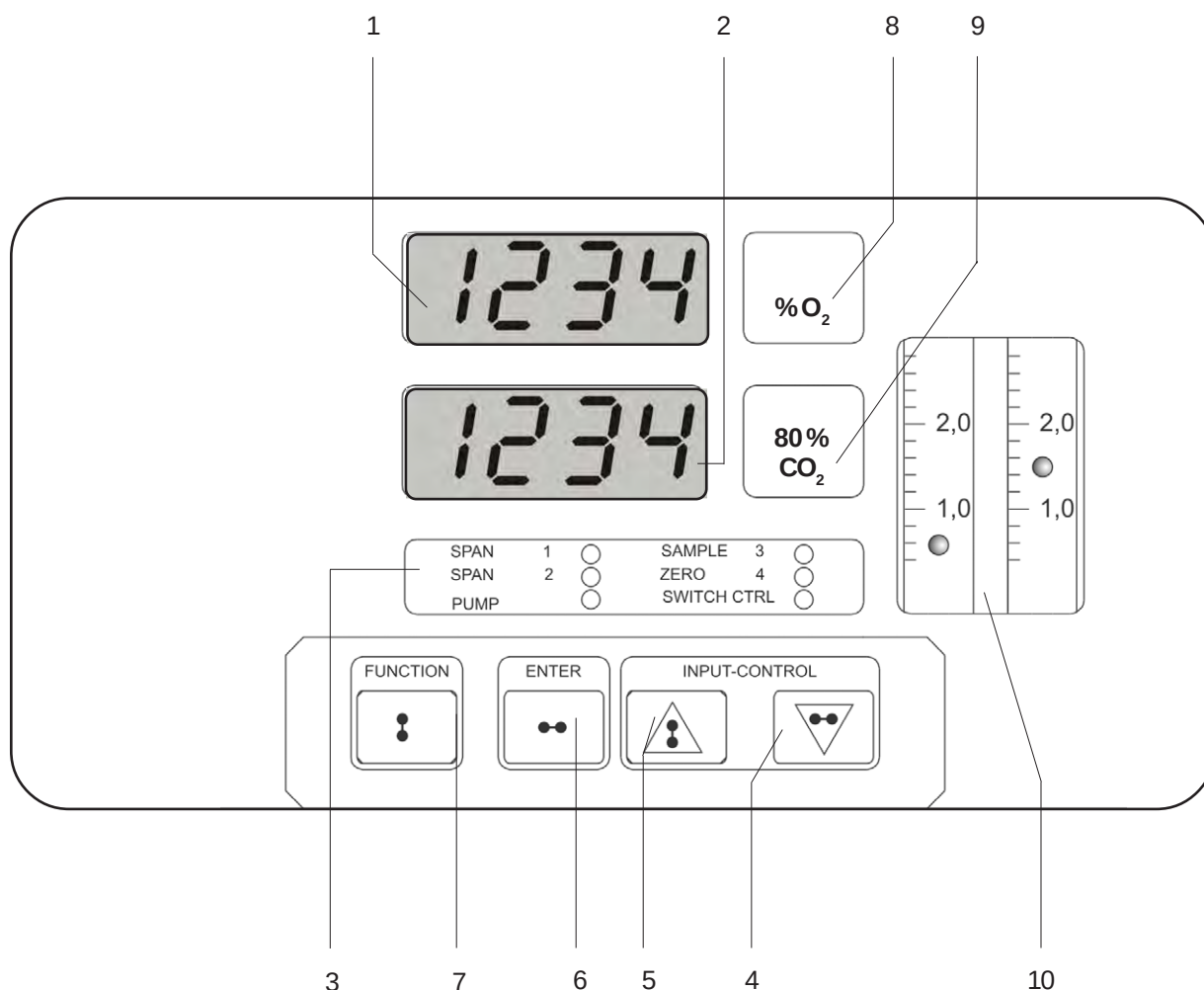


Fig. 1-10: BINOS® 100 F
Magnetically operated front panel (option)

- | | | | |
|---|-------------------------------|----|-----------------------------------|
| 1 | 7 segment display (channel 1) | 6 | Sensor field ENTER |
| 2 | 7 segment display (channel 2) | 7 | Sensor field FUNCTION |
| 3 | LEDs "valves / sample pump" | 8 | Sample gas designator (channel 1) |
| 4 | Sensor field DOWN | 9 | Sample gas designator (channel 2) |
| 5 | Sensor field UP | 10 | Flowmeter (option) |

1-4 BINOS® 100 F

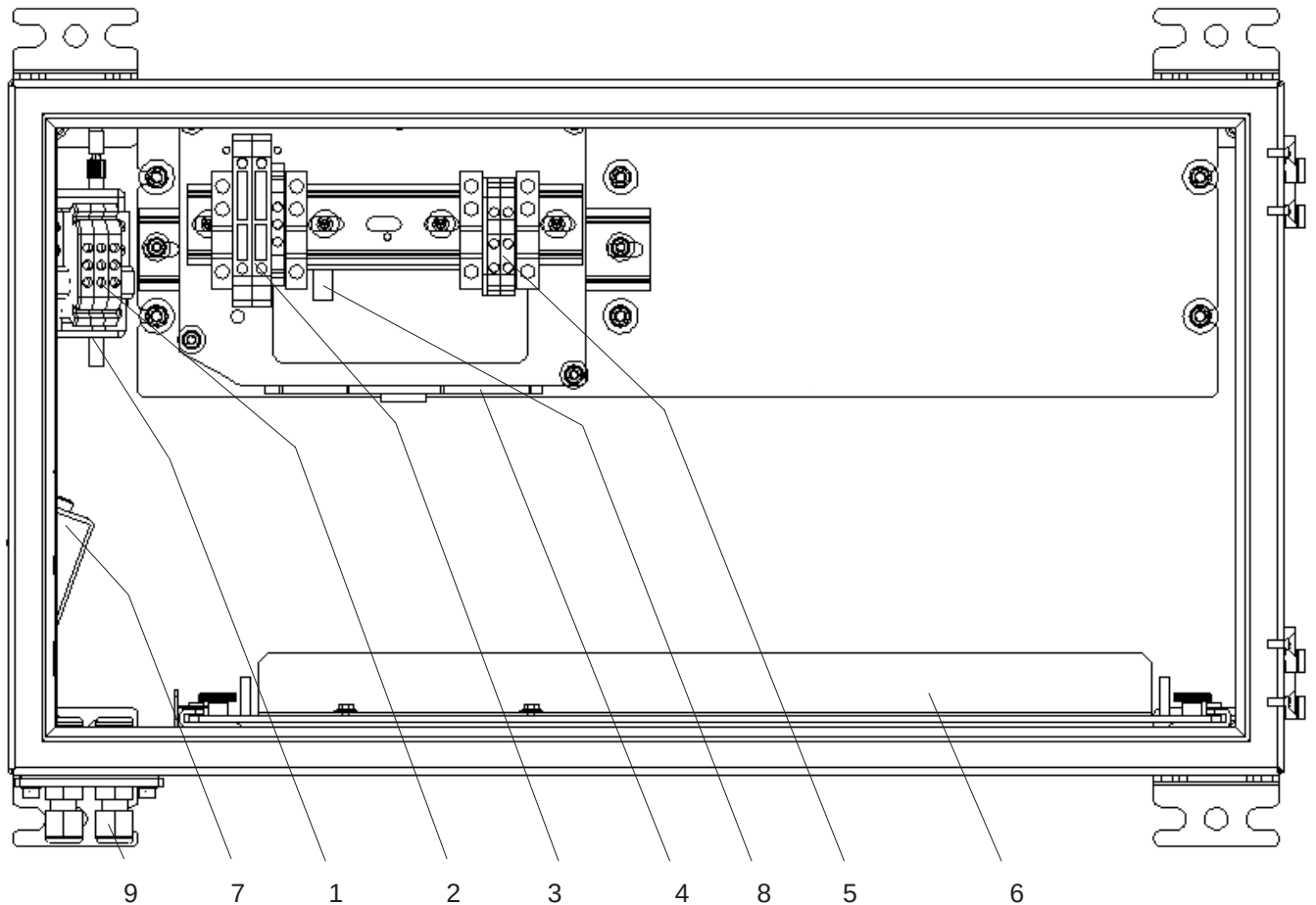


Fig. 1-11: BINOS® 100 F
Interior view (without components)

- | | |
|---------------------------------|---------------------------------|
| 1 Mains EMC filter | 5 Terminals for DC 24 V |
| 2 Mains terminals | 6 Carriage slide for components |
| 3 Mains fuses | 7 Cable glands |
| 4 Power supply (behind barrier) | 8 Mains voltage selector switch |
| | 9 Gas fittings |



Continue with section 1-6
"Internal gas paths", page 1-30

1-5 CAT 100

1-5 CAT 100

The CAT 100 is an analyzer with flameproof enclosure, designed especially for operation in hazardous areas. It is approved to meet the requirements of the European Directive for equipment intended to be used in hazardous areas (commonly known as "ATEX 95") and of the North American Standards (certified by the Canadian Standards Association, CSA International).

The enclosure meets IP 65 (optional: IP 66, tropicalisation), has robust design and is prepared for wall mounting. Therefore the analyzer may be used for general purpose applications requiring outdoor installation, too.

This current instruction manual covers using the CAT 100 analyzer for general purpose applications only!



Installation, startup and maintenance for operation in hazardous areas are described in detail in a separate instruction manual, shipped together with each such analyzer and are not subject of the current instruction manual!



Installed inside the flameproof enclosure an analyzer module holds all measuring components. As the module corresponds to a BINOS® 100 2M with external power supply unit, a CAT 100 analyzer offers almost the same options, e.g. integration of sample pump, solenoid valve blocks and flowmeter.

The analyzer module is big enough to include (depending on application) 2 paramagnetic or thermal conductivity cells (see table 1-5), if required in combination with a software controlled solenoid valve block.

Dual channel variations allow serial and parallel tubing, again in combination with solenoid valve block, if required.

The front panel shows 7 segment displays for both channels and status LEDs for solenoid valves and sample pump (options). Because of the intention to be used in hazardous areas the front panel is located behind a safety glass window. The analyzer is touchless operated by using a magnetically tool, whereat the standard pushbuttons are replaced by sensor fields (see fig. 1-14). The optional sample pump is controlled by a software menu entry.

An optional "FOUNDATION™ Fieldbus" interface enables integration into process control systems.

All interface signals (partial options) are accessible on screw terminal blocks, located in a separate junction box. That applies to mains power supply, too.

Take care of the special installation instructions in chapter 3 "Installation"!

Located below the junction box is a second volume consisting of dome housing, extender housing and base, containing analyzer module, power supply unit, fuses and gas pipes (see fig. 1-12). All three parts of the enclosure are threaded together. Unthreading the parts gives access to the internal components.

For detailed interior views see chapter 8.

1-5 CAT 100

	Variation 1 Variation 2 Variation 3 Variation 4 Variation 5 Variation 6 Variation 7 Variation 8 Variation 9 Variation 10 Variation 11 Variation 12 Variation 13												
IR photometer	1	2							1	1	1		
EO2 sensor			1	2					1			1	
PO2 sensor					1	2				1			1
TC sensor							1	2			1	1	1

Table 1-5: CAT 100
Combinations of measuring principles

CAUTION

Heavy instrument!



CAT 100 analyzers, designed for wall mounting and/or outdoor installation may weigh up to 70 kg, depending on installed options!

Use two persons or a suitable lifting device to move or carry the instrument!

Note!

The analyzers may optionally provide a gas fitting for purging the enclosure (e.g. with nitrogen) when measuring

- low CO₂ concentrations: N₂ minimizes interference by ambient air
- flammable gases: N₂ gives additional safety in the event of leakage inside the instrument

- aggressive components: purging with air gives additional safety in the event of leakage inside the instrument.



Purge medium must be dry, clean and free of corrosive and solvent components!

1-5 CAT 100

WARNING



Connection of mains and interface signals are to be established via internal screw terminal blocks. This requires working inside an opened analyzer, near by potentially live components!



Installation of this analyzer is allowed by qualified personnel only, familiar with the potential risks of working near live components!



The model CAT 100 does not provide a mains switch. A mains switch or circuit breaker must be provided in the building installation. This switch has to be installed near by the analyzer, must be easily operator accessible and has to be designated as disconnecter for the analyzer.

The Junction Box must be protected by fuse supply which has a breaking capacity adjusted to the short circuit of the equipment. The 10 A fuse has to be installed in the building installation.

WARNING

Risk of electric shock!



CAT 100F analyzers provide earth connection terminals. To minimize risk of electric shock the enclosure must be connected to earth! Therefore connecting the analyzer is allowed only by using a 3 pole mains cable providing a separate earthing conductor.

Any interruption of the earthing conductor inside or outside the analyzer as well as loosening the earthing connection may cause serious injury! Intended interruption of earthing connections is not permissible!

1-5 CAT 100

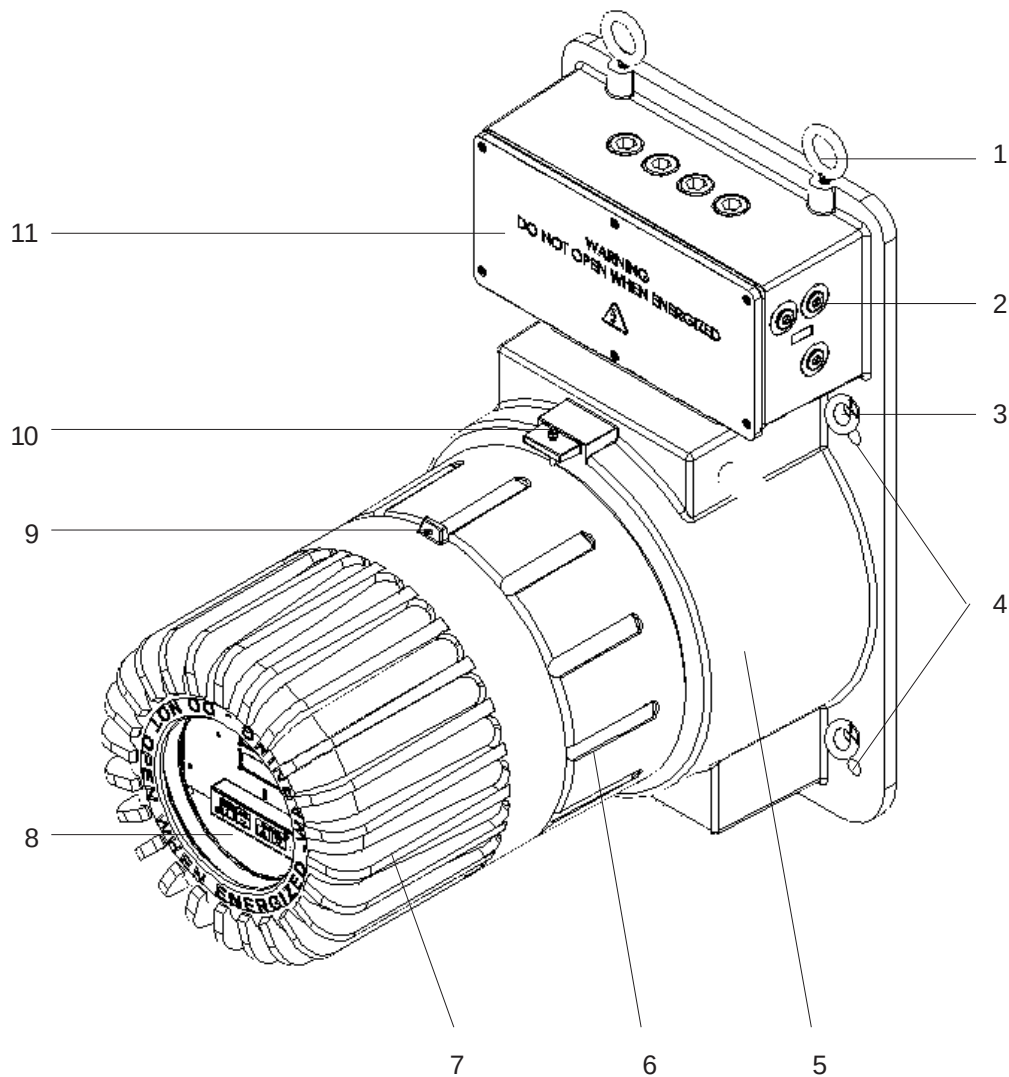


Fig. 1-12: CAT 100
Exterior view

- | | | | |
|---|--|----|---------------------------------|
| 1 | Eye bolts (2 pcs) | 7 | Dome housing |
| 2 | Cable glands for signal cables (3 pcs max) | 8 | Front panel behind safety glass |
| 3 | Eye bolts (2 pcs) | 9 | Interlock screw |
| 4 | Mounting holes (4 pcs.) | 10 | Interlock screw |
| 5 | Base | 11 | Junction box |
| 6 | Extender housing | | |

1-5 CAT 100

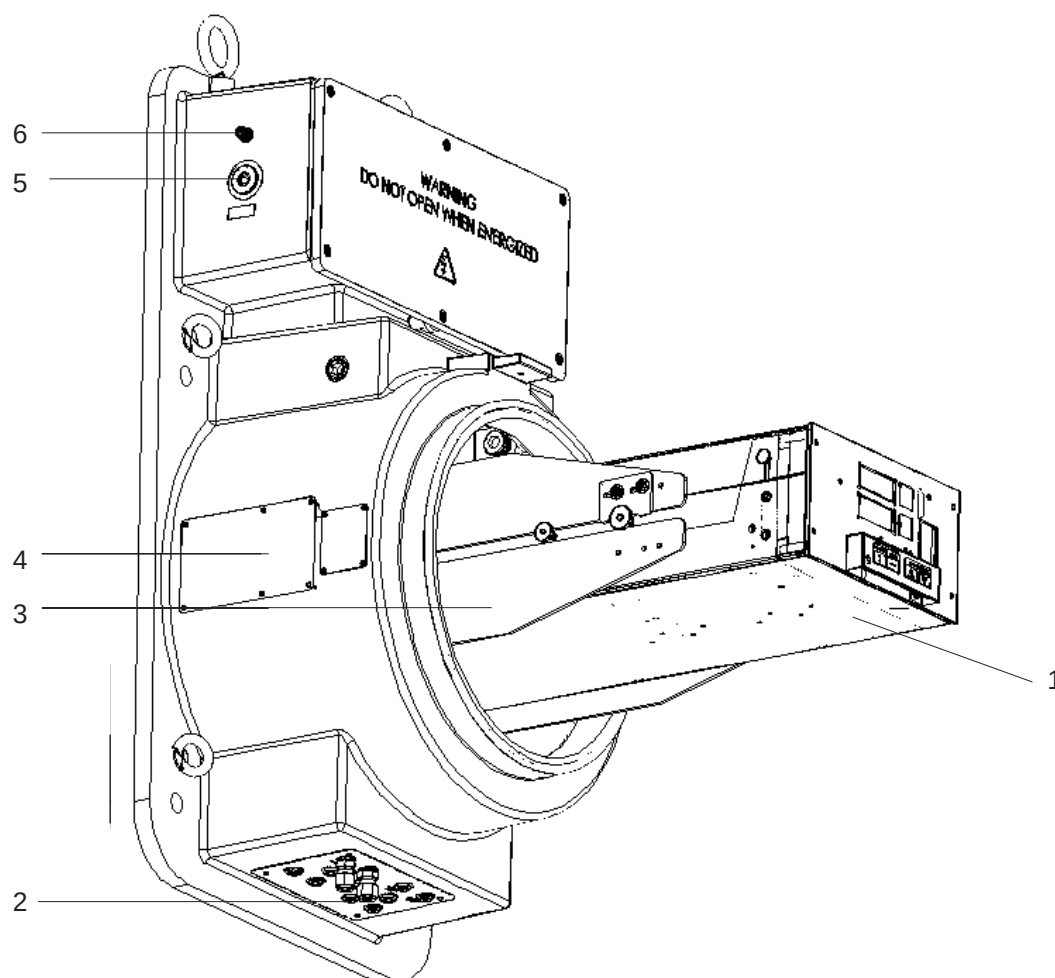


Fig. 1-13: CAT 100
(Dome and extender housing removed)

- | | | | |
|---|--|---|-----------------------------|
| 1 | Analyzer module | 5 | Cable gland for mains cable |
| 2 | Gas fittings with internal flame arrestors | 6 | External earth connector |
| 3 | Bracket for analyzer module | | |
| 4 | Nameplate label | | |

1-5 CAT 100

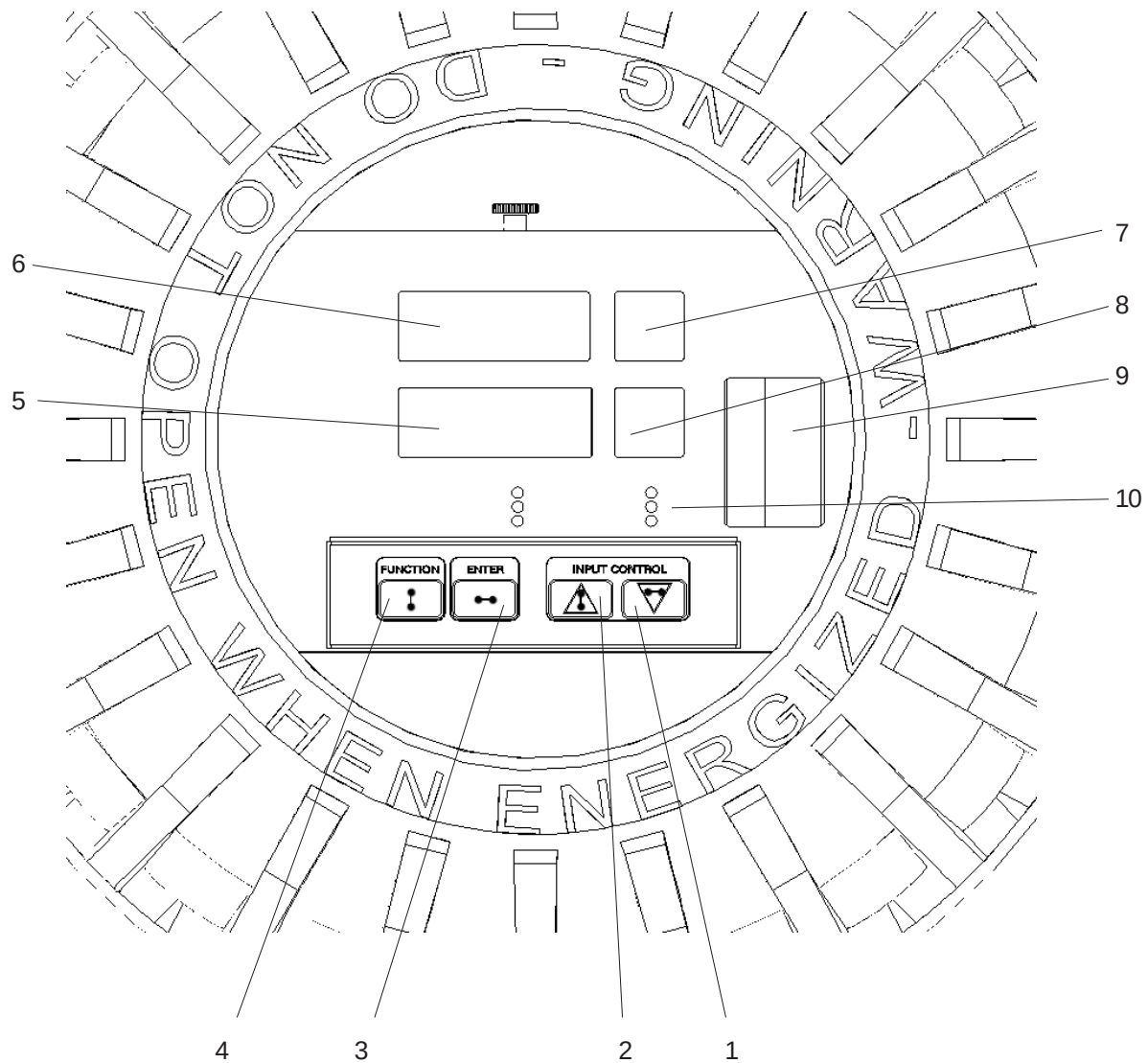


Fig. 1-14: CAT 100
Front panel

- | | | | |
|---|-------------------------------|----|-----------------------------------|
| 1 | Sensor field DOWN | 6 | 7 segment display (channel 1) |
| 2 | Sensor field UP | 7 | Sample gas designator (channel 1) |
| 3 | Sensor field ENTER | 8 | Sample gas designator (channel 2) |
| 4 | Sensor field FUNCTION | 9 | Flowmeter (option) |
| 5 | 7 segment display (channel 2) | 10 | LEDs "valves / sample pump" |

1-5 CAT 100

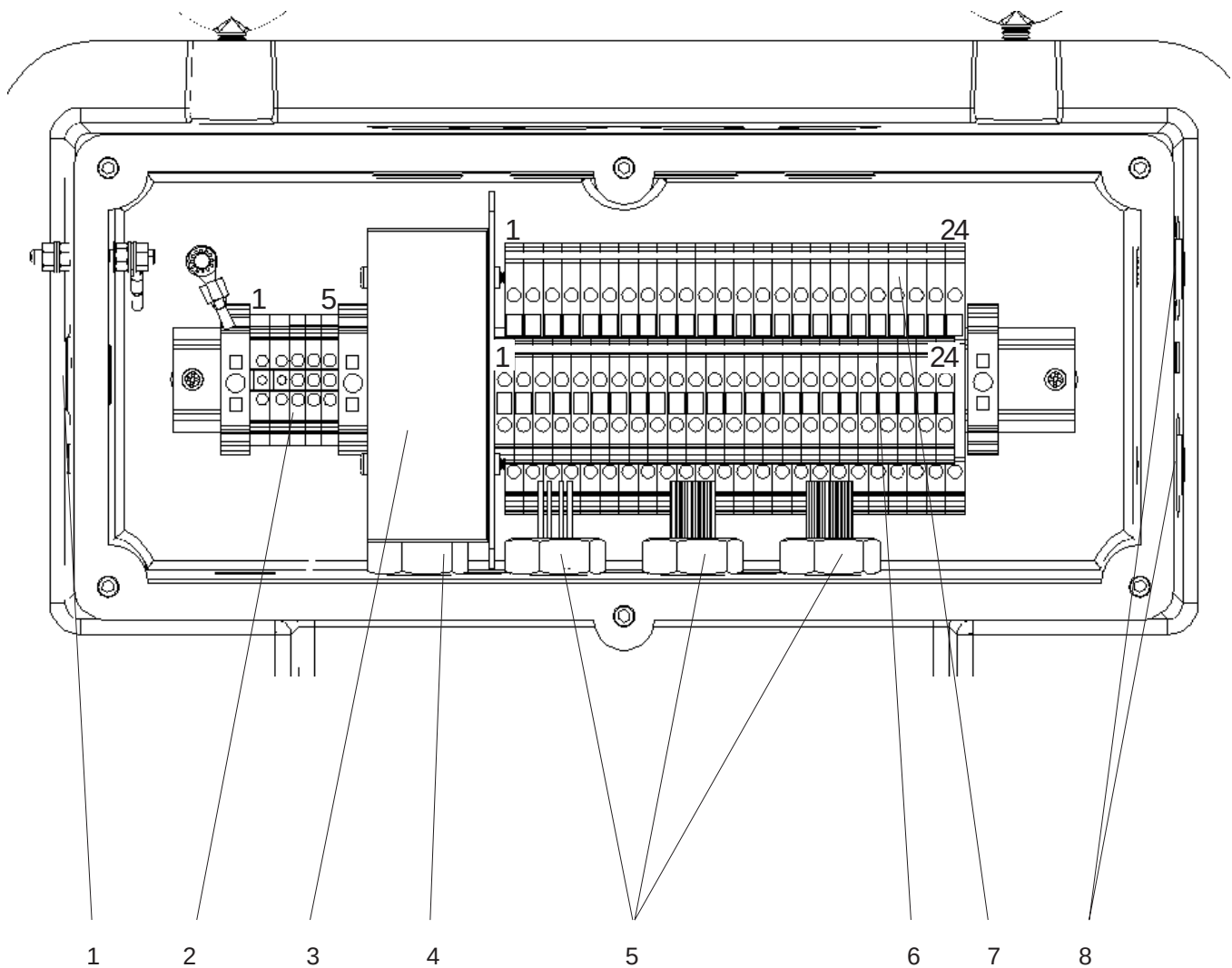
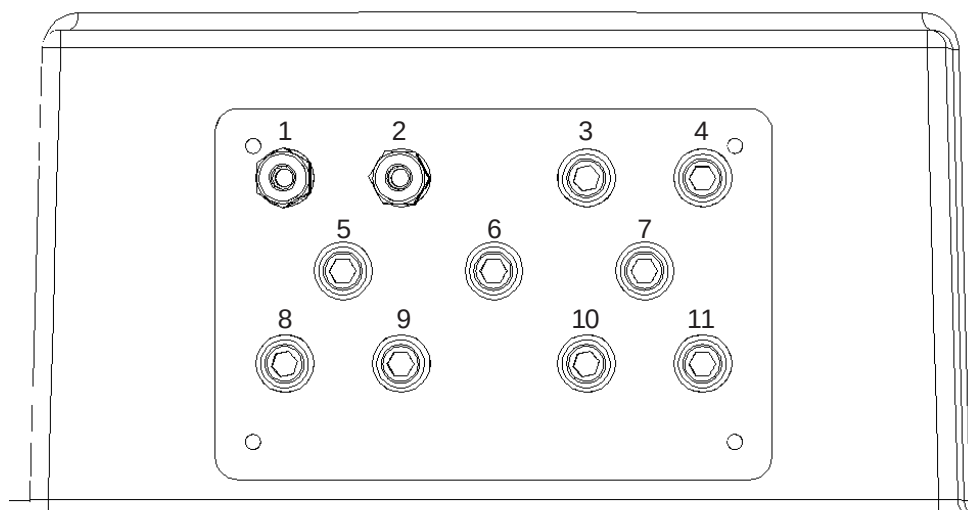


Fig. 1-15: CAT 100
Junction box, interior view

- 1 Cable gland for mains cable
- 2 Mains terminals
- 3 Mains EMC filter
- 4 Feed thru for mains

- 5 Feed thru for mains (3)
- 6 Upper signal terminals (24)
- 7 Lower signal terminals (24)
- 8 Cable gland for signal cables (3)

1-5 CAT 100



Analyzer Configuration	Gas Fitting Assignment			
	1 - Input	2 - Output		
Serial tubing (single,dual, multi channel)				
Parallel tubing	1 - Input (ch 1)	2 - Output (ch 1)	3 - Input (ch 2)	4 - Output (ch 2)
Serial tubing, 1 auto calibration valve	5 - Output (sample, zero, span gas)*	6 - Span gas 1 in	7 - Span gas 2 in	8 - Zero gas in
	9 - Sample gas in			
Parallel tubing 2 auto calibration valves	1 - Input (ch 1)	2 - Output (ch 1)	3 - Input (ch 2)	4 - Output (ch 2)
	6 - Span gas 1 in	7 - Span gas 2 in	8 - Zero gas 1 in	10 - Zero gas 2 in
Optional enclosure purging	11 - Purge gas in			

* Standard: Output = gas out, valve block out is internally connected to analyzer module gas inlet.
Option: valve block output available for external sample handling system (special configuration)

Fig. 1-16: CAT 100
Gas fittings and assignment

1-6 Internal Gas Paths

1-6 Internal Gas Paths

Various materials are available to provide a best possible analyzer adaption to the application. Materials are selected taking into account e.g. diffusion rate, corrosiveness, temperature and pressure of the applied gas.

1-6-1 Gas Paths Materials

Physical and chemical characteristics of applied gases and working conditions (temperature and pressure) affect the available materials.

1-6-2 Safety Filter

All analyzers provide an internal PTFE safety filter. This filter(s) is (are) not a substitute for a dust filter to be installed in the sample handling system!

1-6-3 Fittings

By default all analyzers are equipped with PVDF fittings (ø 6/4 mm), except CAT 100: Swagelok® 1/4" with integrated flame arrestors).

Swagelok® or stainless steel fittings(ø 6/4 mm or 1/4") are available alternatively.

Other fittings are available on request.

1-6-4 Piping

Analyzers are piped with Viton oder PTFE (ø 6/4 mm).

Other materials (e.g. stainless steel) are used optionally, depending on application.

1-6-5 Gas Path Variations

1-6-5 Gas Path Variations

Depending on the application and the selected options several gas paths configurations are

available, as shown in the following figures:

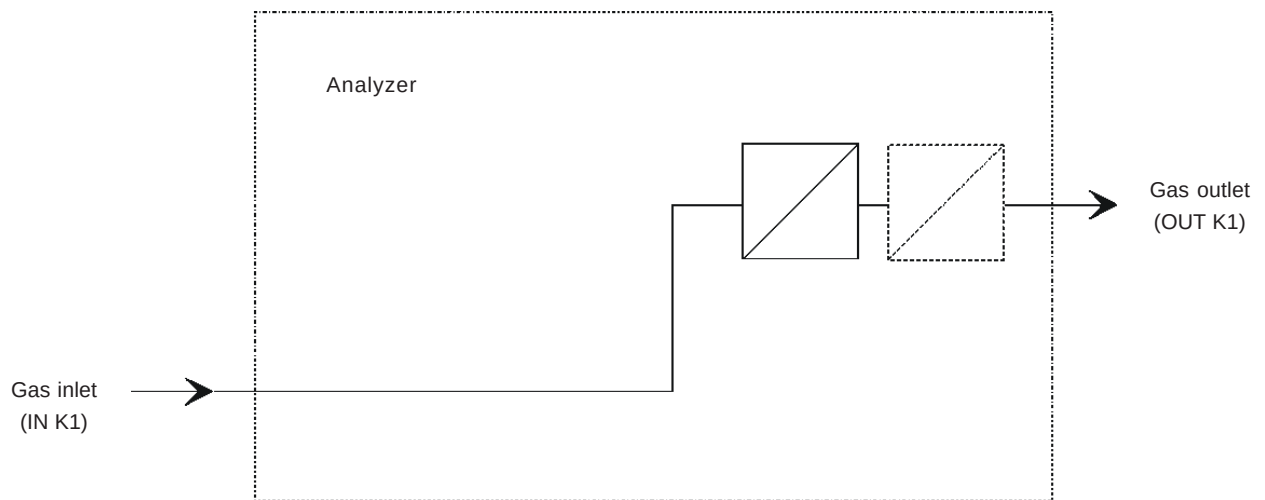


Fig. 1-17: Single channel or serial tubing

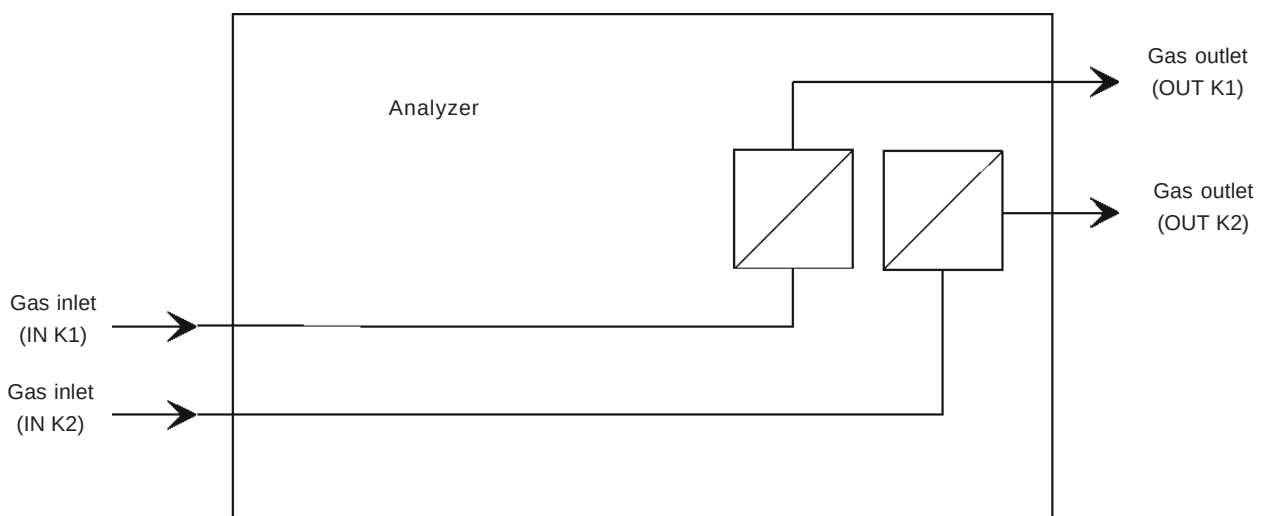
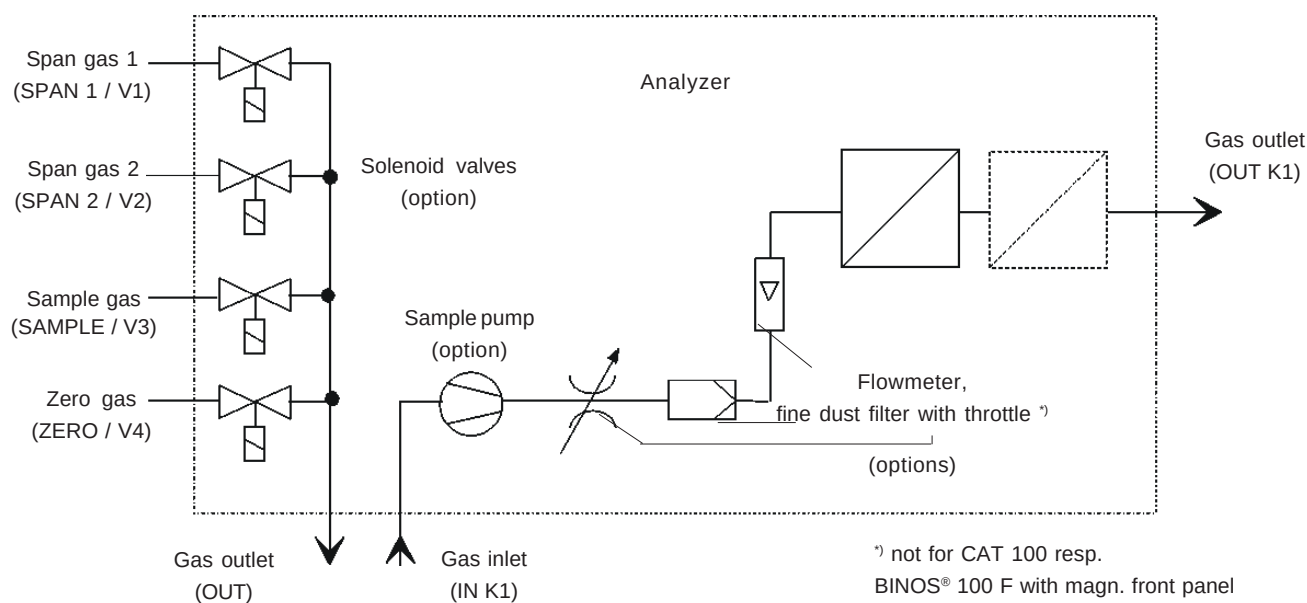


Fig. 1-18: Dual channel parallel tubing

1-6-5 Gas Path Variations



**Fig. 1-19: Single channel resp. serial tubing, with options
(not BINOS® 100 (M), OXYNOS® 100, HYDROS® 100)**

1-6-5 Gas Path Variations

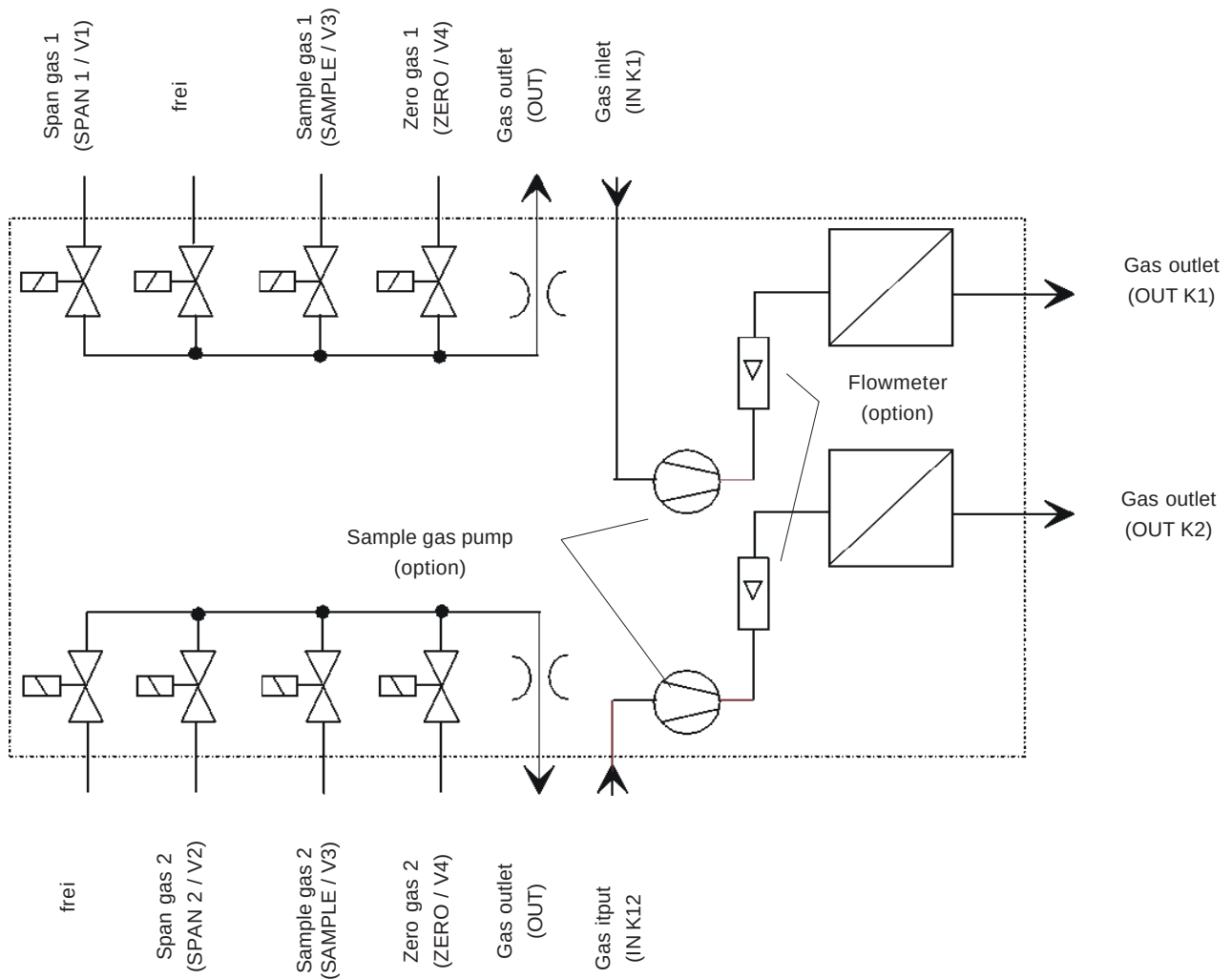


Fig. 1-20: Parallel tubing, with options
(not BINOS® 100 (M), OXYNOS® 100, HYDROS® 100)

1-7 Technical Data

1-7 Technical Data

This section lists all the analyzers technical data, separated into common and model specific data, therefore the user has to select

the appropriate section depending on his analyzer.

1-7-1 Common Technical Data

Site of installation

Permissible ambient temperature (operation):

- OXYNOS/HYDROS 100, BINOS 100 (M/2M/F)

- CAT 100

Permissible storage temperature

Humidity (non condensing)

Pollution degree

Installation category

Altitude

Sourrounding atmosphere

(Analyzers must not be operated in corrosive atmosphere)

+5 °C to +40 °C (+45 °C on request)

0 to +50 °C; -30 °C (GP/ CSA-C/US only)

-20 to +70 °C

< 90 % r. h. at + 20 °C

< 70 % r. h. at + 40 °C

2

II

0 - 2000 m above NN

Analyzers must not be operated in hazardous or flammable atmosphere without additional safety measures.

Compliances

Electrical safety

Europe

CAN / USA

IEC

China

CE, based on EN 61010-1

CAN/CSA-C22.2 No. 1010.1-92

UL 61010A-1

IEC 61010-1

PAC

Electromagnetic compatibility

Europe

Australia

other

CE, based on EN 61326

C-Tick

NAMUR

Purge Option

Purge medium (e.g. for minimizing CO₂ interference, safety in events of internal leakage of flammable or/and aggressive gas etc.) **must be dry, clean and free of corrosive and solvent components and on ambient temperature (at least 20 ... 35 °C) !**

For pressure and flow specification consult factory or your local EMERSON Process Management sales office!

1-7-1 Common Technical Data

Interfaces, signal inputs / outputs

2 analog outputs per channel *)
[(galv. separated:, common ground),
start and end concentration user configurable]
*) *Not for CAT 100 (see section 1-7-2-4) !*

3 analog inputs (option for TC),
for cross compensation of up to 3 interfering
components (via external signals)

8 digital outputs, parallel (galv. separated)

7 digital inputs, parallel
(option)

Serial interface (option, galv. separated)
(not in combination with FOUNDATION™ Fieldbus)

FOUNDATION™ Fieldbus (option)

3 relay outputs (option "status signals")

0 - 10 V and 0 - 20 mA ($R_B \leq 500 \Omega$)
or

2 - 10 V and 4 - 20 mA ($R_B \leq 500 \Omega$)
configurable by keypad

Option: 0 (0.2) - 1 and 0 (4) - 20 mA ($R_B \leq 500 \Omega$)

0 - 10 V, 0 - 20 mA,
2 - 10 V, 4 - 20 mA,
0 - 1 V or 0.2 - 1 V

2 thresholds per channel,
sample gas valve,
zero gas valve,
span gas valve 1,
span gas valve 2 (2 channel analyzer only)
"Open Collector", max. 30 V_{DC} / 30 mA

zero calibration ch1 & ch2,
span calibration ch1,
span calibration ch2,
open zero gas valve,
open span gas 1 valve,
open span gas 2 valve
(2 channel analyzer only)
open sample gas valve /
switch off sample pump

RS 232 or
RS 485 (2- or 4-wire)

(not in combination with serial interface)

"Measuring / calibration"
"Analyzer failure"
"Sample gas pump on / off"
floating contacts, max. 30 V; 1 A; 30 W

1-7-2 Model Specific Specification

1-7-2 Model Specific Data

1-7-2-1 BINOS® 100, BINOS® 100 M, OXYNOS® 100.

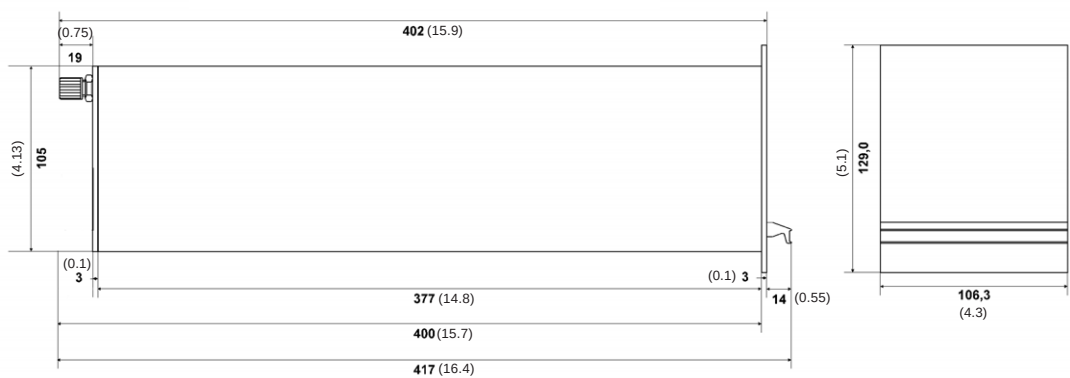


Fig. 1-21: BINOS® 100 (M) / OXYNOS® 100 / HYDROS® 100
Dimensions [approx.; mm (inch)]

Housing

Weight: (depending on analyzer configuration)	approx. 4 - 7 kg
Protection class to EN 60529:	IP 20 for indoor installation Analyzer must not be supposed to dripping or sprayed water.
Gas fittings:	quantity: max. 4 specification: 6/4 mm PVDF optional 6/4 mm or 1/4", stainless steel, other on request

Power Supply

Rated input voltage	24 V _{DC} (+- 5 %)
Rated input power	BINOS® 100 (M) max. 20 W OXYNOS® 100 max. 40 W HYDROS® 100 max. 35 W
Input	3-pole XLR plug (Fig. 1-22)
Fuse (inside analyzer, s. section 8-3, pg. 8-27)	T 4 A / 250 V
External power supply unit	s. power supply options, pg. 1-60

1-7-2-1 BINOS® 100, BINOS® 100 M, OXYNOS® 100, HYDROS® 100 Specification

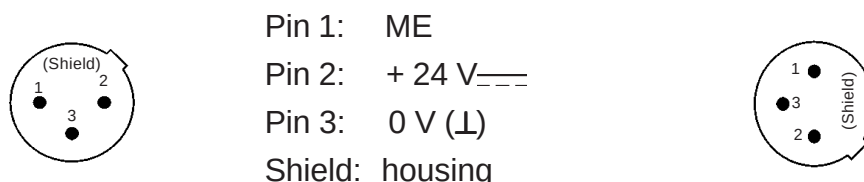


Fig. 1-22: Pin assignment 24 V DC input plug

Signal inputs / outputs

All signal lines need to be connected to 9 pole submin-D-plugs and sockets located at the analyzer's rear panel.

Available signals:

standard:

Analog signal outputs

Digital signal outputs

optional:

Analog inputs (TC; cross compensation)

Serial interface (RS232; RS 485)

Relay status signals

Analog signal outputs

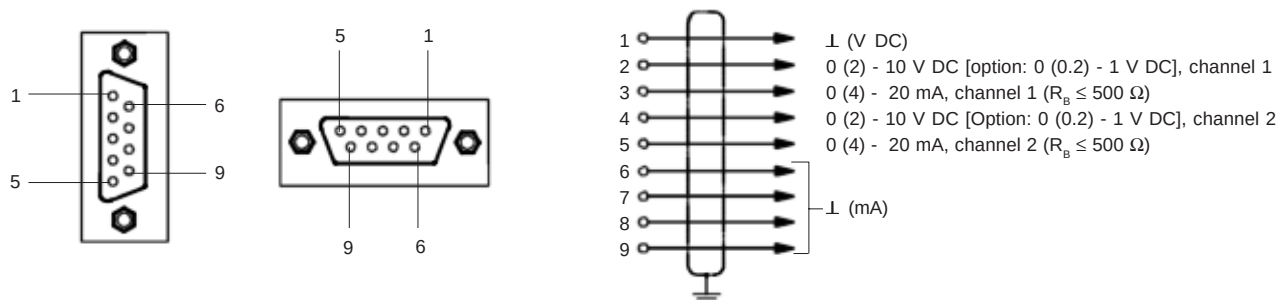


Fig.1-23: Socket X2 pin assignment (analog signal outputs; standard)

Note!

Take care of the special installation instructions in section 3-5!

The current configuration of voltage output (standard: 10 V DC or optional: 1 V DC) is given on order acknowledgement and on nameplate label.

Further configurations: see section 5-4-1

1-7-2-1 BINOS® 100, BINOS® 100 M, OXYNOS® 100, HYDROS® 100 Specification

Digital Outputs

Electrical specification: max. 30 V_{DC}, 30 mA

Configuring outputs: s. section 5-4-2

Note!

Take care of the special installation instructions in section 3-5!

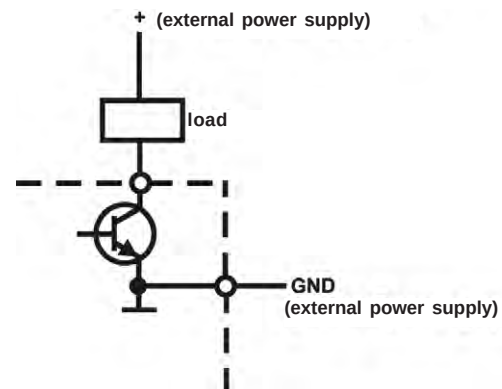


Fig.1-24: Connecting digital outputs

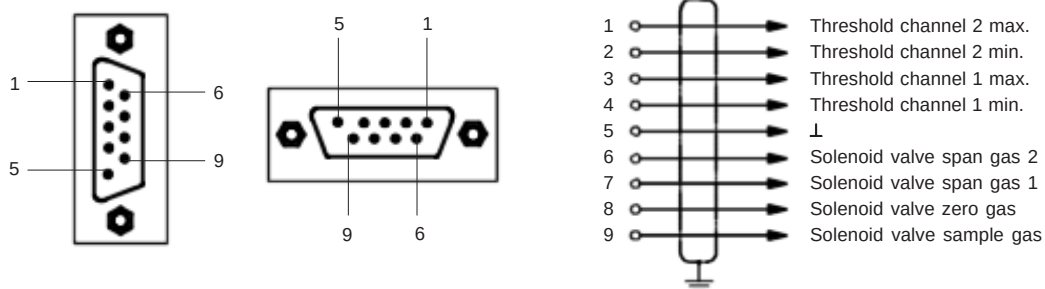


Fig.1-25: Plug X3 pin assignment (digital outputs; standard)

Analog inputs (TC; cross compensation)

Permissible signal ranges: 0 (0.2)- 1 V, 0 (2)- 10 V or 0 (4) - 20 mA, depending on configuration

Functional description: s. section 6-4

Note!

Take care of the special installation instructions in section 3-5!

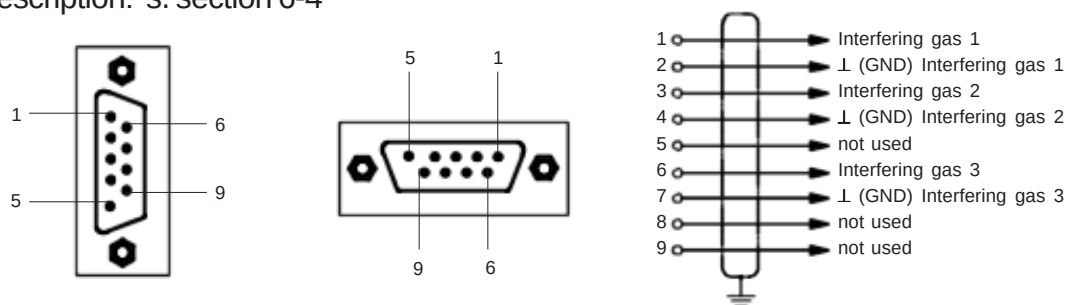


Fig.1-26: Plug Analog IN pin assignment (cross compensation; option)

1-7-2-1 BINOS® 100, BINOS® 100 M, OXYNOS® 100, HYDROS® 100 Specification

Serial Interface

Specification and driving the interface:
see section 6-2

Note!

*Take care of the special installation
instructions in section 3-5!*

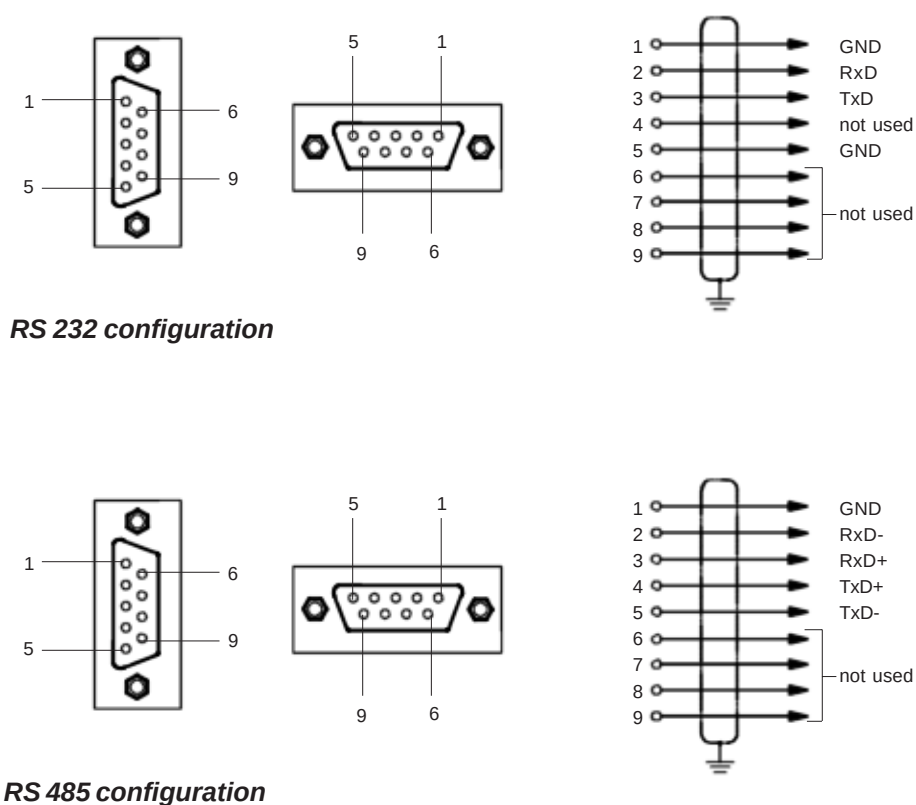


Fig.1-27: Socket X4 pin assignment (serial interface; option)

1-7-2-1 BINOS® 100, BINOS® 100 M, OXYNOS® 100, HYDROS® 100 Specification

Relay status Signals

Design: floating relay contacts

Electrical specification: max. 30 V_—, 1 A, 30 W

Signal configuration: see section 6-1

Note!

Take care of the special installation instructions in section 3-5!

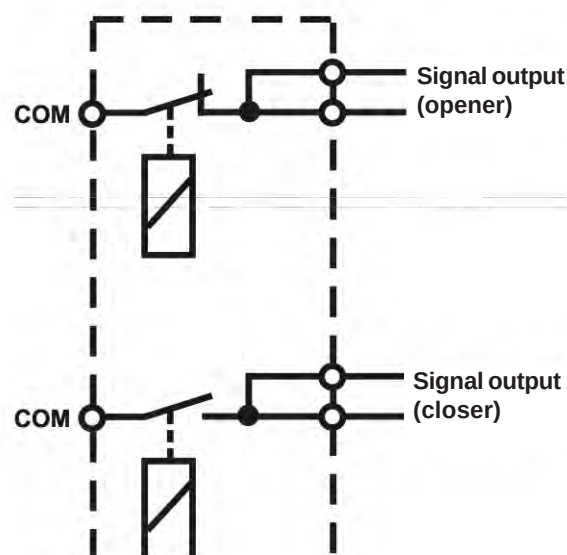


Fig.1-28: Relay status signals, block diagram

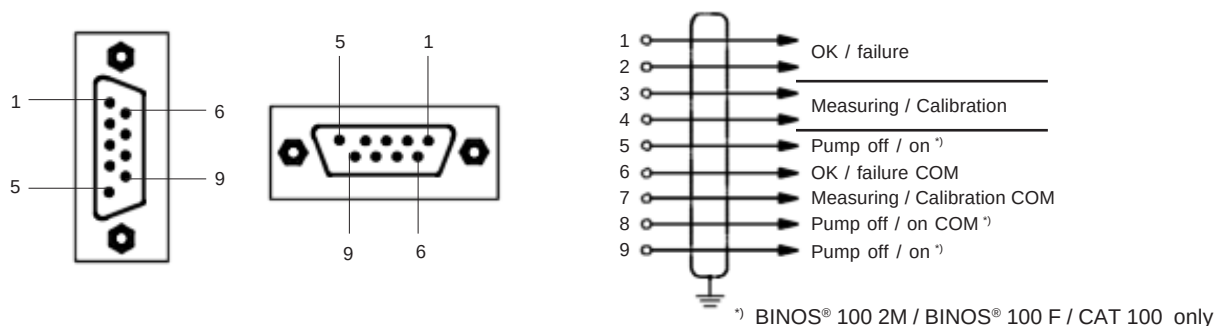


Fig.1-29: Plug X1 pin assignment (relay status signals; option)



Continue with section 1-7-2-5
"Power Supply" on page 1-60

1-7-2-2 BINOS® 100 2M Specification

1-7-2-2 BINOS® 100 2M

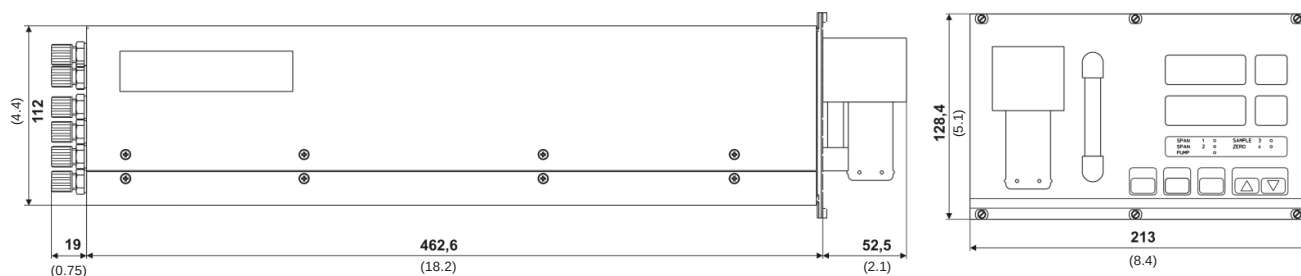


Fig. 1-30: BINOS® 100 2M
Dimensions [approx.; mm (inch)]

Housing

Weight:
(depending on analyzer configuration)

approx. 10 - 12 kg

Protection class to EN 60529:

IP 20 for indoor installation
Analyzer must not be supposed to dripping or sprayed water.

Gas fittings:

quantity: max. 9
specification: 6/4 mm PVDF
optional 6/4 mm or 1/4", stainless steel; other on request

Power Supply

a) with external power supply unit

Rated input voltage

24 V_{DC} (+- 5 %)

Rated input power

max. 100 W

Input

3-pole XLR plug (fig. 1-31a)

Fuse (inside analyzer, s. section 8-3, pg. 8-27)

T 4 A / 250 V

External power supply unit

s. power supply options, section 1-7-2-5

b) with internal power supply unit

Rated input voltage

120 / 230 V_{AC} 50/60 Hz

Input voltage range

99 - 138 / 187-264 V_{AC}

Rated input power

max. 240 VA

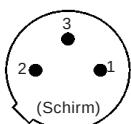
Input

IEC plug (fig. 1-31c)

Fuse (inside analyzer, s. section 8-3, pg. 8-27)

T 4 A / 250 V

1-7-2-2 BINOS® 100 2M Specification



Pin 1: ME
Pin 2: + 24 V ---
Pin 3: 0 V ($\perp$)
Shield: housing

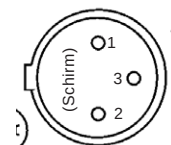


Fig. 1-31a: Pin assignment 24 V DC input plug

Fig. 1-31b: Pin assignment 24 V DC output socket

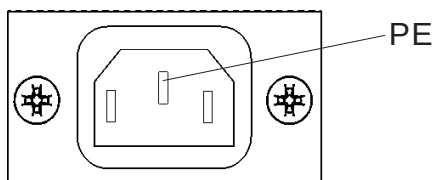


Fig. 1-31c: IEC mains input plug

c) Separate power output for supplying external components

The internal power supply unit provides an additional DC power output enabling the BINOS® 100 2M analyzer to supply external components, e.g. solenoid valves, pump, etc.

Output

3-pole XLR socket (fig. 1-31b)

Rated output voltage

24 V --- (+- 5 %)

Output current

max. 2 A

The DC power output is not fuse protected against overload or short circuit!

In case of overload or failure internal circuitry reduces the output voltage to protect the power supply unit, but this reduces the internal voltage for the analyzer, too, and therefore operation is not longer assured.

We recommend to protect external components by installing an adequate fuse to assure analyzer operation even in case of failure or overload at these components!



1-7-2-2 BINOS® 100 2M Specification

Signal inputs / outputs

Signals accessible at 9-pole submini-d plugs and sockets located at the analyzer's rear panel:

standard:

Analog signal outputs

Digital signal outputs

optional:

Analog inputs (TC; cross compensation)

Serial interface (RS232; RS 485)

Relay status signals

Signals accessible at screw terminal blocks located at the analyzer's rear panel (options):

Digital outputs

FOUNDATION™ Fieldbus

Signal inputs / outputs

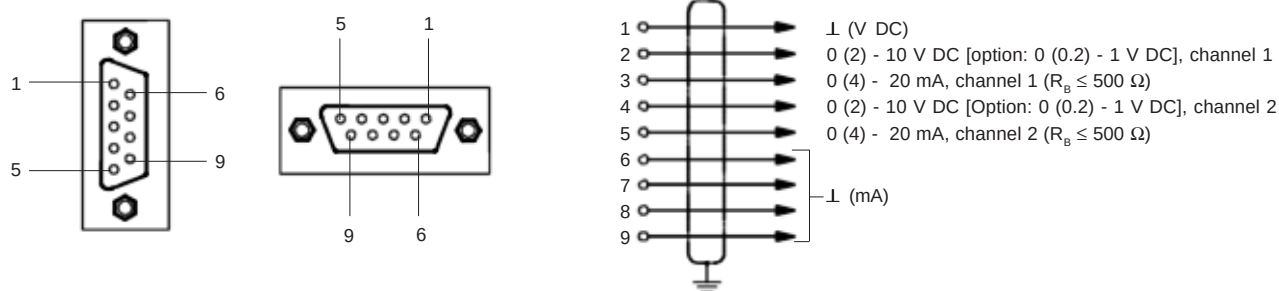


Fig.1-32: Socket X2 pin assignment (analog signal outputs; standard)

Note!

Take care of the special installation instructions in section 3-5!

The current configuration of voltage output (standard:10 V DC or optional: 1 V DC) is given on order acknowledgement and on nameplate label.

Further configurations: see section 5-4-1

1-7-2-2 BINOS® 100 2M Specification

Digital Outputs

Electrical specification: max. 30 V $\overline{\text{---}}$, 30 mA

Configuring outputs: s. section 5-4-2

Note!

Take care of the special installation instructions in section 3-5!

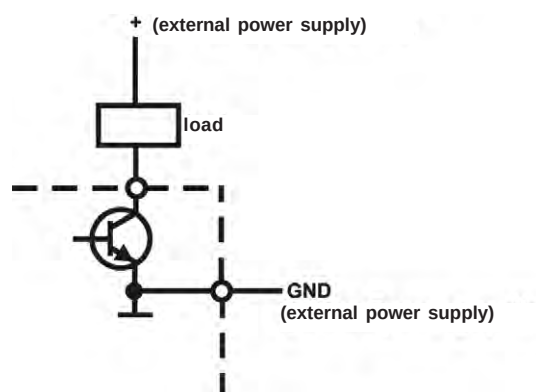


Fig.1-33: Connecting digital outputs

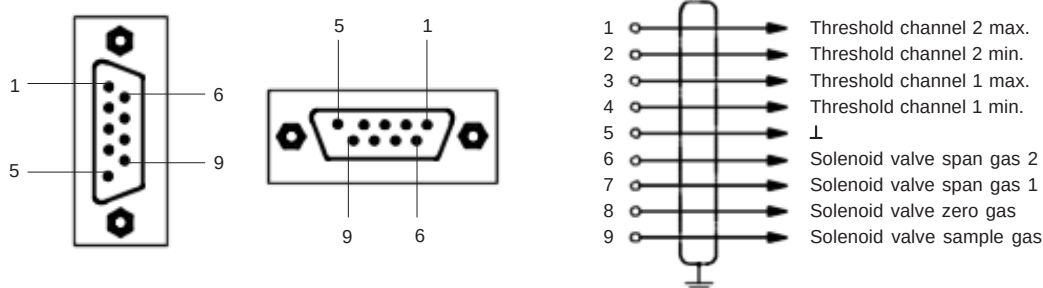


Fig.1-34: Plug X3 pin assignment (digital outputs; standard)

Analog inputs (TC; cross compensation)

Permissible signal ranges: 0 (0.2)- 1 V, 0 (2)- 10 V or 0 (4) - 20 mA.

Functional description: s. Abschn. 6-4

Note!

Take care of the special installation instructions in section 3-5!

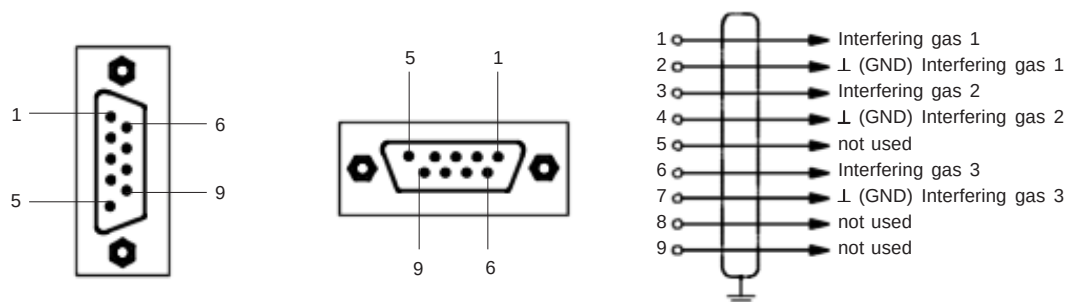


Fig.1-34: Plug Analog IN pin assignment (cross compensation; option)

1-7-2-2 BINOS® 100 2M Specification

Serial Interface

Specification and driving the interface:
see section 6-2

Note!

*Take care of the special installation
instructions in section 3-5!*

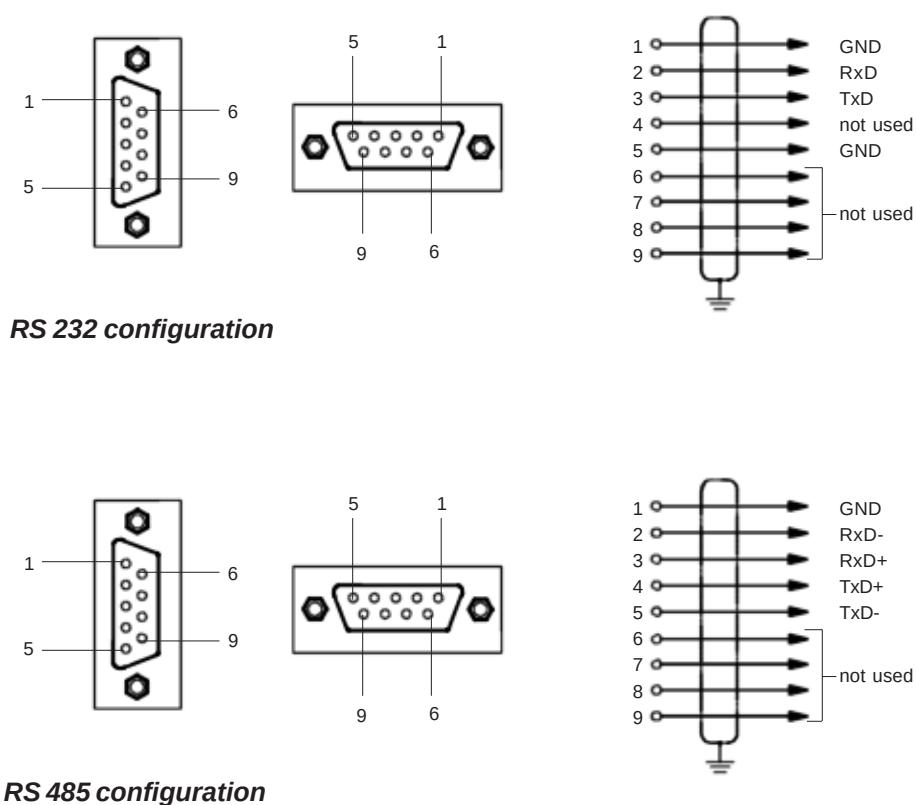


Fig.1-36: Socket X4 pin assignment (serial interface; option)

1-7-2-2 BINOS® 100 2M Specification

Relay status Signals

Design: floating relay contacts

Electrical specification: max. 30 V_~, 1 A, 30 W

Signal configuration: see section 6-1

Note!

Take care of the special installation instructions in section 3-5!

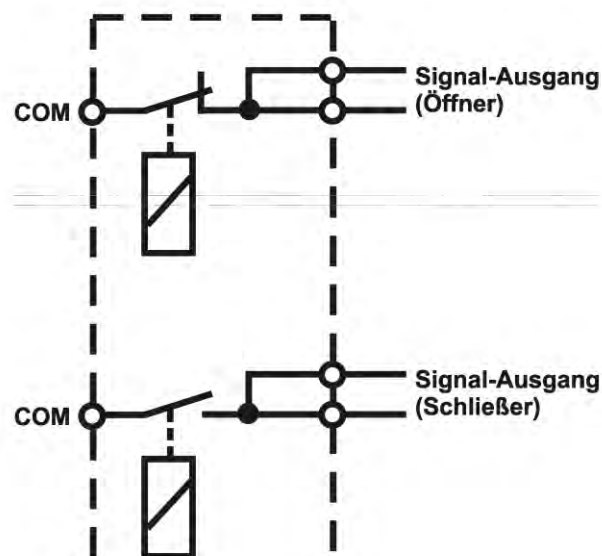


Fig.1-37: Relay status signals, block diagram

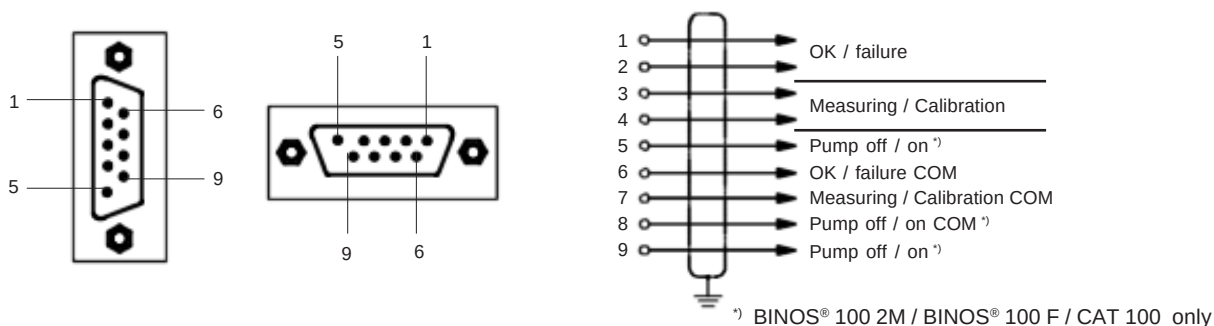


Fig.1-38: Plug X1 pin assignment (relay status signals; option)

1-7-2-2 BINOS® 100 2M Specification

Digital Inputs

Functional description, configuration and electrical specification: see section 6-3

Note!

Take care of the special installation instructions in section 3-5!

V+		+ 5 V DC
V-		⊥
E1		Start zero calibration channel 1 & 2
E2		Start span calibration channel 1
E3		Start span calibration channel 2
E4		Open zero gas valve
E5		Open span gas 1 valve
E6		Open span gas 2 valve
E7		Close sample gas valve / switch sample pump off

Fig.1-39: Digital inputs (option), terminal assignment

FOUNDATION™ Fieldbus

Functional description, configuration and electrical specification: see section 6-3

Note!

Take care of the special installation instructions in section 3-5!

FB +		Fieldbus +
FB -		Fieldbus -
FB +		Fieldbus +
FB -		Fieldbus -

Fig.1-40: FOUNDATION™ Fieldbus (option), terminal assignment



Continue with section 1-7-2-5
"Power Supply" on page 1-60

1-7-2-3 BINOS® 100 F Specification

1-7-2-3 BINOS® 100 F

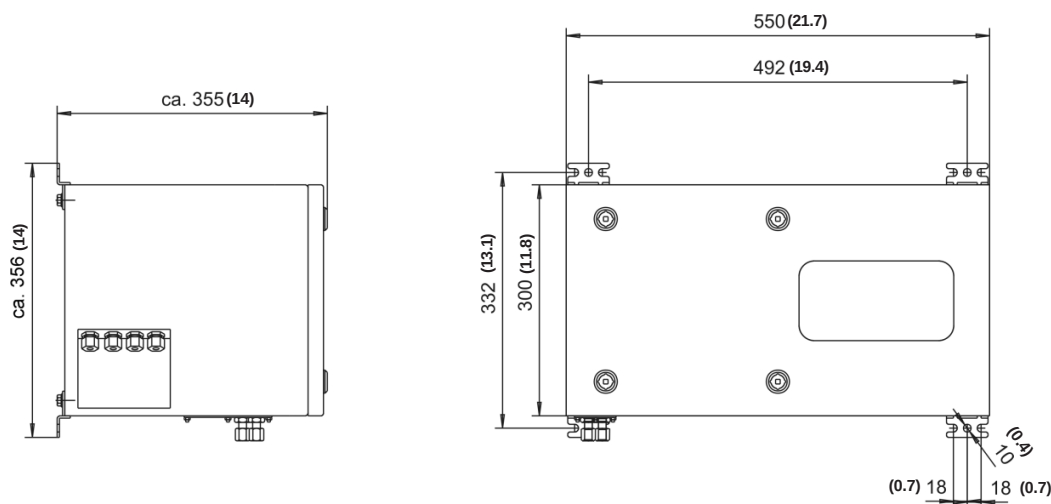


Fig. 1-41: BINOS® 100 F
Dimensions [approx.; mm (inch)]

Housing

Weight:
 (depending on analyzer configuration)

approx. 30 - 35 kg

Protection class to EN 60529:

IP 65

Gas fittings:

quantity: max. 8
 specification: 6/4 mm PVDF
 optional 6/4 mm or 1/4", stainless steel; other on request

1-7-2-3 BINOS® 100 F Specification

Power supply

Rated input voltage

120 / 230 V $\sim$ 50/60 Hz
manual selector switch at powersupply unit

Input voltage range

85 - 132 V / 176 - 264 V $\sim$, 47 - 63 Hz

Rated input power

max. 350 VA

Connection via internal screw terminal near
cable glands, in front of EMC filter (fig. 1-42),
cross section:

max. 4 mm², using conductor sleeves is
not required

Cable entry via

1 cable gland, classified IP 68

Permissible outer cable diameter for mains cable: 7 - 12 mm.

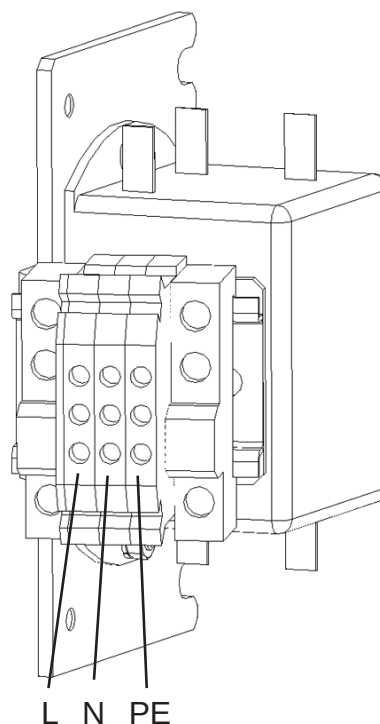


Fig. 1-42: Screw terminals for mains

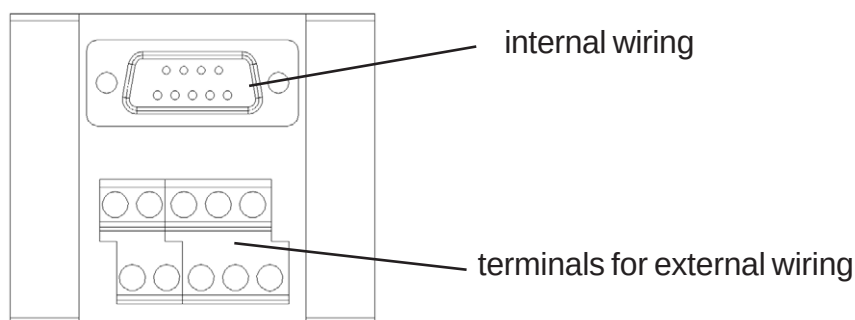
1-7-2-3 BINOS® 100 F Specification

Signal inputs / outputs

Connection via internal screw terminal, located on adaptor elements (fig. 1-43) inside analyzer

Cable entry via up to 3 cable glands, classified IP 68

Permissible outer cable diameter for signal cables: 7 - 12 mm.



**Fig. 1-43: 9-pole adaptor element for signal cables;
submin-d to screw terminals**

Each adaptor is at one side connected to one BINOS® 100 electronic signal output connector and marked accordingly (e.g. signal connector "X1" --> adaptor element "X1").

Refer to the schematics later in this manual for correlations between signals and adaptors.

1-7-2-3 BINOS® 100 F Specification

Available signals

standard:	Analog signal outputs Digital signal outputs
optional:	Analog inputs (TC; cross compensation) Serial interface (RS 232; RS 485) Relay status signals Digital outputs FOUNDATION™ Fieldbus

Analog signal outputs

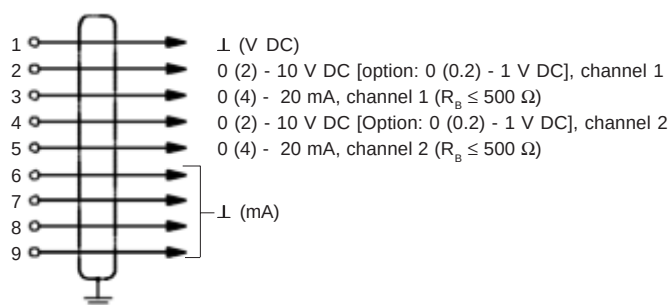


Fig.1-44: Pin assignment adaptor element X 2 (analog signal outputs; standard)

Note!

Take care of the special installation instructions in section 3-5!

The current configuration of voltage output (standard: 10 V DC or optional: 1 V DC) is given on order acknowledgement and on nameplate label.

Further configurations: see section 5-4-1

1-7-2-3 BINOS® 100 F Specification

Digital Outputs

Electrical specification: max. 30 V $\overline{\text{---}}$, 30 mA, "open collector"

Configuring outputs: s. section 5-4-2

Note!

Take care of the special installation instructions in section 3-5!

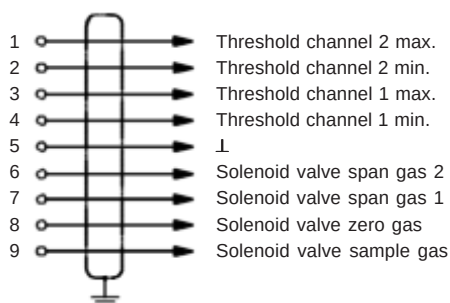


Fig.1-45: Pin assignment adaptor element X3 (digital outputs; standard)

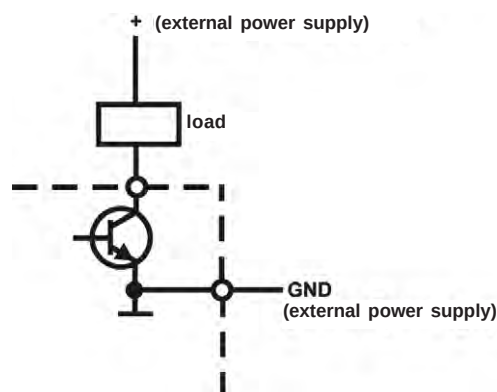


Fig.1-46: Connecting digital outputs

Analog inputs (TC; cross compensation)

Permissible signal ranges: 0 (0.2)- 1 V, 0 (2)- 10 V or 0 (4) - 20 mA, depending on configuration

Functional description: s. section 6-4

Note!

Take care of the special installation instructions in section 3-5!

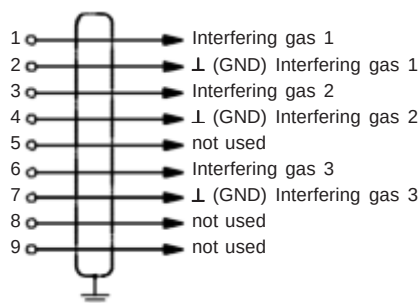


Fig.1-47: Pin assignment adaptor element Analog IN (cross compensation; option)

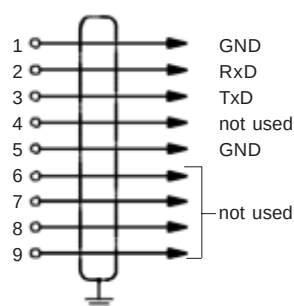
1-7-2-3 BINOS® 100 F Specification

Serial Interface

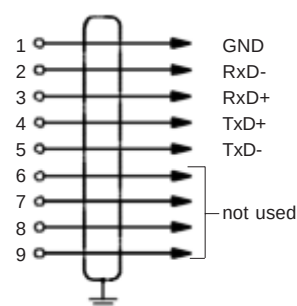
Specification and driving the interface:
see section 6-2

Note!

Take care of the special installation instructions in section 3-5!



RS 232 configuration



RS 485 configuration

Fig.1-48: Pin assignment adaptor element X4 (serial interface; option)

Relay status Signals

Design: floating relay contacts

Electrical specification: max. 30 V $\overline{\text{---}}$, 1 A, 30 W

Signal configuration: see section 6-1

Note!

Take care of the special installation instructions in section 3-5!

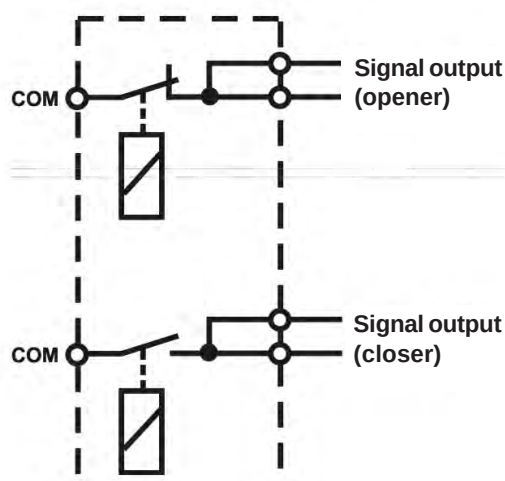
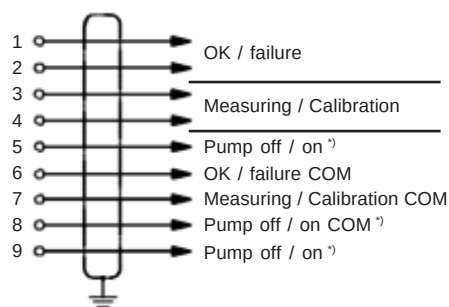


Fig.1-49: Relay status signals, block diagram



*) BINOS® 100 2M / BINOS® 100 F / CAT 100 only

**Fig.1-50: Pin assignment adaptor X1
(relay status signals; option)**

1-7-2-3 BINOS® 100 F Specification

Digital Inputs

Functional description, configuration and electrical specification: see section 6-3

Note!

Take care of the special installation instructions in section 3-5!

V+			+ 5 V DC
V-			⊥
E1			Start zero calibration channel 1 & 2
E2			Start span calibration channel 1
E3			Start span calibration channel 2
E4			Open zero gas valve
E5			Open span gas 1 valve
E6			Open span gas 2 valve
E7			Close sample gas valve / switch sample pump off

Fig.1-51: Digital inputs (option), terminal assignment

FOUNDATION™ Fieldbus

Functional description, configuration and electrical specification: see section 6-3

Note!

Take care of the special installation instructions in section 3-5!

FB +			Fieldbus +
FB -			Fieldbus -
FB +			Fieldbus +
FB -			Fieldbus -

Fig.1-52: FOUNDATION™ Fieldbus (option), terminal assignment



Continue with chapter 2
"Measuring principles", page 2-1

1-7-2-4 CAT 100 Specification

1-7-2-4 CAT 100

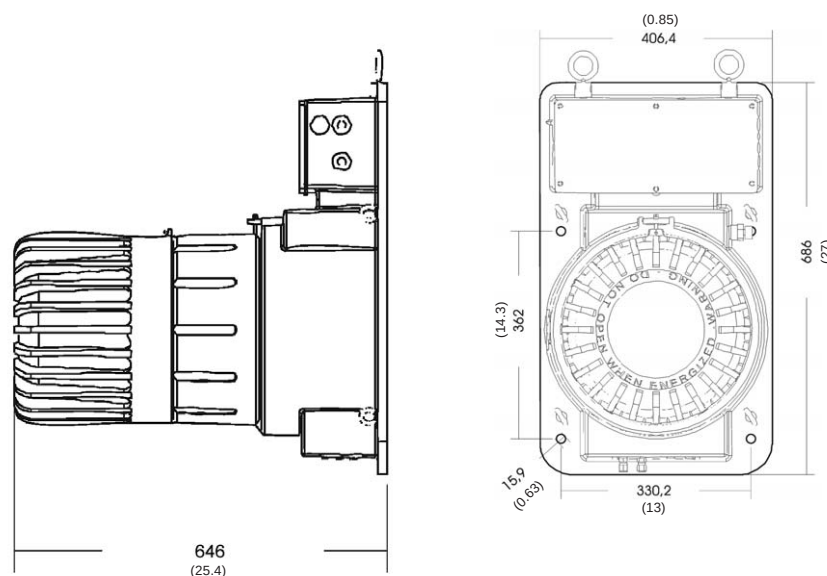


Fig. 1-55: CAT 100

Dimensions [approx.; mm (inch)]

Housing

Weight:
(depending on analyzer configuration)

approx. 70 kg

Protection class to EN 60529:

IP 65 (IP 66 w/o breather), optional with internal tropicalization

Gas fittings:

quantity: max. 11
specification: 1/4" stainless steel

Power Supply

Rated input voltage

115-230 V~ 50/60 Hz

Input voltage range

85 - 264 V~, 47 - 63 Hz

Rated input power

max. 240 W

Connection via screw terminal blocks in junction box (fig. 1-54), cross section:

4 mm² max.

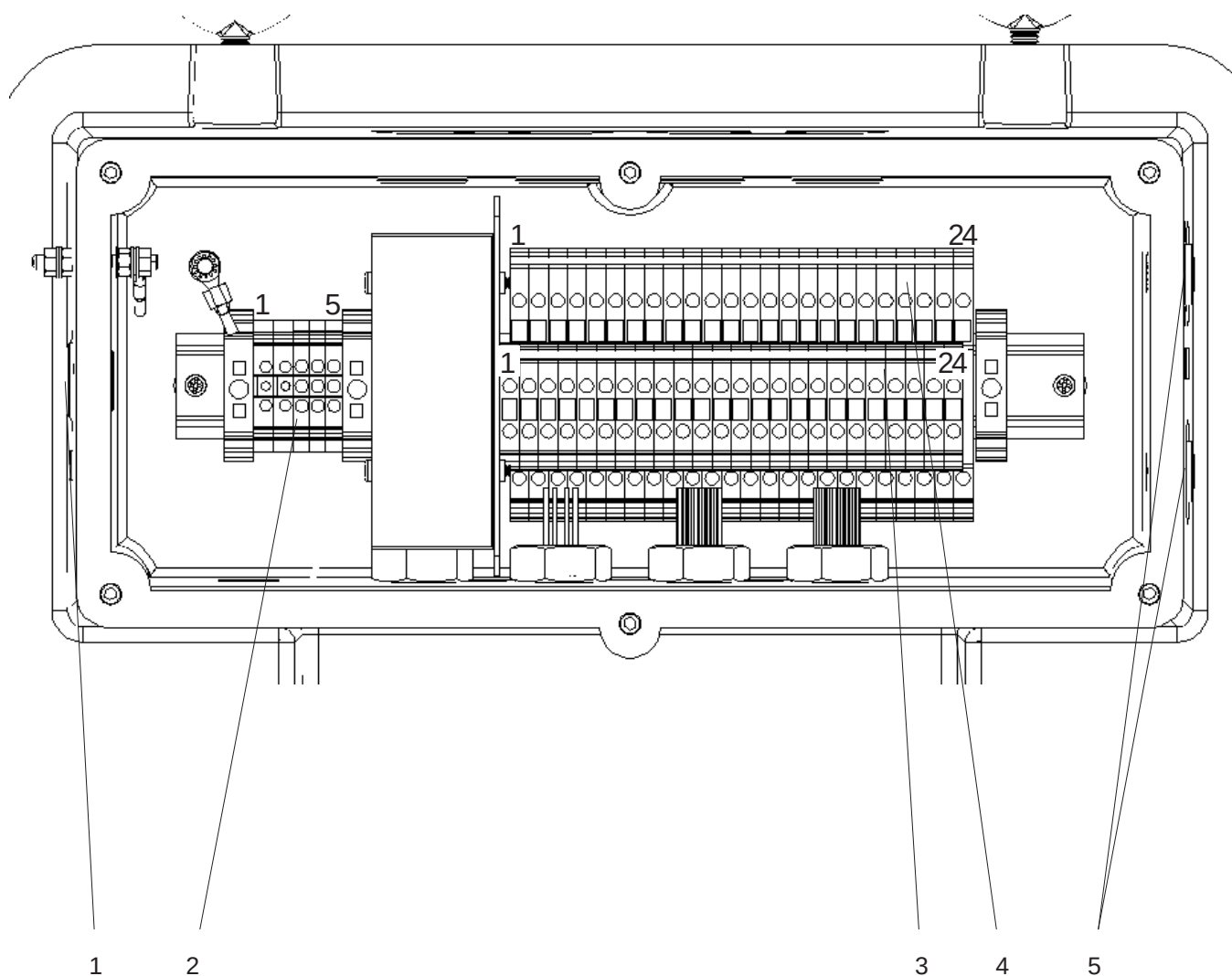
Cable entry:

1 cable gland, to be provided by customer

Cable gland specification:

Threat M20 x 1.5 metric, minimum protection class to meet housing protection class as given on name plate label (IP 65 or IP 66 or better)

1-7-2-4 CAT 100 Specification



**Fig. 1-54: CAT 100
junction box, interior view**

- 1 Cable gland mains
- 2 Mains terminals
- 3 Upper signal terminals (24)

- 4 Lower signal terminals (24).
- 5 Signal cable glands (max. 3)

1-7-2-4 CAT 100 Specification

Signal inputs / outputs

Connection via screw terminal blocks in junction box (fig. 1-54), cross section:

0.2 - 4 mm²

Cable entry:

max. 3 cable glands, to be provided by customer

Cable gland specification:

Threat M16 x 1.5 metric, minimum protection class to meet housing protection class as given on name plate label (IP 65 or IP 66 or better)

Available signals

standard:

2 analog signal outputs

8 digital signal outputs

optional:

3 analog inputs (TC; cross compensation)

3 relay status signals

7 digital inputs

FOUNDATION™ Fieldbus *)

Serial interface (RS 232; RS 485) *) **)

*) FOUNDATION™ Fieldbus not in combination with serial interface!

) Either RS 232 **or RS 485

Note!

Assignment of signals to signal terminals inside junction box varies depending on selected analyzer options. For assignments of your specific analyzer refer to the label on the inner side of the junction box cover.

General terminal assignments (examples) for the different kinds of available signals are given on the following pages.

1-7-2-4 CAT 100 Specification

Analog Signal Outputs

By default two analog outputs are configured to provide 0(4) to 20 mA signals (one for each channel).

Voltage outputs (standard:10 V DC or optional: 1 V DC) may be ordered optionally.

The current analog output configuration for a specific instrument is given on order acknowledgement and on the nameplate label of the analyzer module.

Further configurations: see section 5-4-1.

Note!

Take care of the special installation instructions in section 3-5!

Terminal No.	Upper	Lower
1	Channel 1 signal	GND
2	Channel 2 signal	GND

Table 1-6: Terminal assignments analog outputs

1-7-2-4 CAT 100 Specification

Digital Outputs

Electrical specification: max. 30 V_{DC}, 30 mA,
"open collector"

Configuring outputs: s. section 5-4-2

Note!

*Take care of the special installation
instructions in section 3-5!*

	Terminal No.	Upper	Lower
Thresholds	5		GND
	6	Ch1: upper threshold	Ch1: lower threshold
	7	Ch2: upper threshold	Ch2: lower threshold
Valve controls	8	Ch1: span gas	Ch2: span gas
	9	Zero gas	Sample gas

Table 1-7: Terminal assignments digital outputs

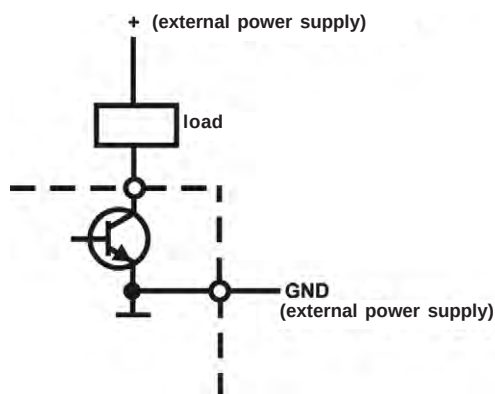


Fig. 1-55: Connecting digital outputs

1-7-2-4 CAT 100 Specification

Analog inputs (TC; cross compensation)

Permissible signal ranges: 0 (0.2)- 1 V,
0 (2)- 10 V or 0 (4) - 20 mA, depending on
configuration

Functional description: s. section 6-4

Note!

*Take care of the special installation
instructions in section 3-5!*

Terminal No.	Upper	Lower
13	Interfering Gas 1	GND 1
14	Interfering Gas 2	GND 2
15	Interfering Gas 3	GND 3

Table 1-8: Terminal assignments analog inputs

Serial Interface

Specification and driving the interface:
see section 6-2

Note!

*Take care of the special installation
instructions in section 3-5!*

RS 232	Terminal No.	Upper	Lower
	3	RxD	GND
RS 485	4	TxD	GND
	Terminal No.	Upper	Lower
	3	RxD +	GND
	4	TxD +	TxD -
	5	RxD -	

Table 1-9: Terminal assignments serial interface

1-7-2-4 CAT 100 Specification

Relay status Signals

Design: floating relay contacts

Electrical specification: max. 30 V_{AC}, 1 A, 30 W

Signal configuration: see section 6-1

Note!

Take care of the special installation instructions in section 3-5!

Terminal No.	Upper	Lower
10	OK / Failure	OK / Failure
11	Meas / Calib	Meas / Calib
12	Pump	Pump

Table 1-10: Terminal assignment relay status signals

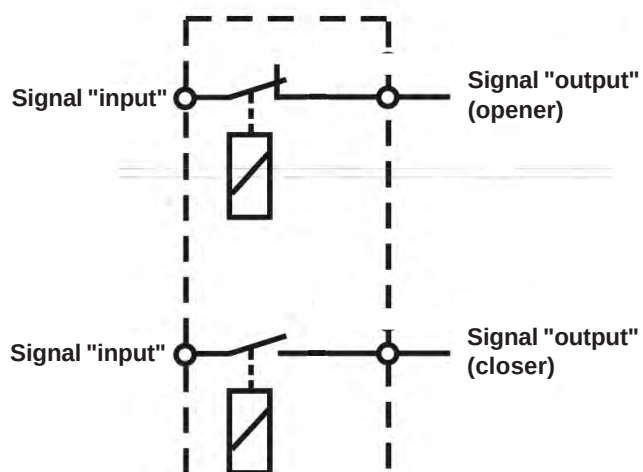


Fig.1-56: Relay status signals, block diagram

1-7-2-4 CAT 100 Specification

Digital Inputs

Functional description, configuration and electrical specification: see section 6-3

Note!

Take care of the special installation instructions in section 3-5!

Terminal No.	Upper	Lower
16	+V5	GND
17	zero cal I&II	span I
18	span II	zero valve
19	span valve I	span valve II
20	sample valve	

Table 1-11: Terminal assignment digital inputs

FOUNDATION™ Fieldbus

Functional description, configuration and electrical specification: see section 6-3

Note!

Take care of the special installation instructions in section 3-5!

Terminal No.	Upper	Lower
3	FB +	FB -
4	FB +	FB -

Table 1-12: Terminal assignment FOUNDATION™ Fieldbus

1-7-2-5 UPS 01 Power Supply Specification
**1-7-2-5 External Power Supply Unit for
Tabletop and Rack Mount Analyzers**

This power supply unit may be ordered for supplying tabletop analyzers and rack mount analyzers not requiring a rail mounted power supply.

Model designation

UPS 01 T

Input

Rated input voltage

120 / 230 V $\sim$ 50/60 Hz

Input voltage range

99 - 138 / 187-264 V $\sim$

Rated input power

max. 240 VA

Input connector

IEC plug (fig. 1-58a)

Output

Rated output voltage

24 V --- (+- 5 %)

Rated output current

max. 5 A

Output connector

3 pole XLR socket (fig. 1-58b)

Protection measures

Overload protection

Current limitation, typ. 110% I_{rated} ,
straight characteristic, output permanent
short-circuit proof

Overtemperature protection

Reduction of output voltage until unit
switches off. Return to normal operation
after cooling down.

1-7-2-5 UPS 01 Power Supply Specification

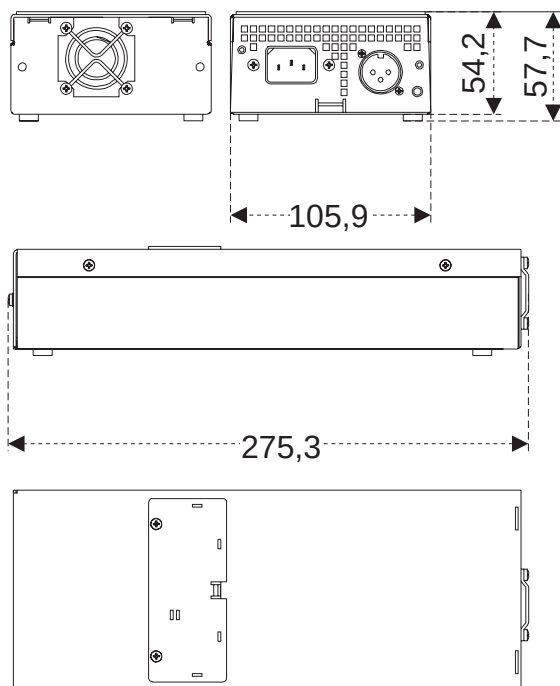


Fig. 1-57: Power supply UPS 01 / 01 T
Dimensions (approx. [mm])

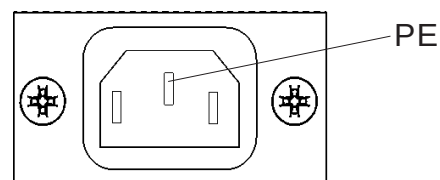


Fig. 1-58a: IEC mains input plug



Pin 1: ME
Pin 2: + 24 V
Pin 3: 0 V (⊥)
Shield: Housing

Fig. 1-58b: 24 V DC output socket
Pin assignment

Weight

approx. 2.5 kg

Compliances

Electrical safety

EN 60950, UL1950, CSA22.2 NO 950-95

Electromagnetic compatibility

EN 50081-1 (emmission)

EN 50082-2 (immunity)

and others

1-7-2-6 5 A Rail Mount Power Supply Specification

1-7-2-6 5 A Power Supply Unit for DIN Rail Installation

This power supply unit may be ordered for supplying analyzers installed in a rack, where mounting the power supply on a DIN rail is required.

Model designation

SL5

Input

Rated input voltage (manual switch)

100-120 / 220-240 V $\sim$ 50/60 Hz

Input voltage range

85 - 132 / 176 - 264 V $\sim$, 47 - 63 Hz

Rated input current

< 2.6 A (switch at 115 V position)
< 1.4 A (switch at 230 V position)

Input

Screw terminals at front (bottom)

Output

Rated voltage

24 V --- (+ 5 %, -1 %)

Output current

max. 5 A

Output

Screw terminals at front (top)

Efficiency

Efficiency

typ. 90 % (230 VAC, 24 V / 5 A)

Losses

typ. 13.3 W (230 VAC, 24 V / 5 A)

Protection Measures

Overload protection

against short circuit, overload and open circuit

Overtemperature protection

Derating > 60° C

1-7-2-6 5 A Rail Mount Power Supply Specification

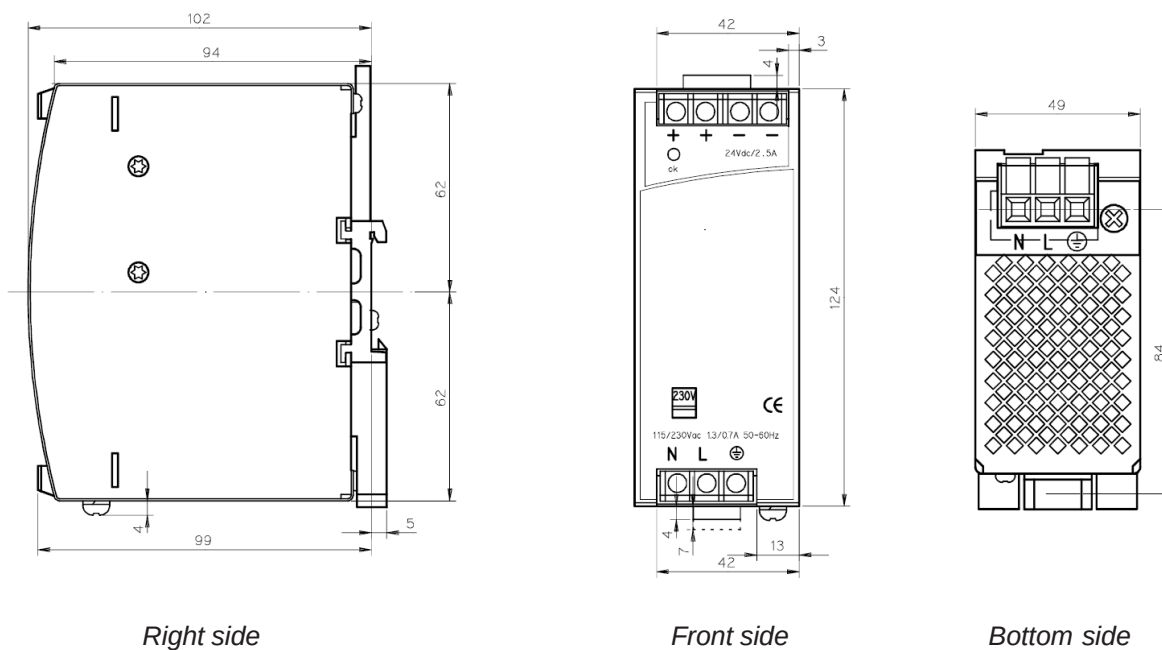


Fig. 1-59: Power Supply SL05
Dimensions (approx. [mm])

Design and Installation

Weight

ca. 0.62 kg

Dimensions (w x h x d)

65 mm x 125 mm x 103 mm

Spacing for cooling

above / below 25 mm each
left / right sides 15 mm each

Compliances

Electrical safety

EN 60950, EN 50178, UL1950,
CSA22.2 Nr. 234-M90

Electromagnetic compatibility

EN 50081-1, class B (emission)
EN 50082-2, class A (immunity)
and others

1-7-2-7 10 A Rail Mount Power Supply Specification

**1-7-2-7 External 10A Power Supply Unit
for DIN Rail Installation**

This power supply unit may be ordered for supplying analyzers installed in a rack, where mounting the power supply on a DIN rail is required.

Model designation

SL10

Input

Rated input voltage (manual switch)

100-120 / 220-240 V $\sim$ 50/60 Hz

Input voltage range

85 - 132 / 176 - 264 V $\sim$, 47 - 63 Hz

Rated input current

< 6 A (switch at 115 V position)
< 2.8 A (switch at 230V position)

Input

Screw terminals at front (left side)

Output

Rated voltage

24 V --- (+ 5 %, -1 %)

Output current

max. 10 A

Output

Screw terminals at front (right side)

Efficiency

Efficiency

typ. 89 % (230 VAC, 24 V / 10 A)

Losses

typ. 29 W (230 VAC, 24 V / 10 A)

Protection Measures

Overload protection

against short circuit, overload and open circuit

Overtemperature protection

Derating > 60° C

1-7-2-7 10 A Rail Mount Power Supply Specification

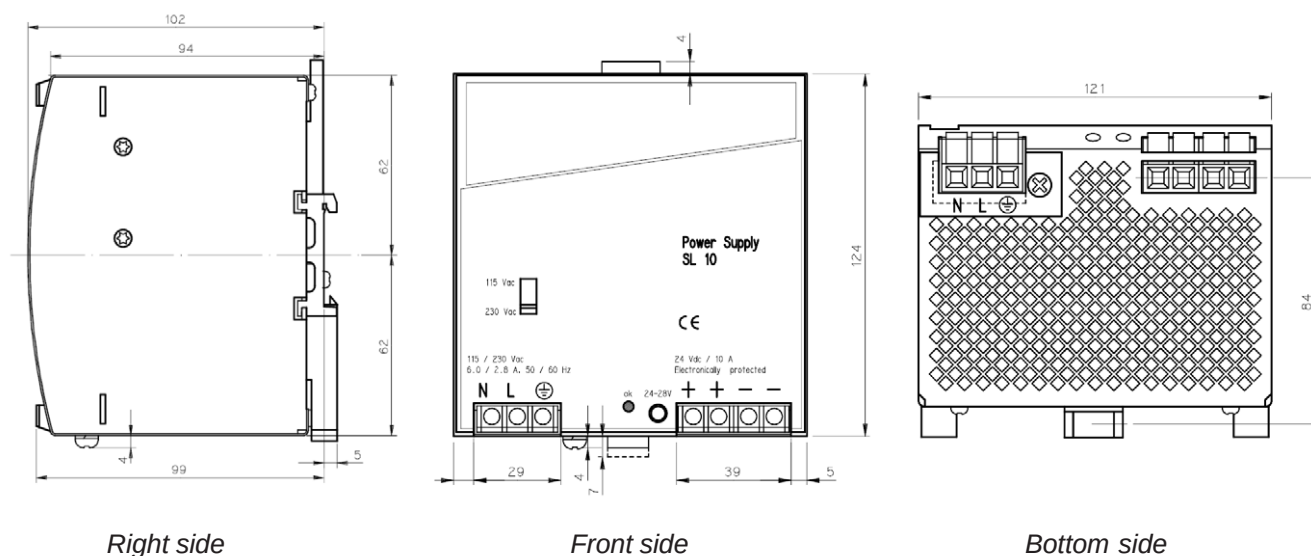


Fig. 1-60: Power Supply SL 10
Dimensions (approx. [mm])

Design and Installation

Weight

approx. 1.2 kg

Dimensions (w x h x d)

120 mm x 124 mm x 102 mm

Spacing for cooling

above / below 25 mm each
left / right sides 15 mm each

Compliances

Electrical safety

EN 60950, EN 50178, UL1950,
CSA22.2 Nr. 234-M90

Electromagnetic compatibility

EN 50081-1, class B (emission)
EN 50082-2, class A (immunity)
and others

1-7-2-8 10 A Tabletop Power Supply Unit

1-7-2-8 10 A Tabletop Power Supply Unit

This power supply unit may be ordered for supplying two tabletop analyzers by one shared power supply unit.

Model designations

A variation for installation in a module rack is available, too, with output currents of 10 A or 5 A

10 A tabletop power supply unit
10 A rack module power supply unit
5 A rack module power supply unit

For electrical input and output data, efficiency, protection measures and compliances (internal power supply unit only) see
section 1-7-2-7 for 10 A versions
section 1-7-2-6 for 5 A version.

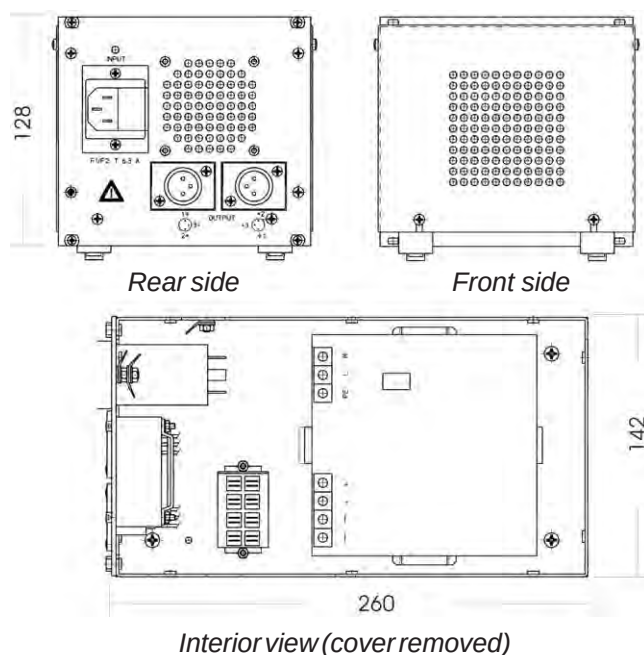


Fig. 1-61: 10 A Tabletop Power Supply
Dimensions (approx. [mm])

Design and Installation

Dimensions, approx. (w x h x d)

142 mm x 260 mm x 128 mm

Spacing for cooling

front / rear side 15 mm each

1-7-3 Specifications at the Nameplate Label

1-7-3 Specifications at the Nameplate Label

The analyzer nameplate label gives important information about the instrument's configuration, installed measuring principle(s), sample gas(es), measuring range(s), power supply and analog output configuration. In case an instrument is built according a suitability approval the related certificate number is given, too.

The nameplate label is located either on the instruments left or right housing side or at the inner side of the front door (BINOS® 100 F). The CAT nameplate label is located at the housings left side whereat the label as it is shown below is affixed to the inner analyzer module.

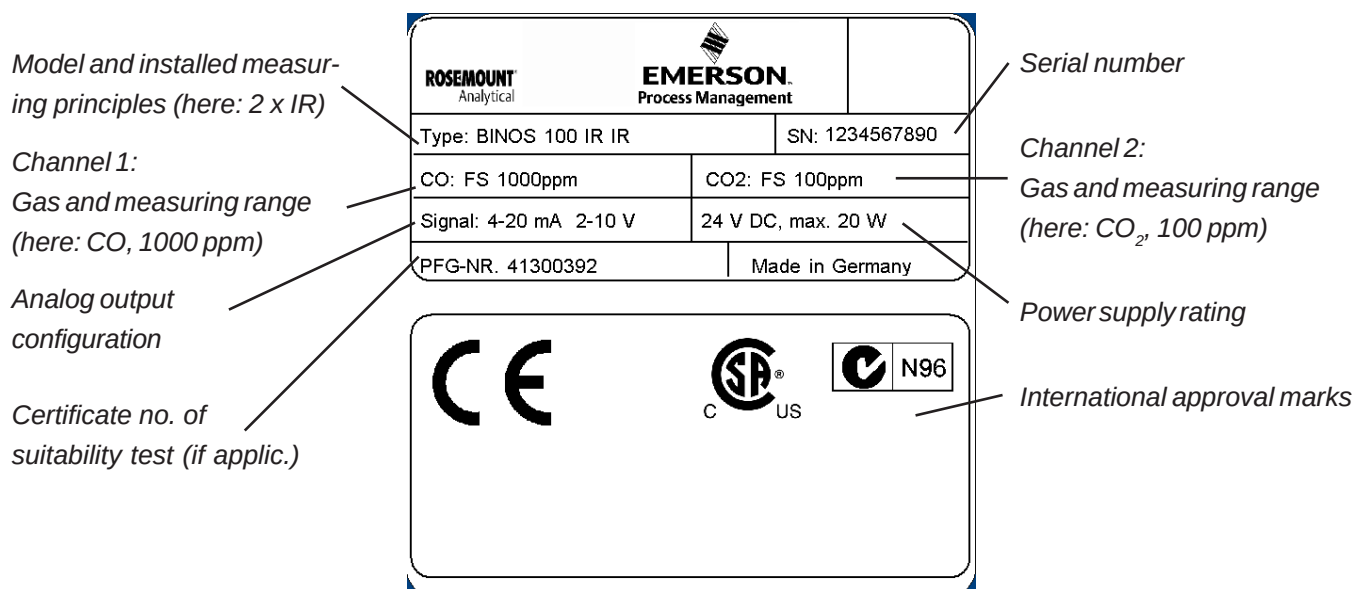


Fig. 1-62: Analyzer Nameplate Label (example)

1-8 Suitability Approvals
1-8 Suitability Approvals

Instrument	Components & Ranges	Approvals & Report No.
BINOS 100 M	CO: 0 - 250 ppm (0 - 250 mg/Nm ³) O ₂ : 0 - 25 % (electrochemical)	TI Air, 13 th BImSchV EN ISO 14956, EN 14181 (QAL 1) <i>TUEV Nord report no. 95 CU 054/B</i>
BINOS 100 2M BINOS 100 F CAT 100	CO: 0 - 250 ppm (0 - 250 mg/Nm ³) O ₂ : 0 - 25 % (electrochemical)	TI Air, 13 th BImSchV <i>TUEV Nord report no. 95 CU 054/B</i>
	O ₂ : 0 - 25 % (paramagnetic)	<i>TUEV Bavaria report no. 156 3793</i>
OXYNOS 100	O ₂ : 0 - 25 % (electrochemical)	TI Air, 13 th BImSchV <i>TUEV Nord report no. 95 CU 054/B</i>
	O ₂ : 0 - 25 % (paramagnetic)	<i>TUEV Bavaria report no. 156 3793</i>

Chapter 2

Measuring Principles

BINOS® 100 series analyzers utilize several measuring principles or combinations of principles, depending on measured gas components. This provides best possible results as the principle always is adjusted to meet the specific characteristics of the particular gas.

The following sections introduce the available measuring principles and show their specific characteristics.

2-1 Infrared Measurement (IR)

This principle makes use of infrared light absorbed by the sample gas. The wave length of the absorbed portion of IR light characterizes the gas component whereas the intensity of absorption is a measure of concentration.

Two different IR measuring principles are available, both comparing concentration dependent and concentration independent signals. The difference of these two signals results in a measure of concentration.

The gas component to be measured and the concentration defines which principle is used in a certain application:

2-1-1 Interference Filter Correlation (IFC)

An undivided analysis cell is alternately passed by light of two different wave lengths, filtered by interference filter out of the spectrum of an IR source: One wavelength covers the absorption band of the measured gas component, the other is selected to cover a region where no absorption occurs.

Fig. 2-1 shows an example of interference filter transmissions and absorption bands of the gas components CO and CO₂. It's easy to see that the interference filters' transmissible spectral bands overlap the absorption bands of the gases, whereas within the bandwidth of the reference filter no absorption takes place. Other gases (CH₄) and HC do not affect the measuring result because they do not absorb IR light of these wavelengths.

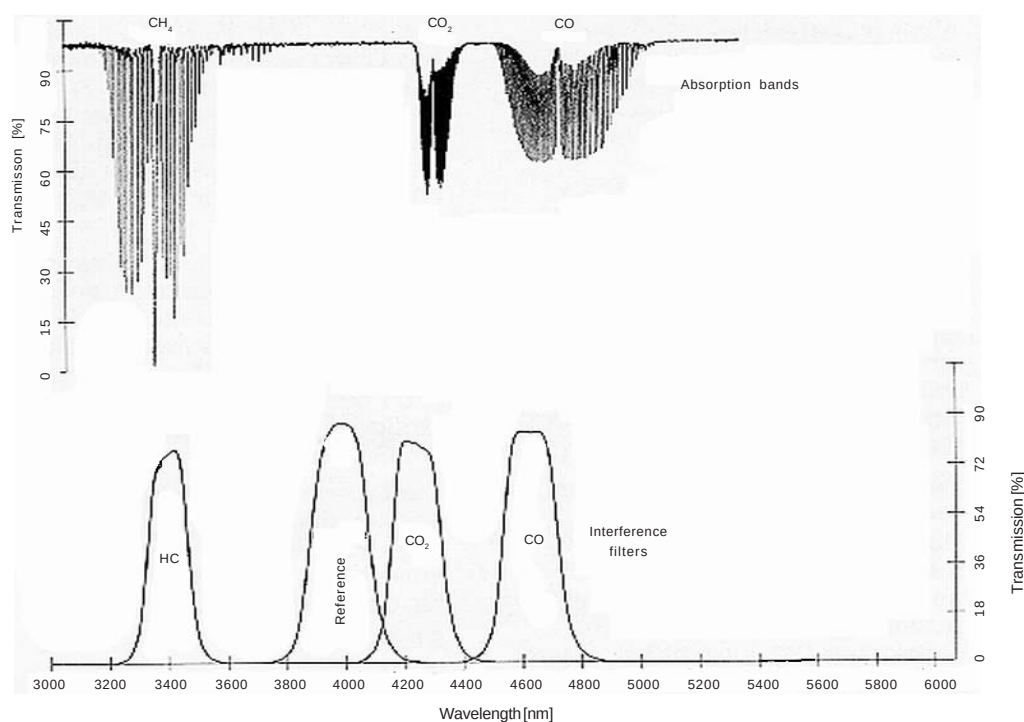
2-1 Infrared Measurement (IR)

Fig. 2-1: Absorption bands of measured gases and transmission of interference filters

A pyroelectrical detector generates a signal using the effect of charge flow caused by heat flow within a piezo crystal:

The IR radiation passing measuring cell and filters results in a change of temperature when arriving at the detector. The IR absorption is different for measuring wavelength and reference wavelength so the crystal is alternately more or less heated. As result the detector gives an alternating voltage signal which is passed to electronics for further processing.

2-1 Infrared Measurement (IR)

2-1-2 Opto Pneumatic Measuring Principle

This principle utilizes a separated analysis cell whose one side is passed by the sample gas. The other side (reference side) is either filled with an inert gas (e.g. nitrogen) or passed by a reference gas stream, depending on application.

Both sides of the cell are alternately irradiated with IR light of same intensity, which afterwards passes a filtering cell before it arrives at the detector.

The opto pneumatical detector, which is used instead of the pyroelectrical detector with the

IFC principle, converts the radiation from sample side and reference side into a voltage signal proportional to the radiation intensity.

The pneumatical detector consists of a gas filled absorption chamber and a compensation chamber, both connected by a flow channel. A micro flow detector is placed within this channel to measure least flows.

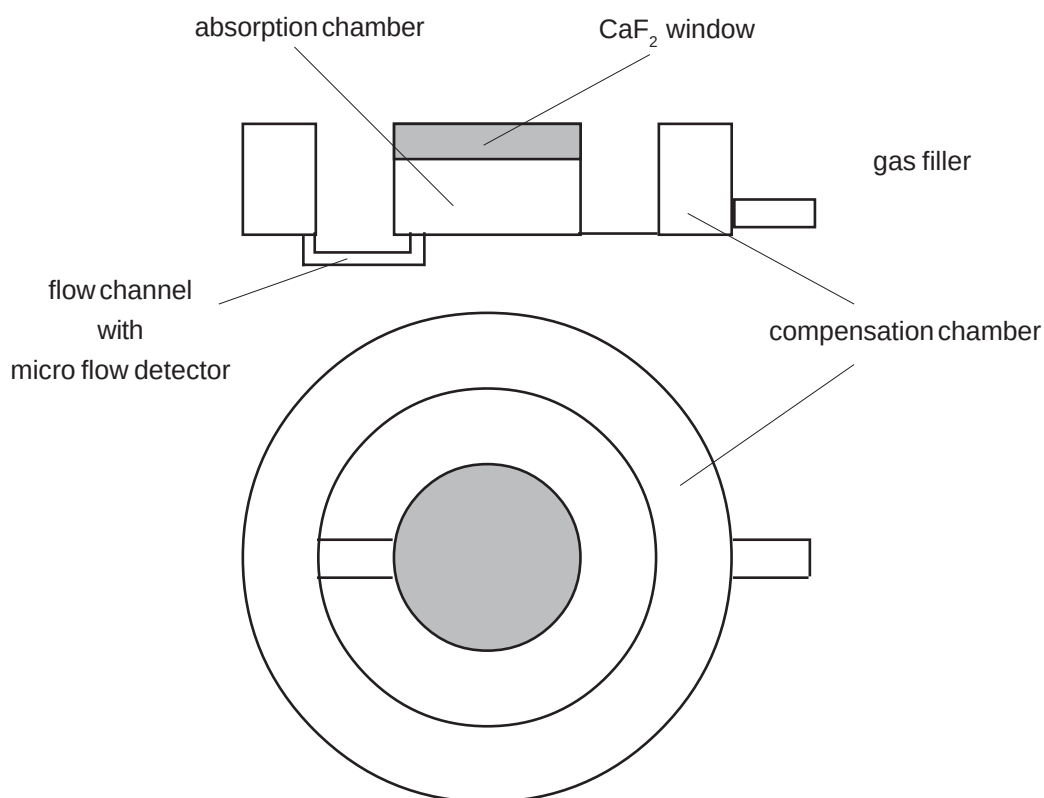


Fig. 2-2: Gas Detector Design Principle

2-1 Infrared Measurement (IR)

The detector is filled with the gas to be measured and therefore sensitive to the related characteristic wavelength band only.

The absorption chamber is sealed with a window transparent for IR radiation [CaF_2 (Calciumfluorid)].

There is no absorption when the IR radiation passes the reference side of the analysis cell, so the intensity is at a maximum when it arrives at the detector. The gas within the detector is heated, therefore expands and flows from absorption chamber through flow channel to the compensation chamber. This flow generates a voltage signal.

When the IR radiation passes the sample side of the analysis cell, part of it is absorbed by the sample gas decreasing the radiation intensity. As an effect the gas within the detector cools down and flows from the compensation chamber back into the absorption chamber. This flow again causes a voltage signal by the micro flow detector which is now reverse to the previous signal.

So the micro flow detector generates alternating signals. The flow channel is designed to not influence the gas flow and therefore the change in signal is proportional to the change in radiation intensity which is proportional to the concentration of gas to be measured.

The flow detector signal is passed to electronics conditioning the signal and converting it into a useful format.

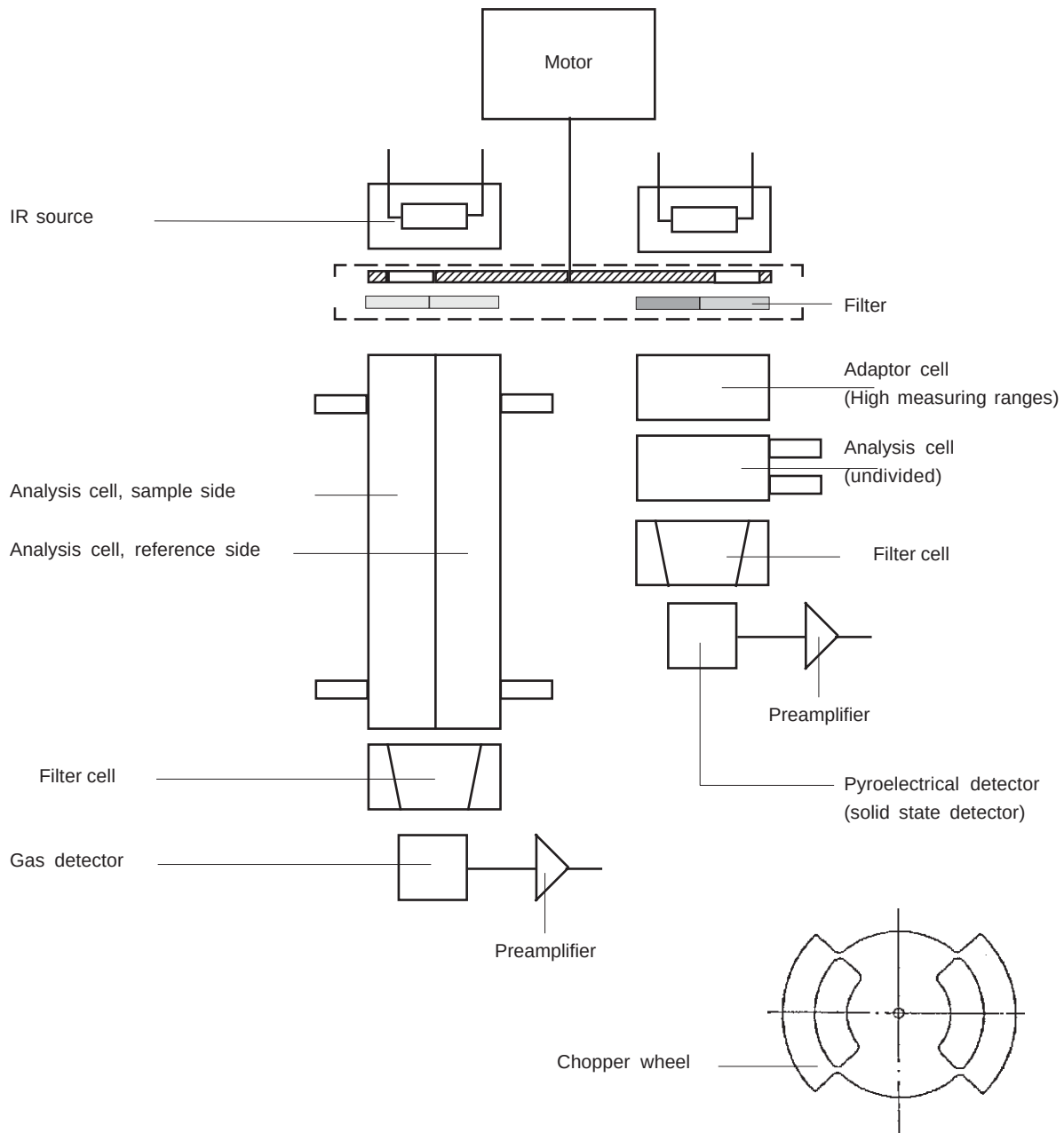
2-1-3 Technical Implementation

The broadband IR radiation emitted by a special source passes a chopper wheel (fig. 2-3 showing a dual channel analyzer). In case of IFC principle it passes additional optical filters, before entering the analysis cell. The radiation leaving the cell is focused to the detector by means of a filter cell. The detector output signal is passed to micro processor controlled electronics conditioning and converting it to reasonable units (Vol-%, ppm, mg/m^3 , etc.) shown on the 7 segment display.

Depending on gas component and measuring range several photometer designs are used with BINOS® 100 series analyzers, differing in analysis cell length, kind of detector and used filters. Optionally the assembly may be sealed against ambience: O-rings between the components prevent ambient air to come in the optical path and interfere the measurement.

The photometer assembly is mounted on the BKS main board, which itself is placed in housings rail.

2-1 Infrared Measurement (IR)



**Fig. 2-3: IR Photometer Assembly Principle,
left side: with gas detector, right side: with pyro electrical detector**

2-1 Infrared Measurement (IR)

Photometer Assembly with Pyro Detector

Fig. 2-4 shows the mechanical design of an pyro detector assembly:

Attached to the chopper are IR source (07) as well as analysis cell (09) with signal detector unit (filter cell 14 / 15) and pyroelectrical detector with integrated pre amplifier (16).

Furthermore the chopper assembly contains the filter aperture (04 / 05) limiting the IR light to the required bandwidth.

The chopper assembly (03) is made of two parts and contains an inner volume where the stepper motor driven chopper wheel is placed. This inner volume is sealed against ambience by means of an o-ring to prevent ambient CO₂ from entering. This avoids pre absorption and drift. An additional absorber is used to remove CO₂ traces existent by diffusion.

Furthermore the chopper assembly contains a light barrier to detect the chopper wheel phasing. A temperature sensor (28) measures the assemblies' temperature. This information is used for the purpose of compensating temperature effects.

The analysis cell is made of an aluminum tube with two gas fittings. This simple design without windows easily allows cleaning if the cell is polluted.

The chopper windows and those at the filter cell remain as only possibly polluted windows. All of them are accessible once the analysis cell is removed.

The filter cell (14 / 15) is made as two-tiered conus system, which is optimized to focus the beam to the active detector area.

High measuring ranges (up to 100 %) require an adaptor cell (10): The analysis cell is in this case the volume between output window of the adaptor cell and input window of the filter cell , covered by a distance ring (08).

2-1 Infrared Measurement (IR)

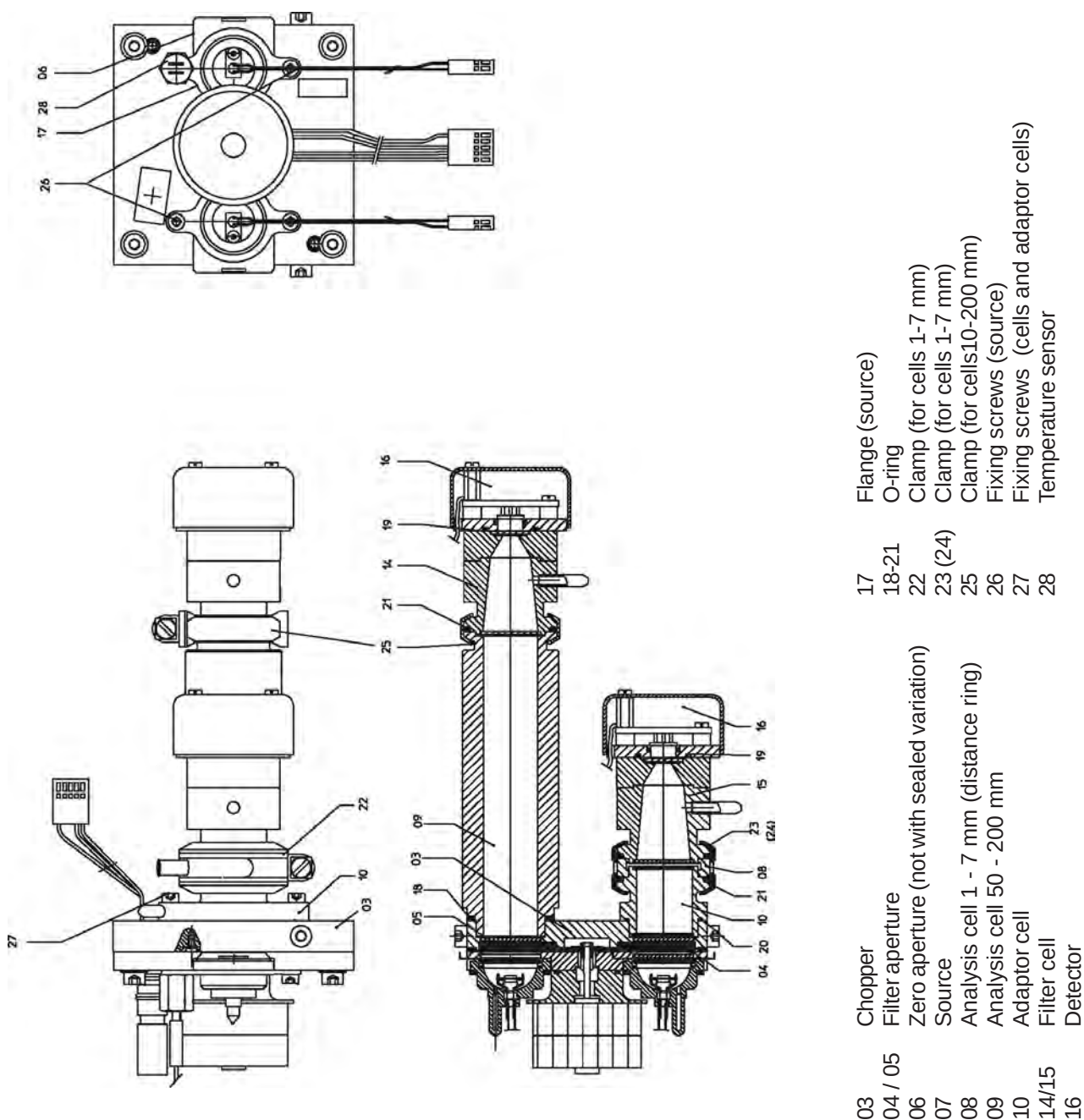


Fig. 2-4: Photometer Assembly with Pyroelectrical Detector, dual channel

2-1 Infrared Measurement (IR)

Photometer Assembly with Gas Detector

The design of a gas detector photometer assembly is on principle similar to the pyro detector design.

One main difference is the analysis cell, that now is lengthwise divided and both ends sealed with windows. This separates the inner volume into a sample side and a reference side. The sample side is passed by the sample gas, while the reference side is either filled with air, neutral gas or reference gas, depending on the application.

To avoid erroneous measurements absorbers may be installed at the reference side to suppress interfering CO₂ traces.

The filter cell is made as single-tier conus system.

The gas detector has separated preamplifier electronics, to which it is connected by a shielded cable.

For lower measuring ranges (long analysis cells) the preamplifier is fixed onto the analysis cell, whereas for higher ranges it is mounted on a special cover plate

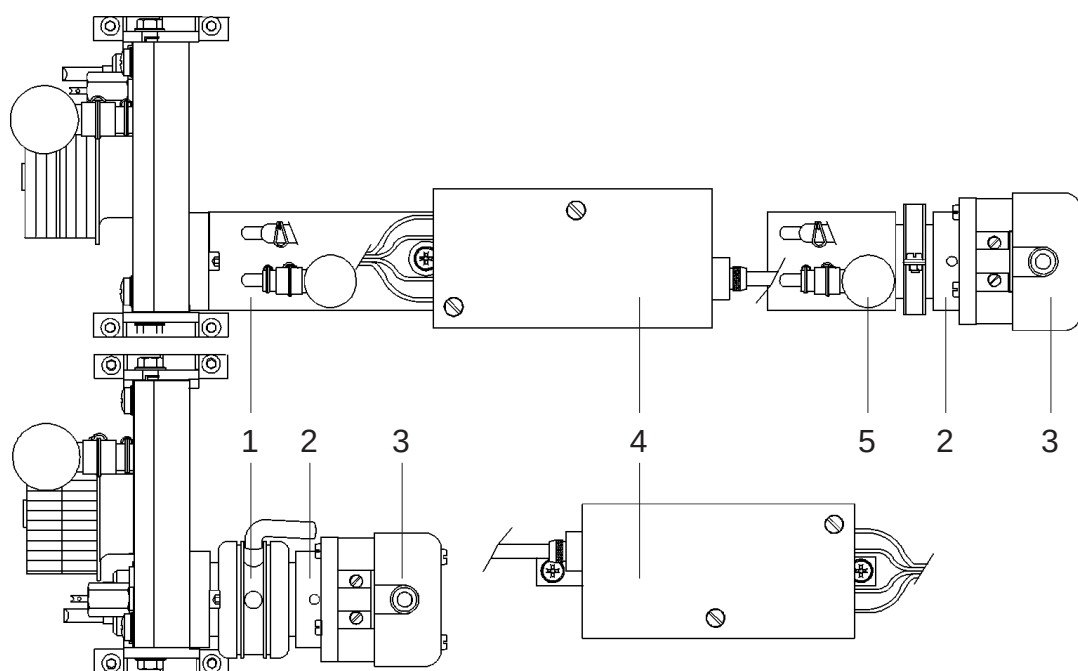


Fig. 2-5: Photometer Assembly with Gas Detector

[Lower example: **higher** measuring ranges, upper example: **lower** measuring ranges]

- | | | | |
|---|---------------|---|--------------|
| 1 | Analysis cell | 4 | Preamplifier |
| 2 | Filter cell | 5 | Absorber |
| 3 | Gas detector | | |

2-2 Oxygen Measurement

2-2 Oxygen Measurement

Two different principles are used for measuring Oxygen concentrations. The currently used principle is given by the channel code (sample gas designator) right beside the 7 segment display (see fig. 1-1):

% O₂ para. = paramagnetic sensor

% O₂ chem. = electrochemical sensor

2-2-1 Paramagnetic Measurement

Oxygen measurement is based on the paramagnetical characteristics of Oxygen molecules:

Two nitrogen filled quartz spheres (N₂ is not paramagnetic) are arranged in a dumbbell configuration and, hinged to a platinum wire, placed inside a cell. Fixed to the wire a small mirror reflects a light beam to a photo detector (fig. 2-6).

The measuring cell is placed inside an inhomogeneous magnetical field generated by a strong permanent magnet of specific design.

Oxygen molecules within the sample gas now due to their paramagnetic characteristics are deflected into the area of highest field strength. This generates different forces on both spheres and the resulting torque turns dumbbell and mirror out of the rest position. This generates a photodetector signal because the beam is deflected, too.

Initiated by the photodetector signal a preamplifier drives a compensation current through a loop surrounding the dumbbell to turn back the dumbbell into the rest position by effect of a magnetic field

So the current compensating the torque affecting the dumbbell is a direct measure for the oxygen concentration within the sample gas.

In addition to measuring cell, permanent magnet, electronics and enclosure the paramagnetic oxygen detector contains a temperature sensor and a heating element to hold the detector at approx. 55 °C.

Several variations are available including corrosion resistant, solvent resistant and/or intrinsically safe (for measuring flammable gases) versions.

Solvent Resistant Paramagnetic Oxygen Sensor: approved solvents (inclusive accompanying disturbing components)			
Component	Concentration	Component	Concentration
Acetone	0.1- 20%	i Butyr acid	0.1- 20%
Acrolein	0.1- 20%	i-Butyr aldehyd	0.1- 20%
Argon	0.1- 20%	i-Propylformiat	0.1- 20%
Aromatics	0.1- 20%	Isopropanol	0.1- 20%
Butadiene	0.1- 20%	Carbon dioxide	0.1- 20%
Butadiene-1	0.1- 20%	Methane	0.1- 20%
Butadiene-2	0.1- 20%	Methanol	0.1- 20%
C2H2	0.1- 20%	Methyl ethyl keton	0.1- 20%
C4H8	0.1- 20%	Methyl acetate	0.1- 20%
C5	0.1- 20%	Methyl kaptane	0.1- 20%
C6H12	0.1- 20%	n-Butane	0.1- 20%
CH3COOH	0.1- 20%	Propadiene	0.1- 20%
Cyclohexane	0.1- 20%	Propane	0.1- 20%
Cyclohexanon	0.1- 20%	Propene	0.1- 20%
Dimethyl sulfide	0.1- 20%	Propylene	0.1- 20%
Acetic acid	0.1- 20%	Propylen oxide	0.1- 20%
Ethane	0.1- 20%	Toluene	0.1- 20%
Ethanol	0.1- 20%	Vinyl acetate	0.1- 20%
Ethene	0.1- 20%	Vinyl acetylene	0.1- 20%
Ethylene	0.1- 20%	Hydrogen	0.1- 20%
Ethylene oxid	0.1- 20%	Xylene	0.1- 20%
Heptane	0.1- 20%	Cyclo hexane	0.1- 20%
Hexane	0.1- 20%		
Conditions			
· Single or summarized concentrations do not exceed 20 %			
· Gas passes gas cooler prior to entering the analyzer			
· Gas dew point at max. 5 °C			
Solvent resistant sensors are consumables!			

**Table 2-1: Solvent Resistant Sensor:
Approved solvents**

2-2 Oxygen Measurement

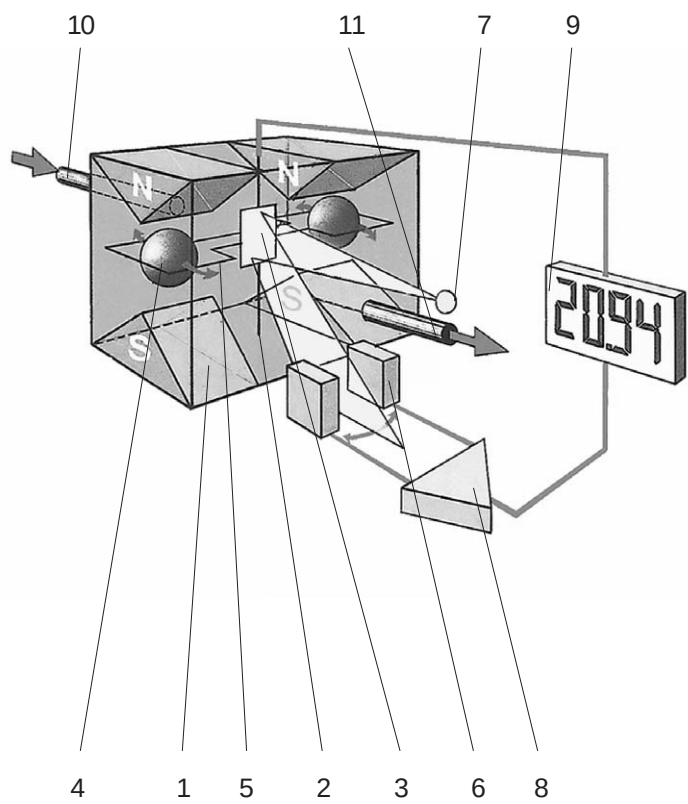


Fig. 2-6: Paramagnetic Oxygen Detector, assembly principle

- | | | | |
|---|-----------------|----|---------------|
| 1 | Permanentmagnet | 6 | Photodetector |
| 2 | Platinum wire | 7 | Light source |
| 3 | Mirror | 8 | Preamplifier |
| 4 | Glass ball | 9 | Display |
| 5 | Loop | 10 | Gas inlet |
| | | 11 | Gas outlet |

Medium affected Materials within Paramagnetic Oxygen Sensor			
Measuring cell type	Standard	Solvent resistant	Corrosion resistant (Chlorine, dry)
Case	SS 1.4571	SS 1.4572	SS 1.4573
Pole nucleus	Tantalum		
Mirror	Glass		
	Rhodium		
Tension band	Platinum alloy		
Loop wire	Platinum alloy		
Supporting wire	Platinum alloy		
Cylinder	Glass		
Cylinder bushing	Ceramics		
Dumbbell	Glass		
Taring	PP	Epoxy	Epoxy
Compound material	Plumb bob, Epoxy	Plumb bob, Epoxy	Epoxy
Seals	FPM	Kalrez	Kalrez

Table 2-2: Medium affected Materials within Paramagnetic Oxygen Sensor

2-2 Oxygen Measurement

2-2-2 Electrochemical Measurement

This sensor uses the principle of galvanic cells, fig. 2-7 shows the design.

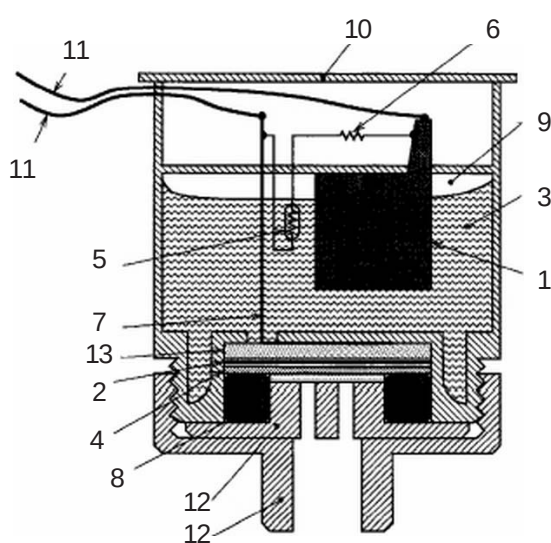


Fig. 2-7: Electrochemical Sensor assembly principle

- | | |
|----|-------------------------------|
| 1 | Anode (lead) |
| 2 | Kathode (Gold) |
| 3 | Electrolyte solution |
| 4 | Membrane |
| 5 | Thermistor |
| 6 | Resistance |
| 7 | Titanium wire |
| 8 | O-Ring |
| 9 | Pressure compensating volumes |
| 10 | Lid |
| 11 | Electrical connections |
| 12 | Lids |
| 13 | Current collector |

The electrochemical oxygen sensor's key components are a lead anode (1) and a gold cathode (2) surrounded by a special acid electrolyte (3).

The gold electrode is integrated solid with the membrane, which is a non-porous fluororesin membrane. Oxygen which barely diffuses through the membrane is electrochemically reduced on the gold electrode.

The temperature compensating thermistor and adjusting resistance are connected between the cathode and anode. The current generated by oxygen reduction is converted into a voltage by these resistances.

The value of the current flowing to the thermistor and resistance varies in proportion to the oxygen concentration of the measuring gases which contact the membrane. Therefore, the voltage at the terminal of the resistances is used for the sensor output to measure the oxygen concentration.

2-2 Oxygen Measurement

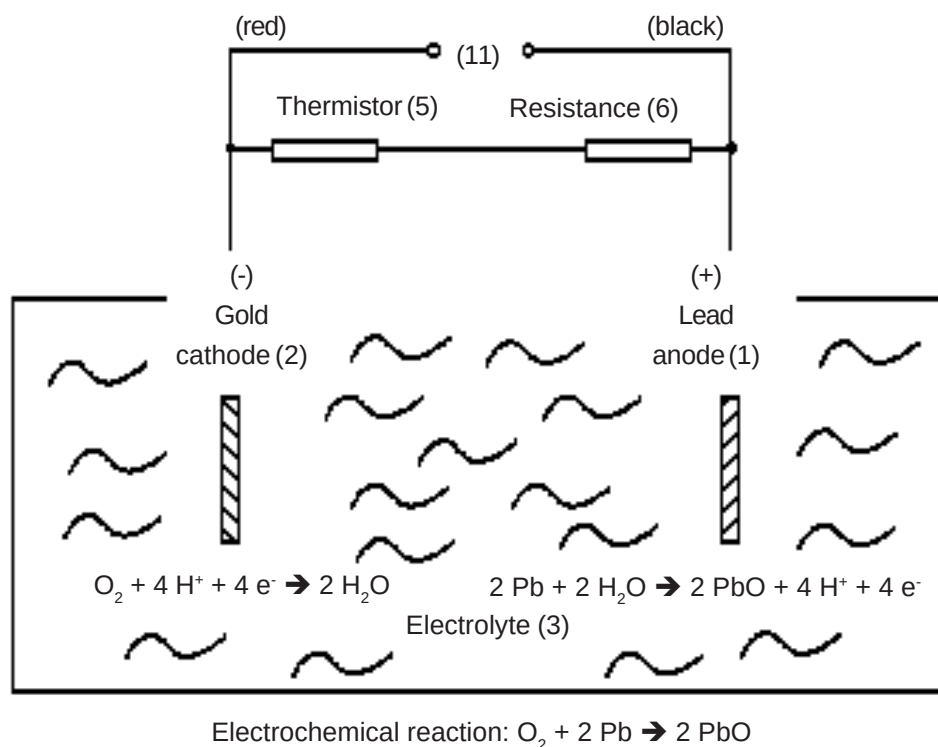


Fig. 2-8: Electrochemical reaction of Oxygen sensor

In consequence of its design the sensor's lifetime is limited and depends on theoretical designed life and oxygen concentration. The sensor output can be taken as a rough criterion for end of lifetime: The sensor is worn when the output in atmosphere is below 70 % of the initial output. The period till this can be calculated by

$$\text{Lifetime} = \frac{\text{designed life (hours)}}{O_2 \text{ concentration (\%)}}$$

The sensor's designed life under constant conditions of 20 °C is approx. **900,000 hrs.**

The lifetime at 21 % oxygen is then calculated to approx. **42,857 hrs, corresponding to approx. 5 years.**

Note!

The given values are for reference only! The expected lifetime is greatly affected by the temperature of the environment in which the sensor is used or stored. Increases or decreases in atmospheric pressure have the same effect as that by increases or decreases in oxygen concentration (Operation at 40 °C halves lifetime).

2-4 Measurement Specification

2-2-3 Special Hints on Oxygen Sensors

Paramagnetic Sensor

The table below shows how accompanying gases interfere the paramagnetic oxygen measurement.

If the concentration of such gases is already given at time of enquiry this interference may be taken into account during factory startup and thus minimized (option).

Cross Inferences for Paramagnetic Oxygen Measurement		
100 % Gas		Zero-level effect % O ₂
Nitrogen	N ₂	+0.00
Carbon Dioxide	CO ₂	-0.27
Hydrogen	H ₂	+0.24
Argon	Ar	-0.22
Neon	Ne	+0.13
Helium	He	+0.3
Carbon Monoxide	CO	+0.01
Methane	CH ₄	-0.2
Ethane	C ₂ H ₆	-0.46
Ethene	C ₂ H ₄	-0.26
Propane	C ₃ H ₈	-0.86
Propene	C ₃ H ₆	-0.55
Nitrogen Oxide	NO	+43.0
Nitrogen Dioxide	NO ₂	+28.0
Nitrous Oxide	N ₂ O	-0.2

Table 2-3: Paramagnetic Oxygen Measurement, cross interference by accompanying gases

Electrochemical Sensor

This sensor is not suitable for anorganic gases containing chlorene or flourene!

In addition is not suitable for sample gases containing ozone, H₂S (> 100 ppm) or NH₃ (> 20 ppm).

Due to the measuring principle the electrochemical oxygen cell requires a minimum internal consumption of oxygen (residual humidity avoids drying of the cell). Supplying cells continuously with dry sample gas of low grade oxygen concentration or with sample gas free of oxygen could result in a reversible detuning of O₂ sensitivity. The output signal will become instable, but response time remains constant.

For correct measurement the cell needs continuously to be supplied with concentrations of at least 0.1 Vol.-% O₂. We recommend to use the cells if need be in alternating mode, means to purge cells with conditioned (not dried, but dust removed) ambient air when measurement pauses.

If it is necessary to interrupt oxygen supply for several hours or days, the cell has to regenerate (supply cell for about one day with ambient air). Temporary flushing with nitrogen (N₂) for less than 1 h (e.g. for analyzer zeroing purpose) has no influence on measuring characteristics.

2-3 Thermal Conductivity Measurement

2-3 Thermal Conductivity Measurement

Thermal Conductivity Measurement primarily is used for measuring concentrations of hydrogen (H_2) and helium (He). These gases are characterized by their specific thermal conductivity, differing clearly from that of other gases (see table 2-1).

Thermal Conductivity of specific gases (100 Vol-%)		
Type of gas		λ in $\mu W / cm \text{ } ^\circ C$
		50 °C
Helium	He	1580
Neon	Ne	516
Argon	Ar	189
Krypton	Kr	102
Xenon	Xe	60
Radon	Rn	26
Hydrogen	H_2	1910
Oxygen	O_2	283
Chlorine	Cl_2	96,8
Sulfur Dioxide	SO_2	113
Nitrogen	N_2	277
Ammonia	NH_3	270
Carbon Dioxide	CO_2	184
Air	N_2 / O_2	276
Hydrochloric Acid	HCl	151
Carbon Monoxide	CO	267
Methane	CH_4	371
Butane	$C_4 H_{10}$	185

Table 2-4: Examples of Specific Thermal Conductivities

2-3-1 Principle of Operation

A Wheatstone bridge, made of 4 temperature sensitive resistors (PT 100 sensors), is surrounded by gas in a way that each 2 sensors are located in the measuring gas stream (R_M) and in a reference gas stream (R_R), see fig. 2-9. The bridge output signal (U_{br}) is adjusted to zero when in rest position (no gas flow).

By default the reference gas path is closed (not flown through by gas). When sample gas is supplied the sensors in the measuring gas path are cooled due to the thermal conductivity effect: The gas absorbs heat and carries it away from the sensors. This tunes the Wheatstone bridge and generates a signal proportional to the thermal conductivity.

Additional electronics linearizes and conditions this signal to provide usefull measuring values.

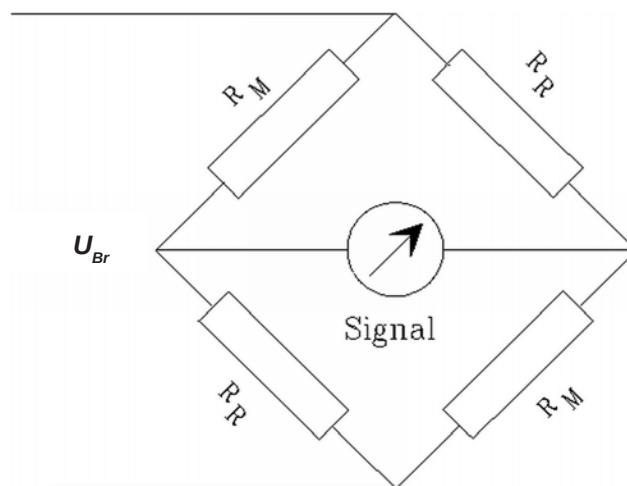


Fig. 2-9: Wheatstone Bridge

Depending on application it is possible to supply a reference gas to the bridge's reference side. The output signal in this case is proportional to the difference of the thermal conductivities of sample and reference gas.

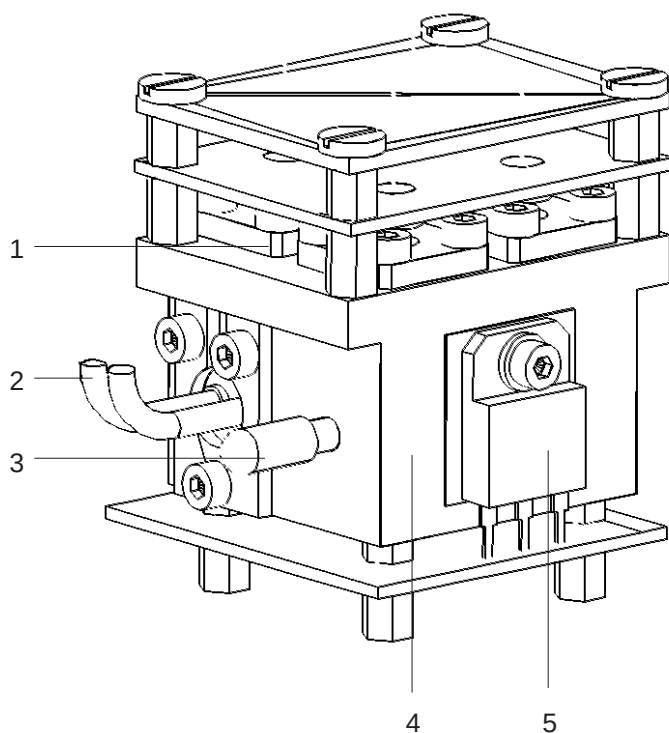
2-3 Thermal Conductivity Measurement

2-3-2 Technical Implementation

A block made either of aluminum, stainless steel or hastelloy contains two gas paths. Both, the volume of the block and the mass of the sensors have been minimized in order to obtain short response times. To suppress

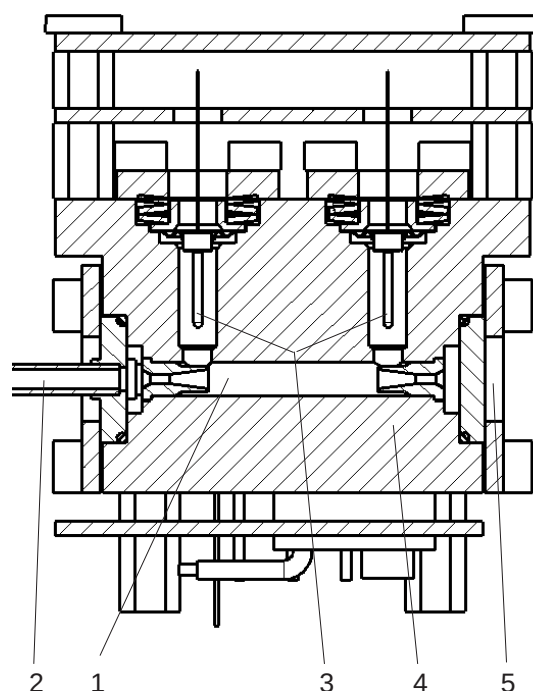
influences by changing ambient temperature the block is thermostatted and isolated against ambience.

The sensors are fully glass packaged to withstand aggressive gases.



**Fig. 2-10a: TC cell, exterior view ,
thermal isolation removed**

- 1 Cover for sensor
- 2 Sample gas inlet and output
- 3 Closed reference side
- 4 Metal block
- 5 Heater for thermostating



**Fig. 2-10b: TC cell,
sectional view**

- 1 Internal gas path
- 2 Sample gas inlet and output
- 3 PT 100 sensors
- 4 Metal block
- 5 Lid

2-4 Measurement Specification

2-4 Measurement Specification

Sample gas components and measuring ranges (standard configurations*)**Note!**

The following table shows generic data. The sample gas(es) and measuring ranges for your specific analyzer are given by the order acknowledgement and on the analyzer's name plate label resp. on the internal CAT analyzer module.

Gas component		Lowest measuring range	Highest measuring range
Carbon dioxide	CO ₂	0 - 100 ppm	0 - 100 %
Carbon monoxide	CO	0 - 200 ppm	0 - 100 %
Ethylene	C ₂ H ₄	0 - 400 ppm	0 - 100 %
Hexane	C ₆ H ₁₄	0 - 500 ppm	0 - 9,000 ppm
Methane	CH ₄	0 - 2 %	0 - 100%
n - Butane	C ₄ H ₁₀	0 - 800 ppm	0 - 100%
Oxygen (electrochemical)	O ₂	0 - 5 %	0 - 25 %***)
Oxygen (paramagnetic)	O ₂	0 - 1%	0 - 100 %
Propylene	C ₃ H ₆	0 - 4,000 ppm	0 - 100 %
Propane	C ₃ H ₈	0 - 1,000 ppm	0 - 100 %
Toluene	C ₇ H ₈	0 - 5,000 ppm	0 - 1.2 %
Water vapor **)	H ₂ O	0 - 1 %	0 - 3 %

*) Other components and configurations on request

**) Dew point below ambient temperature

***) Higher concentrations decrease sensor life time

Table 2-5: Gas components and Measuring Ranges, examples

2-4 Measurement Specification

Measurement Specifications (Values refer to lowest ranges. Standard ranges with better specs.)

	NDIR	Oxygen Sensor (PO ₂ and EO ₂)	Thermal Conductivity
Detection limit	≤ 1 % ^{1) 4)}	≤ 1 % ^{1) 4)}	≤ 2 % ^{1) 4)}
Linearity	≤ 1 % ^{1) 4)}	≤ 1 % ^{1) 4)}	≤ 1 % ^{1) 4)}
Zero-point drift	≤ 2 % per week ^{1) 4)}	≤ 2 % per week ^{1) 4)}	≤ 2 % per week ^{1) 4)}
Span (sensitivity) drift	≤ 1 % per week ^{1) 4)}	≤ 1 % per week ^{1) 4)}	≤ 1 % per week ^{1) 4)}
Repeatability	≤ 1 % ^{1) 4)}	≤ 1 % ^{1) 4)}	≤ 1 % ^{1) 4)}
Total response time (t ₉₀)	< 5 s ^{3) 5)}	< 5 s ^{3) 6)} / Approx. 12 s ^{3) 9)}	3 s ≤ t ₉₀ ≤ 20 s ^{3) 7)}
Permissible gas flow	0.2 - 1.5 l/min	0.2 - 1.0 l/min ⁶⁾ / 0.2 - 1.5 l/min ⁹⁾	0.2 - 1.0 l/min (± 0.1 l/min)
Influence of gas flow	-	≤ 1 % ^{1) 4)}	≤ 1 % ^{1) 4) 13)}
Max. pressure	≤ 1,500 hPa abs.	Atm. pres. ⁶⁾ / ≤ 1,500 hPa abs. ⁹⁾	≤ 1,500 hPa abs.
Influence of pressure	≤ 0.10 % per hPa ²⁾	≤ 0.10 % per hPa ²⁾	≤ 0.10 % per hPa ²⁾
(At constant temperature)	(≤ 0.15 % / hPa at CO ₂) ²⁾		
- With pressure compensation ⁸⁾	≤ 0.01 % per hPa ²⁾ (≤ 0.015 % / hPa at CO ₂) ²⁾	≤ 0.01 % per hPa ²⁾	≤ 0.01 % per hPa ²⁾
Permissible ambient temperature	+ 5 °C to + 40 °C ¹⁰⁾	+ 5 °C to + 40 °C ¹⁰⁾	+ 5 °C to + 40 °C ¹⁰⁾
Influence of temperature			
(at constant pressure)			
- On zero point	≤ 1 % per 10 K ¹⁾	≤ 1 % per 10 K ¹⁾	≤ 1 % per 10 K in 1 h ¹⁾
- On span (sensitivity)	≤ 5 % (+ 5 to + 40 °C) ^{1) 11)}	≤ 1 % per 10 K ¹⁾	≤ 2 % per 10 K in 1 h ¹⁾
Thermostat control	-	Approx. 55 °C ^{6) 12)} / None ⁹⁾	Approx. 75 °C ¹²⁾
Heating-up time	Approx. 15 to 50 minutes ⁵⁾	Approx. 50 minutes ⁶⁾	Approx. 15 minutes

- 1) Related to full scale at system parameter
END = final value set in our factory and OFS = 0
2) Related to measuring value
3) From gas analyzer inlet at 1.0 l/min gas flow
(electr. = 2 s)

- 4) Constant pressure and temperature
5) Dependent on integrated photometer bench
6) Paramagnetic oxygen measurement (PO₂)
7) dependent on measurement range
8) Pressure sensor required

- 9) Electrochemical oxygen measurement (EO₂),
not for use with sample gas containing FCHC's
10) Higher ambient temperatures (45 °C) on request
11) Starting from 20 °C (to + 5 °C or to + 40 °C)
12) Sensor / cell only
13) Gas flow constant within ± 0.1 l/min

Table 2-6: Measurement Specifications

Chapter 3 Installation

This chapter describes in detail how to install the different analyzer variations.

3-1 Abstract

Carefully examine the shipping carton and contents for signs of damage. Immediately notify the shipping carrier if the carton or its contents are damaged. Retain the carton and packing material until the instrument is operational.

WARNING

Electrical shock hazard!



Prior to connecting the analyzer to mains ensure all safety instructions as given in the appropriate chapter at the beginning of this manual and in the following analyzer referred sections are read and understood!



Installation area has to be clean, free from moisture, excessive vibration and frost-protected. Take care to meet the permissible ambient temperatures as given in the technical data section!

Instruments must not be exposed to direct sunlight, fluorescent lamps nor sources of heat.

For outdoor installation it is recommended to mount the instruments into a cabinet. At least sheltering against rain is required.

Do not cover venting openings and take care to mount the instrument in a distance to walls not affecting venting.

3-1 Installation - General

To stay in compliance with regulations regarding electromagnetic compatibility it is recommended to use only shielded cables, as optionally available from Emerson Process Management or equivalent. Customer has to take care that the shield is connected in proper way (see section 3-5). Shield and signal connector enclosure need to be conductively connected, submin-d-plugs and sockets must be screwed on the analyzer.

Using external submin-d-to-terminal adaptor elements (option) affects electromagnetic compatibility. In this case the customer has to take measures to stay in compliance and has to declare conformity, when required by legislation (e.g. European EMC Directive).



Analyzers requiring DC power supply must be supplied by a power supply unit UPS 01 T (option) or equivalent!

(For specifications of power supply units see section 1-7-2-5, page 1-60)

3-2 Gas Conditioning

To ensure trouble-free analyzer operation one has to attach great importance to gas conditioning:

Furthermore the gases must be

- dry
- free of dust
- free of aggressive components affecting gas path materials (e.g. by corrosion).

If moisture can not be avoided take care that the gas' dew point is at least 10 °C below ambient temperature to avoid condensation within the gas path.

Pressure and flow have to be within the limits given in the technical data section (see section 1-7, page 1-34).



All gases must be conditioned before supplying!

When supplying corrosive gases ensure that gas path components are not affected!



Flammable gases must not be supplied without additional protective measures!

It is prohibited to supply explosive gases!

3-3 Installation - Gas Connections

3-3 Gas Connections

WARNING



TOXIC GAS HAZARDS!



Take care that all external gas lines are connected as described and are tight to avoid leaks!



Improperly connected gas lines may cause explosion or death!

Exhaust may contain hydrocarbon and other toxic components (e.g. carbon monoxide)! Carbon monoxide is highly toxic and can cause headache, nausea, loss of consciousness, and death. Avoid inhalation of exhaust!

CAUTION

Do not interchange gas inlet and outlet! All gases must be conditioned before supplying! When supplying corrosive gases ensure that gas path components are not affected!

Max. permissible gas pressure: 1,500 hPa, except instruments with integrated valve blocks (see page 3-4) and/or paramagnetic Oxygen sensor (see table 2-5)!

Exhaust lines must be installed in a descending way, need to be pressureless, frost-protected and in compliance with applicable legislative requirements!

The number of gas fittings as well as their assignment vary depending on analyzer model and selected options.

All gas fittings are labeled and are normally located at the analyzers rear panel, in case of BINOS® 100 F and CAT 100 they are located at the bottom side.

When it is necessary to open gas paths seal the analyzer's gas fittings by using PVC caps to avoid pollution of the internal gas path by moisture, dust, etc.

3-3 Installation - Gas Fittings

The analyzer should be mounted near the sample source to minimize sample transport time. A sample pump may be used to decrease response time, whereat the analyzer

is either operated in bypass mode or protected by an overpressure valve against too high flow and pressure (see fig. 3-1).

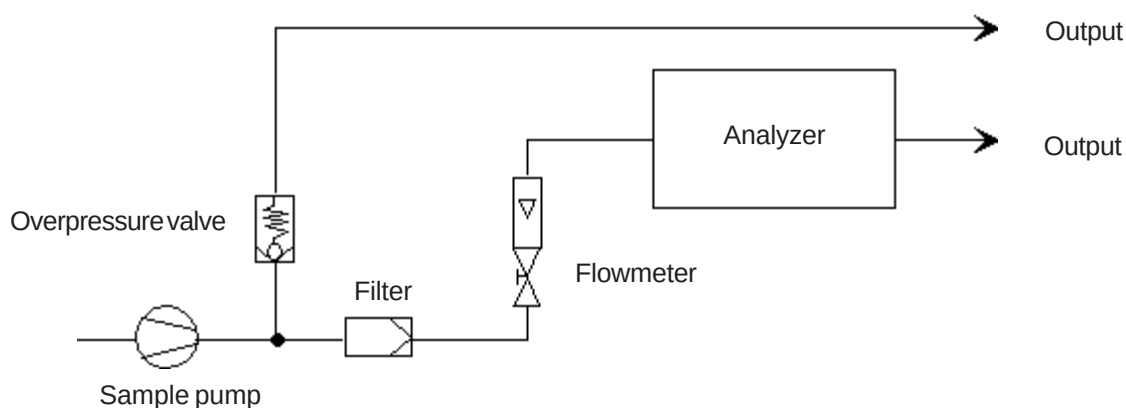


Fig. 3-1: Bypass Mode Installation

Internal Solenoid Valve Block (Option for BINOS® 100 2M / 100 F / CAT 100)

Supply overpressure for all gases is limited to 50 to 500 hPa when an analyzer is equipped with an internal solenoid valve block.

To show the valve status a front panel LED is activated when a valve is open (see fig. 1-4, item 3).

Installation instructions for BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

BINOS® 100 2M

BINOS® 100 F

CAT 100

Hints on wiring signal inputs and outputs

 *Page 3-5*

 *Page 3-9*

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 *Page 3-20*

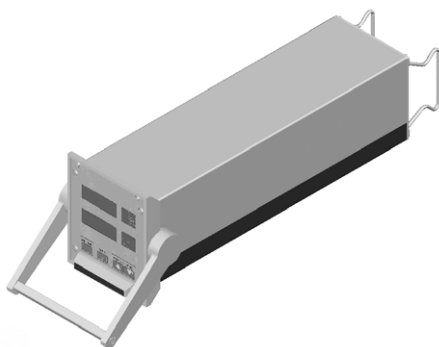
 *Page 3-27*

3-4 Installation - Analyzer Specific Instructions

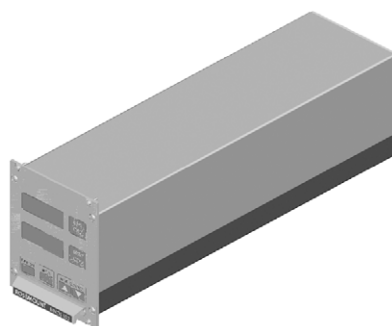
3-4 Analyzer Specific Instructions

3-4-1 1/4 19" Instruments

The analyzer variations **BINOS® 100 (M)**, **OXYNOS® 100**, **HYDROS® 100**, configured as tabletop analyzer provide a handle attached to a front panel frame. The instrument is intended for horizontal or slightly angular (utilizing the handle as support) orientation during operation.



Without front panel frame and handle the analyzers may be mounted into a rack (rack mount version). Four screws located at the front panel are used to fix the analyzer in the rack.



These analyzers need DC 24 V power supply utilizing an external power supply unit. They provide a three pole power input connector located at the rear panel.



The analyzers do not provide a power switch and are operable when connected to power.

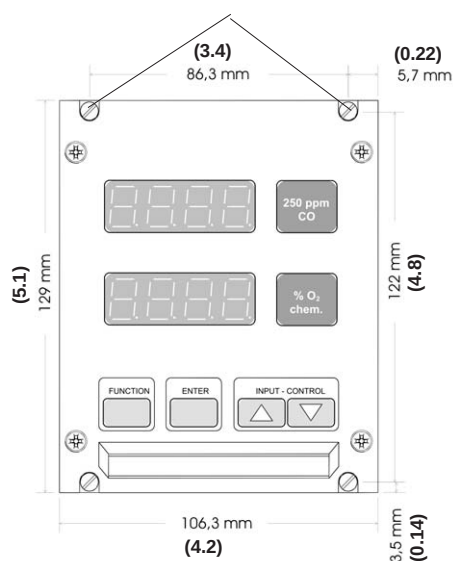


It is recommended to use original Emerson Process Management power supply units and supply cords. If a supplement is used take care of correct pin assignment, as given on the analyzer's rear panel!

Output signals and data are accessible at the rear panel via 9 pole submin-d connectors. Depending on analyzer variation not all signals are available. For information about pin-assignments see section 1-7-2.

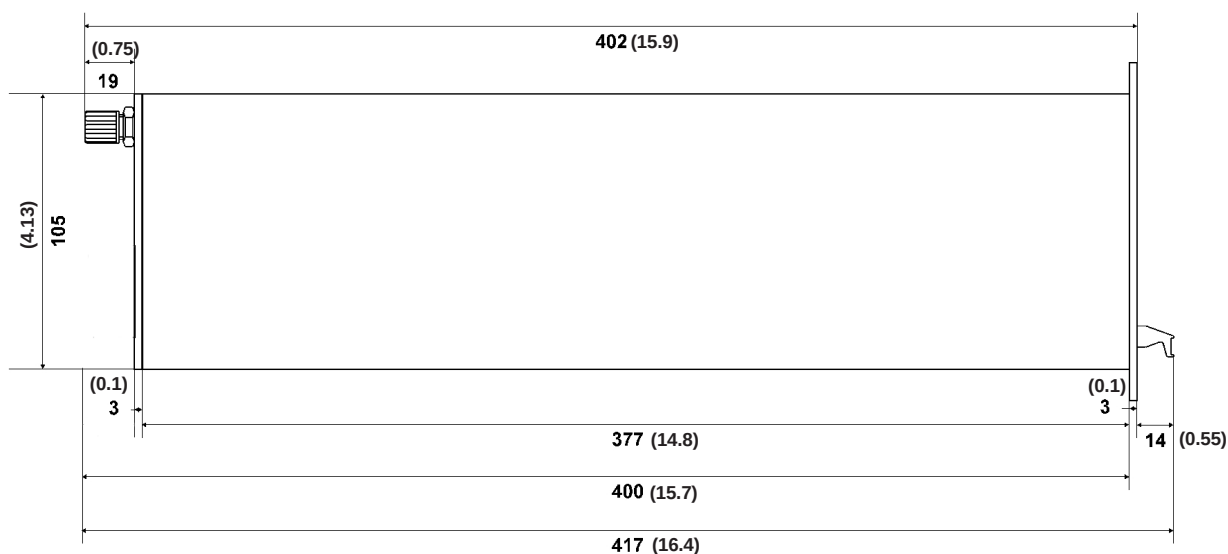
3-4-1 Installation - BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

Screws for rack mounting,
2 x 2 pcs.; M2.5



Front panel 1/4 19" housing

Darstellung/Längenangaben ohne 24 VDC Winkelstecker(+ ca. 30 mm)



**Fig. 3-2: BINOS® 100 (M), OXYNOS® 100, HYDROS® 100,
Dimensions [mm (inch)], w/o front panel frame & handle**

3-4-1 Installation - BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

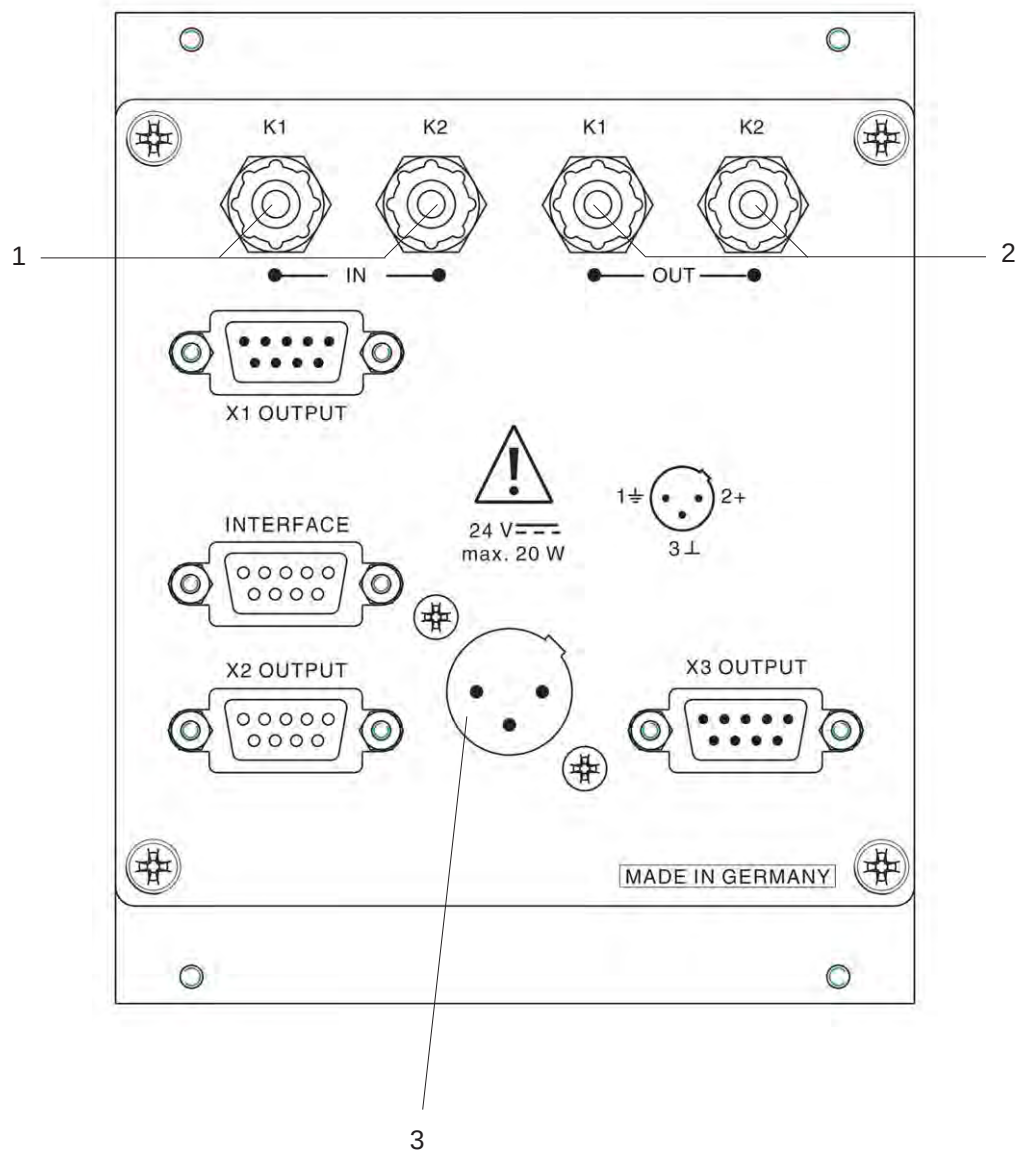


Fig. 3-3: BINOS® 100 (M), OXYNOS® 100 (EO₂), HYDROS® 100 rear panel

- 1 Gas inlets (K1 - channel 1 / input with serial tubing); K2 - channel 2)
- 2 Gas outlets (K1 - channel 1 / output with serial tubing); K2 - channel 2)
- 3 24 V DC power input connector

3-4-1 Installation - BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

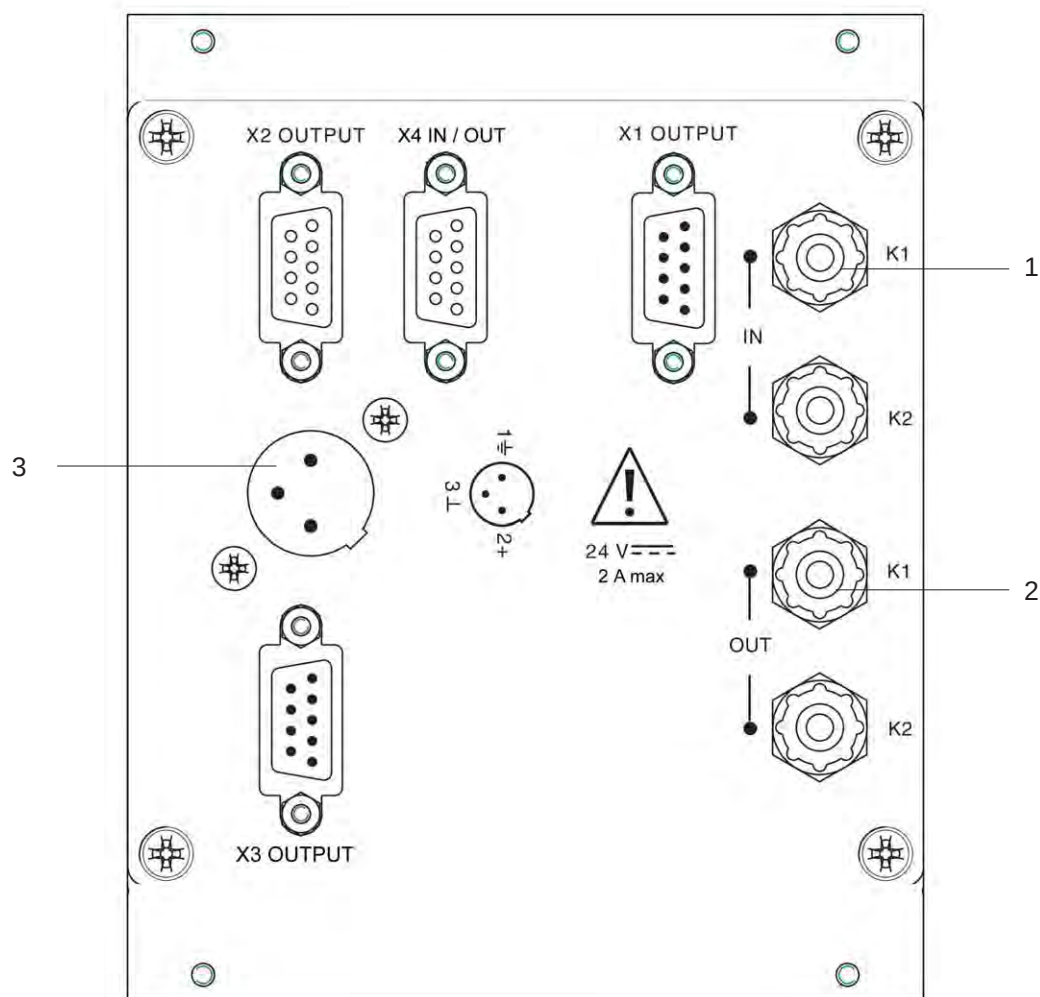


Fig. 3-4: OXYNOS® 100 (PO₂), rear panel

- 1 Gas inlet
- 2 Gas outlet
- 3 24 V DC power supply connector

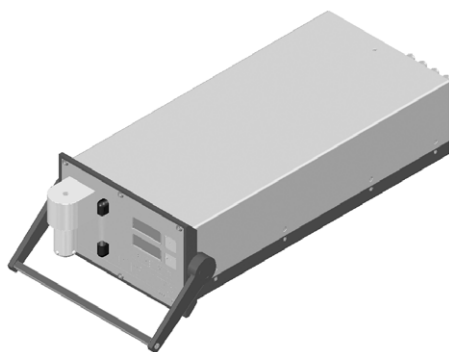


*Continue with section 3-5,
page 3-27*

3-4-2 Installation - BINOS® 100 2M

3-4-2 1/2 19" Instruments

The analyzer variation BINOS® 100 2M, configured as tabletop analyzer provides a handle attached to a front panel frame. The instrument is intended for horizontal or slightly angular (utilizing the handle as support) orientation during operation.



Without front panel frame and handle the analyzer may be mounted into a rack (rack mount version). Six screws located at the front panel are used to fix the analyzer in the rack.



Depending on the internal configuration the BINOS® 100 2M analyzer is either AC operated via an internal power supply unit or 24 V DC operated by an external power supply unit.



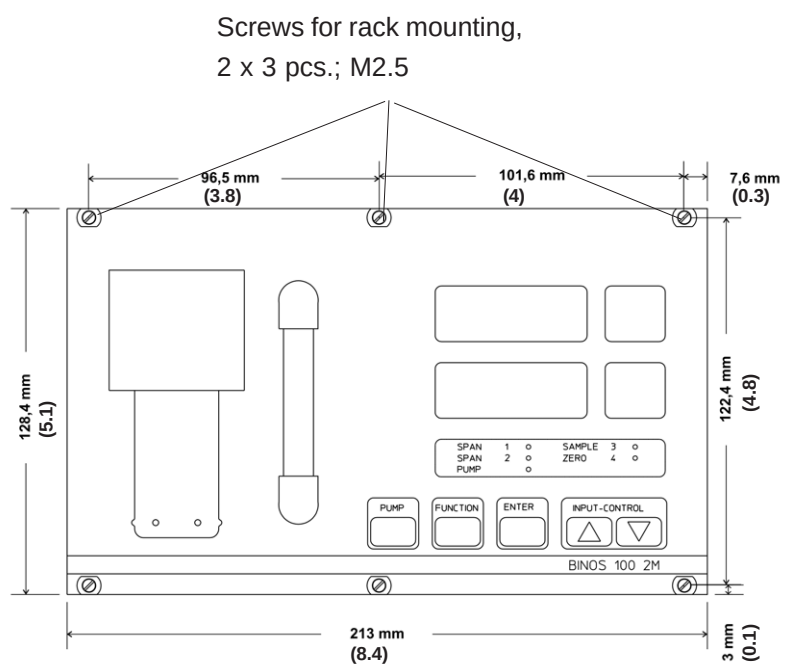
It is recommended to use original Emerson Process Management power supply units and supply cords. If a supplement is used take care of correct pin assignment, as given on the analyzer's rear panel!



The analyzer does not provide a power switch and is operable when connected to power.

The internal power supply unit provides a 24 V DC power supply output for supplying external loads, accessible from the rear panel. Output signals and data are accessible at the rear panel via 9 pole submin-d connectors. Depending on analyzer variation not all signals are available. For information about pin-assignments see section 1-7-2.

3-4-2 Installation - BINOS® 100 2M



Front panel 1/2 19" housing

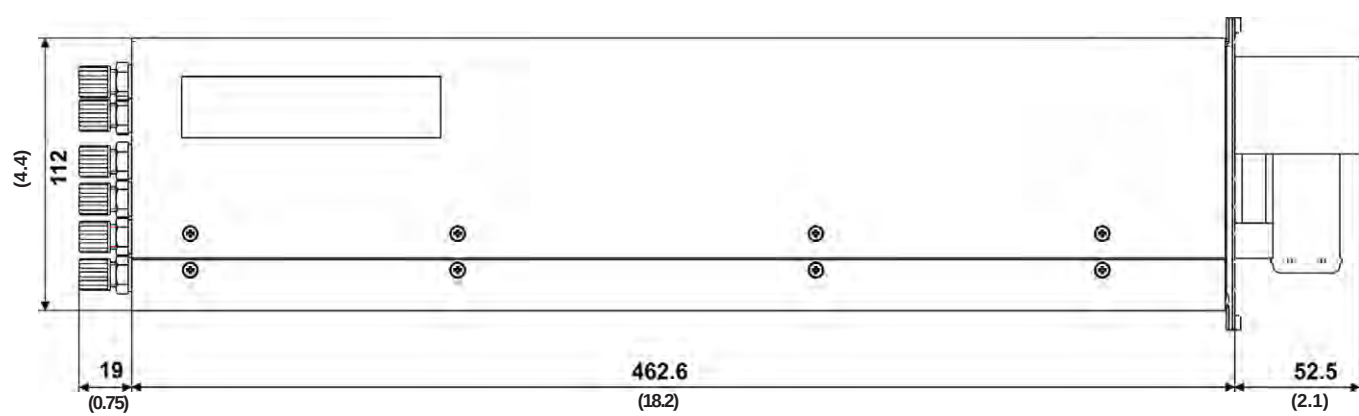


Fig. 3-5: BINOS® 100 2M
Dimensions [mm (inch)], w/o front panel frame & handle

3-4-2 Installation - BINOS® 100 2M

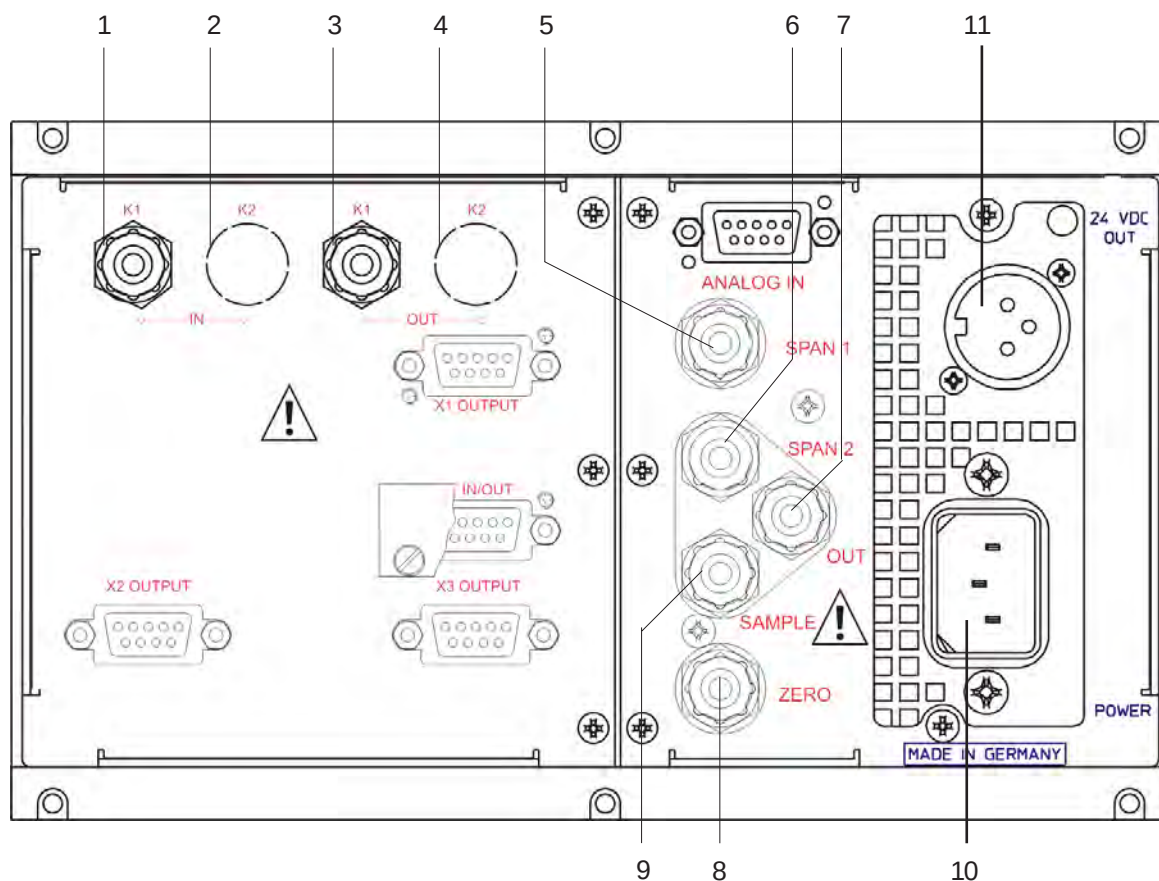
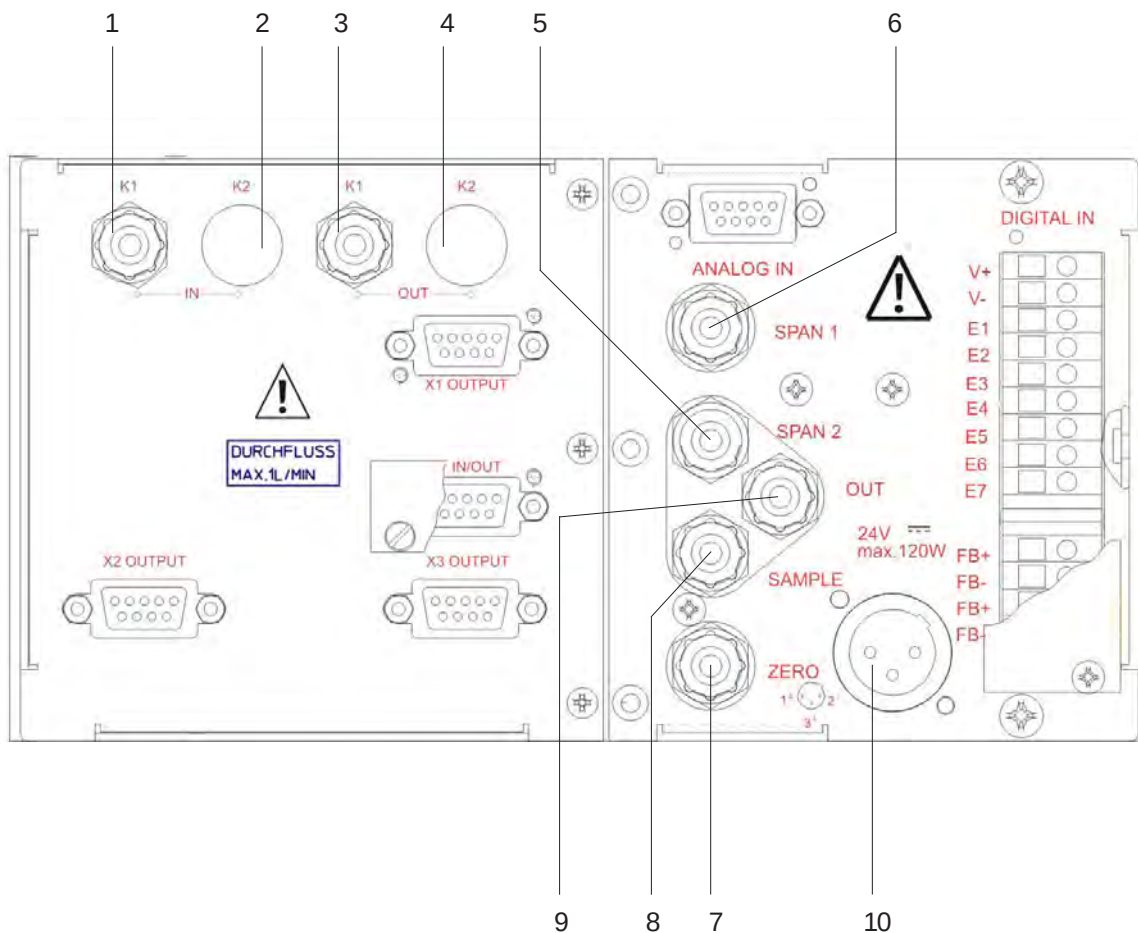


Fig. 3-6: BINOS® 100 2M, variation a (with internal power supply)

Rear panel with all options

- | | | | |
|---|-------------------------------------|----|--|
| 1 | Gas inlet channel 1 | 8 | „Solenoid valves“: Zero gas inlet |
| 2 | Gas inlet channel 2 (option) | 9 | „Solenoid valves“: Sample gas inlet |
| 3 | Gas outlet channel 1 | 10 | Mains input appliance |
| 4 | Gas outlet channel 2 (option) | 11 | 24 V DC output socket for supplying external loads |
| 5 | „Solenoid valves“: Span gas 1 input | | (2 A max., see section 1-7-2-2, page 1-42) |
| 6 | „Solenoid valves“: Span gas 2 input | | |
| 7 | „Solenoid valves“: Common output | | |

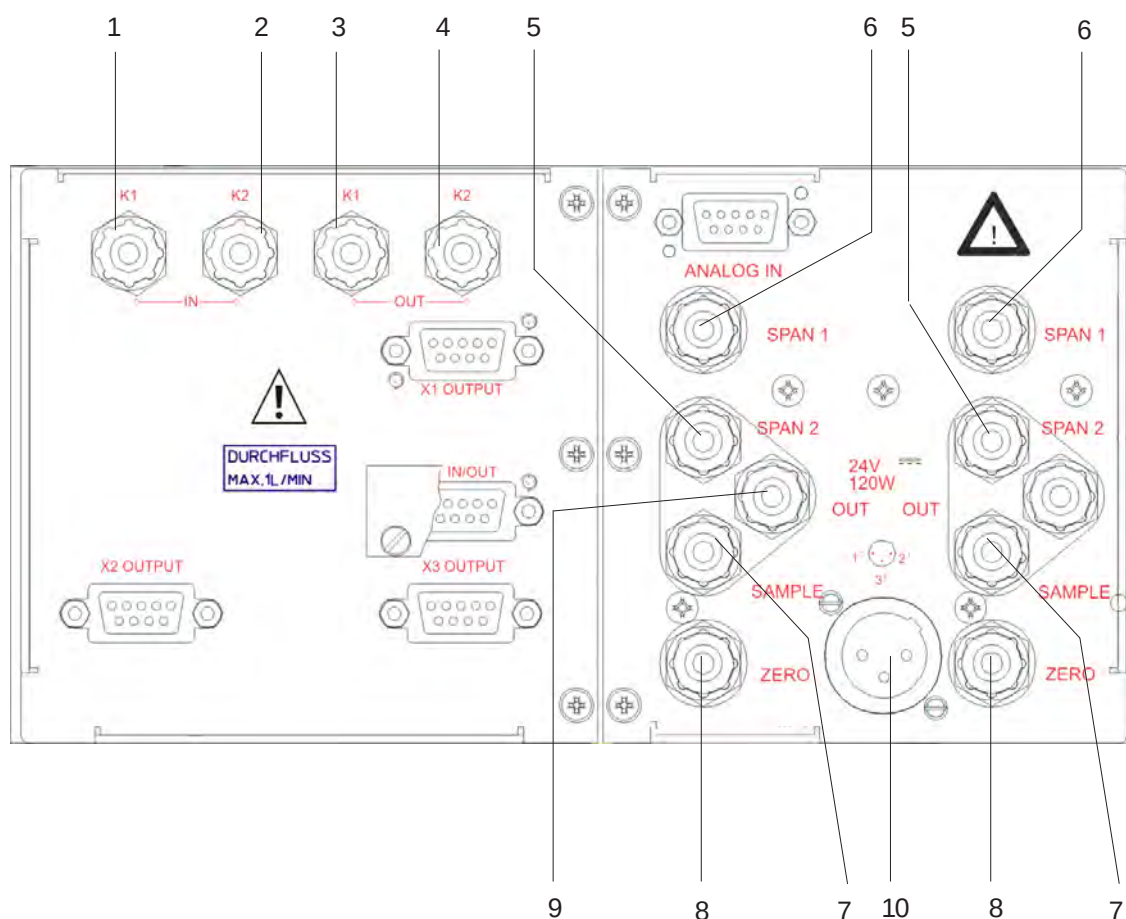
3-4-2 Installation - BINOS® 100 2M



**Fig. 3-7: BINOS® 100 2M, variation b (with external power supply),
Rear panel with all options**

- | | | | |
|---|-------------------------------------|----|-------------------------------------|
| 1 | Gas inlet channel 1 | 6 | „Solenoid valves“: Span gas 1 input |
| 2 | Gas inlet channel 2 (option) | 7 | „Solenoid valves“: Zero gas inlet |
| 3 | Gas outlet channel 1 | 8 | „Solenoid valves“: Sample gas inlet |
| 4 | Gas outlet channel 2 (option) | 9 | „Solenoid valves“: Common output |
| 5 | „Solenoid valves“: Span gas 2 input | 10 | DC power input connector |

3-4-2 Installation - BINOS® 100 2M



**Fig. 3-8: BINOS® 100 2M , special variation,
Rear panel with all options**

- | | | | |
|---|-------------------------------------|----|-------------------------------------|
| 1 | Gas inlet channel 1 | 6 | „Solenoid valves“: Span gas 1 input |
| 2 | Gas inlet channel 2 | 7 | „Solenoid valves“: Sample gas inlet |
| 3 | Gas outlet channel 1 | 8 | „Solenoid valves“: Zero gas inlet |
| 4 | Gas outlet channel 2 | 9 | „Solenoid valves“: Common output |
| 5 | „Solenoid valves“: Span gas 2 input | 10 | DC power input connector |



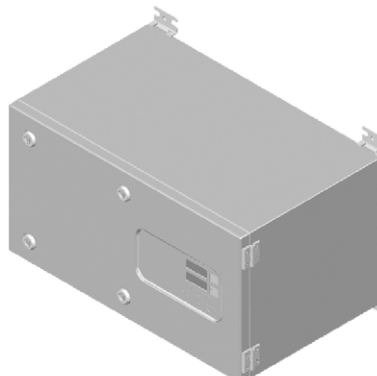
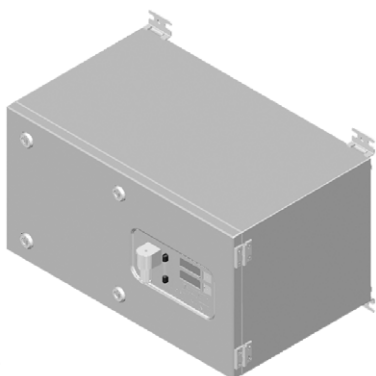
Continue with section 3-5,
page 3-27

3-4-3 Installation - BINOS® 100 F

3-4-3 BINOS® 100 F

The BINOS® 100 F variation is designed for wall mounting and is available in two different versions: One version providing external flow-meter and fine dust filter at the front panel is intended to be installed at protected locations whereas the version without these options may be outdoor installed.

In any case the restrictions for ambient temperatures have to be taken into account, see section 1-7-1, page 1-34!



CAUTION

BINOS® 100 F -- HEAVY INSTRUMENT!

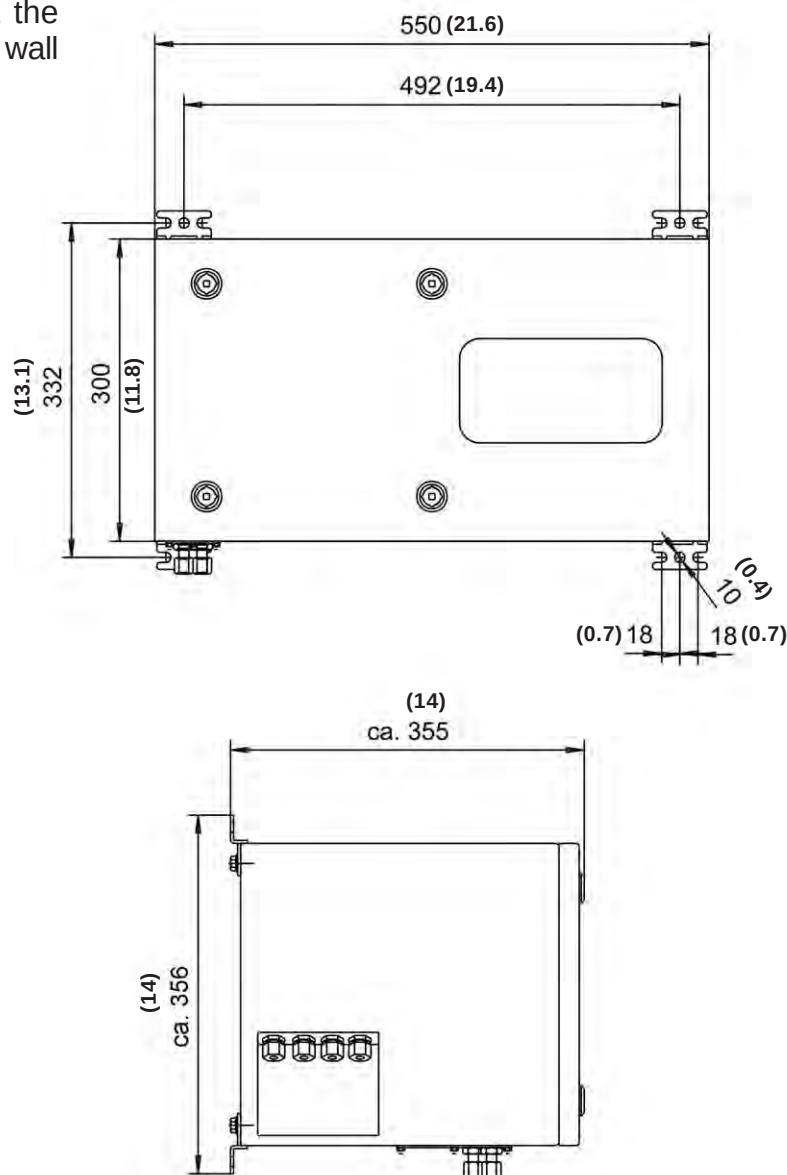


BINOS® 100 F analyzers intended to be wall mounted / outdoor installed weigh up to 30 kg, depending on selected configuration and options!

Analyzer lifting and transportation only by at least two persons or by appropriate tools!

3-4-3 Installation - BINOS® 100 F

Utilizing the four mounting brackets at the analyzer's rear side the instrument can be wall mounted.



CAUTION

Take care to use anchors and bolts specified to be used for the weight of the units!

Take care the wall the unit is intended to be installed at is solid and stable to hold the unit!

Fig. 3-9: BINOS® 100 F
Dimensions [mm (inch)]

3-4-3 Installation - BINOS® 100 F

WARNING**ELECTRICAL SHOCK HAZARD!**

Installation and connecting mains and signal cables are subject to qualified personnel only taking into account all applicable standards and legislative requirements!

Failure to follow may cause warranty invalidation, property damage and/or personal injury or death! Mains and signal cables need to be connected to internal screw terminals requiring to work at open housing near life parts!

Installation of this instrument is subject to qualified personnel only, familiar with the resulting potential risks!

BINOS® 100 F gas analyzers do not provide a mains switch! A mains switch or circuit breaker has to be provided in the building installation. This switch has to be installed near by analyzer, must be easily operator accessible and has to be assigned as disconnecter for the analyzer.

The analyzer provides a protective earth terminal. To prevent electrical shock hazards the instrument must be connected to a protective earth. Therefore the instrument has to be connected to mains by using a three wire mains cable with earth conductor!

Any interruption of the earth connector inside or outside the instrument or disconnecting the earth terminal may cause potential electrical shock hazard!

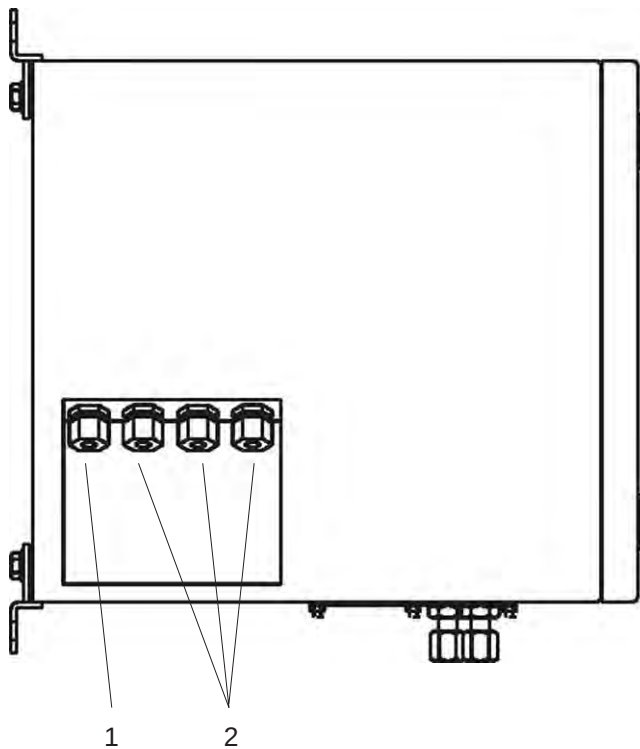
The analyzer does not provide a mains switch and is operable when connected to power.

- Cables entering the analyzer housing should be kept as short as possible inside the enclosure.
- The cable glands are suitable to fix cables with outer diameters of 7 to 12 mm. Special adaptors are available for fixing other or multiple cables in one gland.
- Mains terminals are designed to connect wires up to a cross section of up to 2,5 mm².
- To ensure electromagnetic compatibility it is recommended to use shielded signal cables only!
Take into account the installation instructions for cable glands (page 3-19)!

3-4-3 Installation - BINOS® 100 F

Mains Connection

Insert mains cord into the housing utilizing the cable gland marked in fig. 3-10. Take into account schematic diagram 4.300-5109 (see annex) and connect mains to the terminals located at the inner left side of the housing (fig. 3-11).



**Fig. 3-10: BINOS® 100 F ,
View at cable glands**

- 1 Cable gland for mains cord
- 2 Cable gland for signal lines (up to 3, depending on configuration)

3-4-3 Installation - BINOS® 100 F

The analyzer provides an integrated multi range power supply unit with voltage selector switch, to be set to the appropriate range to meet the nominal voltage (115 V / 230 V) at installation site. The switch is accessible through a safety

barrier when the analyzer's front door is open (fig. 3-11, showing 10 A power supply unit version while a 5 A version is used with BINOS® 100 F).

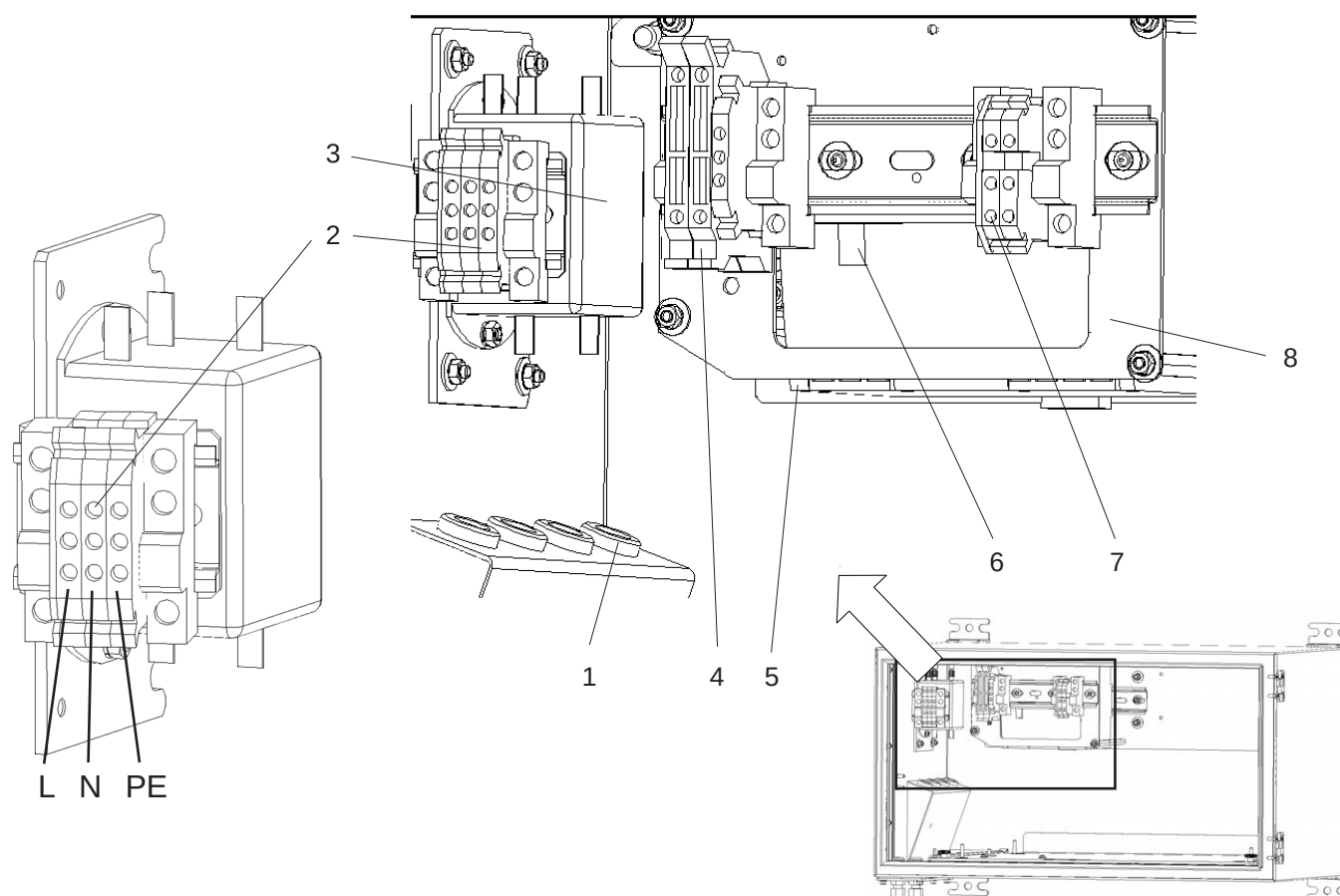


Fig. 3-11: BINOS® 100 F
Power supply and power distribution

- | | |
|------------------------------|--|
| 1 Cable glands | 5 Power supply unit (here: 10 A version) |
| 2 Power input terminals X100 | 6 Voltage selector switch |
| 3 Mains EMC filter | 7 DC 24 V terminals |
| 4 Power input fuses F100 | 8 Safety barrier |

3-4-3 Installation - BINOS® 100 F

Signal and data lines have to be connected to adaptor elements located at the analyzer's inner left side.

These adaptors provide a submin-d connector (instrument side) and screw terminals (out going side) to connect external cables. Inside the analyzer signal distribution utilizes ribbon cable connections between adaptors and BKS (main board) submin-d connectors. Adaptor terminals are one-to-one connected to the submin-d pin of same designation: Terminal 1 is connected to pin 1, terminal 2 to pin 2 etc. (fig. 3-12).



Cables for external data processing must be double insulated for mains voltage when used inside the instrument!

If double insulation not possible signal cables inside the analyzer must be installed in a way that a distance of at least 5 mm is ensured permanently (e.g. by utilizing cable ties).

The number of installed adaptor elements varies depending on analyzer configuration, see schematic 4.300-5109 in annex.

For detailed pin designations see section 1-7-2-3.

Cable gland assembly instructions



1. Strip cable insulation
2. Uncover shield mesh
3. Feed cable through gland nut and into fixing element
4. Put the shielding mesh over the element the way that it covers the o-ring 2 mm.
5. Stick the fixing element into the neck and fix the gland.

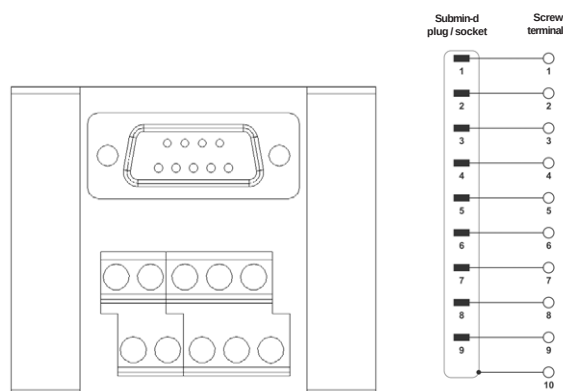
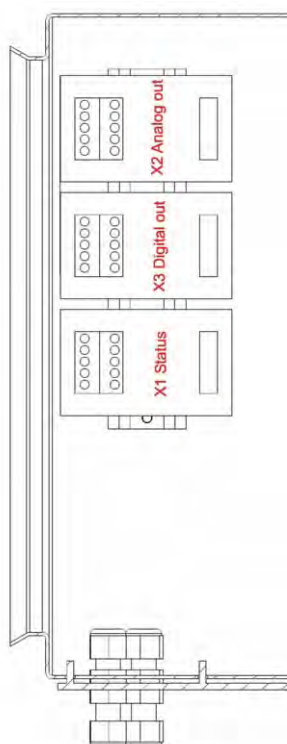
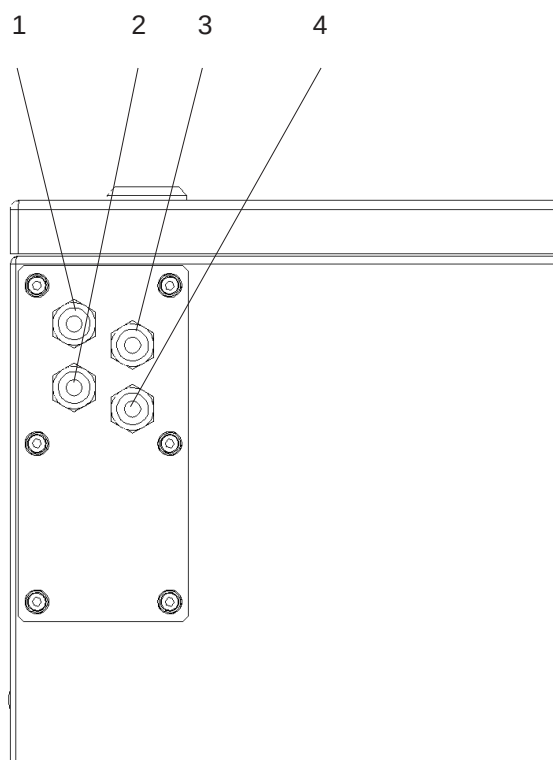


Fig. 3-12: Adaptor element, 9 pole

3-4-3 Installation - BINOS® 100 F



**Fig. 3-13a: BINOS® 100 F ,
Location of adaptor elements**



**Fig. 3-13b: BINOS® 100 F ,
Gas fittings at analyzer's bottom side
(other configurations possible,
see labels at instrument)**

- 1 Gas inlet channel 1
- 2 Gas inlet channel 2
- 3 Gas outlet channel 1
- 4 Gas outlet channel 2

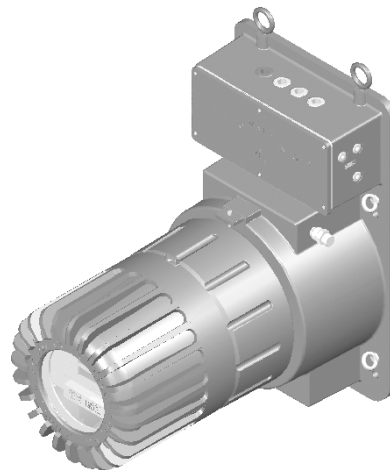


*Continue with section 3-5,
page 3-27*

3-4-4 Installation - CAT 100

3-4-4 CAT 100

CAT 100 analyzers are designed for wall mounting and intended to be outdoor installed taking into account the ambient temperature limits as given in the technical data section of this manual (s. section 1-7-1).



ACHTUNG

CAT 100 -- HEAVY INSTRUMENT!



CAT 100 analyzers, intended to be wall mounted / outdoor installed weigh up to 70 kg, depending on selected configuration and options!

Analyzer lifting and transportation only by at least two persons or by appropriate tools!

The instrument provides two eye bolts located at the rear side's top edge to crane the analyzer!

3-4-4 Installation - CAT 100

WARNING**ELECTRICAL SHOCK HAZARD!**

Installation and connecting mains and signal cables are subject to qualified personnel only taking into account all applicable standards and legislative requirements!



Failure to follow may cause warranty invalidation, property damage and/or personal injury or death! Mains and signal cables need to be connected to internal screw terminals requiring to work at open housing near live parts!

Installation of this instrument is subject to qualified personnel only, familiar with the resulting potential risks!



CAT 100 gas analyzers do not provide a mains switch! A mains switch or circuit breaker has to be provided in the building installation. This switch has to be installed near by analyzer, must be easily operator accessible and has to be assigned as disconnecter for the analyzer.

The junction box must be protected by fuse supply (10 A fuse) which has a breaking capacity adjusted to the short circuit of the equipment.

The analyzer provides a protective earth terminal. To prevent electrical shock hazards the instrument must be connected to a protective earth. Therefore the instrument has to be connected to mains by using a three wire mains cable with earth conductor!

Any interruption of the earth connector inside or outside the instrument or disconnecting the earth terminal may cause potential electrical shock hazard!

The analyzer does not provide a mains switch and is operable when connected to power.

3-4-4 Installation - CAT 100



To ensure protection against water and dust use cable glands with minimum IP 65 classification!

The junction box provides at its left side one threat M20x1.5 and at its right side up to three threats M16x1.5 for installing cable glands.

The left gland is the mains power cord entry, the right side glands for signal cables.

Not used threats must be sealed by approved sealing plugs (factory delivery condition).

This analyzer model is supplied by an international wide range power supply, automatically adapting to the voltage at site (s. sect. 1-7-2-4).

All cables (mains power cord and signal cables) are connected via screw terminals located inside a junction box right above the analyzer dome (fig. -14).

See schematics in appendix and the label on the junction box cover for detailed terminal assignments.

Cables entering the junction box should be kept as short as possible when inside the box.

Mains terminals are capable to connect cables up to 4 mm² cross section.

WARNING

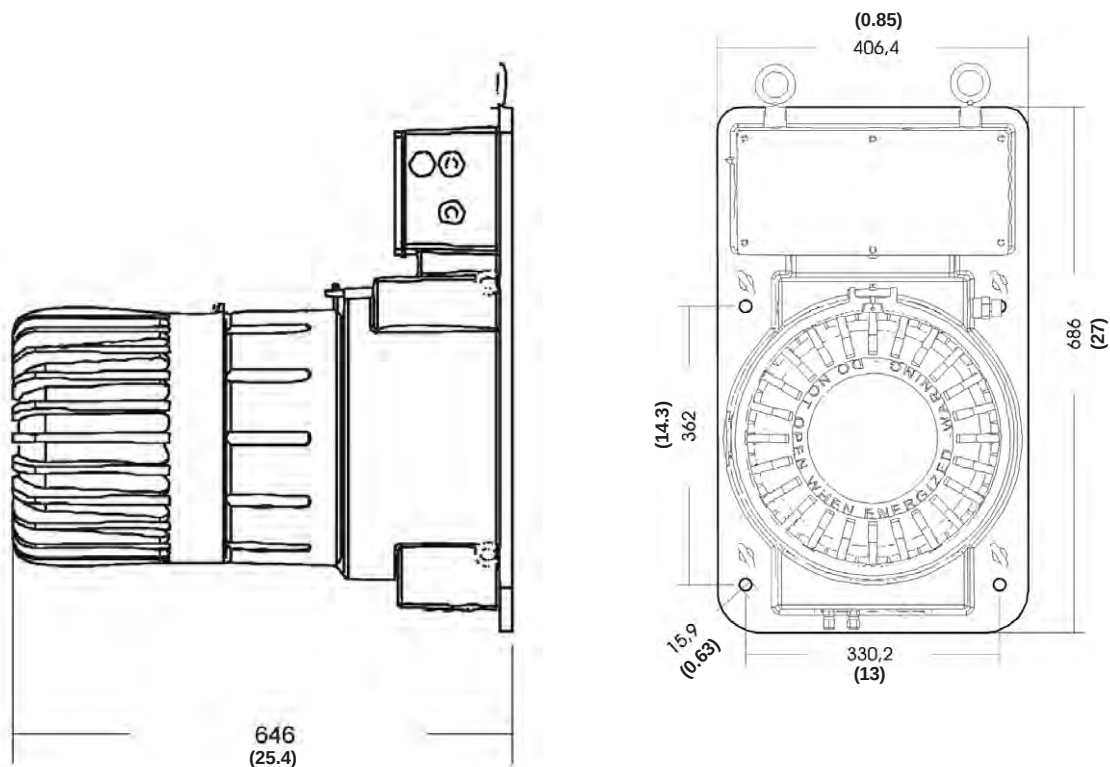
EXPLOSION HAZARD!



Instruments covered by an approval for hazardous areas (identifiable by a term „ATEX" on the nameplate label) MUST be equipped with cable glands, marked EEx e and ATEX!

This applies, too, when the instrument currently is NOT installed in a hazardous area!

3-4-4 Installation - CAT 100



Mount CAT 100 analyzers to a wall or stand by utilizing 4 through holes of $\varnothing$ 15.9 mm (0.63 in) each.

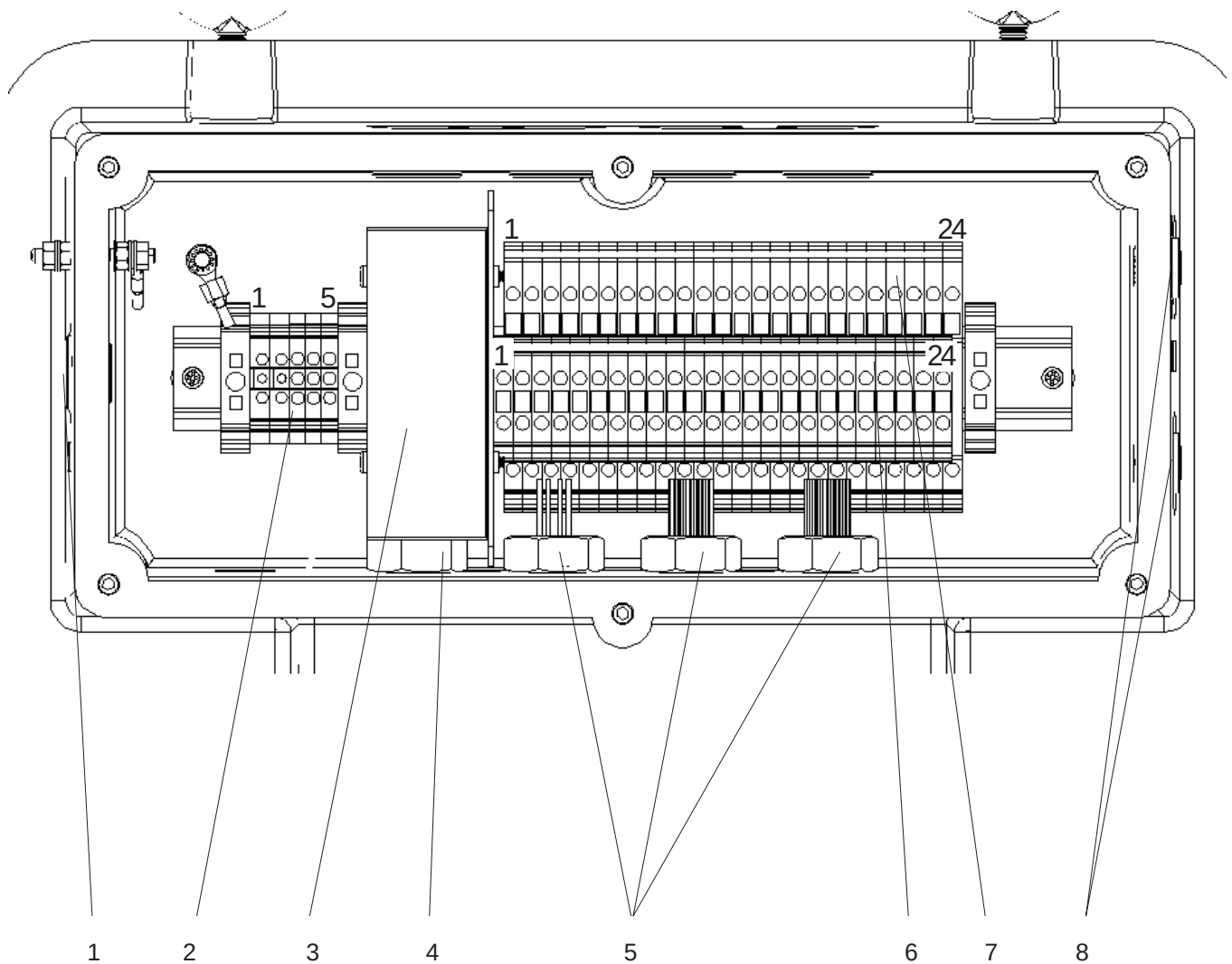
CAUTION

Take care to use anchors and bolts specified to be used for the weight of the units!

Take care the wall or stand the unit is intended to be installed at is solid and stable to hold the units!

*Fig. 3-14: CAT 100,
Dimensions [mm (inch)]*

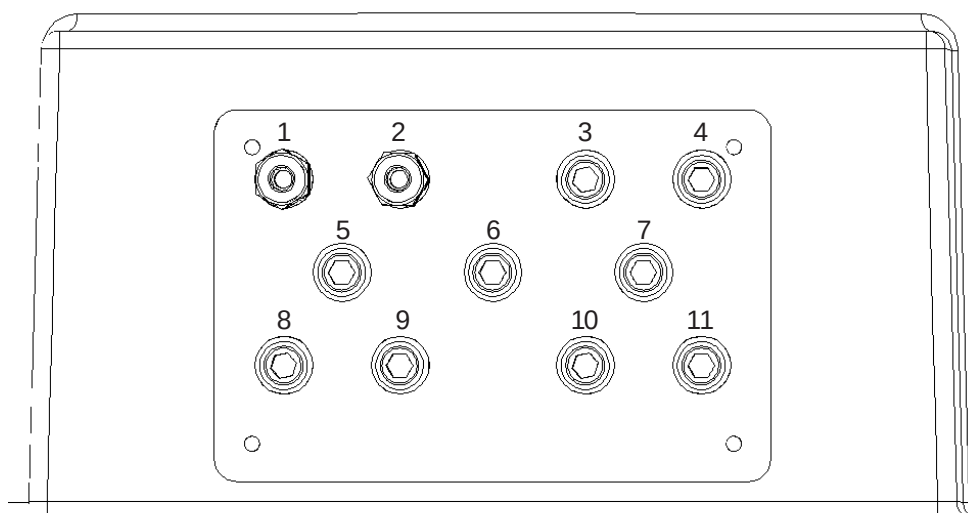
3-4-4 Installation - CAT 100



**Fig. 3-15: CAT 100,
Junction box, interior view**

- | | | | |
|---|-----------------------------------|---|---|
| 1 | Threat for mains cord cable gland | 5 | Flameproof feed thru, signals (3x) |
| 2 | Mains screw terminals | 6 | Upper signal screw terminals (24) |
| 3 | Mains EMC filter | 7 | Lower signal screw terminals (24) |
| 4 | Flameproof feed thru, mains | 8 | Threats for signal cables cable glands (3x) |

3-4-4 Installation - CAT 100



Analyzer Configuration	Gas Fitting Assignment			
Serial tubing (single or dual channel instrument)	1 – Gas in	2 – Gas out		
Parallel tubing (dual channel instrument)	1 – Gas in (ch 1)	2 – Gas out (ch. 1)	3 – Gas in (ch. 2)	4 – Gas out (ch. 2)
Serial tubing (single or dual channel instru.) 1 auto calibration valve	5 – Gas out (Sample, zero & span gas)*	6 – Span gas 1 in	7 – Span gas 2 in	8 – Zero gas in
	9 – Sample gas in			
Parallel tubing (dual channel instrument) 2 auto calibration valves	1 – Gas in (ch 1)	2 – Gas out (ch. 1)	3 – Gas in (ch. 2)	4 – Gas out (ch. 2)
	6 – Span gas 1 in	7 – Span gas 2 in	8 – Zero gas 1 in	10 – Zero gas 2 in
CAT housing purging (option)	11 – Purge gas in			
* Standard: Gas out = gas exhaust; valve block outlet internally connected to BINOS gas inlet; Option: Valve block outlet may be used for external sample handling system (special configuration).				

**Fig. 3-16: CAT 100,
Assignment of gas fittings and analyzer's bottom view at gas fittings**

3-5 Wiring Signal Terminals

3-5 Wiring Signal Terminals

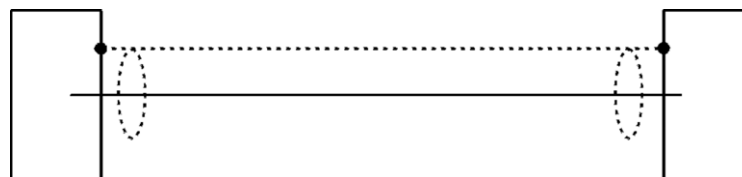
Emerson Process Management has taken any effort during the BINOS® 100 series development process to ensure electromagnetically compatibility (EMC; concerning emission and immunity), stated by EMC measurements according EN 61326.

Nevertheless EMC is not only influenced by the instrument's design, but widely by the installation procedure at site, too. Take care of the following sections and measures described within to ensure safe and trouble-free analyzer operation!

3-5-1 Electrical Connections in General

To minimize electromagnetically interferences by the analyzer's environment it is necessary to carefully execute all electrical connections between the analyzer and other instruments:

- It is recommended to use shielded cables for signal lines, only! Shield has to be connected to the housing at both ends of one connection (fig. 3-17).



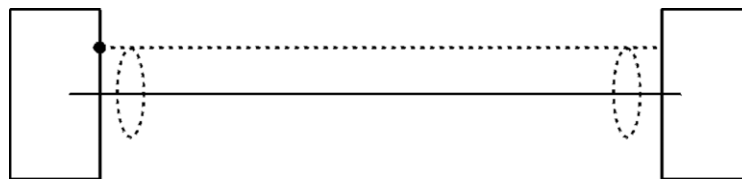
**Fig. 3-17: Shielded Signal Cable,
shield connected at both ends**

3-5-1 Electrical Connections in General

Local on-site conditions usually differ from test conditions and may require special measures. This is when strong fields are expected, potentially generating high parasitic currents on the cable shield. Such currents result in differences of potential between connected housings.

Two possible measures to avoid parasitic currents are described, whereat installation personnel familiar with EMC problems has to decide about the use of either measure:

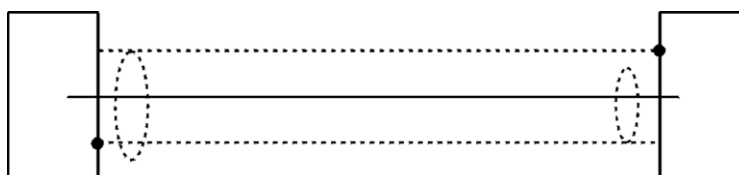
- Shield is connected at one side of the cable only (recommended to the analyzer's housing): Protection against external disturbances is increased but influence by parasitic currents is prevented due to opening the ground loop.



**Fig. 3-18: Shielded Signal Cable,
shield connected at one end**

- Using double-shielded cables:
In this case one shield is connected to the analyzer's housing while the other shield is connected to the external equipment. This gives an advantage when both instruments are supplied by different supply networks (e.g. when installed in different buildings).

This measure is more costly but offers best immunity against disturbances from outer fields and from parasitic currents.



**Fig. 3-19: Double-shielded Signal Cable,
shields connected at both sides**

3-5-2 Wiring Inductive Loads

3-5-2 Wiring Inductive Loads

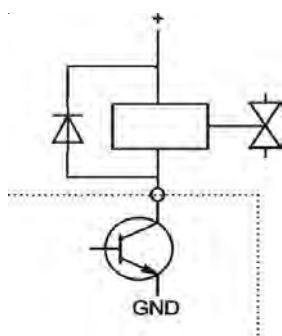
Switching inductive loads is a standard application generating electromagnetic disturbances:

The moment an inductive load (e.g. relay, valve, etc.) is switched off, it's magnetic field defies the change of current flow, generating high voltages (up to hundreds of volts) at the coil's contacts. This impulse reproduces on connected wires and may influence electrical equipment nearby or destroy signal inputs and/or outputs on electronic boards.

A simple measure helps to avoid such effects:

- Shunt a silicon diode to the inductive load's contacts shorting the voltage impulse just at it's source.
The diode's cathode needs to be connected to the positive side of the coil, the anode to the negative side (fig. 3-20).

Suitable filter components are available on request for standard valves.



**Fig. 3-20:Suppressor Diode
for Inductive Loads**

3-5-3 Driving Multiple Loads

Another popular application is driving multiple loads within one system by multiple outputs, whereat the supply voltage for the loads is taken from one common source.

To minimize load switching generated disturbances special care is required when wiring the system:

- **AVOID** to „serial" wire the loads' power supplies at which the power supply line starts at the source and successively connects all loads (fig. 3-21):

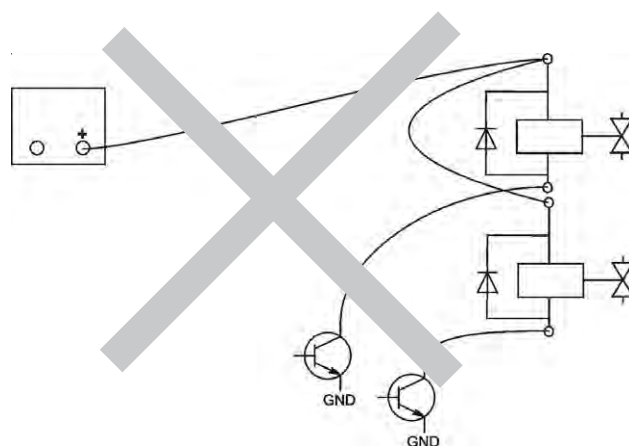


Fig. 3-21:„Serial" Wiring

- **It's better** to apply „parallel" wiring at which each single load is supplied by a separate connection starting from a distribution point: Both „+" and „-" wire of any load are run together, starting at the

3-5-4 Driving High Current Loads

point of distribution and ending at the load (see fig. 3-22). The effect of minimizing disturbances is intensified when using two-wire drilled cables.

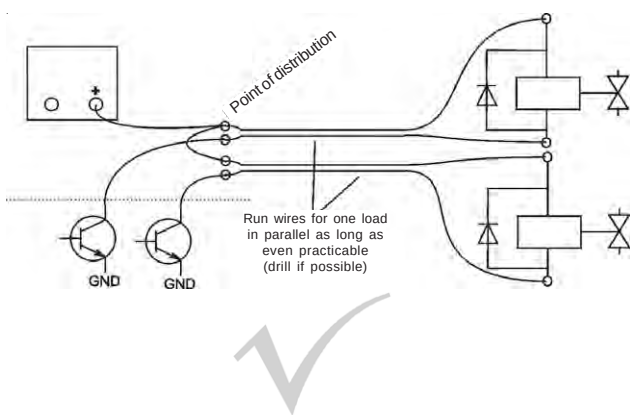


Fig. 3-22: Running Supply Lines "Parallel"

3-5-4 Driving High Current Loads

Loads with currents exceeding the rated currents specified for BINOS® 100 series analyzers outputs ($>30\text{ mA}$ resp. $>1\text{ A}$) must not be driven directly by digital or relay outputs. Driving such loads requires external relays acting as decoupling devices: The BINOS® 100 output drives the external relay, which itself drives the load.

It is recommended to use separate supplies for analyzer and high current loads to minimize interferences (fig. 3-23).

As described before using suppressor diodes for inductive loads is strongly recommended!

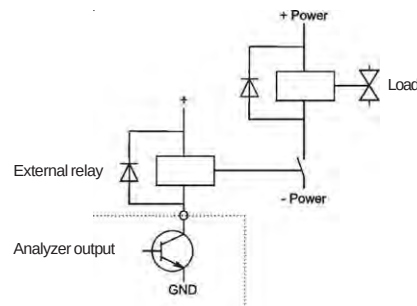


Fig. 3-23: Driving High Current Loads

Chapter 4 Operation

4-1 Abstract

BINOS® 100 series analyzers are operated by four push-buttons, located at the analyzer's front panel (fig. 4-1).

Two variations (BINOS® 100 2M and BINOS® 100 F) provide an additional push-button "PUMP" (fig. 4-2), only operable when the optional internal sample pump is selected.

The field housing variation BINOS® 100 F optionally may be ordered with a magnetically operated frontpanel (fig. 4-3), whereat the push-button "PUMP" is replaced by a separate software menu entry.

CAT 100 analyzers utilize the magnetically operated front panel, only.

The operator is informed about the actual analyzer status by messages on two 4-digit LED displays.

Parameters changed by the operator are stored in a memory with battery backup to prevent data loss caused by power failures.

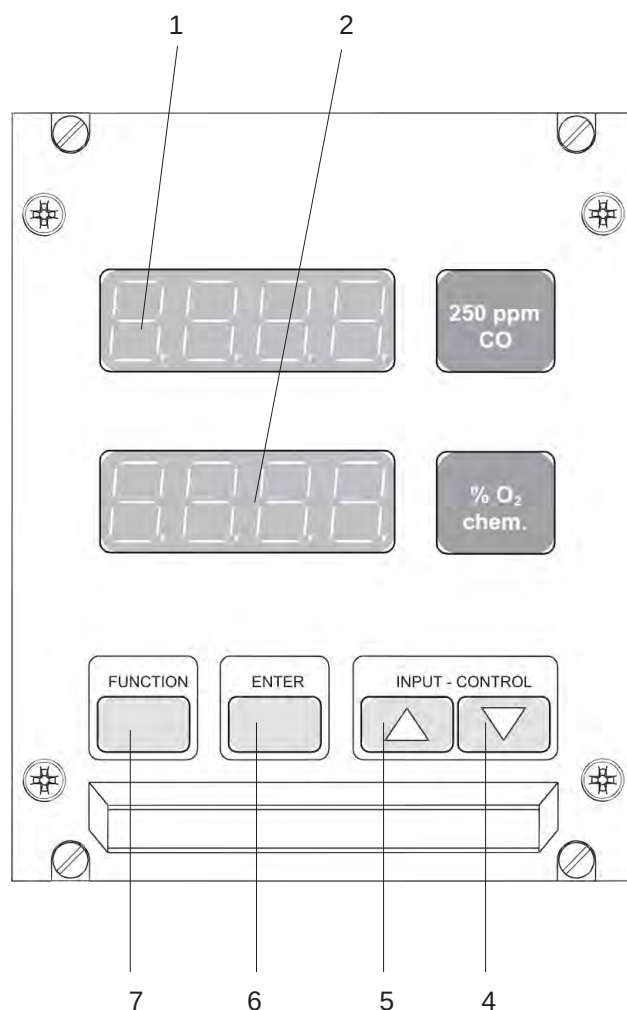


Fig. 4-1: BINOS® 100 (M), OXYNOS® 100, HYDROS® 100, front side view

- 1 7 segment display (channel 1)
- 2 7 segment display (channel 2)
- 4 Pushbutton **DOWN**
- 5 Pushbutton **UP**
- 6 Pushbutton **ENTER**
- 7 Pushbutton **FUNCTION**

4-1 Operation

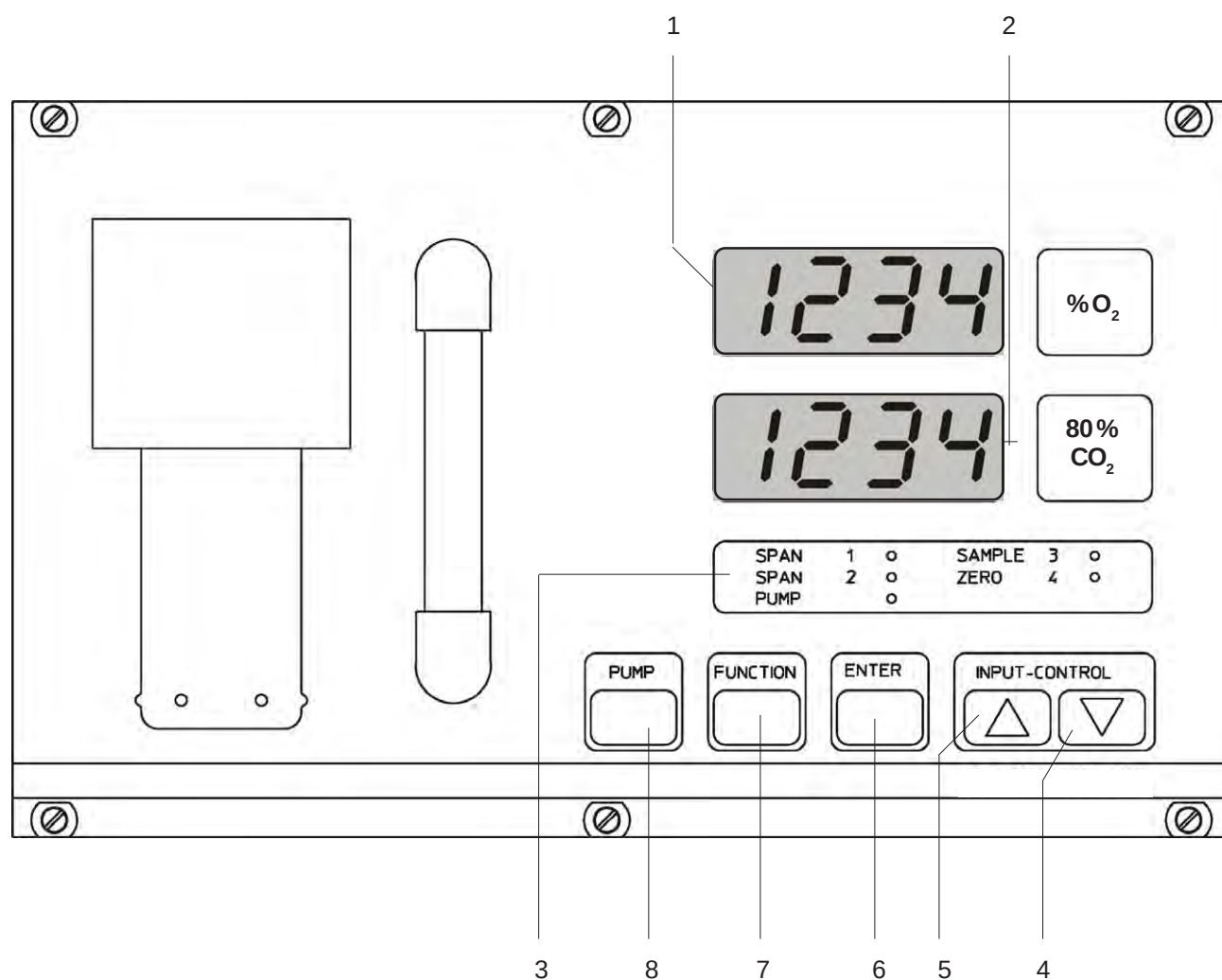


Fig. 4-2: BINOS® 100 2M (standard version)
front side view

- | | | | |
|---|-------------------------------|---|-------------------------------------|
| 1 | 7 segment display (channel 1) | 5 | Pushbutton UP |
| 2 | 7 segment display (channel 2) | 6 | Pushbutton ENTER |
| 3 | LEDs "valves / sample pump" | 7 | Pushbutton FUNCTION |
| 4 | Pushbutton DOWN | 8 | Pushbutton for optional sample pump |

4-1 Operation

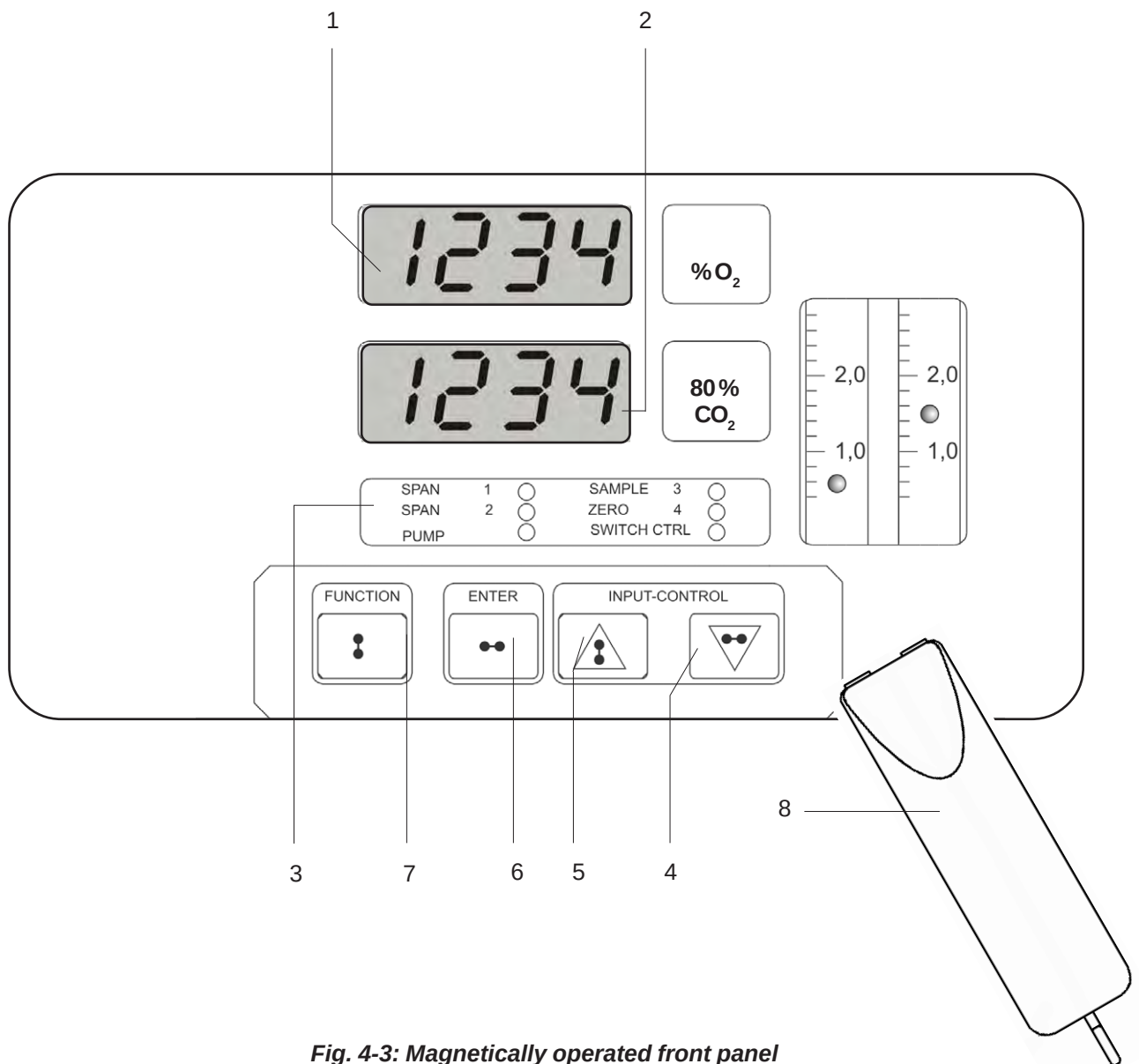


Fig. 4-3: Magnetically operated front panel

- | | | | |
|---|-------------------------------|---|------------------------------|
| 1 | 7 segment display (channel 1) | 5 | Sensor field UP |
| 2 | 7 segment display (channel 2) | 6 | Sensor field ENTER |
| 3 | LEDs "valves / sample pump" | 7 | Sensor field FUNCTION |
| 4 | Sensor field DOWN | 8 | Magnetic actuator |

4-2 Button "FUNCTION"

4-2 Button "FUNCTION"



Starting at the measuring display the BINOS® 100 series analyzer software offers several submenus, each correlating with a specific analyzer function.

The "FUNCTION" push-button (item 7 in fig. 4-1 to 4-3) is used to scroll through the submenus in rotation, starting and ending with the measuring display (see page 4-5).

To select a submenu and perform any action it is not only required to scroll to, but also to enter the submenu by pressing the "ENTER" push-button (sect. 4-3).

The BINOS® 100 series analyzers provide 8 digits only for displaying measuring values, submenu headers, messages and parameters. Therefore identifiers had to be abbreviated and detailed descriptions are not available on the display.

All abbreviations used by the analyzer software are described in the following sections as well as by fig. 4-5.

Hints on the magnetically operated front panel:

To avoid unintended activation the front panel is locked by an additional access code. This code has to be entered each time to leave the measuring display:



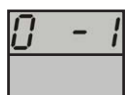
The upper display shows "CODE" after the FUNCTION sensor field has been activated. Now the correct code has to be entered within 20 seconds to get access to scrolling through submenus.

If the code has not been entered within that time slice the sensor fields keep locked and the analyzer returns to the measuring display!

The access code is **DOWN - UP - ENTER - FUNCTION.**

Starting with the measuring display the FUNCTION push-button scrolls through the following submenus:

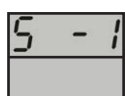
4-2 Button "FUNCTION"



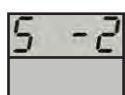
Zero (0) calibration channel 1



Zero (0) calibration channel 2



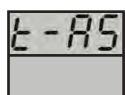
Span calibration channel 1



Span calibration channel 2

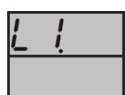


Interval time for automatic time controlled zero (0) calibration

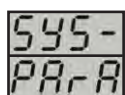


Interval time for automatic time controlled span calibration

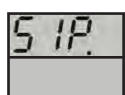
Available only in combination with internal solenoid valves or digital outputs in combination with external solenoid valves. Requires parameter Auto set to "1"



Concentration thresholds (Limits)



System parameters



Serial interface parameters

with optional RS 232 / 485 interface only!

Pressing FUNCTION once more returns to measurement display.

Fig. 4-4: BINOS® 100 Software Menus

4-3 Button "ENTER"

4-3 Button "ENTER"



Push-button **ENTER** (item 8 in fig. 4-1 to 4-3) is used to

- enter a submenu (select a function or parameter)
- pass a value to the currently selected parameter
- start actions, e.g. zero calibration, span calibration

The first time ENTER is pressed while scrolling through the menus (by FUNCTION button) the following message appears:



Entering submenus is locked by an access code to prevent unauthorized parameter changing.

Once a valid code has been entered access to submenus is permitted as long as the measuring display is not selected. Entering the code is required when again entering submenus from the measuring display.

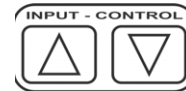
Entering an invalid code keeps the text CODE in the upper display and the lower line is changed to "0" to give a new try.



Access code factory setting is "1" !

It is strongly recommended to change this code during startup.

4-4 Buttons "INPUT - CONTROL"



These push-buttons (item 9 & 10 in fig. 4-1 to 4-3) are used to change parameter values. Single activation changes the given value by 1 digit:

UP		increase value by 1
DOWN		decrease value by 1

Once the limit is reached by changing the value in one direction, it is set to the start value.

Keeping a push-button pressed increases the rate of changing in two steps to facilitate changing higher values.

Once one of the limit values is reached, the value is set to the opposite limit and begins to change in the same direction as before.

Changing parameters is limited to parameter dependent useful values only. This range can not be exceeded. In addition any entered value is checked for plausibility to prevent maloperation.

After a time period of approx. 60 to 120 seconds without user activity the instrument returns to the measuring display !

4-5 Button "PUMP"

4-5 Button "PUMP"



This push-button is used for switching the optional internal sample gas pump (BINOS® 100 2M, CAT 100 and BINOS® 100 F only).

The button toggles the current pump mode: A switched off pump is switched on and vice versa.

Alternatively the pump may be switched on/off by utilizing

- a related software menu [system parameter "PUMP" (sect. 5-3-17)]
- software commands [RS 232/485-interface option *) (sect. 6-2)]
- FOUNDATION™ fieldbus commands [fieldbus-option **) (sect. 6-3)] or
- digital inputs [option **) (sect. 6-3)]

Analyzers with magnetically operated front panel do not provide the PUMP button, select one of the listed alternatives.

A green control light at the front panel (PUMP, item 3 in fig. 4-2 & 4-3) is activated when the pump is switched on. Optionally the status can be output by a digital status signal (relay output, sect. 6-1).

*) not in combination with FOUNDATION™ Fieldbus

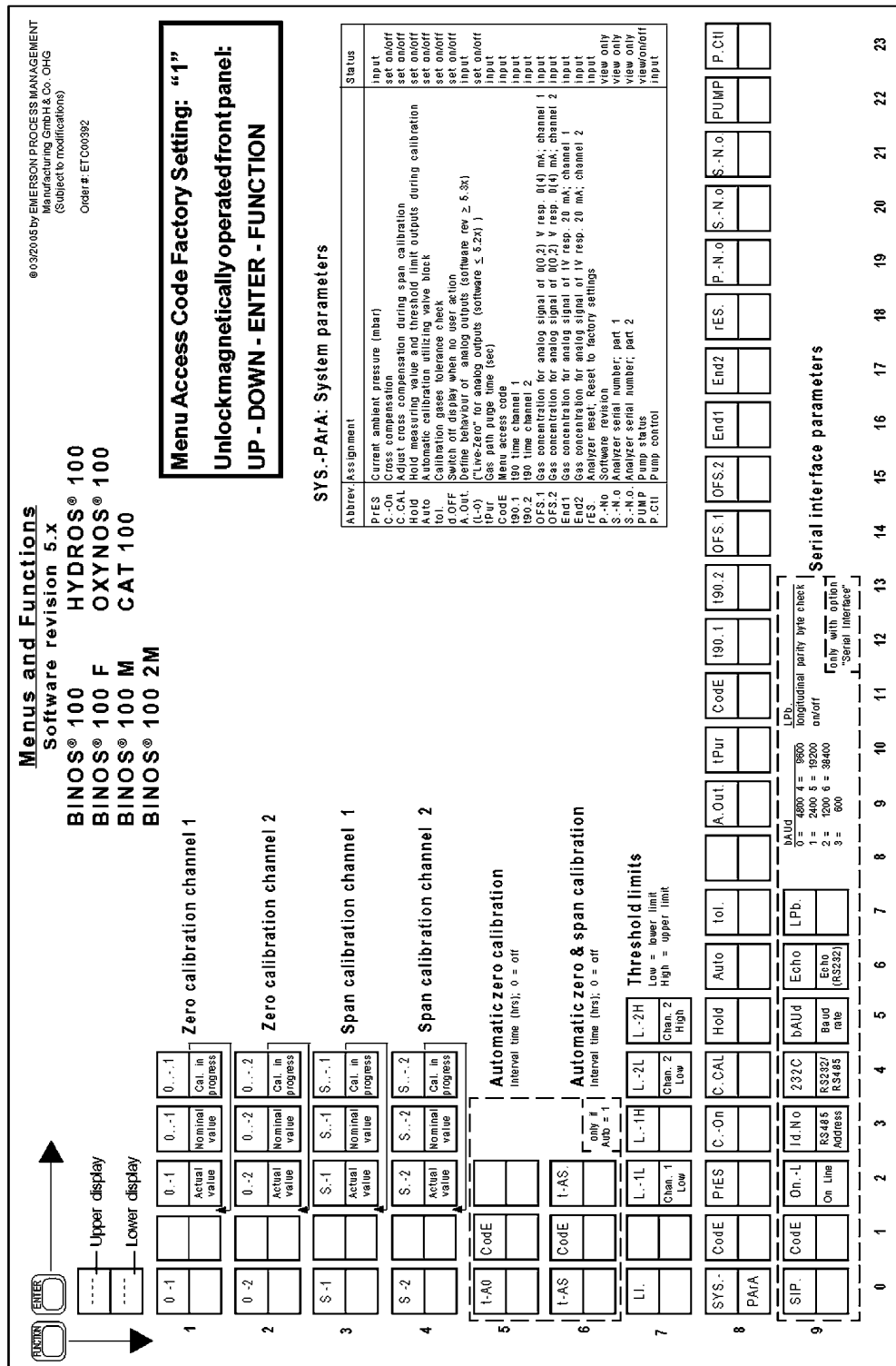
**) not in combination with RS 232/485 interface

4-6 Software Menu Structure

Fig. 4-5 (next side) shows the BINOS® 100 series software menu structure with all submenus. This table is provided as a laminated version with every analyzer. Additional copies may be ordered (part # ETC00392) at our local sales office.

4-6 Software Menu Structure

Fig. 4-5:
BINOS® 100 Series
Software Menu Structure



4-7 Magnetically Operated Front Panel

4-7 Magnetically Operated Front Panel

Analyzers intended either to be used in hazardous area or for outdoor usage are provided with a magnetically operated touch panel (in part optionally). This special version ensures ingress protection (IP) and protection

against mechanical shocks. On account of its design some special measures have to be considered to operate the associated analyzer in a safe manner, so read the following instructions carefully!

4-7-1 Front Panel Elements

Whereas standard front panels are using keys for operating the analyzer the magnetically operated touch panel uses contactless technology instead. This results in a slightly different design:

Each standard panel key is replaced by a reed relay located behind the front panel design foil.



**Fig. 4-6: Magnetically operated frontpanel
(CAT 100 version)**

4-7-2 Accessory

To operate the magnetically operated touch panel a special tool (actuator) is required. The following picture shows this tool as it is provi-

ded together with your analyzer. If it is missing please contact your local sales office.

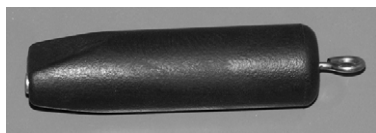
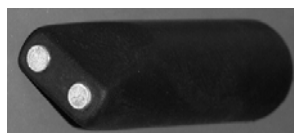


Fig. 4-7: Magnetic actuator



**Fig. 4-8: Magnetic actuator,
view at magnets**

4-7 Magnetically Operated Front Panel

4-7-3 Operating the Front Panel

The touch panel is operated by activating the reed relays utilizing the actuator. This is done by holding that actuator's end in front of the key that is equipped with 2 small magnets. As the reed relays are installed alternating vertically and horizontally to prevent accidentally activation of an adjacent key, the actuator has to be aligned the same way.

The design foil shows for each key how the actuator has to be aligned using the symbols

„●—●“ and „●⊥“ (Fig. 4-9).

A green light („switch ctrl“) is turned on when the actuator has the correct position and activates the reed relay. It is located right above the input control keys (fig. 4-10 and 4-11 showing the BINOS® 100 F front panel version).

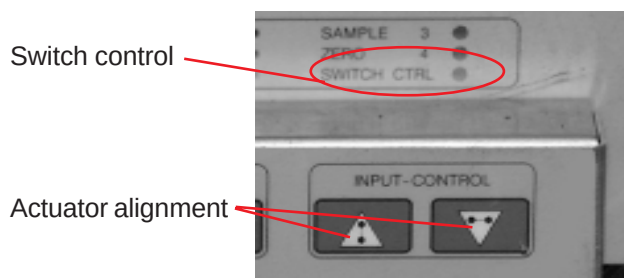


Fig. 4-9: Sensor details



Fig. 4-10: Vertically aligned, activating UP key



Fig. 4-11: Horizontally aligned, activating DOWN key

4-7 Magnetically Operated Front Panel

4-7-4 Storing the actuator tool

The tool is provided with a chain and a special fixture to be installed at one of the analyzer's mounting screws (s. fig. 4-12).

It is recommended to place the actuator tool on the fixture as shown to avoid unintended interference while not in use.



Fig. 4-12: Actuator stored when not in use

Chapter 5 Startup

5-1 Abstract

After having installed the instrument according to the instructions given in chapter 3 "Installation" it is powered by connecting to mains utilizing the mains power plug (tabletop / rack mount variations) or the mains disconnect switch (wall mounted instruments).

Activated displays indicate correct power supply.

Once powered the analyzer performs a power on self test (POST):

First display shows the installed software revision:



Analyzer
without fieldbus



Analyzer
with fieldbus

(Your specific instrument software revision may differ!)

When POST has finished the instrument shows first measuring results or gives error messages, depending on the analyzer status. A flashing **batt** indicates that due to a battery failure analyzer data have been reset to factory default settings. The text is cleared by pressing any button.



For best measuring results allow 15 to 50 minutes warm-up time (depending on detector) after switching on the analyzer.

After warm-up the analyzer is operable but before starting measurements it is recommended to

- check system parameters (sect. 5-3) and
 - perform a first calibration (sect. 5-4).
- The following sections describe in detail how to perform both tasks.

Note!

All the following sections show figures of displays containing numbers like "123" or "1234". These numbers are exemplary only and will differ on your analyzer!

5-2 Software Menu Structure

5-2 Software Menu Structure

Although already described in section 4-6 here is once more the BINOS® 100 series software menus and functions overview.

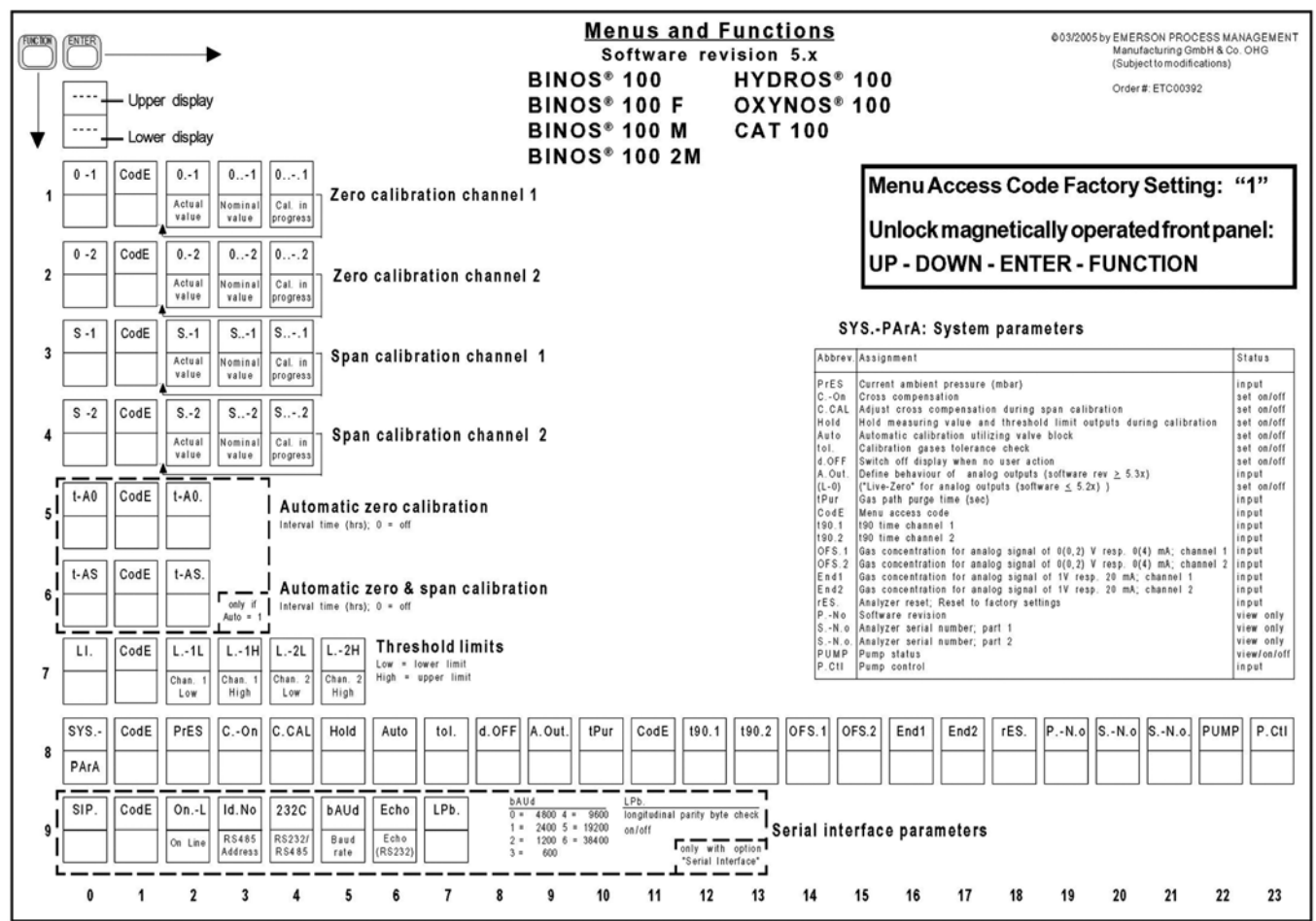


Fig. 5-1: Software Menu Structure

5-3 Checking System Parameter Settings

5-3 Checking System Parameter Settings

During first startup the correct system parameter configuration should be checked and adjusted to your needs, where required. System parameters are set within the menu "System parameters".

The following sections give detailed information about the system parameter submenus. Descriptions are given step-by-step, whereat always pressing the ENTER button completes one step and continues with the next submenu if nothing different is mentioned.

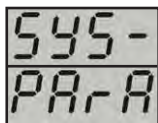
5-3-1 Menu "System Parameter"

Press button



several times

until the following screen appears



Press button



The following message appears if the menu access code has not already been entered



Select access code utilizing



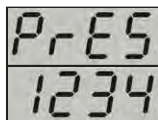
Enter code pressing



Next screen shows ... (to be cont'd on the next side)

5-3-2 Menu "Pressure Compensation"

5-3-2 Menu "Druckkorrektur"



To avoid faulty measurements as a result of atmospheric pressure variations this menu allows to enter a fixed pressure value (hPa; mbar) within a range of 800 to 1,300 hPa. This value should meet the currently measured atmospheric pressure and will be used to correct the analyzer's measuring values:

Enter value using



and

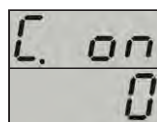


As an option an internal pressure sensor (operating range 660 to 1,100 hPa, higher ranges on request) may be used to permanently measure the atmospheric pressure and automatically correct the measuring values, so significantly decreasing interference by changing ambient pressure (see technical data).



In case there is an internal sensor installed, manually entering a pressure is prohibited and the menu shows the currently measured ambient pressure.

5-3-3 Menu "Internal Cross Compensation"



This menu allows to enable internal cross compensation.

Cross compensation is used with dual channel analyzers to minimize cross compensation for both internally measured gases (e.g. CO₂ and CO).

Option 0: Internal cross compensation disabled

Option 1: Internal cross compensation enabled

Enter value using

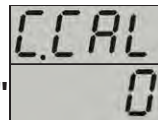


and



5-3-4 Menu "Calibrate Cross Compensation"

5-3-4 Menu "Calibrate Cross Compensation"



To have cross compensation work properly, it is necessary to determine so called cross compensation factors during a span calibration.

Pure span gases are required for determining the factors.

Once the cross compensation factors have been calculated, **mixed span gases** may be used for future span calibration, too.

Option 0: Perform span calibration with **mixed span gases**

Option 1: Calibrate cross compensation factors during span calibration
(**pure span gas**)

Enter value using



and



To calibrate cross compensation perform as follows:

- Carry out zero calibration for both measuring channels (see sect. 5-5).
- Carry out span calibration for both measuring channels as described in section 5-5.
- Again perform a span calibration for the measuring channel, only, that has been span calibrated **first** in step above.



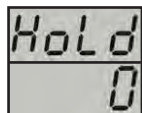
To perform a cross compensation calibration the parameter values entered according 5-3-3 and 5-3-4 MUST be "1" !

Pure span gases are required !

Using mixed span gases requires C.Cal = 0 !

5-3-5 Menu "Hold"

5-3-5 Menu "Hold"



System parameter **HOLD** allows to lock the analog outputs and threshold outputs during calibration to hold the last measured value.

- Option 0: Outputs remain unlocked
- Option 1: Outputs are locked during calibration

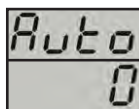
Enter value using



and



5-3-6 Menu "Automatic Calibration"



Enables automatic time controlled or remote calibration utilizing external or internal solenoid valve blocks.

- Option 0: Disable time controlled / remote calibration
- Option 1: Enable time controlled / remote calibration

Enter value using

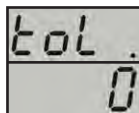


and



5-3-7 Menu "Tolerance Check"

5-3-7 Menu "Tolerance Check"



Enter this menu to enable / disable tolerance checking while calibration.

With tolerance check enabled during calibration the analyzer checks that the entered (setpoint) values for zero gas and span gas are reasonable compared to the currently connected gas. If the currently connected gas concentration differs more than 10 % of measuring range from zero (during zero calibration) or span (during span calibration), calibration is aborted.

When calibration is aborted due to tolerance check a failure message is displayed (s. list of failures, sect. 7), which by default is reset after 2 - 3 minutes. Setting the tolerance check parameter to "2" disables failure message auto-reset: To reset the message it is required to perform either a valid calibration or to reset the analyzer.

Option 0: Tolerance check is disabled

Option 1: Tolerance check is enabled, failure message auto-reset after 2-3 minutes

Option 2: Tolerance check is enabled, failure message auto-reset disabled

Enter value using



and



5-3-8 Menu "Display Off"

5-3-8 Menu "Display Off"



Enables / disables display standby mode: Option "1" enables standby mode whereat the display is turned of after a time period to 1-2 minutes without operator activity.

Pressing any button turns on the display without any other action.

- Option 0: Disables display turn off (display always on)
- Option 1: Enables display auto turn off

Enter value using

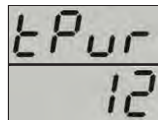


and



5-3-9 Menu "Gas Path Purge Time"

5-3-9 Menu "Gas Path Purge Time"



Before starting a calibration it must be verified that the gas paths have been sufficiently flown through by the calibration gas and do not contain other components. The time period that has to pass by to ensure the gas path is flown through before the calibration starts, is entered in this menu "Gas Path Purge Time".

Acceptable values are 0 to 99 seconds.

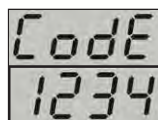
Enter value using



and



5-3-10 Menu "Access Code"



To prevent system parameters from unauthorized changing an access code is required to enter software submenus. The factory default setting is "1".

It is strongly recommended to change this code to an individual code! This code may be entered using the menu "Access Code".

Enter value using



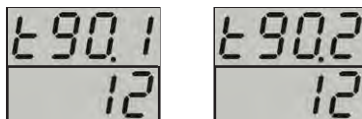
and



Store the changed code at a safe place! Submenus are not accessible if correct code is not entered!

5-3-11 Menus "Response Time t_{90} "

5-3-11 Menus "Response Time t_{90} "

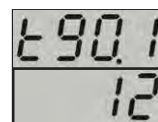


Measuring values on display and at analog outputs may dither, depending on application. Adding an electronic damping smoothens the signal by calculating average values of several sequential measuring values before changing the output. This affects the analyzer response time t_{90} .

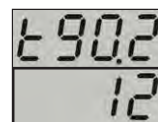
BINOS® 100 series analyzer allow to affect signal smoothing by defining an electronic t_{90} time individually for each measuring channel to optimize application handling.

Electronic response time input range is **2 - 60 seconds** !

Enter channel 1 response time when display shows



Enter channel 2 response time when display shows



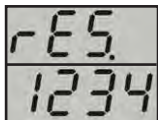
Enter value using



and



5-3-12 Menu "Reset"



5-3-12 Menu "Reset"

This menu is used to perform an analyzer reset to load factory default settings.

Activating the reset utilizing this menu is equivalent to removing power supply and disabling RAM battery buffering by removing the battery jumper J7 (s. page 8-32).



All parameters and calibration factors entered by the operator are lost when performing a reset!

To prevent accidentally resetting the analyzer it is required to enter the access code:

Enter code using



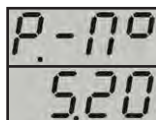
and



After reset the analyzer restarts by performing the initial startup sequence (POST), see page 5-1.

5-3-13 Menu "Software Revision"

5-3-13 Menu "Software Revision"



The lower display shows the installed software revision.

Press



to leave this screen.

5-3-14 Menus "Analyzer Serial Number"



Two screens are required to show the analyzer serial number (8 digits). Serial numbers are required to receive service and maintenance information from factory.

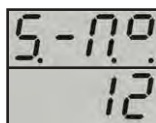
When first part of serial number is shown,

press



to switch to second part

Second part of serial number:



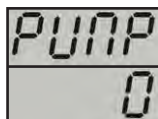
Press



to leave this screen.

5-3-15 Menu "Pump Status"

5-3-15 Menu "Pump Status" *)



This menu shows the current internal sample pump status.

If pump control is set to "1" ("P.Ctl.", see 5-3-18) to enable pump control via system parameter, this menu is used to switch the pump on / off, too.

Display "0" or change parameter to "0":	Sample pump is off / is switched off
Display "1" or change parameter to "1":	Sample pump is on / is switched on

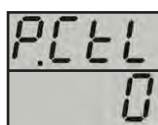
Enter value using



and



5-3-16 Menu "Pump Control" *)



This menu is used to define how the internal sample pump is controlled:

Option 0:	Enables frontpanel push-button for pump control
Option 1:	Enables system parameter "PUMP STATUS" for pump control (s. 5-3-17)
Option 2:	Enables digital inputs (option) for pump control (page 6-16).

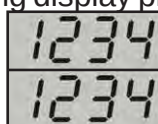
Enter value using



and



To leave this screen and return to measuring display press



till screen shows measuring values

*) BINOS® 100 2M / BINOS® 100 F / CAT 100 option only!

5-4 Analog Outputs

5-4 Analog Outputs

To enable external data acquisition any analyzer provides two galvanically isolated analog outputs (1 for each measuring channel), offering both voltage and current signals at a time.

Analog output factory setting is 0-20 mA (current signal) resp. 0-10 V (voltage signal), which optionally may be set to a range of 0-1 V.

Tabletop / rack mount instruments provide a 9 pole submin-d socket (X2) at their rear panel to connect external signal lines.

Wall mountable analyzers require external signal lines to be connected to either internal adaptor elements (BINOS® 100 F) or internal

screw terminals inside the junction box (CAT 100).

Fig. 5-2 shows the submin-d socket pin assignment. The adaptor element as it is used with BINOS® 100 F is internally connected through one by one (terminal 1 is connected to submin-d pin 1, terminal 2 to pin 2, etc.). Assignment of terminals within the CAT 100 junction box is given on a label attached to the inner side of the junction box cover.

The following sections 5-4-x describe analog output software settings.

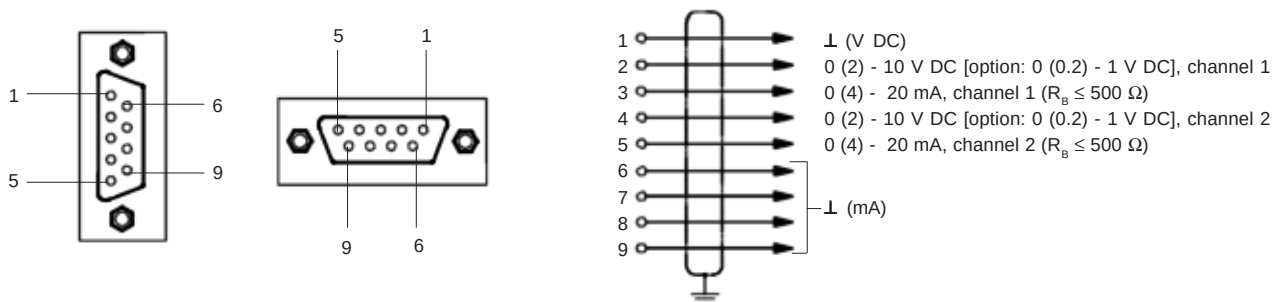


Fig.5-2: Pin Assignment Socket X 2
(analog outputs)

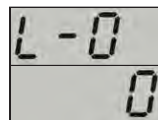
Note!

Take care of the hints about wiring signal outputs, given in section 3-5!

Your analyzer's voltage output configuration (Standard: 10 V DC oder optional: 1 V DC) is marked on the analyzer's nameplate label.

5-4-1 Menu "Life-Zero for Analog Outputs"

5-4-1 Menu "Life-Zero for Analog Outputs"



Note!

Settings described in this section refer to software revisions prior to rev. V5.3 only!

Menu "L-0" was replaced by menu "A.Out." for software revisions V5.3 and later (see sect. 5-4-2)!

By default a gas concentration of "0" generates an similar analog output of "0" (0 V resp. 0 mA).

As a signal cable break results in a "0" signal, too, an external data acquisition is not capable of detecting such a failure and accepts a gas concentration signal of "0".

To avoid this situation analog outputs can be set to an operation mode called "Life-Zero": Here a concentration corresponding to the lower measuring range limit (e.g. "0") is assigned to an analog signal of 2 (0.2) V resp. 4 mA. Thus a failure like cable break is clearly detectable by a signal of 0 V resp. 0 mA.

Option 0: Signal range 0 - 10 VDC (option: 0 - 1 VDC) / 0 - 20 mA

Option 1: Signal range 2 - 10 VDC (option: 0.2 - 1 VDC) / 4 - 20 mA ("**Life - Zero**")

Enter value using



and



5-4-2 Menu "Adjusting Analog Outputs"**5-4-2 Menu "Adjusting Analog Outputs"****Note!**

Settings described in this section refer to software revisions V5.3 and later, only!

Software revisions till V5.2 show a menu "L-0", described in section 5-4-1.

By default a gas concentration of "0" generates an similar analog output of "0" (0 V resp. 0 mA).

As a signal cable break results in a "0" signal, too, an external data acquisition is not capable of detecting such a failure and accepts a gas concentration signal of "0".

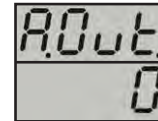
Such an analog output operation mode is selected by setting the **A.Out.** parameter to "0" and corresponds to parameter **L-0** set to "0" for software revisions prior to V5.3.

The commonly used methode to detect cable breaks is driving the analog outputs in LIFE-ZERO mode: a concentration corresponding to the lower measuring range limit (e.g. "0") is assigned to an analog signal of 2 (0.2) V resp. 4 mA. Thus a failure like cable break is clearly detectable by a signal of 0 V resp. 0 mA.

This LIFE-ZERO mode is selected when setting parameter **A.Out.** to "1" and corresponds to parameter **L-0** set to "1" for software revisions prior to V5.3.

Operation Modes corresponding to NAMUR Recommendation 43 (NE 43)

Both modes described above do not provide a signal capable to detect a failure within the measuring system. Here the output signal behaviour is undefined: either it keeps the last value or is set to an arbitrary value. Measuring system failure are not detectable by an external data acquisition system.



NE 43 gives recommendations how to setup analog outputs to avoid above situation and the BINOS® 100 series analyzers consider NE 43 with software revisions V5.3 and later:

Setting **A.Out.** to values of 2 ... 5 defines specific signal levels for analog outputs in case of measuring system failures. During normal operation these values are not output, so an data acquisition system is capable to distinguish between

- cable break ("0" signal),
- failure (signal outside accepted range, but differing from "0")
- valid measuring value(signal within accepted range)

In case there is no failure but only a measuring range overrun or underrun the output signal will increase / decrease until it is equal to the limit given in table 5-1 and then keeps this value until the measuring signal is back within the measuring range.

Special feature for single channel instruments!

To stay compatible to software revisions prior to V5.3 the analog output of channel 2 provides a "0" signal (0 V resp. 0 mA) when **A.Out.** is set to 0 or 1.

Operating modes **A.Out.** = 2 ... 5 adjust the channel 2 analog output to a value below the accepted range but above the failure limit. This ensures that a data acquisition system recognizes the signal as a non valid measuring signal and does not give an alarm signal due to a missing measuring channel.

5-4-2 Menu "Adjusting Analog Outputs"

A.Out. operation modes in comparison:

A.Out.	Operation Mode	Failure Signal Level acc. NE 43	Valid Measuring Signal Range	Measuring Signal Exceeds Lower Range Limit	Measuring Signal Exceeds Upper Range Limit	Measuring System Failure	Cable Break	Channel 2 when not equipped
0	Dead-Zero	-	0 ... 10 V 0 ... 20 mA (0 ... 1 V, option)	< -9,5 V < -19 mA (< -0.95 V)	> 10.5 V > 21 mA (> 1.05 V)	undefined	0 V 0 mA (0 V)	0 V 0 mA (0 V)
1	Life-Zero	-	2 ... 10 V 4 ... 20 mA (0.2 ... 1 V, option)	< -9,5 V < -19 mA (< -0.95 V)	> 10.5 V > 21 mA (> 1.05 V)	undefined	0 V 0 mA (0 V)	2 V 4 mA (0.2 V)
2	similar Dead-Zero	below	0 ... 10 V 0 ... 20 mA (0 ... 1 V, option)	- 0,1 V -0.2 mA (-0.01 V)	10.25 V 20.5 mA (1.025 V)	-1 V -2 mA (-0.1 V)	0 V 0 mA (0 V)	-0.1 V -0.2 mA (-0.01 V)
3	similar Life-Zero	below	2 ... 10 V 4 ... 20 mA (0.2 ... 1 V, option)	1,9 V 3.8 mA (0.19 V)	10.25 V 20.5 mA (1.025 V)	1 V 2 mA (0.1 V)	0 V 0 mA (0 V)	1.9 V 3.8 mA (0.19 V)
4	similar Dead-Zero	above	0 ... 10 V 0 ... 20 mA (0 ... 1 V, option)	- 0,1 V -0.2 mA (-0.01 V)	10.25 V 20.5 mA (1.025 V)	>10.5 V >21 mA (>1.05 V)	0 V 0 mA (0 V)	-0.1 V -0.2 mA (-0.01 V)
5	similar Life-Zero	above	2 ... 10 V 4 ... 20 mA (0.2 ... 1 V, option)	1,9 V 3.8 mA (0.19 V)	10.25 V 20.5 mA (1.025 V)	>10.5 V >21 mA (>1.05 V)	0 V 0 mA (0 V)	1.9 V 3.8 mA (0.19 V)

Select operation mode by
entering the related number using

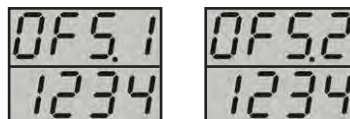


and



5-4-3 Menu "Analog Output Offset"

5-4-3 Menu "Analog Output Offset"



Enter this menu to define a gas concentration associated with the lower analog output limit, commonly called "offset".

Example:

Analyzer measuring range: 0 - 25 %

Expected measuring values: 10 - 25 %

Entering an offset of 10 defines that such a gas concentration (10 %) generates an analog output signal of 0 V (0 mA) resp. 2 V (0.2 V; 4 mA) in life-zero mode (see. 5-4-1).

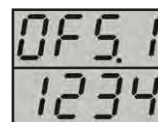


It's only possible to enter numeric values, the measuring unit (% , ppm, etc.) results from the measuring channel configuration.

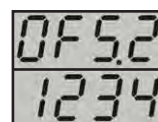
Defining an analog output offset does NOT affect the front panel measuring values!

Offset concentrations can be defined for each channel separately:

Enter the channel 1 offset into the following screen



the channel 2 offset can be entered when screen shows



Enter values using



and



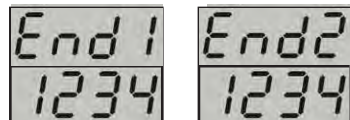
Note !

*Specifications given in the technical data section are based on **OFS. = 0** and **END = factory set default measuring range upper limit** (see 5-4-3)!*

Values are not checked for plausibility so it's operator's responsibility to enter reasonable values only!

5-4-4 Menu "Analog Output Upper Limit"

5-4-4 Menu "Analog Output Upper Limit"



Similar to the menu described before (sect. 5-4-2) this menu allows to define a gas concentration associated with the upper analog output limit.

Example:

Analyzer measuring range: 0 - 25 %

Expected measuring values: 0 - 15 %.

Entering a value of 15 defines that such a gas concentration (15 %) generates an analog output signal of 10 V (1 V) resp. 20 mA.

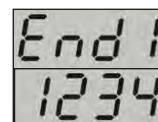


It's only possible to enter numeric values, the measuring unit (% , ppm, etc.) results from the measuring channel configuration.

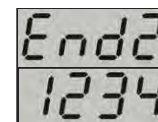
Defining an analog output upper limit does NOT affect the front panel measuring values!

Analog output upper limit can be defined for each channel separately:

Enter the channel 1 analog output upper limit into the following screen



the channel 2 analog output upper limit can be entered when screen shows



Enter value using



and



Note !

*Specifications given in the technical data section are based on **OFS. = 0** (see 5-4-2) and **END = factory set default measuring range upper limit** !*

Values are not checked for plausibility so it's operator's responsibility to enter reasonable values only!

5-5 Digital Outputs

5-5 Digital Outputs

Every analyzer by default offers digital outputs with the following functionalities:

- Concentration thresholds: 1 upper and 1 lower threshold for each channel
- Control of 4 external solenoid valves (span gas channel 1 / channel 2, zero gas, sample gas).

These outputs are accessible via rear panel connector X3 or internal screw terminals, depending on analyzer variation (s. chapter 1).

5-5-1 Concentration Threshold Operation Modes

One upper and one lower concentration threshold can be set for each channel, freely configurable within -110 % and +110 % of the associated measuring range (using values exceeding 100 % suppresses threshold alarms). The moment the measured concentration exceeds one of the thresholds the related digital output is set and the rightmost decimal point of the related display starts flashing.

In addition the operation mode of the alarm outputs may be factory set to one out of two modes separately for each of the four thresholds utilizing the parameter "Alarm value":

- Alarm value set to "1": Activated alarm outputs a HIGH level signal.
- Alarm value set to "0": Activated alarm outputs a LOW level signal. This mode is commonly known as "fail-safe mode".

By setting the threshold levels the operator has the choice of different operation modes:

- Window mode: If the concentration exceeds the limits defining a concentration window an alarm is activated.
- High and high-high alarms: This mode allows to set a prealarm level and a main alarm level for increasing concentrations
- Low and low-low alarms: This mode allows to set a prealarm and a main alarm for decreasing concentrations.

Refer to the following paragraphs and figures 5-3 to 5-5 for detailed information on alarm settings!

Note!

The operation mode by default is set to "1" and the threshold levels to "lower measuring limit" resp. "upper measuring limit" unless customer has ordered different settings. Refer to the analyzer startup documentation sheet for information about "Analyzer settings" and "alarm value" setup for your specific instrument.

Note!

Alarm value settings affect the digital output behavior only! Decimal point operation mode remains unchanged!

5-5 Digital Outputs

• Defining a window

Defining a window between the high and the low threshold level (fig. 5-3) gives an alarm when the current concentration either exceeds the upper limit (area "D") or falls below the lower limit (area "B").

Maximum one alarm is activated at a time for each channel!

The digital output signal depends on the alarm value settings:

"1" means if an alarm is activated, the corresponding alarm outputs a "High" level signal.

Setting the alarm value to "0" reverses the alarm signal level:

If an alarm is activated, the corresponding alarm outputs a "Low" level signal.

Note!

Alarm value parameter factory setting is "1" unless otherwise specified at time of order.

As long as any alarm is activated the rightmost decimal point at the related display starts flashing.

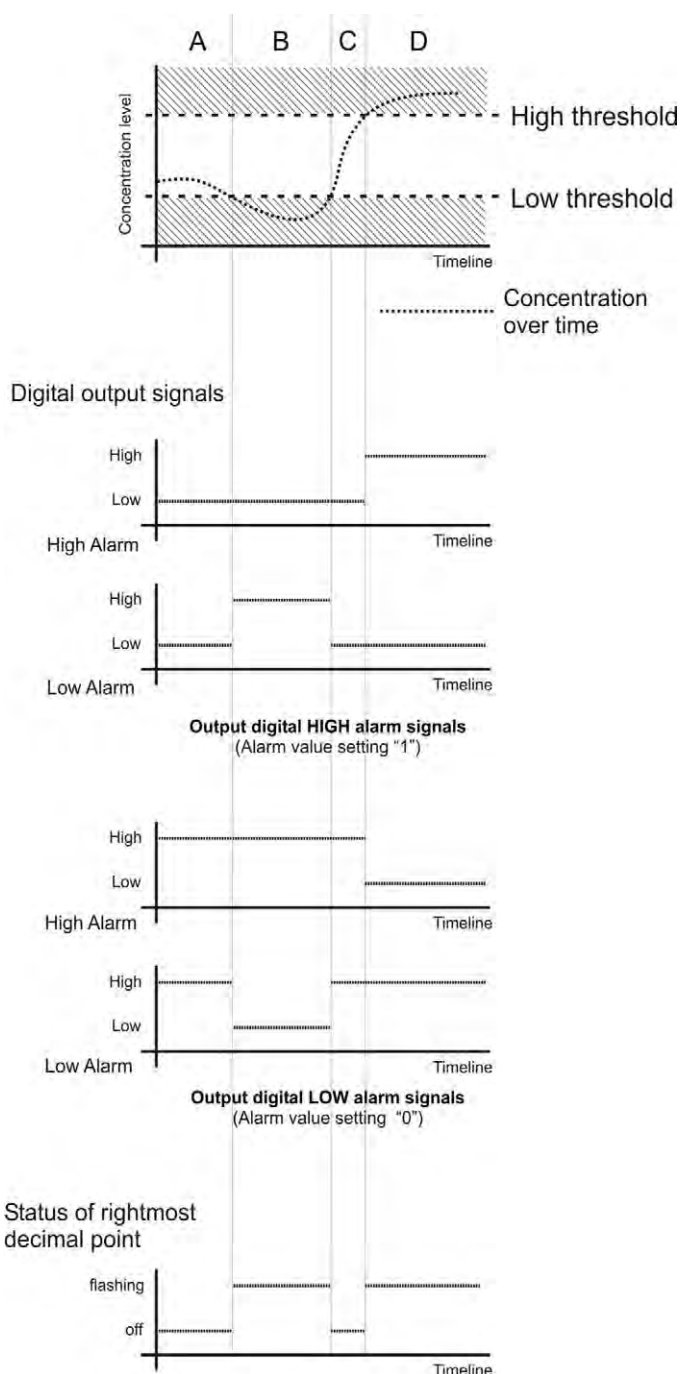


Fig. 5-3: Thresholds defining a window

5-5 Digital Outputs

• Setting "HIGH" and "HIGH-HIGH" alarms

Setting the lower threshold above the allowed level and the upper threshold again above the lower threshold (fig. 5-4) results in an operation mode which gives a prealarm ("HIGH alarm") when the current concentration exceeds the "high" threshold (area "B"). Also, a main alarm ("HIGH-HIGH alarm") is activated if no corrective action was performed and the current concentration exceeds the "high-high" threshold (area "C").

Up to two alarms can be activated at a time for each channel!

For achieving "High" level alarm signals it is necessary to set the alarm value parameter for the lower limit to "0" and for the higher limit to "1".

For achieving "Low" level alarm signals it is necessary to set the alarm value parameter for the lower limit to "1" and for the higher limit to "0".

Note!

Alarm value parameter factory setting is "1" for both unless otherwise specified at time of order.

As the rightmost decimal point setting is not affected by the alarm value parameter its operation mode remains unchanged: Non-flashing when the current concentration is between the limits and flashing when it is outside the limits (see fig. 5-4).



This mode does not comply with the output settings and therefore the decimal point may not be used as an indicator for alarms!

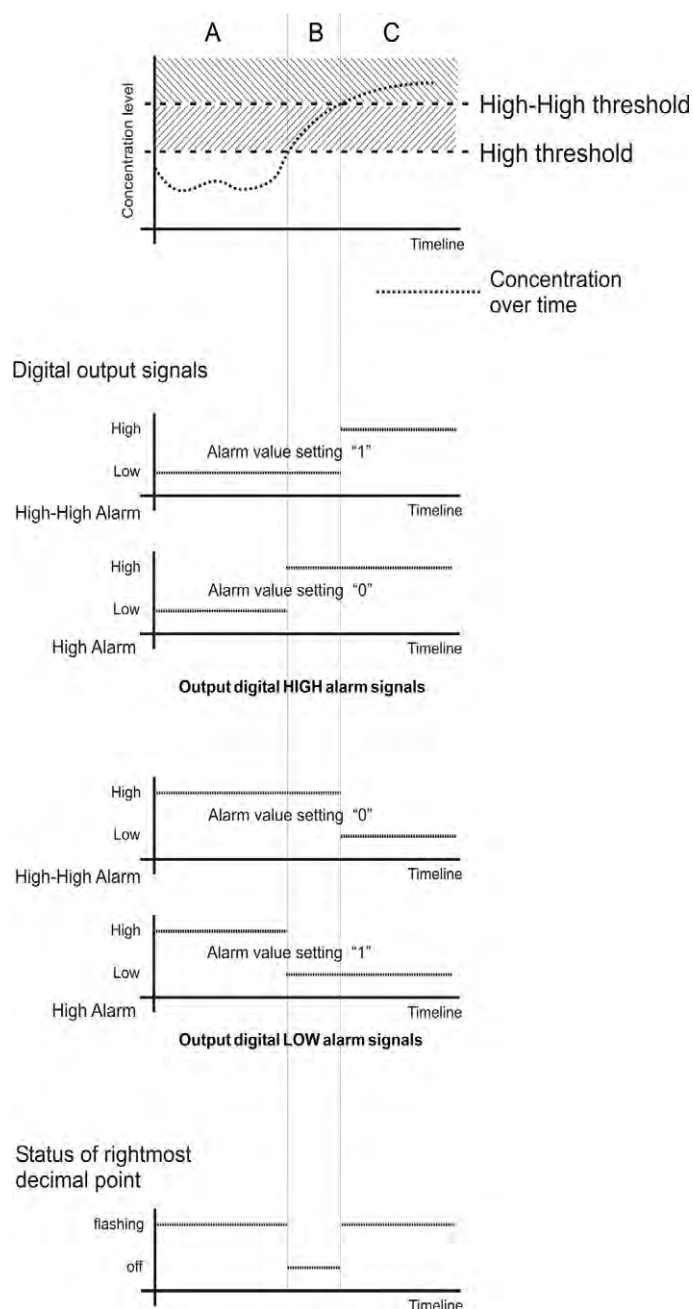


Fig. 5- 4: HIGH and HIGH-HIGH alarm mode

5-5 Digital Outputs

• Setting "LOW" and "LOW-LOW" alarms

Setting the upper threshold below the allowed level and the lower threshold again below the upper threshold (fig. 5-5) results in an operation mode which gives a prealarm ("LOW alarm") when the current concentration falls below the "low" threshold level (area "B"). Also, a main alarm ("LOW-LOW alarm") is activated if no corrective action was performed and the current concentration falls below the "low-low" threshold level (area "C").

Up to two alarms can be activated at a time for each channel!

For achieving "High" level alarm signals it is necessary to set the alarm value parameter for the lower limit to "0" and for the higher limit to "1".

For achieving "Low" level alarm signals it is necessary to set the alarm value parameter for the lower limit to "1" and for the higher limit to "0".

Note!

Alarm value parameter factory setting is "1" for both unless otherwise specified at time of order.

As the rightmost decimal point setting is not affected by the alarm value parameter its operation mode remains unchanged: Non-flashing when the current concentration is between the limits and flashing when it is outside the limits (see fig. 5-5).



This mode does not comply with the output settings and therefore the decimal point may not be used as an indicator for alarms!

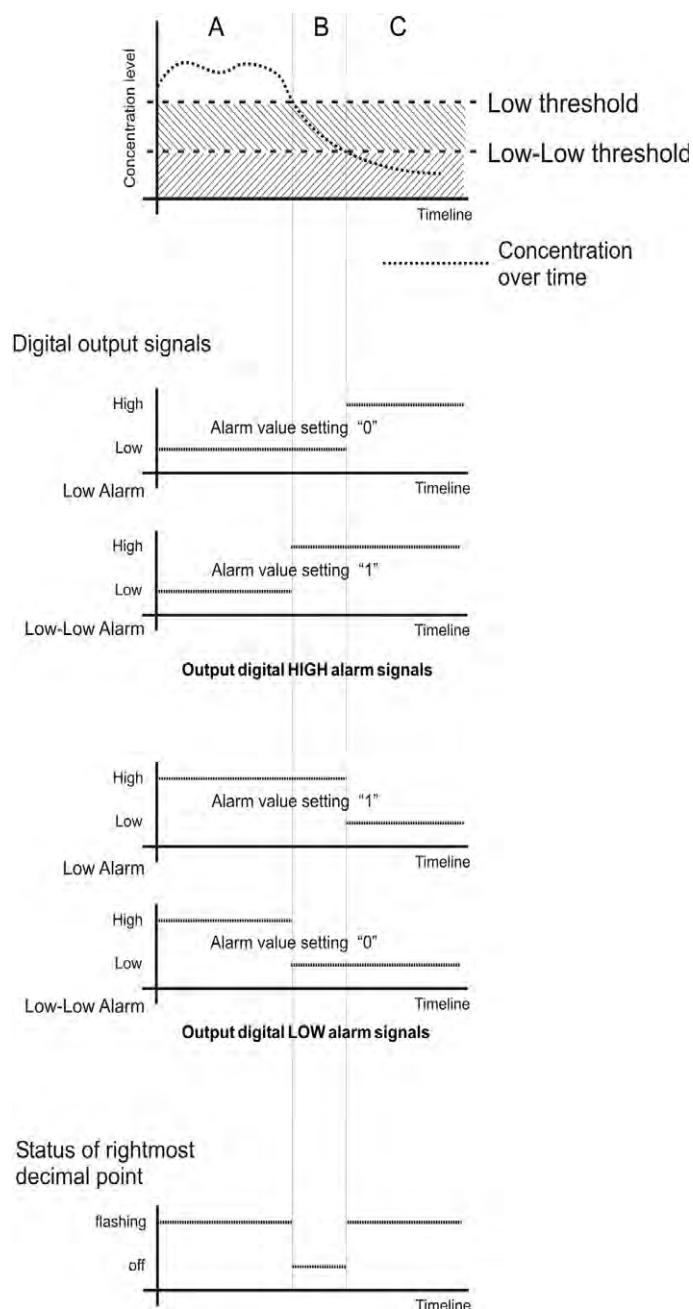


Fig. 5- 5: LOW and LOW-LOW alarm mode

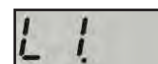
5-5 Digital Outputs

5-5-2 Setting Concentration Thresholds

Press the FUNCTION button several times



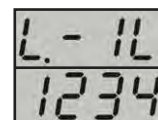
until the display shows



Pressing ENTER opens the access code query (if code not already entered)



Now enter the **lower** concentration threshold of **channel 1**



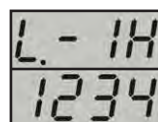
Adjust threshold utilizing



and enter the value using

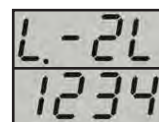


Repeat the steps above for the **upper** threshold of **channel 1**,

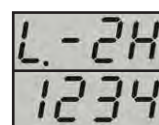


5-5-1 Configuring Concentration Thresholds

for the **lower** concentration threshold of **channel 2** and



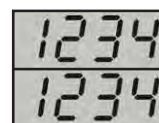
for the **upper** threshold of **channel 2**



Pressing button FUNCTION several times



returns to the measuring display:



5-5-2 Valve Control

5-5-3 Valve Control

Optional external solenoid valves can be controlled when connected to the digital outputs plug X3 (e.g. external valves are required for automatic calibration procedures).

Setting system parameters is required as described in section 5-6 to control the valves

Note!

Take care of the hints about wiring signal outputs, given in section 3-5!

5-5-4 Wiring Digital Outputs

Fig. 5-3 shows pin assignments for X3, exemplary by means of the rear panel submini-d plug, as it is used with tabletop / rackmount instruments. Digital outputs provide optocoupler controlled "open collector" outputs with a maximum load of 30 V DC / 30 mA. External wiring has to comply with fig. 5-4.

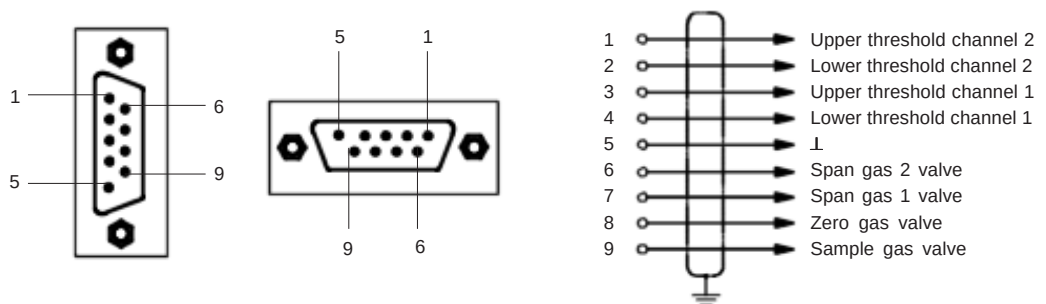


Fig. 5-6: Pin Assignment X3
(digital outputs)

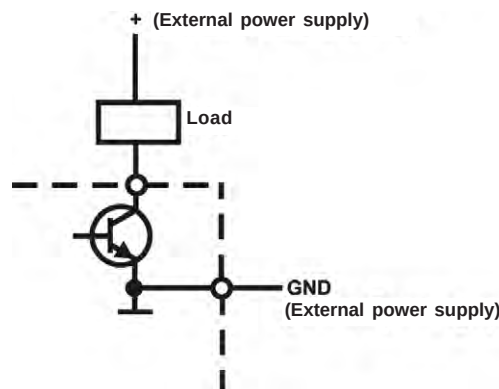


Fig. 5-7: Digital Outputs - External Wiring

5-6 Calibration Procedures

5-6 Calibration Procedures

To achieve correct measuring results it is recommended to carry out zero and span calibrations on a regular weekly basis, whereat a zero calibration must precede a span calibration.

Zero calibration

To perform a zero calibration supply either Nitrogen (N₂) or an other suitable zero gas [conditioned ambient air or industrial air (NOT for Oxygen measurement!)] to the gas path.

Span calibration

Supply span gases with concentrations of 80 % to 110 % of the upper measuring range limit to the gas path. Using lower concentrations may decrease accuracy when measuring above the span gas concentration! If the Oxygen concentration is known ambient air may be used for an Oxygen channel span calibration.



Supply all calibration gases with the same flow as the sample gas (recommended approx. 1 l/min), pressureless and utilizing the right gas fitting (see sect. 3-4).



Ensure the warm-up time after switching on has passed by!

Warm-up time is 15 to 50 minutes depending on measuring system!

Note!

Typically a calibration procedure is carried out manually by supplying the gases sequentially by hand and activating the procedures via the front panel keys.

Utilizing optional internal or external solenoid valves allow to carry out calibration procedures automatically, controlled by the analyzer's software.

The following sections describe in detail how to carry out a manual calibration (sect. 5-5-1) or an automatic calibration (sect. 5-5-2).

5-6-1 Manual Calibration

5-6-1 Manual Calibration

5-6-1-1 Zero Calibration

During a zero calibration the currently measured concentration is declared to be "0". The procedure of a manual zero calibration is as follows:

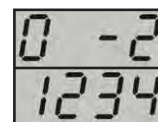
Press button



until the one of these screens appear, depending on which channel to calibrate:



(Channel 1
screens this
column)



(Channel 2
screens this
column)

After pressing



the access code inquiry appears



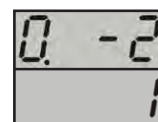
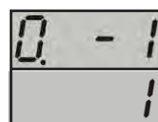
Enter the access code



and confirm by pressing



The display now shows (depending on the channel selected before) one of the following
The lower line shows the current zero level (actual value).



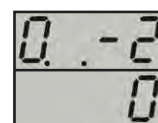
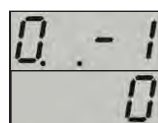
Note the number of decimal points in the upper line to distinguish between these and the following screens!

5-6-1 Manual Calibration

To proceed it's necessary to press
after the adjusted purge and t_{90} times have
elapsed (button is locked until)



The display now shows the zero setpoint value

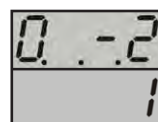
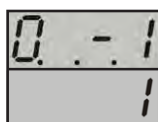


If both values match press FUNCTION to leave
this menu.

If the values differ press



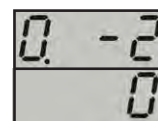
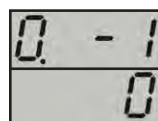
to switch to the next display once more showing
the actual value for zero



and start the calibration procedure by pressing



As soon as the calibration procedure has fin-
ished, the display changes to show the right
now adjusted actual value for zero



Zero calibration procedure has finished and the front panel key pad is unlocked after another purge
time and t_{90} time have elapsed. If HOLD is set to "1" (s. 5-3-5) analog outputs and thresholds are
unlocked together with the keypad.

To leave the zero calibration menu press



5-6-1 Manual Calibration

5-6-1-2 Span Calibration

Performing span calibrations on a regular basis is essential for achieving accurate measuring results!

Always perform a zero calibration before starting a span calibration.

During a span calibration the measured concentration is set to the known span gas concentration set point.

Using gas mixtures for span calibration requires setting the C.Cal parameter to "0" (s. 5-3-4)!

To make use of pressure compensation even if no sensor is built-in, the current ambient pressure must be set before starting the calibration (s. 5-3-2).

Note:

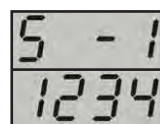
Supply span gases with concentrations of 80 % to 110 % of the upper measuring range limit to the gas path. Using lower concentrations may decrease accuracy when measuring above the span gas concentration! If the Oxygen concentration is known, ambient air may be used for an Oxygen channel span calibration.

Performing a span calibration:

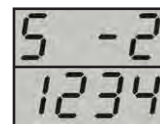
Several times press button



until the one of these screens appear, depending on which channel to calibrate:



(Channel 1
screens this
column)

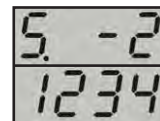
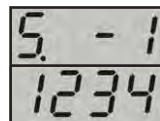


(Channel 2
screens this
column)

After pressing ENTER
the access code inquiry appears (if code has not already been entered) or



the screen changes to show (depending on the channel selected before) one of the following



The lower line shows the current concentration value.

Note the number of decimal points in the upper line to distinguish between these and the following screens!

5-6-1 Manual Calibration

To proceed it's necessary to press
after the adjusted purge and t_{90} times have
elapsed (button is locked until)

Screen changes to show the setpoint value

If differing from the setpoint value given on the
span gas bottle's certificate, change the value
accordingly using

and confirm with

Current measuring value is shown

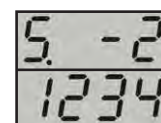
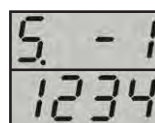
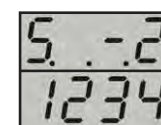
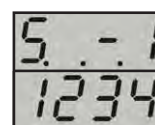
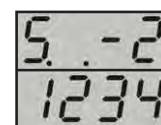
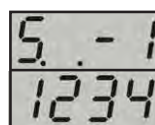
If this value is equal the certificate's value the
calibration procedure may be aborted by
pressing FUNCTION.

To proceed with the span calibration press

As soon as the calibration procedure has
finished the screen again shows the **current
concentration**.

Span calibration procedure has finished and the front panel key pad is unlocked after another purge
time and t_{90} time have elapsed. If HOLD is set to "1" (s. 5-3-5) analog outputs and thresholds are
unlocked together with the keypad.

To leave the span calibration menu press



Continue with section 5-8
"Measuring", page 5-32

5-6-2 Time Controlled Calibration

5-6-2 Time Controlled Calibration

Time controlled calibration requires internal (option) or external solenoid valves controlled by the analyzer's digital outputs!

The system parameter "automatic calibration" (see 5-3-6) must be set to "1" (enabled).

Time controlled calibration means to automatically perform calibration procedures after preset time periods. The time period is user configurable and monitored by the analyzer's micro processor. When the time period has elapsed the micro processor

activated the calibration procedure and at it's end re-starts the time period.

To setup the time controlled calibration select the submenus **t - AO** and **t - AS**.

Note !

*Gases **must** be supplied by analyzer controlled valves (external or internal) zu ensure on-time feeding of calibration gases!*

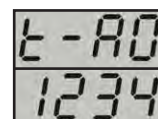
When changing span gas concentrations update the analyzer's setpoint value (page 5-26) before performing a calibration!

5-6-2-1 Zero Calibration

Several times press



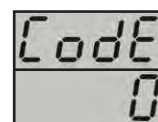
till this screen appears



After pressing ENTER



the access code inquiry appears (if code has not already been entered)



Select access code using

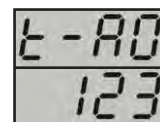


and confirm pressing



5-6-2 Time Controlled Calibration

Screen changes to show



Select a time period (in hrs) to elapse before a time controlled zero calibration has to be performed.

Reference time is time of setting this parameter.

Valid range: **0 - 999** (hrs)

Note !

Entering "0" (zero) disables time controlled calibration!

Select time period using



and confirm pressing



Once confirmed the countdown starts immediately.



*Continue with section 5-8
"Measuring", page 5-32*

5-6-2 Time Controlled Calibration

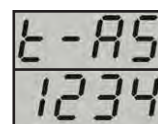
5-6-2-2 Combined Zero and Span Calibration

Use this menu to setup a time controlled zero calibration automatically followed by a span calibration.

Several times press button



until this screen appears:

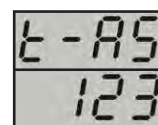


After pressing ENTER



the access code inquiry appears (if code has not already been entered) or

screen immediately changes to show



Select a time period (in hrs) to elapse before a time controlled calibration has to be performed.

Reference time is time of setting this parameter.

Valid range: **0 - 999** (hrs)

Note !

Entering "0" (zero) disables time controlled calibration!

5-7 Remote Calibration

Select time period using



and confirm pressing



Once confirmed the countdown starts immediately.



Continue with section 5-8
"Measuring", page 5-32

5-7 Remote Calibration (Option)

Optional remote calibration requires internal (option) or external solenoid valves controlled by the analyzer's digital outputs!

Verify correct calibration gas setpoint values are set within the analyzer's system parameters (page 5-25).

The system parameter "automatic calibration" (see 5-3-6) must be set to "1" (enabled).

Depending on the software command remote calibration may include separate zero and/or span calibration for each channel (as described in sect. 5-6-1) or a combined zero and span calibration as described in sect. 5-6-2-1.

Remote calibration can be initialized by sending the command utilizing one of the following options

- serial interface (section 6-2)
- digital inputs (section 6-3)
- FOUNDATION™ Fieldbus (section 6-3)

Refer to the listed sections for detailed information about software commands or how to configure the digital inputs.

5-8 Measuring

5-8 Measuring

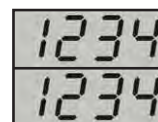
Once the analyzer has been calibrated it is prepared to start measuring:

Connect your sample gas to the sample input fitting, switch on the sample gas pump (optional), adjust the flow rate.

If not already in the measuring display, press



several times to switch to the measuring display.



For best measuring results allow 15 to 50 minutes warm-up time (depending on detector) after switching on the analyzer.

Note !

After 60 to 120 seconds without operator activity the analyzer returns to the measuring display regardless the currently selected submenu.

The analyzer keeps the measuring display until the operator selects another function.

If system parameter "display off" is set to "1" display is turned off after 60 to 120 seconds without operator activity!

5-9 Putting Out of Operation

Special hints on electro chemical Oxygen sensors !

Due to the measuring principle the electrochemical oxygen cell requires a minimum internal consumption of oxygen (residual humidity avoids drying of the cell). Supplying cells continuously with dry sample gas of low grade oxygen concentration or with sample gas free of oxygen could result in a reversible detuning of O₂ sensitivity. The output signal will become instable, but response time remains constant.

For correct measurement the cell needs continuously to be supplied with concentrations of at least 0.1 Vol.-% O₂. We recommend to use the cells if need be in alternating mode, means to purge cells with conditioned (not dried, but dust removed) ambient air when measurement pauses.

If it is necessary to interrupt oxygen supply for several hours or days, the cell has to regenerate (supply cell for about one day with ambient air). Temporary flushing with nitrogen (N₂) for less than 1 h (e.g. for analyzer zeroing purpose) has no influence on measuring characteristics.

5-9 Taking Out of Operation

If you intend not to use the analyzer for a longer period or to take it out of operation, purge all gas paths with Nitrogen (N₂) or ambient air for at least 5 minutes.



Purge gas paths of analyzers with electro chemical Oxygen sensors with conditioned ambient air only, before sealing gas fittings!

- apply Nitrogen or ambient air to sample input fitting,
- optionally switch on the sample gas pump,
- adjust the gas flow.

After at least 5 minutes:

- Switch off the analyzer by disconnecting from power.
- Disconnect all gases.
- Immediately seal all gas fittings to avoid internal gas path pollution.

Chapter 6 Description of Options

As mentioned in chapter 1 all analyzers may be equipped with several options to best possible match the requirements of your specific analytical application. Available options are described in detail in this chapter.

Note!

For a description of the magnetically operated front panel (option for BINOS® 100 F, but standard for CAT 100) refer to section 4-7 (page 4-9).



Take care that not all described options are available for all instruments!

Furthermore selecting some options may exclude the availability of other options. These limitations are noticed within the concerned section.

<i>For description of</i>	<i>digital status signals</i>		<i>page 6-2</i>
	<i>serial interface</i>		<i>page 6-3</i>
	<i>digital inputs</i>		<i>page 6-16</i>
	<i>FOUNDATION™ Fieldbus</i>		<i>page 6-18</i>
	<i>cross compensation (TC)</i>		<i>page 6-19</i>

6-1 Option Relay status Signals

6-1 Relay Status Signals

Three relay status outputs are available to digitally monitor the analyzer status. These outputs are accessible at the rear panel plug X1 (table top / rack mount versions) resp. on internal screw terminals (wall mount instruments).

The status relays provide floating contacts with a maximum load of 30 V / 1 A / 30 W !

Each contact provides 3 terminals:

- 1 input (COM)
- 2 outputs, connected in parallel, design is either normally closed (NC) or normally open (NO)

The operation mode (NC / NO) is NOT user configurable and has to be defined at time of placing the analyzer order (external wiring see fig. 6-1).

Exemplary fig. 6-2 shows the pin assignment of the rear panel plug X1, as it is used with tabletop / rack mount analyzers.

Note!

Take care of the hints about wiring signal outputs, given in section 3-5!

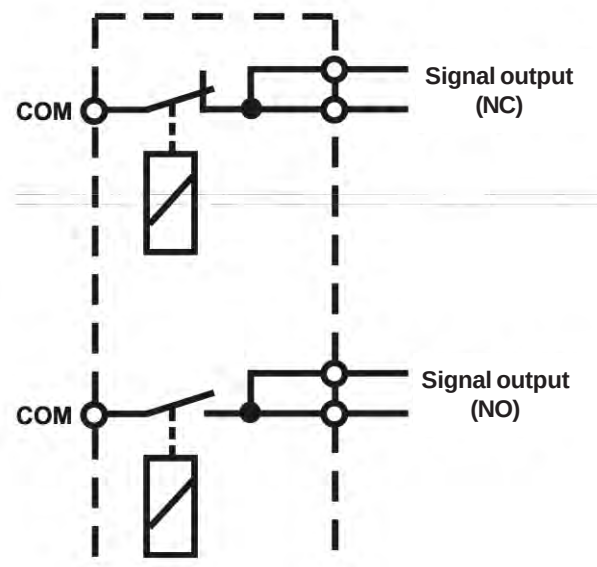


Fig.6-1: Relay status Signals, schematic

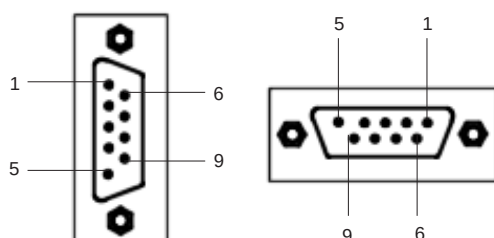
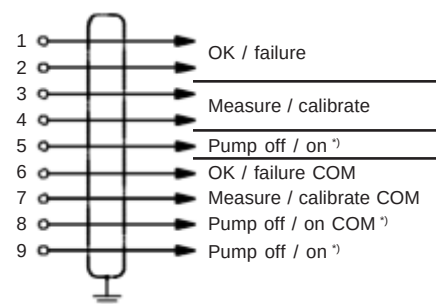


Fig.6-2: Pin assignment X1 (relay status signals)



*) BINOS® 100 2M / BINOS® 100 F / CAT 100 only

6-2 Option Serial Interface

6-2 Serial Interface

Note!

Option "serial interface" is not available in combination with Foundation Fieldbus!

The analyzers may be equipped with a serial interface to provide communication with an external host (e.g. data acquisition system). The interface supports parameter data transmission, changing of parameters and initializing functions (procedures).

The simplest serial interface variation is of type RS 232 and does NOT provide galvanic isolation against analyzer electronics.

Interfaces with galvanic isolation are optionally available, too. These versions are of type RS 232 or RS 485. An RS 485 interface allows to connect not only one analyzer but to built up a network of several instruments, whereat each instrument is identified within the network by a unique ID.

The interface is accessible via rear panel submin-d plug X4 (tabletop / rack mount instruments; pin assignment s. fig. 6-3) resp. internal screw terminals (wall mount versions).

Read the following instructions carefully to obtain failure-free data communication!

RS-232 Interface:

Acceptable burden range is 3 kΩ to 7 kΩ!

New definitions within EIA/TIA-232-D now reference a maximum permittable cable capacitance instead of a maximum cable length.

The maximum cable length therefore is calculated taking into account the specific stray capacitance, given in the cable data sheet:

$$L = 2200 \text{ pF} / C_x$$

with

L: cable length in m
Cx: cable stray capacitance in pF / m (see datasheet)

RS-485 Interface:

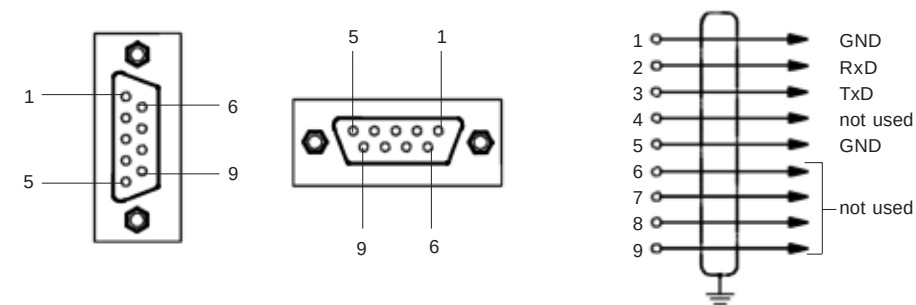
This interface is designed to overcome large distances.

Setting up the connection by using twisted pair cables and considering sound engineering practice allows to achieve cable lengths of 1,000 m and more, whereat universally valid values can not be given.

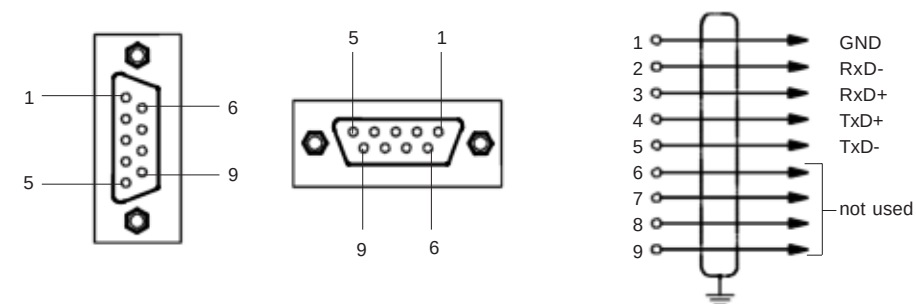
Furthermore a connection needs to be terminated at both ends, means the first and the last instrument within a chain require a termination, the instruments between not.

BINOS® 100 series analyzers are shipped without termination, unless order specifies something different. Termination can not be set by operator, because this requires opening the enclosure and configuring solder bridges on electronic boards. Contact your EMERSON Process Management sales office if you want to change termination after instrument has been shipped!

6-2 Option Serial Interface



RS 232 Configuration



RS 485 Configuration

Fig. 6-3: Socket X4 - Pin Assignment
(Serial interface)

6-2 Option Serial Interface

6-2-1 Enabling / Disabling Serial Interface

The instrument's serial interface may be switched "on-line" or "off-line" utilizing either the front panel key pad or a software command.

Toggling via key pad:

SIP - Parameter

On. - L. = 1 on-line (communication enabled)

On. - L. = 0 off-line (communication disabled)

disabled)
Toggling via software command:

command # 6: on-line (communication enabled)

command # 7: off-line (communication disabled)

When set to "**off - line**" mode, the analyzer accepts a command # 6 only, all other software commands are ignored and replied by a corresponding status telegram.

6-2-2 Configuring the Interface

Failure free communication requires all connected interfaces to use the same parameters.

The analyzer software allows to setup the following interface parameters:

- ID - Nr.: 0 to 99 (instrument-ID for RS 485 interface)
- Baud rate: 600 / 1,200 / 2,400 / 4,800 / 9,600 / 19,200 / 38,400 bits/s

- Echo: on / off (reply on received commands with same command)

- LPB test: on / off (longitudinal parity test)

The following parameters are not changeable:

- Data bits: 8
- Stop bits: 2
- Parity: none

Enter the submenu "SIP." to set interface parameters (see next side):

Set selected parameter using



and submit using

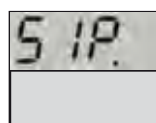


6-2 Option Serial Interface

Starting at the measuring display press



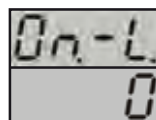
several times to see this screen



After pressing ENTER
the access code inquiry appears (if code has
not already been entered) or



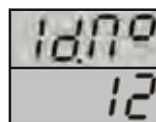
the screen changes to show the first interface
parameter menu (set on-line / off-line)



0: off-line
1: on-line

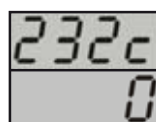
followed by pages to set

the instrument ID,



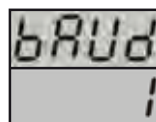
0 ... 99

the type of interface,



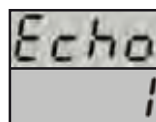
1: RS 232
0: RS 485

baud rate,



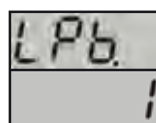
0: 4,800 1: 2,400
2: 1,200 3: 600 4: 9,600
5: 19,200 6: 38,400

echo mode



0: aus
1: an

and longitudinal parity test.



0: aus
1: an

6-2 Option Serial Interface

6-2-3 Communication

Any interface communication has to be activated by the host: the analyzer remains passive until the host sends a request for information or action.

Communication takes place utilizing so called telegrams with a syntax defined by a fixed protocol.

Any telegram starts with a start sign "\$", followed by a 3 digit command number.

Further code within one telegram is separated by ";".

A "CR" END marker concludes any telegram.

After receipt of an END marker the analyzer tries to interpret the telegram as valid. If the telegram syntax is valid the analyzer sends a reply telegram consisting of **start sign**, **command number**, **data** (if required), **block parity bit** and **END marker**. If the telegram is not valid the analyzer replies with a status telegram containing a failure information. So receiving a END marker always results in an analyzer reply message.

To detect submission failures the host may include a longitudinal parity byte before sending the END marker, which can be verified by the analyzer. The analyzer itself is always sending a longitudinal parity byte.

There is no time-out between start sign and END marker; the time is not limited.

Telegrams sent by the host before the analyzer has sent its reply on a earlier telegram are not stored in the input buffer and therefore ignored by the analyzer.

The analyzer software is designed to send a new telegram at least every 150 ms.

6-2 Option Serial Interface

6-2-3-1 Telegram Syntax

A telegram is composed as follows:

Start sign ("\$" = Hex 24)

Missing start sign result in a related status telegram sent out by the analyzer.

End marker ("CR" = Hex 0D)

Without END marker the telegram is not decoded, the analyzer does not respond, no reply is sent.

Command number

Each command is identified by a 3 digit number. If the sent number has less than 3 digits or contains non numeric characters the analyzer generates a status telegram.

Not used or not assigned command numbers generate a status telegram, too.

In RS 232 mode the command number follows directly after the start sign while in RS 485 mode the instrument ID is sent first, followed by the separator (" ; ") and the command number.

Separator(";" = Hex 3B)

Multiple elements within one telegram line are separated by a " ; ". A missing " ; " results in a miss interpretation and is answered with a status telegram.

6-2 Option Serial Interface

6-2-3-2 Status Telegram

Status telegrams are sent when the syntax of a received telegram is faulty or if the analyzer can't perform an action.

Possible status telegrams:

\$ID;000;S100;LPB<CR>	Unknown command number
\$ID;000;S101;LPB<CR>	LP byte faulty
\$ID;000;S102;LPB<CR>	Start sign faulty
\$ID;000;S103;LPB<CR>	Input buffer overflow
\$ID;xxx;S104;LPB<CR>	Analyzer off - line
\$ID;xxx;S105;LPB<CR>	Command line too long
\$ID;xxx;S106;LPB<CR>	Unknown command
\$ID;xxx;S107;LPB<CR>	Invalid integer number
\$ID;xxx;S108;LPB<CR>	Number out of range
\$ID;xxx;S109;LPB<CR>	Invalid failure / status no.
\$ID;xxx;S110;LPB<CR>	Command currently not executable
\$ID;xxx;S111;LPB<CR>	Character failure during transmission
\$ID;xxx;S112;LPB<CR>	Zero calibration active
\$ID;xxx;S113;LPB<CR>	Span calibration active
\$ID;xxx;S114;LPB<CR>	Invalid real number
\$ID;xxx;S115;LPB<CR>	Automatic function disabled
\$ID;xxx;S116;LPB<CR>	Parameter out of range
\$ID;xxx;S117;LPB<CR>	Purging active

Legend:

xxx : Command number
ID : Instrument ID (in RS 485 mode)
LPB : Longitudinal parity byte
<CR> : END marker

6-2 Option Serial Interface

6-2-3-3 Number Formats

A telegram may contain integer and real numbers with the following limitations:

Integer:

- maximum number is $2^{16} - 1$
- positive values only
- no decimal point

Real:

- 6 digits maximum
- no alpha characters (e.g. 2.2E-6)
- analyzer sends real numbers with 6 digits only

6-2-3-4 Longitudinal Parity Test

The host may include a longitudinal parity byte (LPB) before sending the END marker. The LPB consists of 2 characters.

A LPB is a XOR function of all characters sent in the command line before.

LPB format is hexadecimal: a value of "13" is

sent as the two characters 0D (030H and 044H).

LPB testing may be enabled or disabled (see 6-2-2).

6-2-3-5 Command Syntax

Legend of used symbols:

EP :	Receiving parameter analyzer accepts value
SP :	Sending parameter analyzer sends value
EK :	Receiving command
k :	Channel number (0 to 2)
m :	Measuring range number
w :	Value
<ID>:	Instrument ID (RS 485 mode) after start sign
LPB :	Longitudinal parity byte
<CR>:	END marker

All command numbers not listed in the following section are replied to with status 106.

Not yet used numbers are spare for future extensions (if necessary).

6-2 Option Serial Interface

6-2-3-6 List of Available Commands

Command syntax:	Description:
\$ID;001;k;LPB<CR>	EK stand by
\$ID;002;k;LPB<CR>	EK sample gas valve open
\$ID;003;k;LPB<CR>	EK zero gas valve open
\$ID;005;m;k;LPB<CR>	EK span gas valve open
\$ID;006;LPB<CR>	EK on-line
\$ID;007;LPB<CR>	EK off-line
\$ID;008;LPB<CR> *)	SP pump status *)
\$ID;009;w;LPB<CR> *)	EK pump status *)
\$ID;011;m;k;LPB<CR>	SP measuring range limit value
\$ID;013;k;LPB<CR>	SP t_{90} - time
\$ID;014;w;k;LPB<CR>	EP t_{90} - time
\$ID;017;k;LPB<CR>	SP purge time
\$ID;018;w;k;LPB<CR>	EP purge time
\$ID;019;k;LPB<CR>	SP purge time
\$ID;020;w;k;LPB<CR>	EP purge time
\$ID;023;k;LPB<CR>	SP concentration
\$ID;028;m;k;LPB<CR>	SP span gas concentration
\$ID;029;m;w;k;LPB<CR>	EP span gas concentration
\$ID;030;LPB<CR>	SP status messages
\$ID;031;t;LPB<CR>	SP serial number (t=0, max. 10 characters), channel code (t=1: channel 1 / t=2: channel 2)
\$ID;603;k;LPB<CR>	SP gas component
\$ID;604;k;LPB<CR>	EK automatic zero calibration
\$ID;605;k;LPB<CR>	EK automatic span calibration
\$ID;606;0;LPB<CR>	EK automatic zero and span calibration
\$ID;626;LPB<CR>	SP E-Prom / software revision
\$ID;627;LPB<CR>	SP failure message (possible " batt " failure is reset during readout)
\$ID;645;0;LPB<CR>	SP current pressure value
\$ID;646;w;LPB<CR>	SP valve status (w=1: sample gas valve, w=2: zero gas valve, w=4: span gas valve ch. 1, w=8: span gas valve ch. 2)

6-2 Option Serial Interface

6-2-3-7 Reply Telegrams

Reply telegrams contain the same syntax as the related command (SP) telegrams but with additional parameters (s. 6-2-3-5).

6-2-3-7-1 Status Messages (Command 030)

The reply telegram for command "SP status messages" \$ID;030;LPB<CR>

looks like \$ID;030;a;b;c;LPB<CR>

Characters a, b, c stand for:

a: Value of variable "OK status" 0 = Relay powered off 1 = Relay powered

b: Value of variable "Calibration" 0 = Relay powered off >0 = Relay powered

b	Function
0	No calibration
1	Zero calibration channel 1
2	Zero calibration channel 2
3	Zero calibration channel 1 & 2
4	Span calibration channel 1
5	Span calibration channel 2
6	Start zero calibration channels 1 & 2, continue with span calibration channels 1 & 2
7	Span calibration channels 1 & 2
8	reserved
9	reserved
10	Wait for purge time and t_{90} -time to elapse

c: Sample gas pump *) 0 = Relay powered off (pump off) >0 = Relay powered (pump on)

*) BINOS® 100 2M / BINOS® 100 F/ CAT 100

6-2 Option Serial Interface

6-2-5-7-2 Pump Status (Command 008) *)

The reply telegram for command "SP pump status" \$ID;008;LPB<CR>

looks like \$ID;008;w;LPB<CR>

with w: pump status 0 = pump off 1 = pump on

*) BINOS® 100 2M / BINOS® 100 F/ CAT 100

6-3 Option Digital Inputs / Foundation™ Fieldbus

6-3 Option Digital Inputs / FOUNDATION™ Fieldbus

Note!

This option is available for BINOS® 100 2M, BINOS® 100 F and CAT 100 only.

The option FOUNDATION™ Fieldbus is not available in combination with serial interface!

6-3-1 Digital Inputs

BINOS® 100 2M, BINOS® 100 F and CAT 100 optionally provide digital inputs to activate valves or start calibration procedures by simply supplying a voltage. Therefore digital inputs require internal valves or external valves connected to digital outputs to work properly (see sect. 6-1).

Digital inputs are accessible at screw terminals at the analyzer's rear panel (BINOS® 100 2M; see fig. 1-6) or at internal screw terminals (see schematics in annex).

Fig. 6-4 shows that inputs E1 to E3 are used to start calibration procedures while inputs E4 to E7 are used to activate valves resp. to switch a pump (E/).

Working Mode

A selected input is activated by applying an external HIGH level control voltage for at minimum 2.5 seconds. To detect a HIGH signal, the dc voltage must be ≥ 4.5 V, to detect a LOW level the dc voltage has to be ≤ 1.5 V, measured against the reference terminal labeled "V-".

An auxiliary DC +5 V supply is available at terminal "V+" to enable using floating relay contacts to activate digital inputs.

To avoid problems caused by HIGH level signals at multiple inputs, each input is assigned a priority: The activated input with the highest priority is evaluated only.

Priorities are assigned as follows:

Input	E1 - Priority 1
	E2 - Priority 2
	E3 - Priority 3
	E4 - Priority 4
	E5 - Priority 5
	E6 - Priority 6
	E7 - Priority 7

Electrical data

Inputs:

LOW: $U_{in} \leq 1.5$ V

HIGH: $U_{in} \geq 4.5$ V

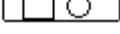
Input resistance: 57.5 kΩ

Input protected against overvoltage up to approx. 40 V. Open input is at LOW level.

Auxiliary supply voltage:

DC voltage +5 V; 2.5 mA

short circuit protected

V+		+ 5 V DC
V-		⊥
E1		Start zero calibration channels 1 & 2
E2		Start span calibration channel 1
E3		Start span calibration channel 2
E4		Open zero gas valve
E5		Open span gas 1 valve
E6		Open span gas 2 valve
E7		Close sample gas valve / sample gas pump off

**Fig. 6-4: Digital Inputs,
Pin assignments for BINOS® 100 2M**

6-3 Option Digital Inputs / Foundation™ Fieldbus

6-3-2 Starting a Calibration Procedure

Apply a HIGH level voltage to one of the inputs E1 to E3 for at least 2.5 seconds to start a calibration procedure. When HIGH level voltages are supplied to multiple inputs the inputs are evaluated in decreasing order of priority: highest priority is assigned to E1, lowest priority to E3.

Refer to table 6-1 for the array:

Action / Input	E1	E2	E3
Start zero calibration	H	X	X
Start span calibration ch 1	L	H	X
Start span calibration ch 2	L	L	H
	H: HIGH		
	L: LOW		
	X: don't care		

Table 6-1: Digital Inputs E1-E3, array

6-3-3 Valve Control

Inputs E4 to E7 are evaluated in decreasing order of priority: highest priority within this group is assigned to E4, lowest priority to E7.

Applying a HIGH signal to one of these inputs opens the associated valve and closes all other valves, regardless of the inputs of lower priority. E7 has a converted input logic: applying a LOW signal **opens** the valve, closing the valve requires a HIGH level signal.

Action / Input	E4	E5	E6	E7
Open zero gas valve	H	X	X	X
Open span gas 1 valve	L	H	X	X
Open span gas 2 valve	L	L	H	X
Open sample gas valve	L	L	L	L
	H: HIGH			
	L: LOW			
	X: don't care			

Table 6-2: Digital Inputs E4-E7, array

6-3-4 Pump Control

Input E7 may be used to control the optional internal sample gas pump: applying a LOW signal switches on the pump, a HIGH level signal switches off.

Pump control depends on the E7 input only and is independent on inputs E1 to E6!

Note!

Pump control via digital input E7 requires system parameter "P.Ctl" set to "2" (s. page 5-13, section 5-3-18).

6-3 Option Digital Inputs / Foundation™ Fieldbus

6-3-5 Foundation™ Fieldbus

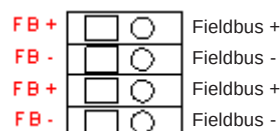
Note!

Option "FOUNDATION™ Fieldbus" is not available in combination with serial interface!

Option "FOUNDATION™ Fieldbus" requires software revision 5.x1 (examine via system parameter menu, see page 5-1).

FOUNDATION™ Fieldbus communication is provided via rear panel screw terminals (BINOS® 100 2M; fig. 6-5) resp. via internal screw terminals (see schematics in annex).

For a detailed description of FOUNDATION™ Fieldbus refer to instruction manual ETC00624, shipped together with all instruments providing FOUNDATION™ Fieldbus communication!



**Abb.6-5: FOUNDATION™ Fieldbus,
terminal assignment BINOS® 100 2M**

WARNING

EXPLOSION HAZARD

FOUNDATION™ Fieldbus installations may be intrinsically safe (Ex i) for use in hazardous areas !



Foundation™ Fieldbus provided by CAT 100 is NOT intrinsically safe (NOT Ex i) and therefore is NOT allowed to be connected to Ex i installations! Violation may cause explosions, damage to installations, severe injury and death !

Installations within hazardous areas must be carried out by personnel familiar with the special requirements of such locations and instructed about the risks and legislative and local installation / safety requirements !

6-4 Option Cross Compensation (TCD)

6-4 Cross Compensation (TCD)

Note!

Option "Cross Compensation (TCD)" is available for thermal conductivity measurement only!

Thermal conductivity measurement has one disadvantage compared to e.g. infra red measurement when measuring complex gas mixtures:

As long as a gas mixture is composed of at maximum two components with differing thermal conductivity (so called "binary mixtures"), the gas analyzer can be calibrated to show the concentration of one of these components. The same is valid for quasi binary mixtures composed of more than two components but with only one component changing its concentration while the accompanying mixture remains constant.

Typically the gas mixture to be analyzed is composed of several components of varying proportions. In this case the analyzer's measuring result is not selective and therefore not assignable to the component of interest. To get a solution anyhow, it is required to measure the accompanying components by using additional instruments and to supply their results electronically to the TC analyzer for cross compensation needs. The result of such a electronic cross compensation is a

measuring value directly assigned to the component of interest.

BINOS® 100 series analyzers offer electronic cross compensation of up to three accompanying components as an option for TC measurement only. These component have to be measured by separate instruments and the results to be provided as electronic signals.

Acceptable signals are 0 (0.2)- 1 V, 0 (2)- 10 V or 0 (4)-20 mA, whereat the corresponding measuring ranges must be defined (e.g. 4-20 mA correspond to 0-20 % CO₂).

Cross compensation signals are supplied by rear panel connector "Analog In" (9-pole submin-d; table top / rackmount instruments) or internal screw terminals (wall mount instruments). Fig. 6-6 exemplary shows the pin assignment of the rear panel connector. For terminal assignment of wall mount instruments refer to schematics in annex.

Note!

Take care of the hints about wiring signal outputs, given in section 3-5!

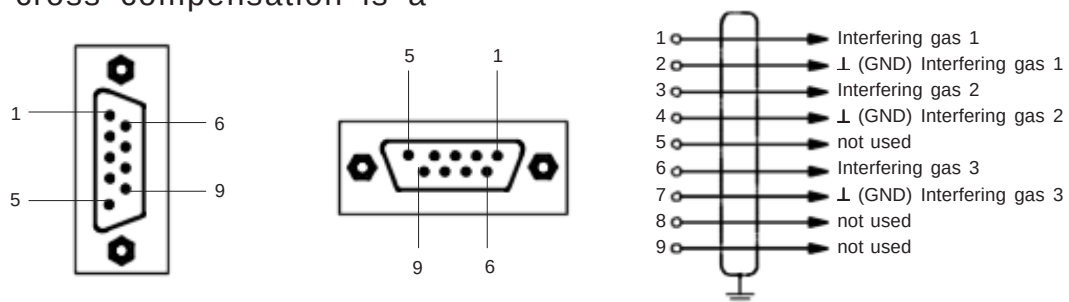


Fig.6-6: Plug "Analog In", pin assignment (cross compensation)

6-5 Option Fine Dust Filter / Needle Valve

6-5 Fine Dust Filter / Needle Valve

Note!

Option "Fine Dust Filter / Needle Valve" is available for variations BINOS® 100 F (not in combination with magnetically operated front panel) and BINOS® 100 2M only. An additional flow meter is mounted onto the front panel.

Depending on application and pollution the filter element has to be replaced on a regular basis (see "Maintenance" in section 9-1, page 9-2).

6-5-1 Fine Dust Filter

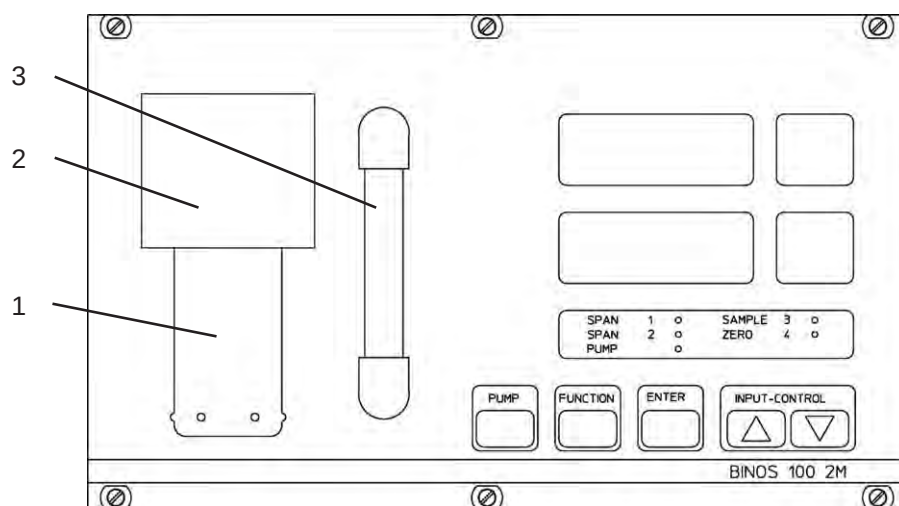
The optional fine dust filter may be mounted on the front panel providing additional protection for the gas paths against pollution by small particles (fig. 6-7).

A filter element with a pore size of 2 µm is placed within a cap made of glass. Gas is supplied to the filter carrier and flowing from the outside through the filter element to the inner side, making it easy to recognize pollution.

6-5-2 Needle Valve

In addition to the filter element the filter carrier contains a needle valve.

By utilizing a screwdriver and a screw on top of the filter carrier (fig. 6-8) the gas flow may be adjusted between "0" and "max". The currently adjusted flow can be monitored by means of the flowmeter on the front panel (fig. 6-7 and section 6-6).



- 1 Fine dust filter (Option)
- 2 Filter carrier with needle valve
- 3 Flowmeter

Fig.6-7: Fine Dust Filter with Needle Valve at BINOS® 100 2M Front Panel

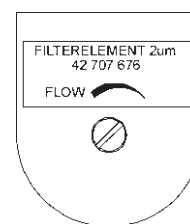


Fig.6-8: Filter carrier with needle valve, view at screw for flow adjustment

Chapter 7 Bug Fixing

7-1 Error Codes

Some failures possibly occurring during normal operation are indicated by error codes on the displays.

In case of an failure the measuring display alternatly is replaced by an error code in form of

E. XX (E = error (failure), XX = code).

Optionally a digital status signal "Failure" may be generated (section 6-1, page 6-2).

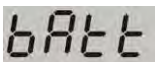
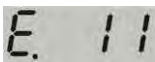
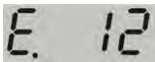
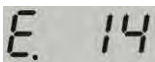


Take care of safety instructions when working on the instrument!

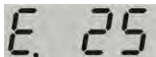
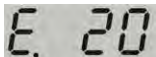
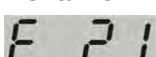
The following table describes the different error codes, gives hints on potential causes and describes how to solve the problems.

Error Code	Potential Cause	How to Solve
Display dark	1. Display switched off 2. Power supply missing BINOS® 100 F 3. Connection front panel / BKS faulty	1. Press any key Check parameter dOFF (s. section 5-3-8). 2. Check mains connection Check external power supply Check mains supply Check fuse on electronics board BKS (s. 8-3-1, page 8-27) Check internal fuses F1, F2 3. Check connection front panel board BKB to BKS (X1) (fig. 8-25, page 8-27).

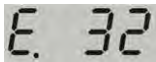
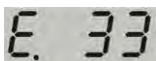
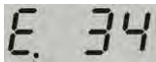
7-1 Error Codes

Error Code	Potential Cause	How to Solve
 flashing	1. Battery buffer faulty 2. EPROM default values loaded	Check if jumper J7 is set correctly (s. 8-4, page 8-32). replace battery, if battery voltage is below 3.5 V (BKS jumper J7 set) Error code is reset by pressing any key or by sending software command \$ 627
 Channel 1  Channel 2 A/D conversion finished signal missing	Only one code appears: 1. A/D converter of the associated channel faulty Both codes appear: 2. Positive or negative reference voltage is missing 3. Light barrier signal missing 4. IR channel: chopper motor not turning 5. Supply voltage missing (internal 6 V DC)	1. Switch instrument off and on again 2. Check reference voltages (see 8-2-1-2 & 8-2-1-3). 3. Check connection light barrier / X9 (s. fig. 8-25, page 8-27). Check measuring point acc. 8-2-1-5 4. Check connection chopper motor / X2 (s. fig. 8-25, page 8-27). Check measuring point acc. 8-2-1-5 5. Check measuring point acc. 8-2-1-1
 Temperature compensation faulty	1. Temperature channel A/D conversion failed 2. Supply voltage missing (internal 6 V DC)	1. Switch instrument off and on again 2. Check measuring point acc. 8-2-1-1

7-1 Error Codes

Error Code	Potential Cause	How to Solve
 Channel 1  Channel 2 Tolerance failure	1. Span gas setpoint value exceeds linearisation table (above upper range limit!)	1. Correct span gas setpoint value
 Channel 1  Channel 2 Tolerance failure Zero gas differs more than 10 % of measuring range from zero	1. Wrong zero gas used 2. IR channel: Photometric components polluted 3. Instrument not yet calibrated	1. Check zero gas 2. Check photometric components Clean photometric components 3. Switch off tolerance check before starting a calibration procedure
 Channel 1  Channel 2 Tolerance failure Span gas differs more than 10 % from setpoint value	1. Wrong setpoint value 2. Wrong span gas used 3. IR channel: Photometric components polluted 4. Instrument not yet calibrated	1. Check setpoint value 2. Check span gas 3. Check photometric components Clean photometric components 4. Switch off tolerance check before starting a calibration procedure
 Channel 1  Channel 2 Tolerance failure Measuring value more than 10% above full scale	1. Sample gas concentration too high (more than 10 % above full scale)	1. Check sample gas concentration 2. Use another analyzer suitable for the sample gas concentration

7-1 Error Codes

Error Code	Potential Cause	How to Solve
 Channel 1	Time-out for XON (serial interface)	Missing XON when driving serial interface (time-out > 60 sec.).
 Channel 1  Channel 2 Threshold limit entry faulty	Threshold limit entered is faulty due to a wrong decimal point	Set threshold limit to acceptable value utilizing submenu "LI."
 Channel 1  Channel 2 Wrong span gas value	Span gas value entered is faulty due to a wrong decimal point	1. Perform span calibration to correct setpoint value 2. Send command "\$029" to serial interface to correct setpoint value
 Channel 1  Channel 2 Wrong offset value	Offset value entered is faulty due to a wrong decimal point	Correct offset value "OFS.x"

7-1 Error Codes

Error Code	Potential Cause	How to Solve
 Faulty pressure measurement	1. Measuring range exceeded 2. Connection faulty 3. Pressure sensor defective	1. Current pressure exceeds sensor measuring range 2. Check connection at Sensor 3. Replace pressure sensor
 EPROM checksum faulty	1. EPROM defect 2. BKS defective	1. Replace EPROM (see service manual) 2. Replace BKS
 SRAM test faulty	SRAM / BKS defective	Replace BKS

7-2 Other Problems

7-2 Other Problems

Problem	Potential Cause	How to Solve
Analog output signal missing	BKS defective	Replace BKS
Fluctuating or invalid readout	1. Leak in gas path 2. Ambient air contains high concentration of measured gas component 3. Fluctuating gas pressure 4. Oxygen sensor / signal detector not connected	1. Perform leak test (sect. 9-2, page 9-3) 2. Check absorber (at chopper / cells) and replace if need be Replace photometer by sealed version (option). Purge instrument with neutral gas 3. Check gas path before and behind cell and sensor Remove restriction behind gas outlet Reduce pump rate or gas flow 4. Check connections: X5 / detector channel 1 (Oxygen sensor or IR detector) X6 / IR detector channel 2 (s. fig. 8-25, page 8-27)

7-2 Other Problems

Problem	Potential Cause	How to Solve
Fluctuating or invalid readout (cont'd)	5. electrochemical Oxygen sensor worn 6. IR channel: Source not connected or defective 7. Analog preamplifier of affected channel defective 8. Gas path(s) polluted	5. Check sensor and replace if need be (see 9-4, page 9-5) 6. Check connections: X3 (1/2) / source channel 1 X3 (4/5) / source channel 2 (fig. 8-25, page 8-27) If source housing is cold: Exchange both source in case of dual channel analyzer / replace source if need be (see service manual) 7. Check measuring point acc. 8-2-1-6 8. Check analysis cell and windows for pollution Clean polluted parts (see service manual) Check gas paths for pollution and clean gas paths if need be

7-2 Other Problems

Problem	Potential Cause	How to Solve
Fluctuating or invalid readout (cont'd)	9. Influence of ambient pressure 10. Temperature in gas path(s) below dew point 11. A/D converter of affected channel defective	9. Set ambient pressure to proper value (see 5-3-2) Sensor failure (see error code E37) 10. Check temperature of gas path(s) remove all sources of condensation Keep all temperatures at least 10 °C above sample gas temperature 11. Replace BKS
Readout response time too long (t_{90} time)	1. Wrong response time settings (t_{90} time). 2. Pump rate too low 3. Gas path(s) polluted	1. Check t_{90} time (see 5-3-11) 2. Distance between sampling point and analyzer too long Replace pump by external model with higher pump rate (operate in bypass mode; see 3-4) 3. Check gas path and sample handling system for pollution Clean gas path, replace filter elements

7-2 Other Problems

Problem	Potential Cause	How to Solve
No gas flow	1. Needle valve closed (option for BINOS® 100 2M / F) 2. Sample gas pump (option for BINOS® 100 2M / F / CAT 100) switched off 3. Membrane of sample gas pump defective 4. Sample gas pump defective 5. Solenoid valves (option) not opened / defective 6. Gas path(s) polluted	1. Open needle valve (option, see 6-5-2) 2. Switch sample gas pump on (see 4-5) 3. Replace sample pump membrane 4. Replace sample gas pump 5. External valves: Check connection between valves and digital outputs For valve control via serial interface or digital inputs: Any valve activated? Check valve seat and replace if need be Replace solenoid valves 6. Check gas path and sample handling system for pollution Clean gas path, replace filter elements

Chapter 8 Troubleshooting on Electronics

Several electronic boards used within the analyzers provide measuring points supporting troubleshooting. To get access to these measuring points opening the analyzer housing is required.

The following sections describe how to open the different analyzer housings provided the instruments are fully accessible, e.g. are taken out of the rack.



Take care of the safety instructions (page S-2) while working at and inside the instruments!



Working inside the instruments is subject to briefed personnel familiar with potential risks!

Read how to open 1/4 19" housings (BINOS® 100 (M),

OXYNOS® 100, HYDROS® 100)



page 8-2

1/2 19" housing (BINOS® 100 2M)



page 8-8

Field housing (BINOS® 100 F)



page 8-13

Flameproof housing (CAT 100)



page 8-15

8-1 Getting Access to Internal Analyzer Components

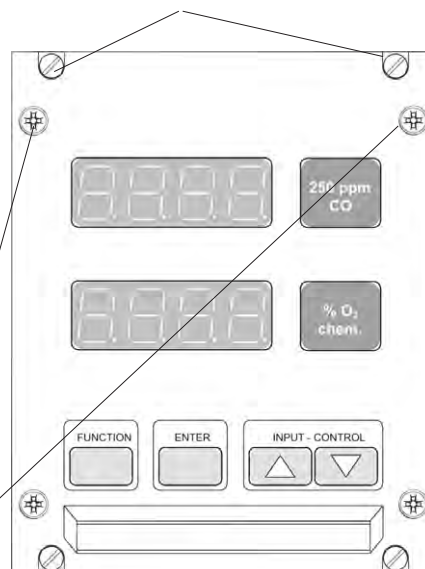
8-1 Getting Access to Internal Analyzer Components

8-1-1 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

- Disconnect analyzer from power supply
- Table top instruments:
 - remove rubber feet
 - loosen fixing screws of front panel frame (fig. 8-1) and remove frame with handle towards rear panel
- Housing cover can be removed after loosening the upper Phillips screws at front and rear panel (fig. 8-1)
- Loosening the lower Phillips screws at front and rear panel (fig. 8-1) allows to remove these panels, too
- Repeat above steps in reverse order to assemble the housing

Fig. 8-2 to 8-6 exemplary show several internal configurations of analyzers with 1/4 19" housing.

Fixing screws for front panel frame
(2 pcs. each at top and bottom)



Fixing screws for housing cover

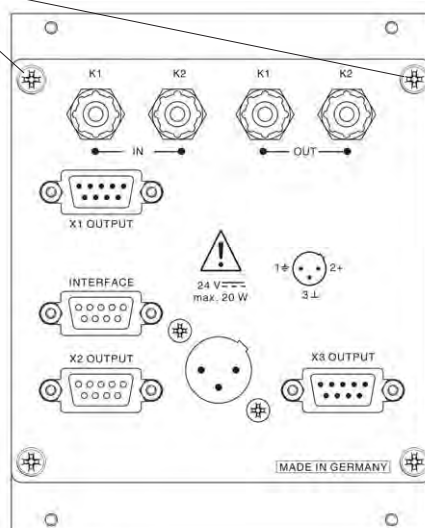


Fig. 8-1: 1/4 19" Housing
Fixing screws for housing

8-1-1 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

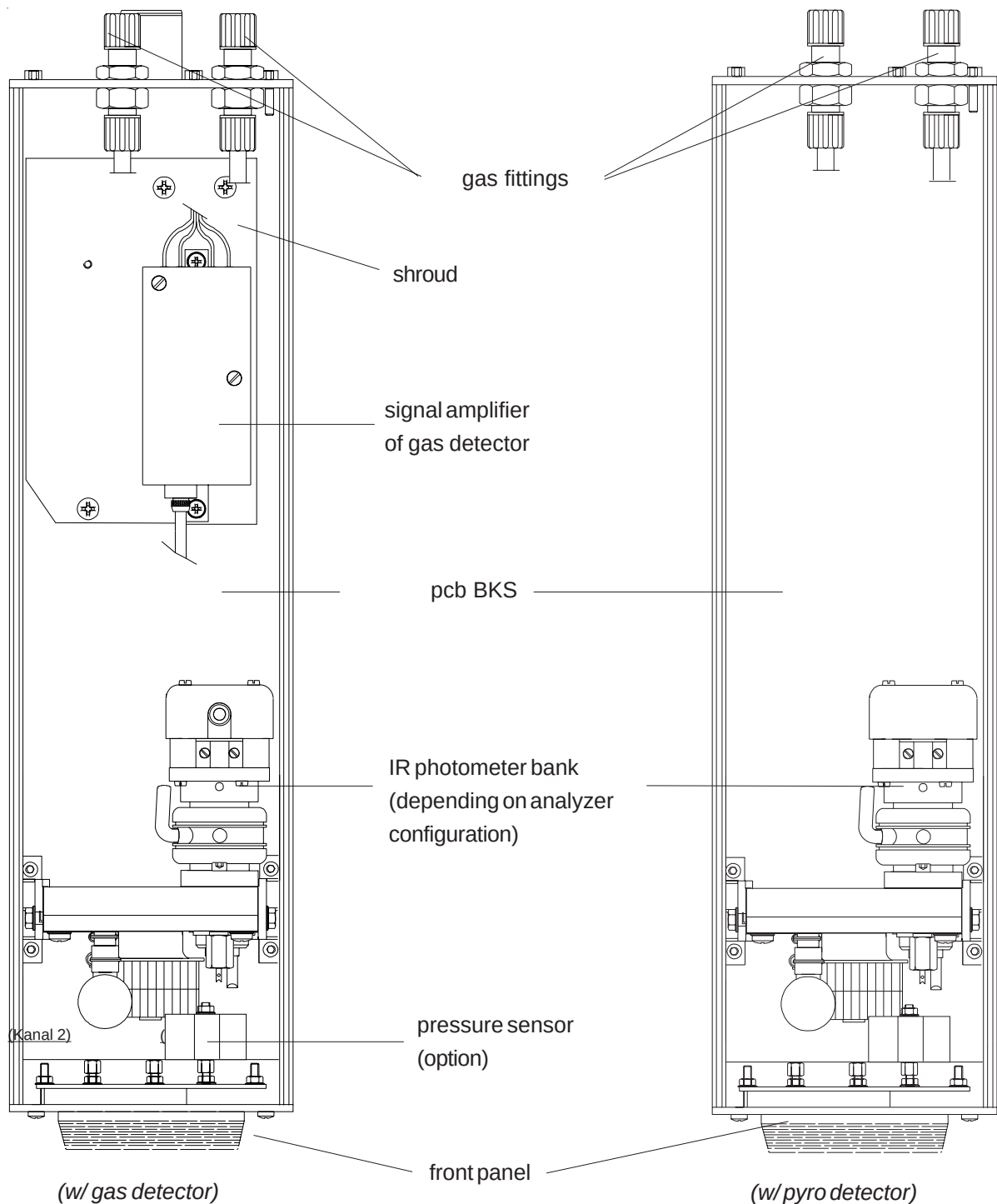


Fig. 8-2: BINOS® 100, interior view (1 IR channel configuration)

8-1-1 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

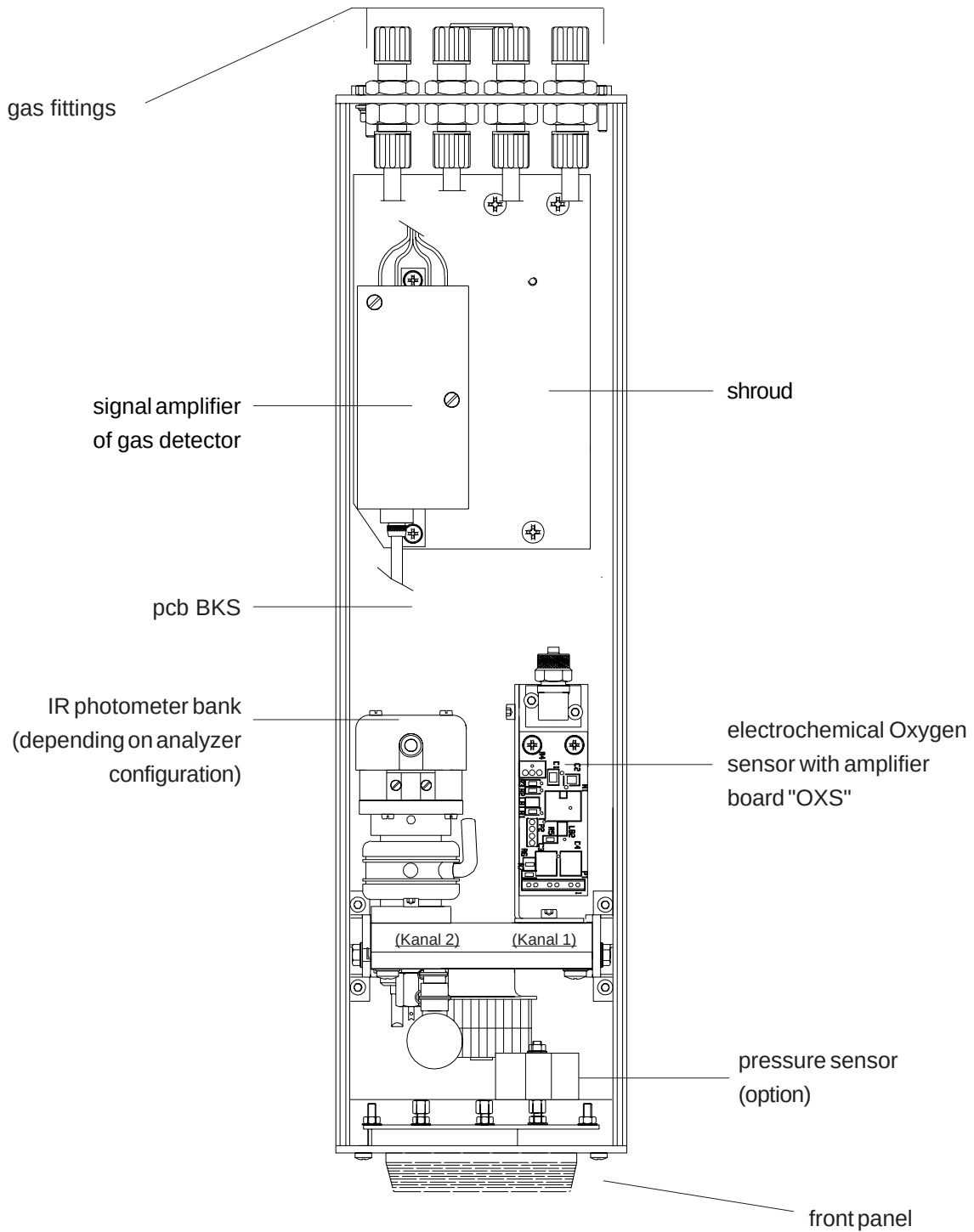


Fig. 8-3: BINOS® 100 M, interior view (IR channel & electrochemical Oxygen sensor)

8-1-1 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

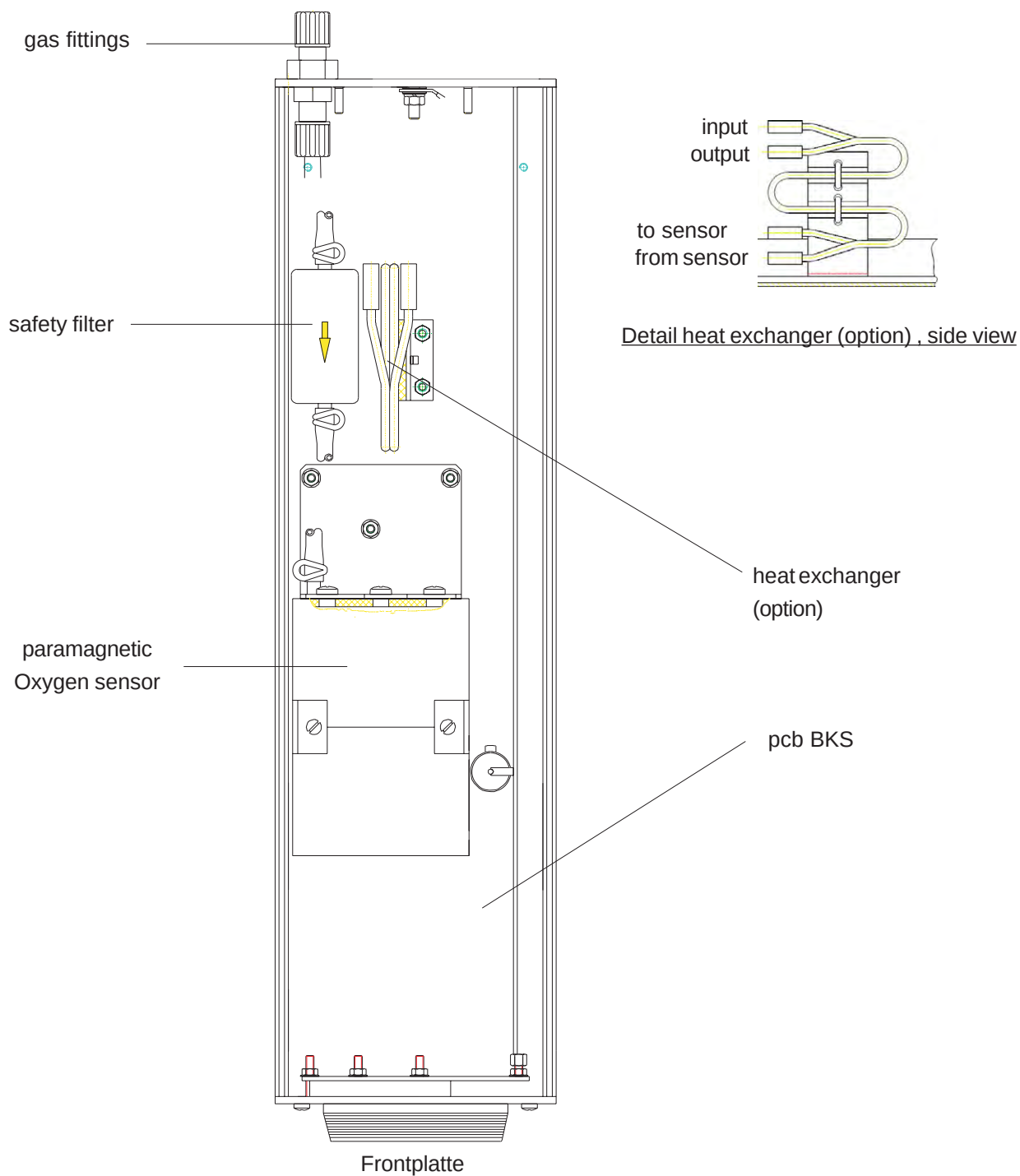


Fig. 8-4: OXYNOS® 100, interior view (w/ paramagnetic Oxygen sensor)

8-1-1 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

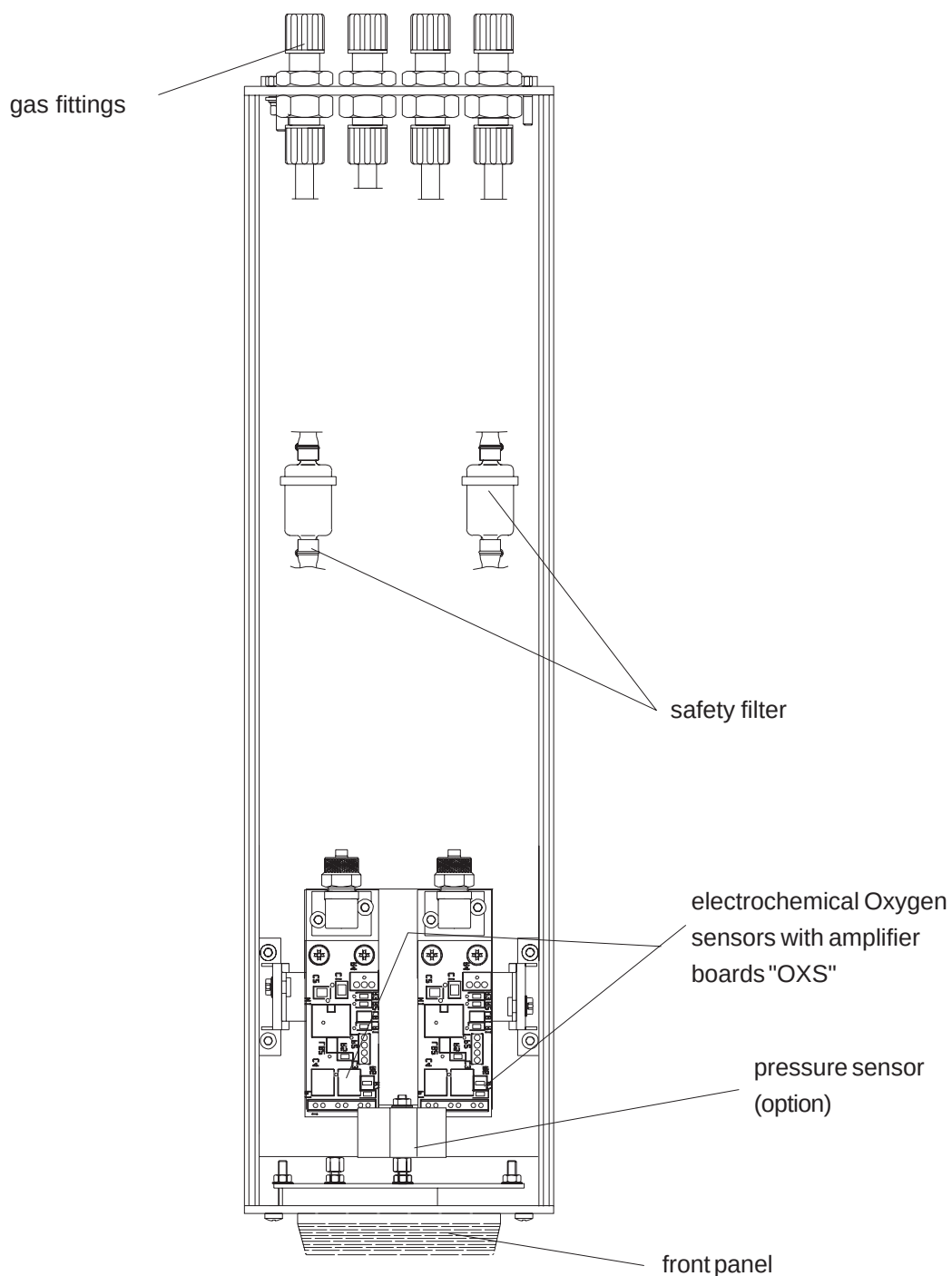


Fig. 8-5: OXYNOS® 100, interior view, (dual channel w/ electrochemical Oxygen sensors)

8-1-1 BINOS® 100 (M), OXYNOS® 100, HYDROS® 100

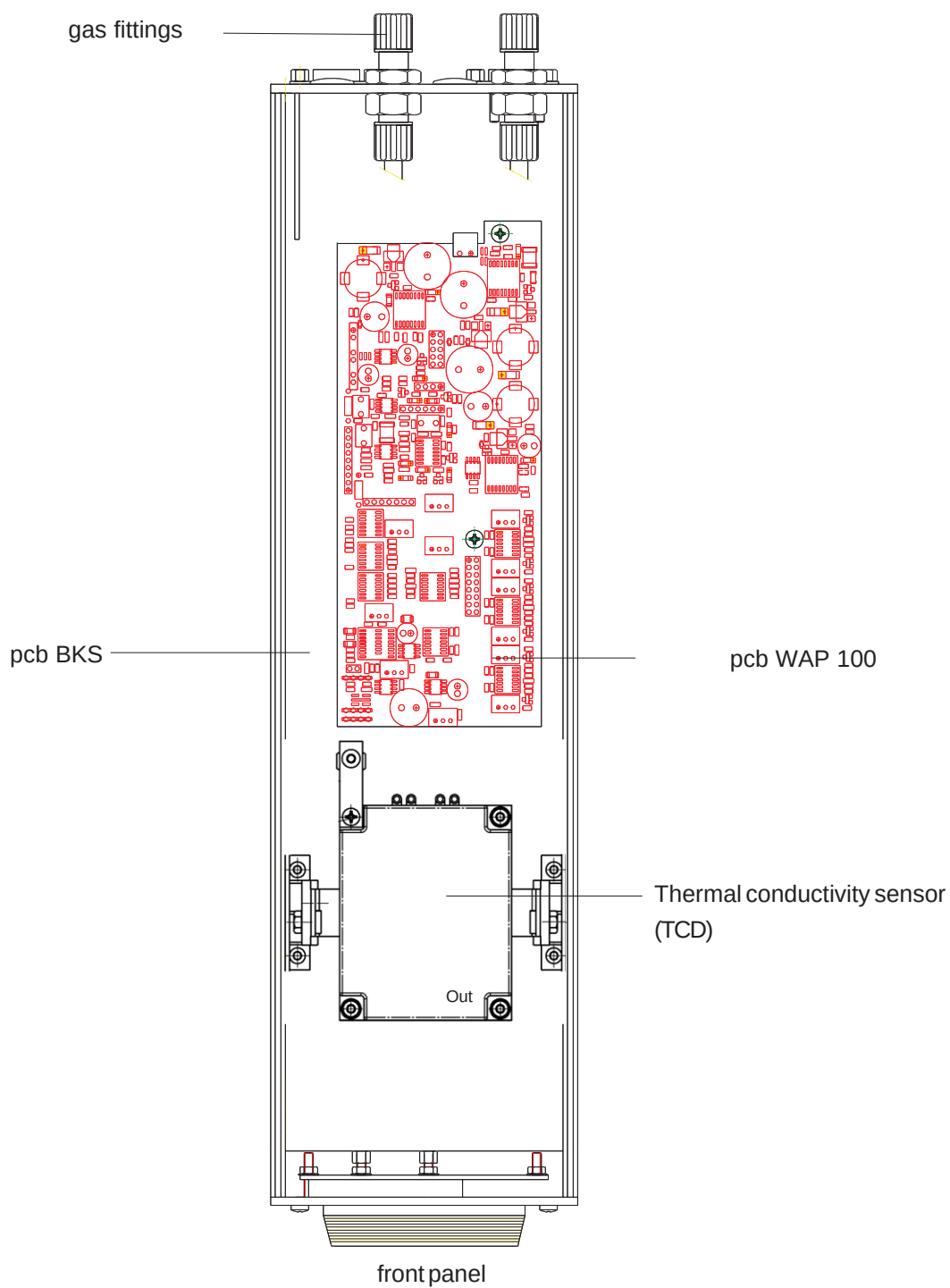


Fig. 8-6: HYDROS® 100, interior view

8-1-2 BINOS® 100 2M

8-1-2 BINOS® 100 2M

- Disconnect analyzer from power supply
- Table top instruments:
 - optionally loosen fixing screws of front panel frame (fig. 8-7) and remove frame with handle towards rear panel
- Housing cover can be removed after loosening the 8 Phillips screws at left and right side of cover (fig. 8-7) and one screw at the top of the cover
- Loosening the 6 Phillips screws at the internal front panel frame (fig. 8-8) allows to remove the front panel
- The rear panel is divided into a panel for the physics side and one for the electronics side, each fixed by 6 Phillips screws
- Repeat above steps in reverse order to assemble the housing

Fig. 8-9 to 8-12 exemplary show several internal configurations of a BINOS® 100 2M analyzer.

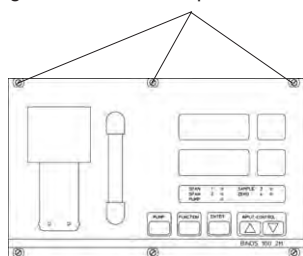


Optionally internal components may be fixed to the front panel, e.g. fine dust filter, flowmeter (see fig. 8-2)!



Take care not to squeeze internal electrical connections and gas lines when assembling the housing!

Fixing screws for front panel frame (6 pcs)



Fixing screws for housing cover (8 pcs)

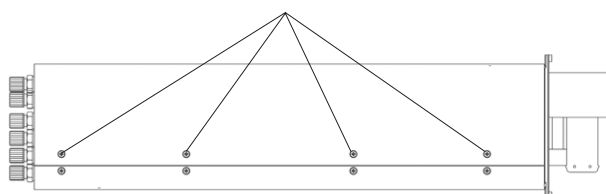


Fig. 8-7: BINOS® 100 2M
Housing Fixing Screws

Fixing screws for front panel (6 pcs)

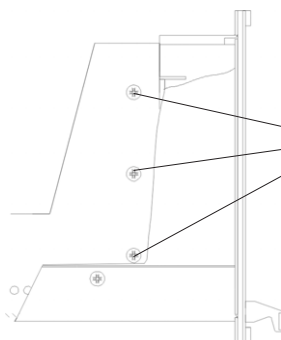


Fig. 8-8: BINOS® 100 2M
Fixing screws for front panel

8-1-2 BINOS® 100 2M

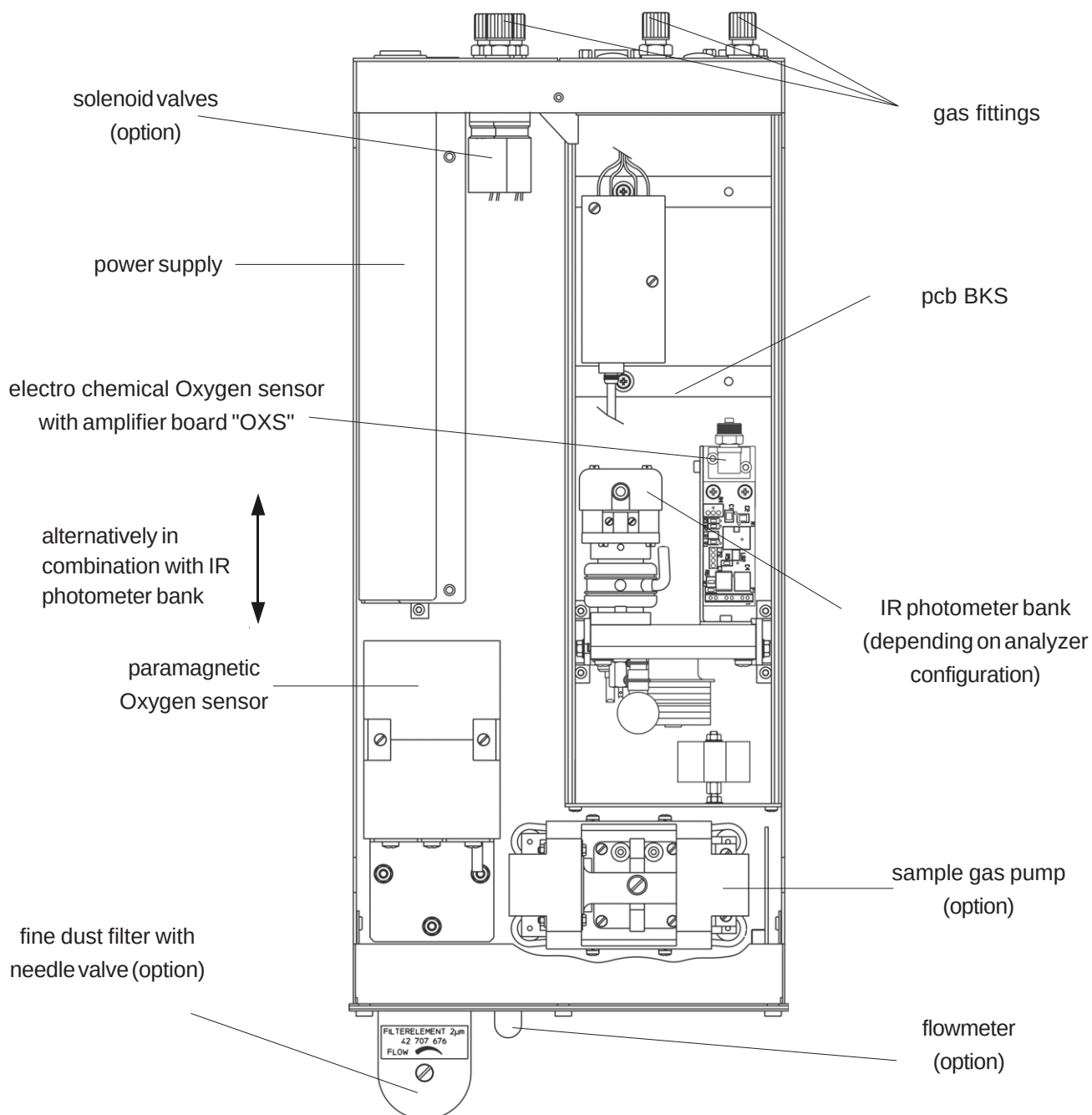


Fig. 8-9: BINOS® 100 2M (variation A), dual channel, interior view
(IR channel & Oxygen sensor [paramagnetic or electrochemical])

8-1-2 BINOS® 100 2M

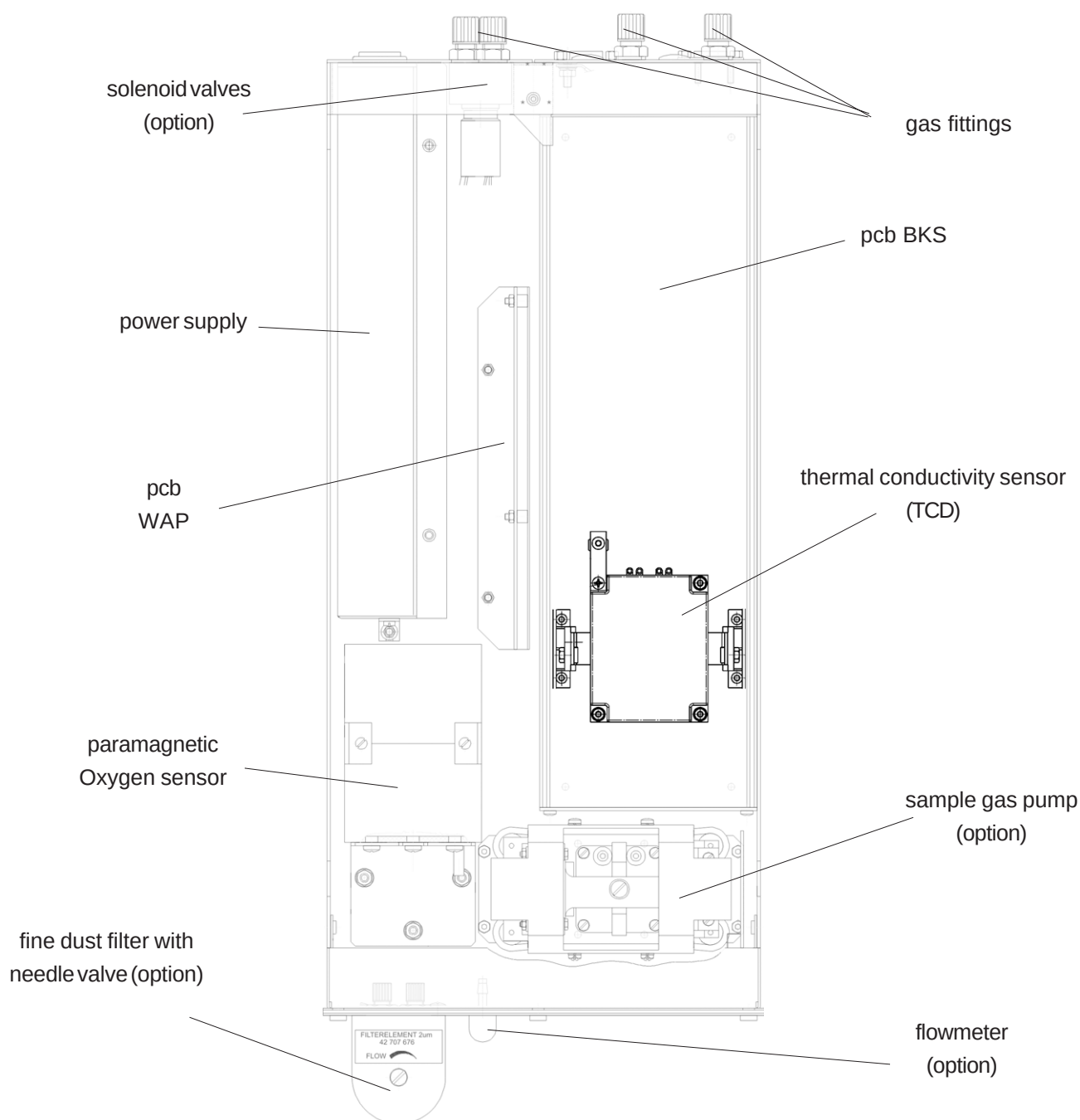


Fig. 8-10: BINOS® 100 2M (variation A), dual channel, interior view
(TCD & paramagnetic Oxygen sensor)

8-1-2 BINOS® 100 2M

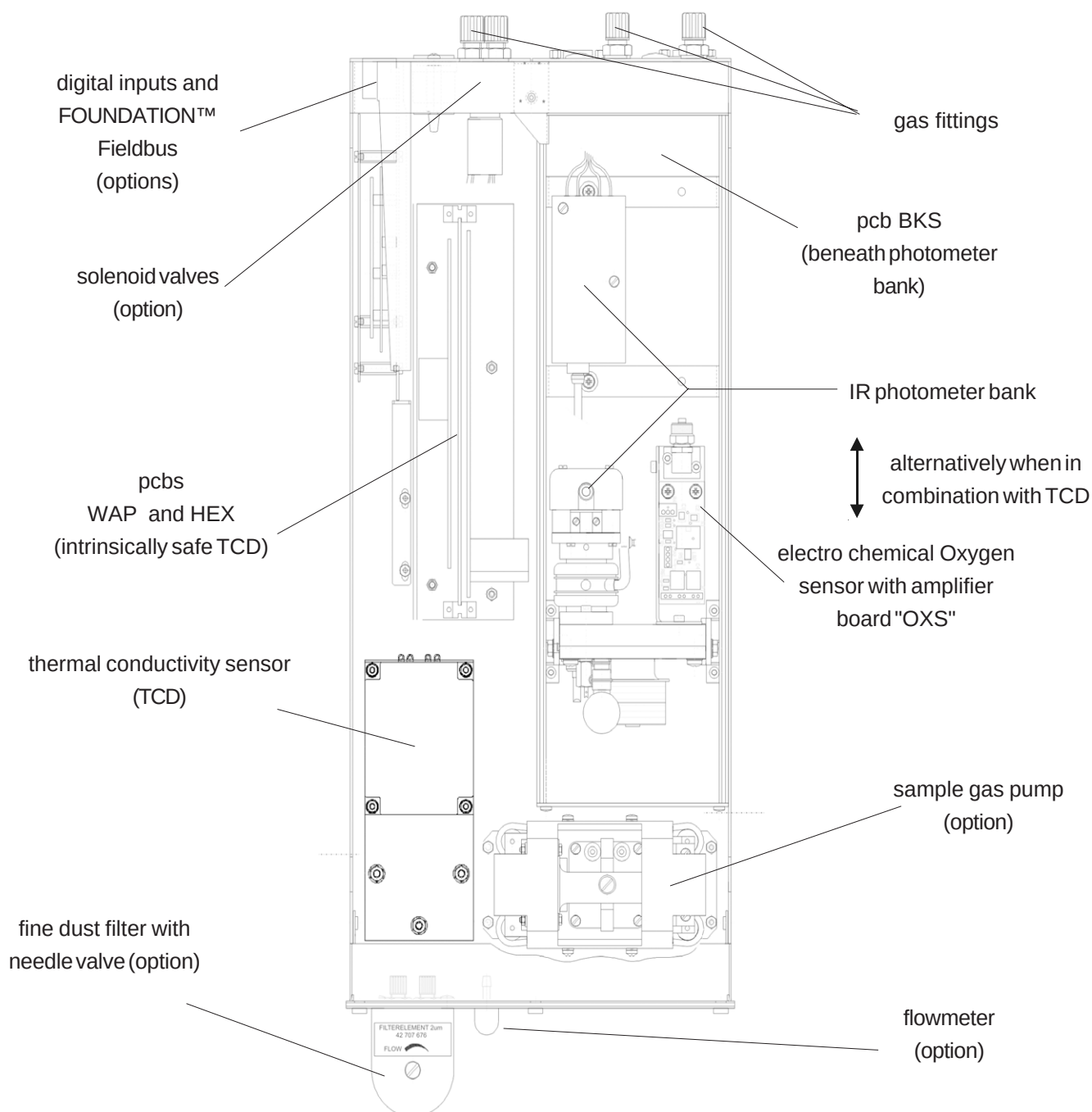


Fig. 8-11: BINOS® 100 2M (variation B), dual channel, interior view
(TCD, intrinsically safe & IR channel or electrochemical Oxygen sensor)

8-1-2 BINOS® 100 2M

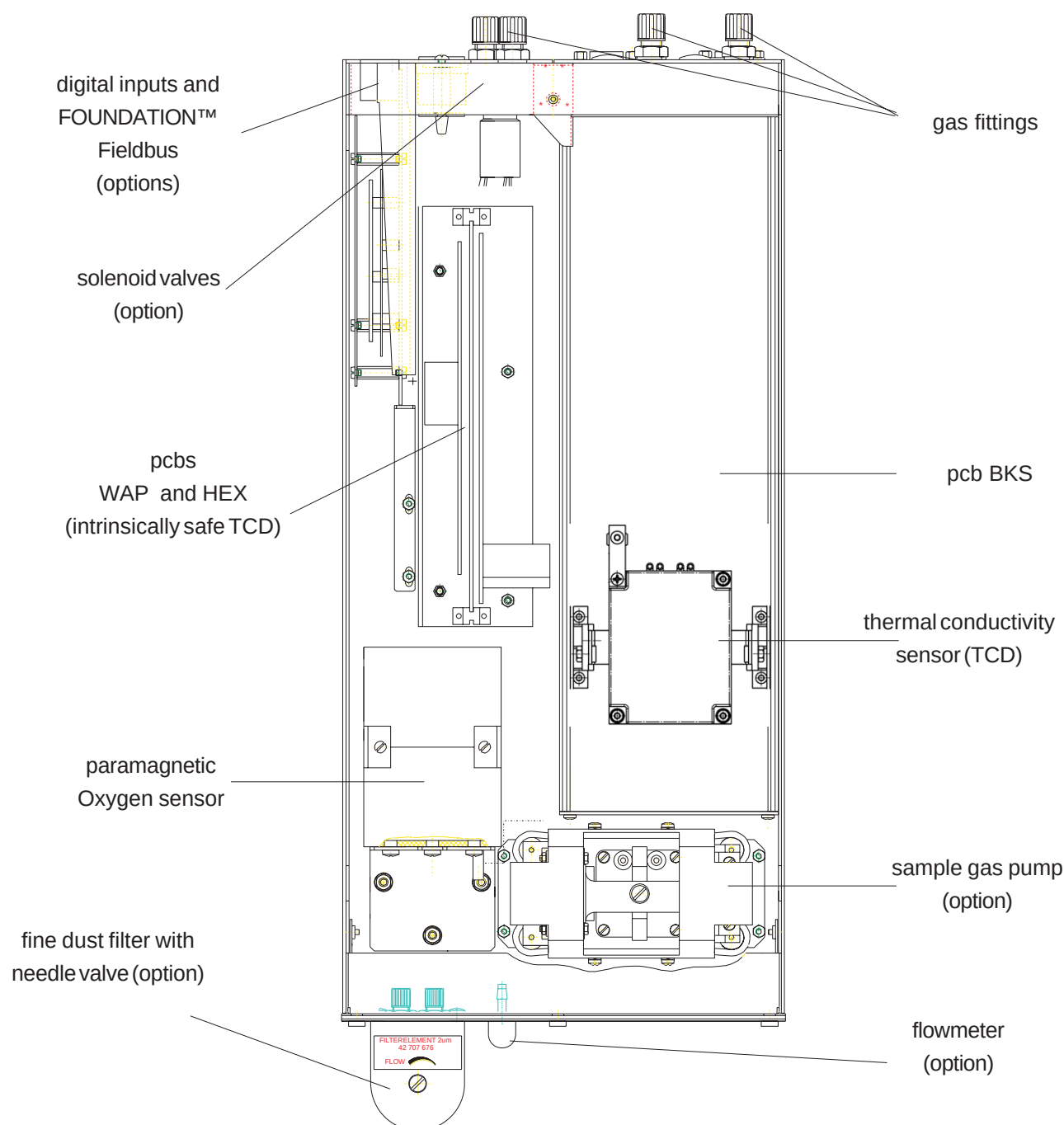


Fig. 8-12: BINOS® 100 2M (variation B), dual channel, interior view
(paramagnetic Oxygen sensor & intrinsically safe TCD)

8-1-2 BINOS® 100 2M

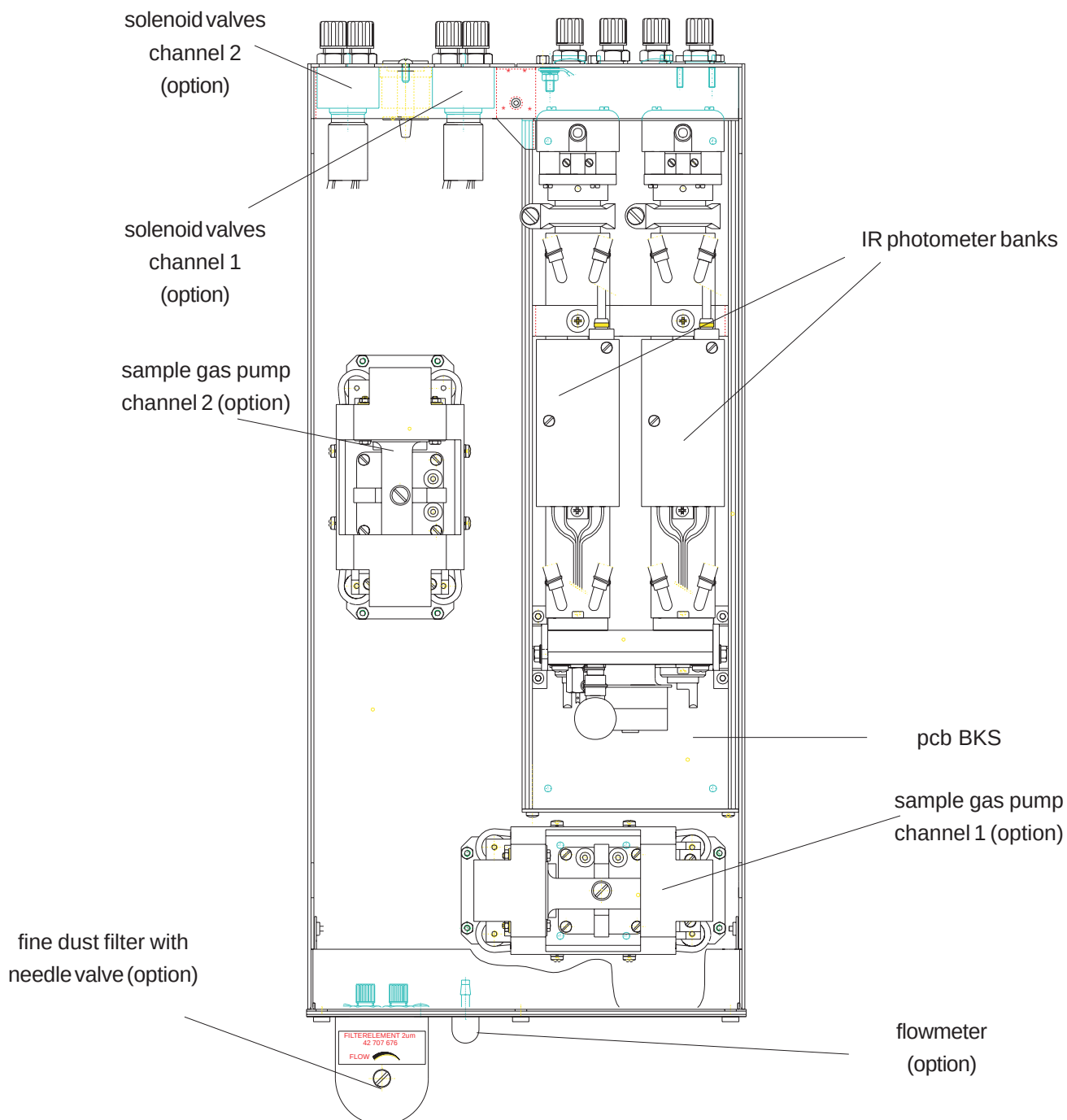


Fig. 8-13: BINOS® 100 2M (special version), dual channel, interior view
(2 IIR channels, 2 sample gas pumps & 2 solenoid valve blocks)

8-1-3 BINOS® 100 F

8-1-3 BINOS® 100 F

WARNING

EXPLOSION HAZARD!



Opening Instruments installed in hazardous areas is permitted only taking into consideration the special conditions for such areas !

Refer to the special instruction manual for operating instruments in hazardous areas to detailed information about handling such equipment !

WARNING

ELECTRICAL SHOCK HAZARD!



Live parts are accessible when working at open instruments!

- The front door is closed utilizing 4 sash fasteners (fig. 8-14). To open use a four-square Allen key shipped together with the instrument and turn the fastener according the markings printed onto the front door.
- Open the door to the right side.
- Most components are located on a photometer slice carriage which can be pulled out to the front after removing 2 knurled head screws (fig. 8-15). Store the screws on bolts placed nearby the screws. Pulling out the sliding carriage in a first step is possible only to a mechanical stop (fig. 8-15). Push the stop aside to completely pull out the carriage slide of the housing.

Note!

Before taking the slide out of the housing disconnect all electrical and gas path connections!

Be careful when working at photometer components: HOT parts!

- Repeat above steps in reverse order to assemble the housing.

To identify the different components installed within a BINOS® 100 F gas analyzer see fig. 8-9 to 8-13 showing several BINOS® 100 2M configurations, also valid for BINOS® 100 F.

8-1-3 BINOS® 100 F

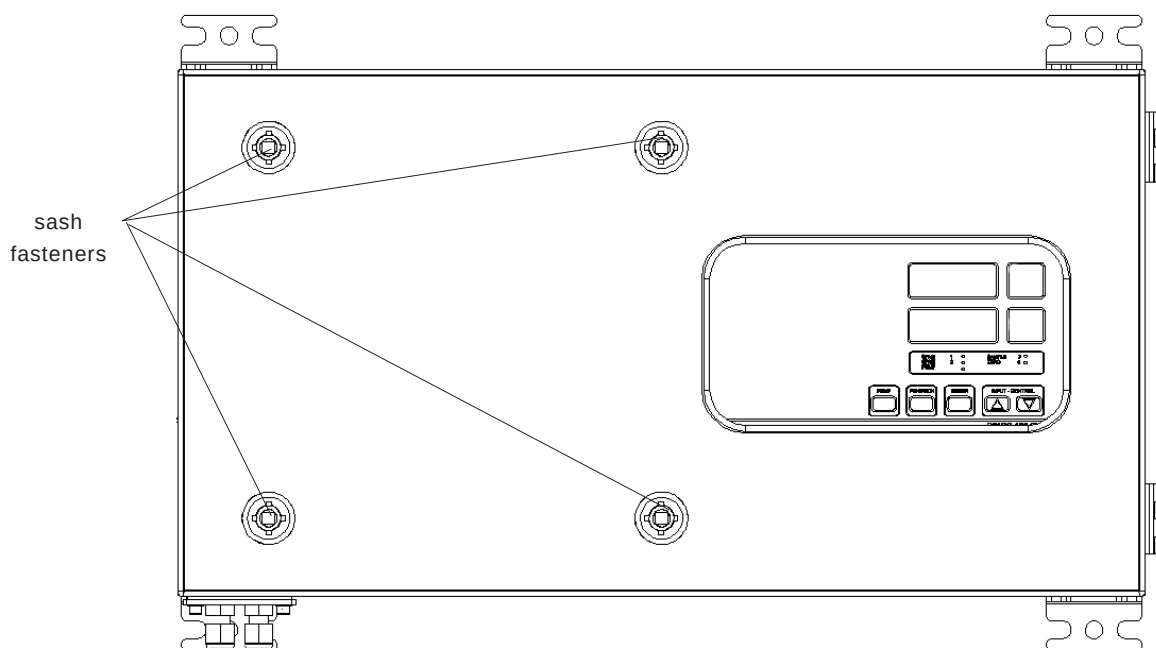


Fig. 8-14: BINOS® 100 F, front side view

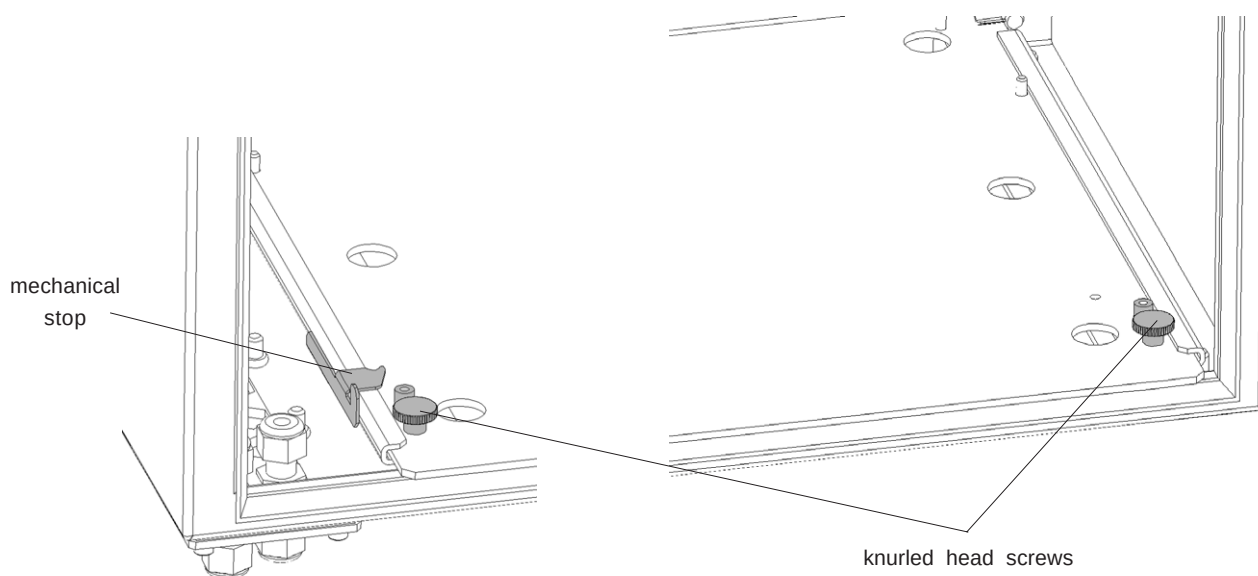


Fig. 8-15: BINOS® 100 F, photometer carriage slide

8-1-4 CAT 100

8-1-4 CAT 100

WARNING

EXPLOSION HAZARD!



Opening Instruments installed in hazardous areas is permitted only taking into consideration the special conditions for such areas !

Refer to the special instruction manual for operating instruments in hazardous areas to detailed information about handling such equipment !

WARNING

ELECTRICAL SHOCK HAZARD!



Live parts are accessible when working at open instruments!

CAUTION

HEAVY PARTS!



Individual housing components weigh up to 27 kg !

Dome components slip off when reaching end of thread !

Use suitable tools or work in pairs !

The CAT 100 housing consists of two main parts:

- junction box
- dome

The junction box contains an encapsulated EMI mains filter and all the terminals for connecting mains supply and signal lines. The dome covers all other components, e.g. power supply unit and analyzer module.

Junction Box

- Open the junction box by loosening the 6 Allen screws (fig. 8-16) utilizing the (imperial) Allen key shipped with the instrument.
- Remove the cover.

Note!

Assemble the cover utilizing ALL 6 screws to ensure proper IP rating against dust and water!

Dome

To get access to internal components and the analyzer module the dome's components have to be removed (fig. 8-17):

- loosen the outer screw
- turn the dome counterclockwise to remove it
- loosen the inner screw
- turn the extender ring counterclockwise to remove it

8-1-4 CAT 100

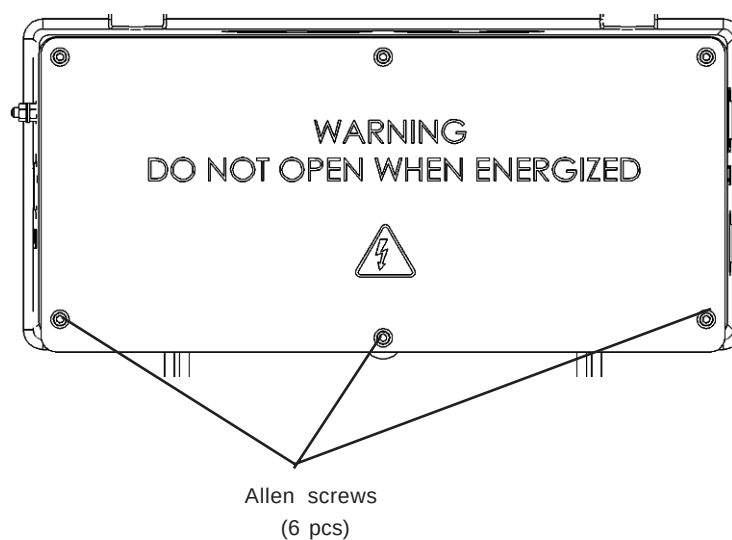


Fig. 8-16: Junction box

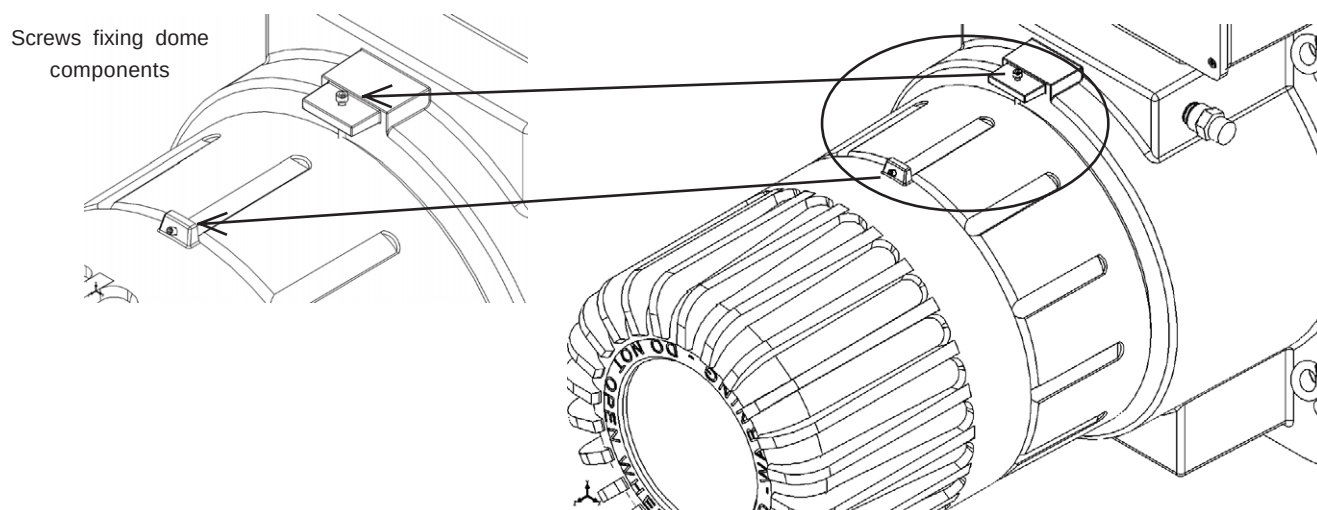


Fig. 8-17: Dome, fixing screws

8-1-4 CAT 100

- When dome is removed unscrew the knurled head screws at the analyzer module's sides (fig. 8-18) and pull it to the front until the last screw locks into the detent (fig. 8-19).

Note!

Mark or remember the detent where the first knurled head screw was placed, to place it there again when assembling the unit!

Otherwise it may happen the analyzer is not magnetically operable any more because the distance between front panel and glass is too great.

- It's now possible to pivot the module downwards and to unscrew three additional knurled head screws to remove the module cover.

Furthermore it is possible to take the module out of the CAT housing after disconnecting all gas path and electrical connections. This gives access to the components installed inside the cast iron housing (fig. 8-20).

The analyzer module used within the CAT 100 is identical to a BINOS® 100 2M analyzer without internal over supply unit, except it is magnetically operated (fig. 8-21).

For this reason see fig. 8-3 to 8-13 when identifying the different components installed within a CAT 100 analyzer module: These figures show several BINOS® 100 2M configurations, also valid for the CAT 100 module.

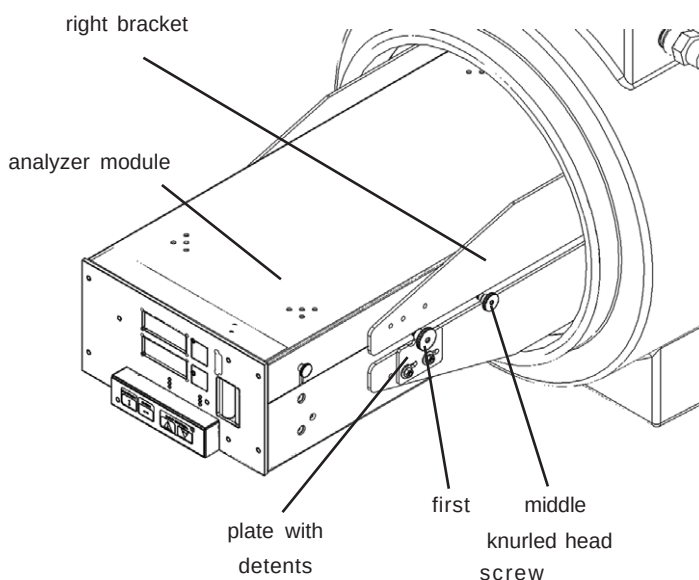


Fig. 8-18: View at analyzer module

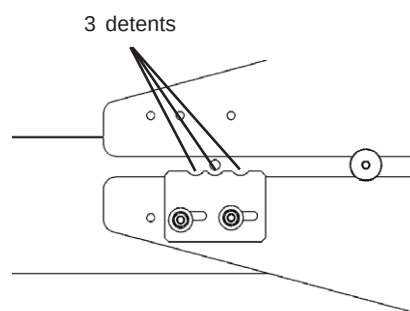


Fig. 8-19: Plate with detents

8-1-4 CAT 100

Repeat the steps in reverse order to assemble the housing.

When having assembled the dome perform a first test to ensure the front panel sensor keys can be activated. If all sensor fields are accessible use the screws on top of the dome and extender ring to fix both units (fig. 8-17).

8-1-4 CAT 100

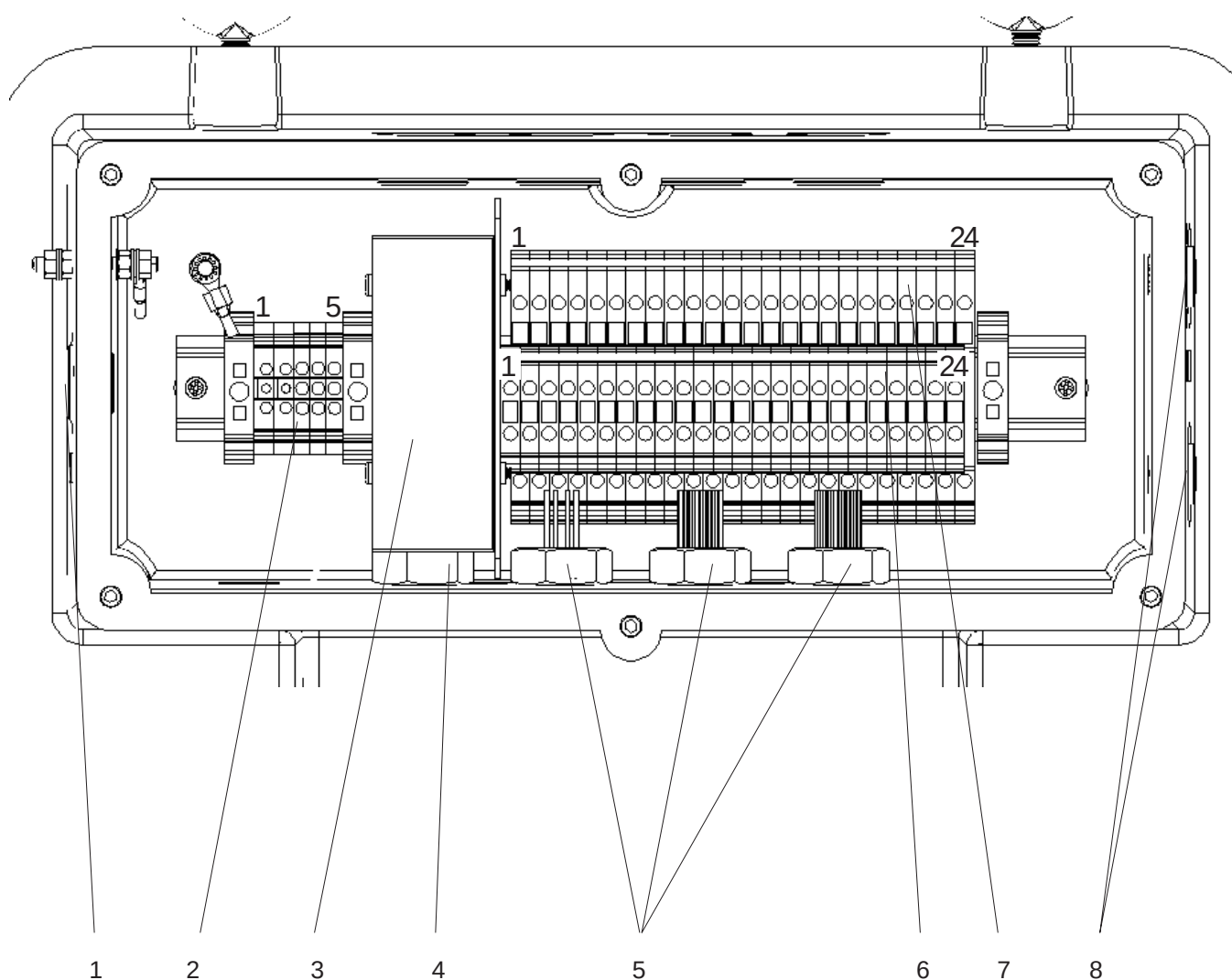
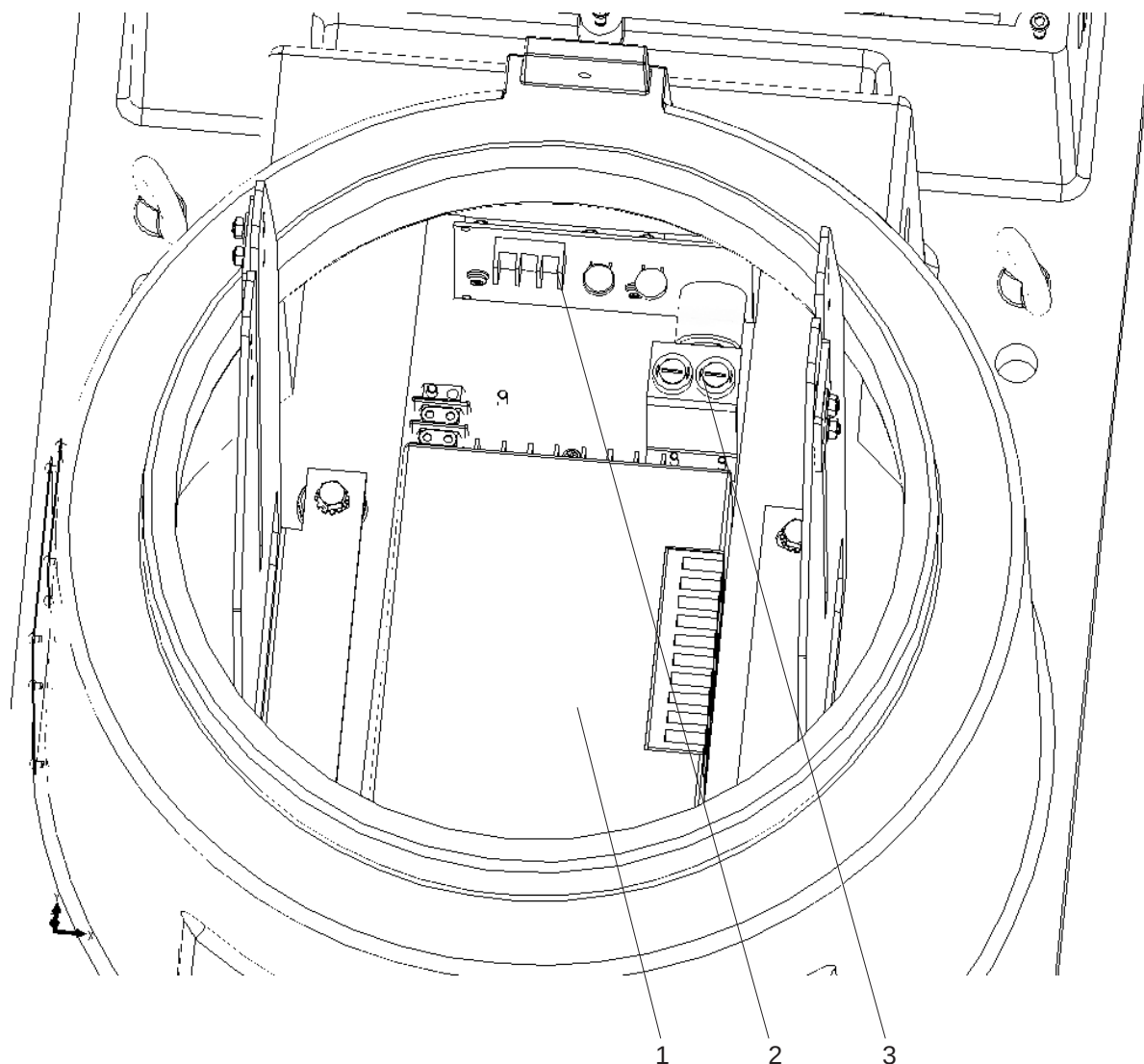


Fig. 8-20: CAT 100, Junction Box, Interior View

- 1 Mains cord cable gland
- 2 Mains terminals
- 3 Mains EMI filter
- 4 Mains feedthrough

- 5 Signal cable feedthroughs (3x)
- 6 Upper signal terminals (24)
- 7 Lower signal terminals (24)
- 8 Signal cable glands (3x)

8-1-4 CAT 100



**Fig. 8-21 CAT 100,
Cast Iron housing, Interior View**

- 1 Power supply unit
- 2 Filter pcb
- 3 Fuses for power supply unit

8-1-4 CAT 100

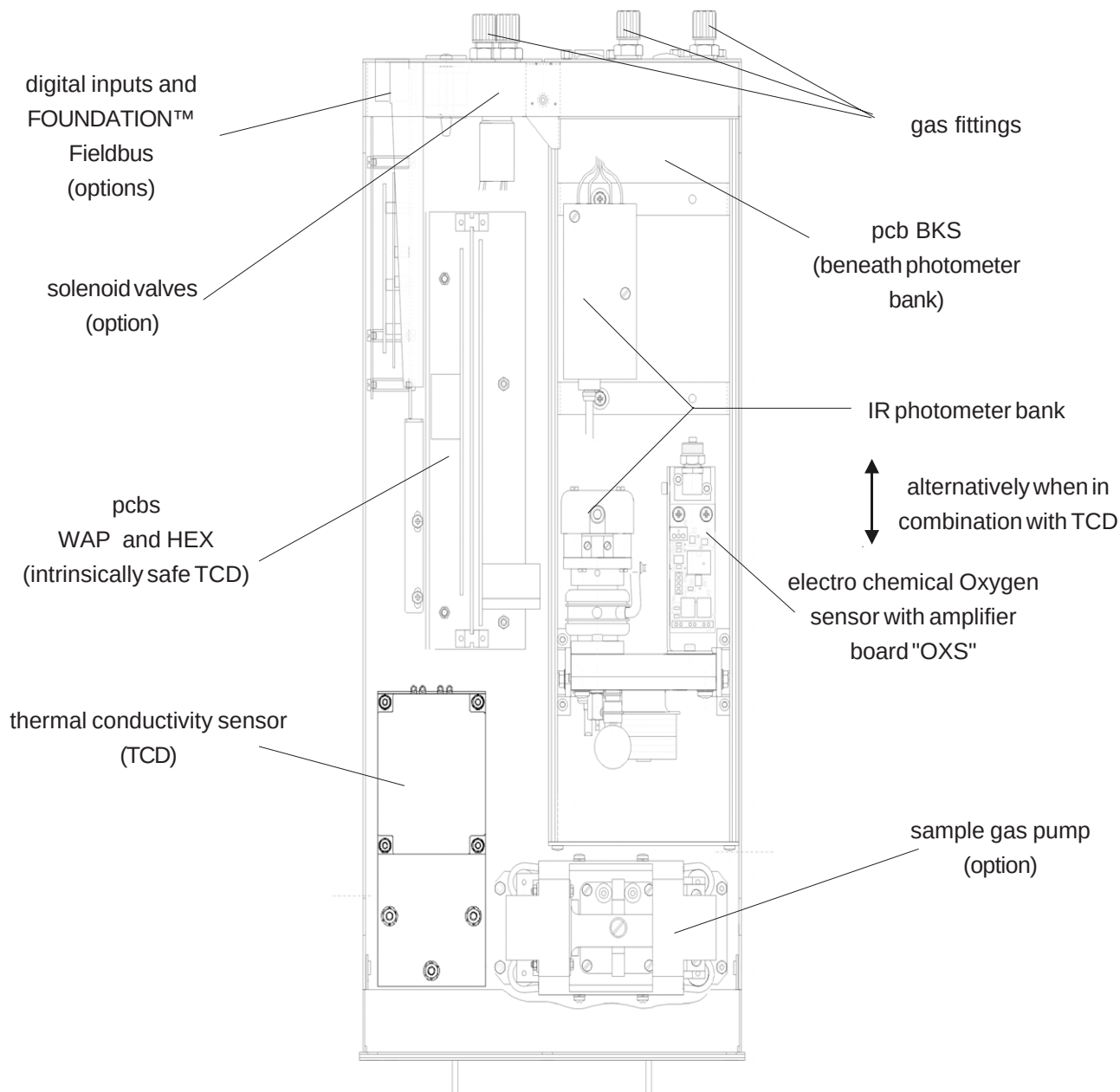


Fig. 8-22: CAT 100, internal analyzer module (BINOS® 100 2M)
exemplary, TCD (intrinsically safe) &
IR channel or electro chemical Oxygen sensor (alternatively)

8-2 Measuring Points

8-2 Measuring Points

8-2-1 Measuring Points at BKS

(Pin 1 of all connectors are marked within the silk screen or by use of square pads)

Measure all voltages against GND, available at X11 and X28 !



Observe all safety instructions given at the beginning of this manual!

8-2-1-1 Supply voltage +6 V

Measuring Point: X14

Measuring instrument: DVM

Signal: + 6 V DC (+10 mV / -200 mV)
(adjust with R90 if need be)

Failure: No or wrong signal

Potential cause:

- Supply voltage missing
- Supply voltage connected wrong way round or is < 9 V
- Fuse F2 broken (fig 8-26)
- Pcb. BKS defective

8-2-1-2 Positive reference voltage

Measuring Point: X5, pin 6

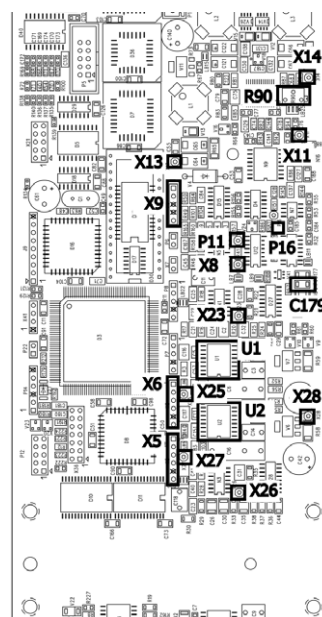
Measuring instrument: DVM

Signal: + 5,535 V DC (± 60 m)

Failure: No or wrong signal

Potential cause:

- +6V voltage faulty
- +3 V Reference voltage faulty
(measure at C179, pin 1
(left pin in fig. 8-22))



Frontplattenseite

**Fig. 8-23: Pcb. "BKS" (section),
measuring points**

How to solve:

- Connect supply voltage
- Check polarity or replace power supply unit
- Replace F2 (by same type !)
- Replace pcb BKS

How to solve:

- Check supply voltage (s. 8-2-1-1)
- Replace pcb BKS

8-2 Measuring Points

8-2-1-3 Negative reference voltage

Measuring Point: X5, pin 4

Measuring instrument: DVM

Signal: inverse positive
reference voltage

Failure: Both reference voltages differ by
more than 10 mV

$$(U_{\text{ref. pos.}} + U_{\text{ref. neg.}} \leq 10 \text{ mV}) !$$

How to solve:

Replace pcb BKS

8-2-1-4 Temperature sensor

Measuring Point: X8

Measuring instrument: DVM

Signal: approx. $0 \pm 500 \text{ mV DC}$
(at ambient temperature)

Potential cause:

- IR measurement or paramagnetic
Oxygen measurement
 - a) sensor not connected (see 8-3)
 - b) sensor defective
 - c) sensor cable broken
 - d) pcb BKS defective
- elektrochemical Oxygen measurement:
 - a) sensor not connected (see 8-3-2)
 - b) pcb OXS defective
 - c) pcb BKS defective

How to solve:

- a) connect temperature sensor
 - b) replace temperature sensor
 - c) replace temperature sensor
 - d) replace pcb BKS
-
- a) connect temperature sensor
 - b) replace pcb OXS
 - c) replace pcb BKS

8-2 Measuring Points

8-2-1-5 Light barrier signal

Measuring Point: X9, pin 2

Measuring instrument: Oscilloscope

Signal: rectangle (level see below)
frequency = 24 Hz (± 0.1 Hz)

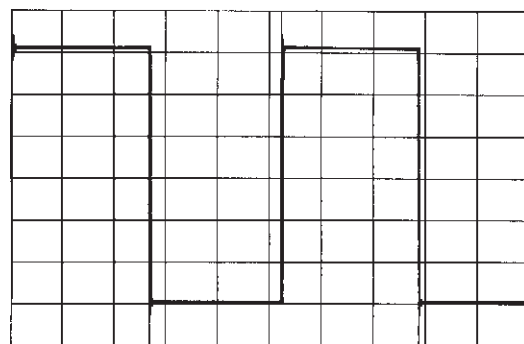


Fig. 8-24: Light barrier signal

Failure: No or faulty signal

Potential cause:

- IR measurement:
Signal level set point: $U = 6 V_{SS} (\pm 0.3 V)$

How to solve:

- a) Chopper not connected
- b) Chopper not turning
- c) Light barrier not connected
- d) Light barrier defective
- e) Cable broken
- f) BKS defective

- a) Connect chopper (see 8-3)
- b) Switch off analyzer and on again
- c) Connect light barrier (see 8-3)
- d) Replace chopper
- e) Replace chopper
- f) Replace pcb BKS

- Oxygen measurement w/o IR channel:
Signal level set point:
LOW $\leq 0.45 V$; HIGH $\geq 2.4 V$ (TTL logic)

- a) μP not working
- b) BKS defective
- c) solder bridge **LB 18** not configured properly (pins 1-2 connected)

- a) Switch off analyzer and on again
- b) Replace pcb BKS
- c) **Configure LB 18 properly** (connect pins 1 & 2). LB 18 is accessible at the pcb's solder side and requires dismantling the pcb.

8-2 Measuring Points

8-2-1-6 Analog Preamplifier

a) Paramagnetic Oxygen Measurement

Measuring Point: X 25 (channel 1)

Measuring instrument: DVM

Signal when connected to

zero gas: 0 V dc (± 50 mV)

ambient air (approx. 21 Vol. - % O₂):

100 % O₂ sensor: approx. 840 mV

25 % O₂ sensor: approx. 3.36 V

(Type of sensor: see separate nameplate label)

Failure: No signal or wrong measuring values

Potential cause:

- a) Oxygen sensor not connected
- b) Oxygen sensor defective
- c) BKS defective

How to solve:

- a) Connect Oxygen sensor (s. 8-3)
- b) Replace Oxygen sensor
- c) Replace pcb BKS

b) IR Measurement

Measuring Point:

X 25 (channel 1; not for instruments with Oxygen measurement)

X 27 (channel 2)

Measuring instrument: DVM

Signal when connected to

zero gas: 0 V dc (± 100 mV)

Zero point voltage and span voltage must differ at least by 600 mV (at ranges < 1000 ppm difference should be 500 mV min.)

Failure: No signal or wrong measuring values

Potential cause:

- a) Detector not connected
- b) Detector defective
- c) BKS defective

How to solve:

- a) Connect detector
- b) Replace detector
- c) Replace pcb BKS

8-2 Measuring Points

8-2-2 Measuring Points at OXS (Electrochemical Oxygen Measurement)

8-2-2-1 Sensor Signal

Measuring Points: Tp 1 (Signal)
Tp 2

Measuring instrument: DVM

Signal when connected to ambient air (approx.
21 Vol. - % O₂): approx. 3.36 V with new cell

Failure: No signal or wrong measuring values

Potential cause:

- Sensor not connected to pcb OXS
- "OXS" defective
- Sensor defective or weared
- BKS defective

How to solve:

- Connect sensor to pcb OXS
- Replace OXS
- Replace sensor
- Replace BKS

Note!

*Sensor is weared when connected to ambient
air the output voltage is less than 2.8 V:
Replace sensor!*

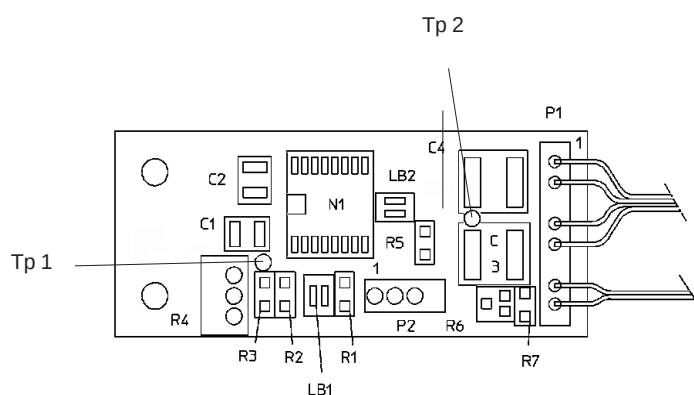


Fig. 8-25: Pcb OXS, assembled, top view

8-3 Connectors on Electronics Boards

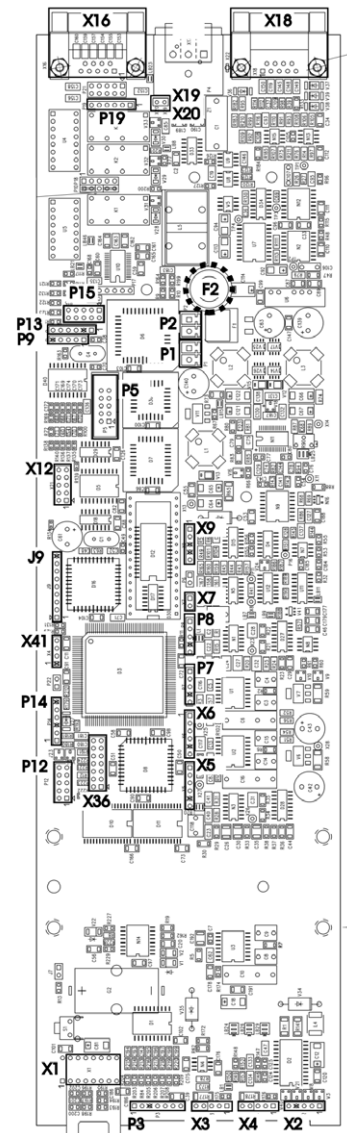
8-3 Position and Assignment of Connectors on Electronics Boards



Observe all safety instructions given at the beginning of this manual!

8-3-1 BKS

- X 19 (+) 24 V DC supply voltage input
- F 2** Fuse for DC 24 V input
Type: Wickmann 372
Rating: T 4 A / 250 V
- P 1 24 V DC distribution for
P 2 optional internal loads(e.g.
thermostat control for para-
magnetic O₂ Sensors)
- X 1 Front panel board BXF (P 8)
- X 16 Digital outputs
- X 18 Analog outputs
- J 9 Status signals and serial
interface



Front panel side

Fig. 8-26: "BKS", Position of Connectors

8-3 Connectors on Electronics Boards

The next sections describe assignment of connectors for several analyzer configurations. For position of connectors see fig. 8-26.

8-3-1-1 IR Measurement w/o Oxygen Sensor

- X 2 Chopper
- X 3
(pins 1& 2) IR source channel 1
- X 4
(pins 1& 2) IR source channel 2
- X 5 IR detector (channel 1)
- X 6 IR detector (channel 2)
- X 7 Temperature sensor (chopper)
- X 9 Light barrier (chopper)

8-3-1-3 Combination of IR and Oxygen Measurement

- X 2 Chopper
- X 4
(pins 1& 2) IR source channel 2
- X 5 Oxygen sensor
- X 6 IR detector (channel 2)
- X 7 Temperature sensor (chopper)
- X 9 Light barrier (chopper)

8-3-1-2 Oxygen Measurement w/o IR

- X 5 Oxygen sensor (channel 1)
[electrochemical Measurement:
"OXS" (cable P1, 5 pole plug)]
- X 6 Oxygen sensor (channel 2)
[electrochemical Measurement:
"OXS" (cable P1, 5 pole plug)]
- X 7 Temperature sensor
[electrochemical Measurement:
"OXS" (cable P1, 2 pole plug)]

8-3-1-4 TCD w/o IR channel

IN PREPARATION!

8-3 Connectors on Electronics Boards

8-3-1-5 Combined IR and TCD Measurement

- X 2 Chopper
- X 4
(pins 1& 2) IR source channel 2
- X 5 TC sensor (from WAP 100,
cable P15, 5 pole plug)
- X 6 IR detector (channel 2)
- X 7 Temperature sensor (chopper)
- X 9 Light barrier (chopper)

8-3-1-6 Combined Oxygen and TCD Measurement

IN PREPARATION !

8-3 Connectors on Electronics Boards

8-3-2 OXS (Electrochemical Oxygen Measurement)

P2	Oxygen sensor
P1, 5 pole plug	BKS, X5 (Sensor signal)
P1, 2 pole plug	BKS, X7 (temperature sensor; not in combination with an IR channel)

Detail A:

Pcb OXS, assembled

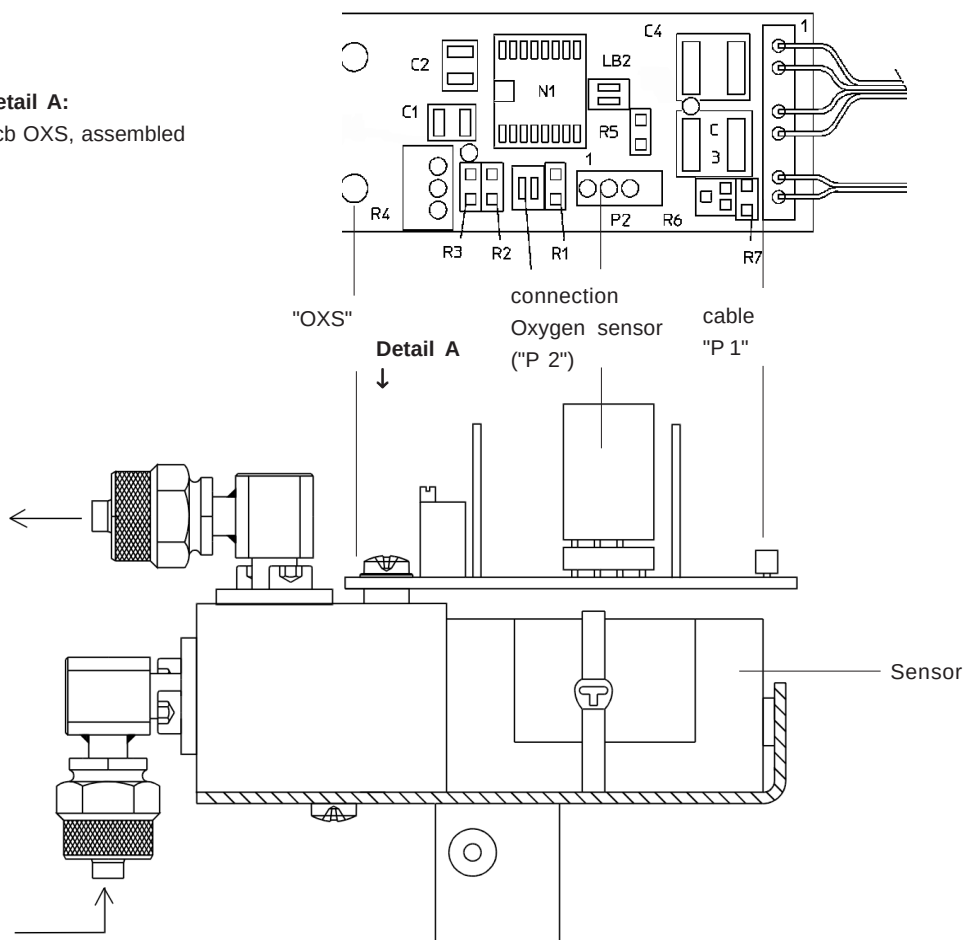


Fig. 8-27: Pcb "OXS" (Electrochemical Oxygen Measurement)

8-3 Connectors on Electronics Boards

8-3-3 WAP 100 (TC Measurement)

P 2	24 V DC supply from BKS (P1 or P2)
P 18	to sensor
P 15	BKS, X5 (sensor signal)
P 10	Analog inputs for cross compensation
P 16	Thermostate control for sensor block

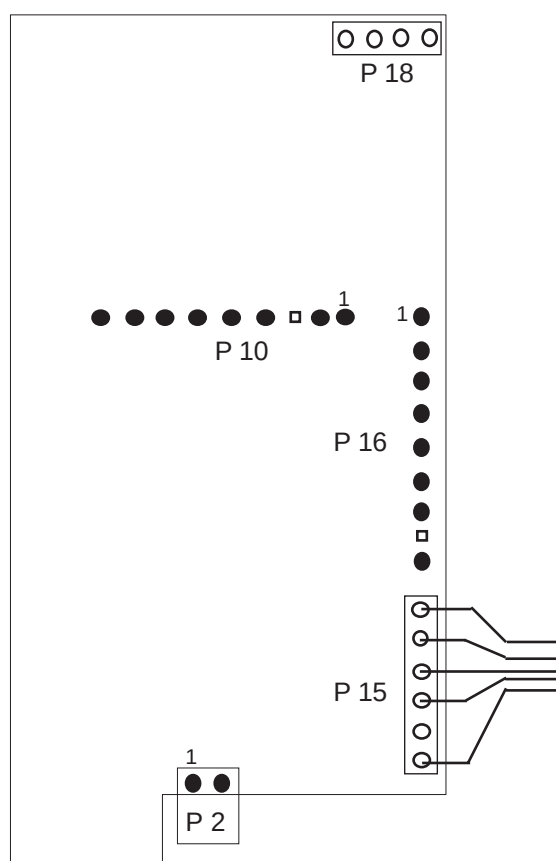


Fig. 8-28: Pcb "WAP 100"

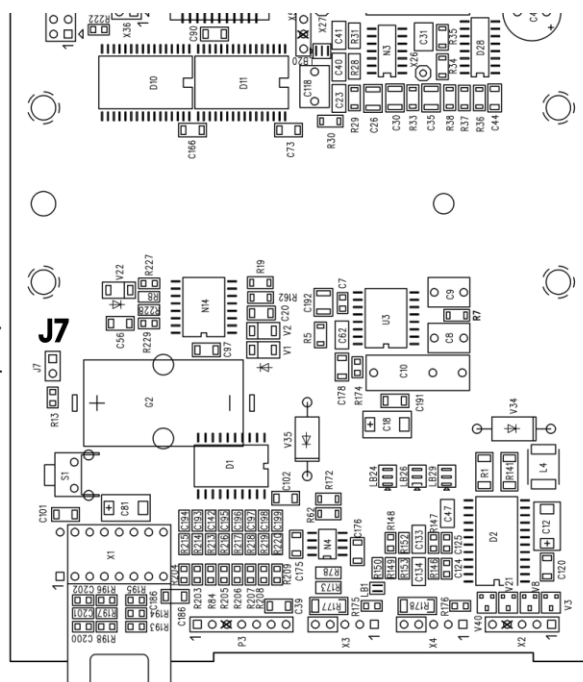
8-4 BKS 20 Jumper Configuration

8-4 BKS 20 Jumper Configuration

There is only one single jumper on the BKS board: J7 is for enabling SRAM battery buffering. With buffering enabled (jumper set) all operator configured data is stored within the SRAM memory and protected against power supply failure.

If J7 is not set properly all data gets lost and the analyzer is reset to factory default settings the moment the analyzer is disconnected from power supply!

J7
Battery buffer jumper



Front panel side

Fig. 8-29: "BKS" (section)

Chapter 9 Maintenance

Maintenance carried out on a regular basis ensures long-term efficiency of your EMERSON Process Management gas analyzer!

For a description of how to *replace the fine dust filter*



page 9-2

perform a leak test



page 9-3

clean the analyzer's outside



page 9-4

*replace the electrochemical
Oxygen sensor*



page 9-5

9-1 Replacing the Fine Dust Filter

9-1 Replacing the Fine Dust Filter

The filter element of the optional fine dust filter (fig. 9-1) needs to be checked for pollution on a regular basis, depending on the application and to be replaced if need be.



Before opening gas paths they must be purged with ambient air or neutral gas (N₂) to avoid hazards caused by toxic, flammable, explosive or harmful to health sample gas components!

Procedure

- Remove glass cap by turning it counter-clockwise (switch off the sample gas pump, if installed) and remove the filter fixing element.
- Replace the filter element by a new clean element (order no.: 42 707 676).

Filter elements are consumables!

Never try to clean polluted elements! Use only new original spare parts for replacing polluted elements!

Dispose polluted filter elements taking into account the local legislation!



- Insert filter fixing element and glass cap (fig. 9-1)
- Perform a leak test (see 9-2)

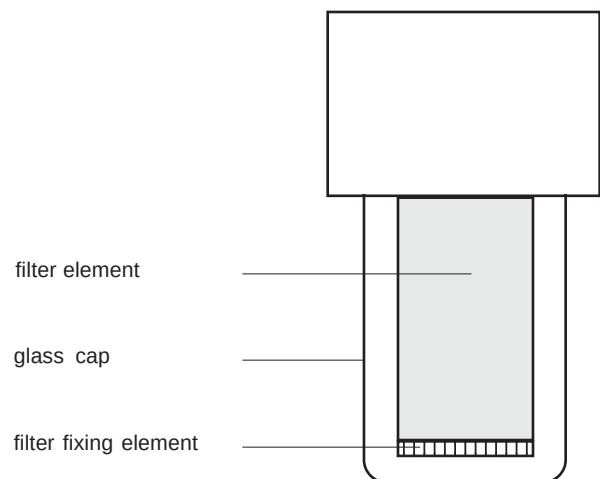


Fig. 9-1: Fine Dust Filter

9-2 Leak Testing

9-2 Leak Testing

The gas path system should be leak tested at least on a bimonthly basis and after maintenance, replacement or repair of gas path parts.



Before opening gas paths they must be purged with ambient air or neutral gas (N₂) to avoid hazards caused by toxic, flammable, explosive or harmful to health sample gas components!

Required tools

- U-turn manometer for max. 500 hPa
- Stop valve

Procedure

- Connect the water filled u-turn manometer to the analyzer's sample gas output (disconnect external gas lines).
- Install the stop valve between gas input fitting and a Nitrogen (N₂) supply.
- Open the stop valve until the internal gas path is under pressure of approx. 50 hPa (corresponding to 500 mm water column)
- Close the stop valve. After a short time for the water to balance the water level must not change over a time period of approx. 5 minutes!

Note!

It is recommended to include external equipment (e.g. cooler, dust filters, etc.) into a leak test!

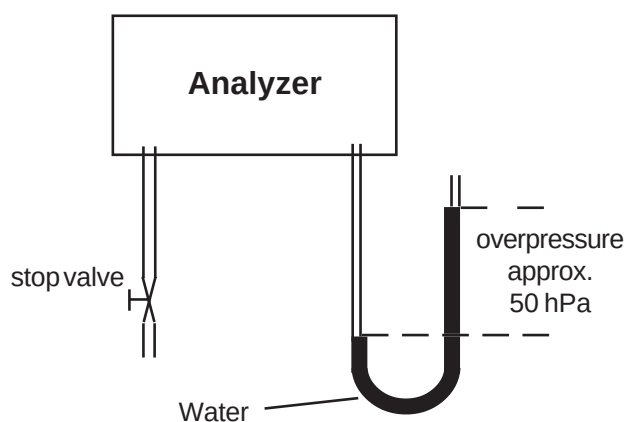


Fig. 9-2: Leak Testing with u-turn manometer



Max. pressure 500 hPa!

Analyzers with parallel tubing require separate leak tests for each gas path !

9-3 Cleaning the Analyzer's Outside

9-3 Cleaning the Analyzer's Outside

Use a liquid general purpose detergent and a lint-free cloth for cleaning the analyzer's outside.

Procedure

- Disconnect instrument from mains!



Before opening gas paths they must be purged with ambient air or neutral gas (N₂) to avoid hazards caused by toxic, flammable, explosive or harmful to health sample gas components!



If opening the gas paths is required seal the open analyzer's gas fittings utilizing PVC caps to avoid pollution of inner gas path.

- Moisten the lint-free cloth with a mixture of 3 parts of water and 1 part of the general purpose detergent.



Do NOT drench the cloth, just moisten it to avoid liquid entering the housing!

- Clean the analyzer housing outside with the moistend cloth
- If required dry the housing after cleaning

9-4 Replacing the Electrochemical Oxygen Sensor**9-4 Replacing the Electrochemical Oxygen Sensor**

In consequence of its design the sensor's lifetime is limited and depends on theoretical designed life and Oxygen concentration. The sensor output can be taken as a rough criterion for end of lifetime: The sensor is wearied when the output in atmosphere is below 70 % of the initial output. The period till this can be calculated by

$$\text{Lifetime} = \frac{\text{designed life (hours)}}{\text{O}_2 \text{ concentration (\%)}}$$

The sensor's designed life under constant conditions of 20 °C is approx. **900,000 hrs.**

The lifetime at 21 % Oxygen is therefore calculated to approx. **42,857 hrs, corresponding to approx. 5 years.**

Note!

The given values are for reference only! The expected lifetime is greatly affected by the temperature of the environment in which the sensor is used or stored. Increases or decreases in atmospheric pressure have the same effect as that by increases or decreases in Oxygen concentration. (Operation at 40 °C halves lifetime).

Note!

Due to the measuring principle the electrochemical Oxygen cell requires a minimum internal consumption of Oxygen (residual humidity avoids drying of the cell). Supplying cells continuously with dry sample gas of low grade Oxygen concentration or with sample gas free of Oxygen could result in a reversible detuning of O₂ sensitivity. The output signal will become instable, but response time remains constant.

For proper measurement results the cell needs to be supplied continuously with concentrations of at least 0.1 Vol.-% O₂. We recommend using the cell if need be in alternating mode, means to purge the cell with conditioned ambient air (not dried, but dust removed) when measurement pauses.

If it is necessary to interrupt Oxygen supply for several hours or days, the cell has to regenerate (supply cell for about one day with ambient air). Temporary flushing with Nitrogen (N₂) for less than 1 h (e.g. for analyzer zeroing purpose) has no influence on measuring characteristics.

9-4 Replacing the Electrochemical Oxygen Sensor

Depending on the current analyzer configuration choose one out of the two procedures described in the following sections to disassemble the sensor.

9-4-1 Analyzer with Oxygen Measurement only, no IR Channel

- Open the housing (chapter 8).
- Disconnect all electrical connections between pcb's OXS and BKS (X5, X6 and X7).
- Disconnect all gas path connections at the sensor holder.
- Unscrew the two Allen screws (fixing screws in fig. 9-3).
- Take the entire holder out of the instrument.
- Disconnect sensor signal connector P2 from pcb OXS (fig. 9-4 and 9-5).
- Cut the cable tie.
- Remove the Phillips screws from the sensor holder (fig. 9-4)
- Disassemble the sensor holder with sensor from mounting bracket.

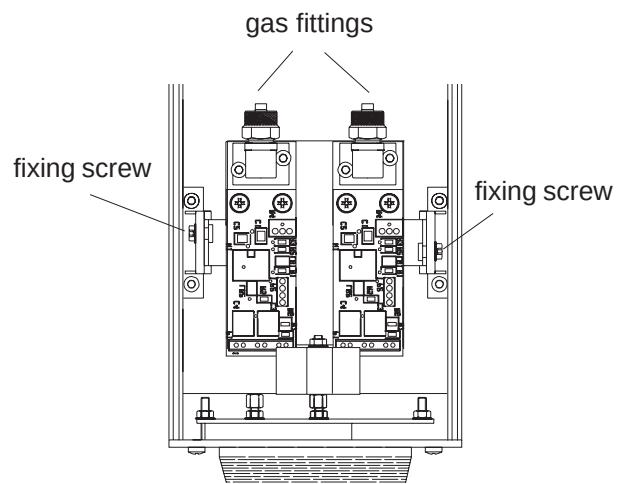


Fig. 9-3: Internal configuration, dual channel electrochemical Oxygen measurement

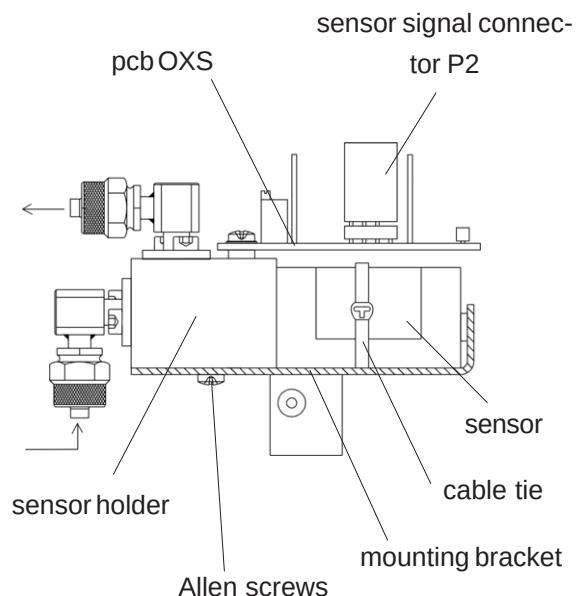


Fig. 9-4: Sensor holder with pcb, gas fittings and sensor

9-4 Replacing the Electrochemical Oxygen Sensor

9-4-2 Analyzer with Combined IR and Oxygen measurement

- Open the housing (chapter 8).
- Disconnect sensor signal connector P2 from pcb OXS (fig. 9-4 and 9-5).
- Unscrew the two Allen screws at the sensor holder (fig. 9-5; screws may alternatively be aligned to the instrument's outside).
- Remove the sensor holder with sensor from the mounting bracket (fig. 9-4).

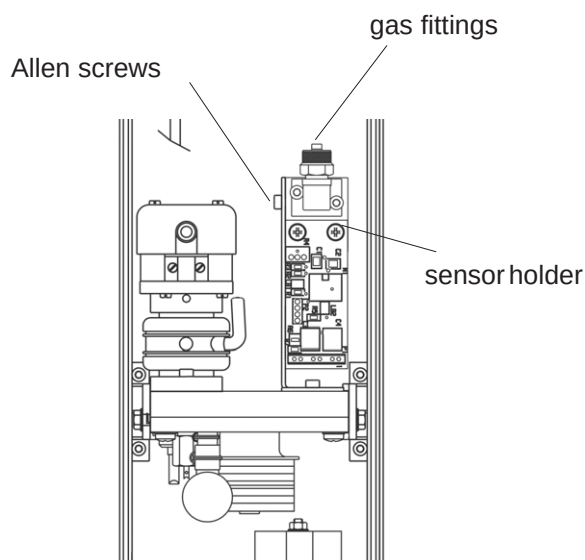


Fig. 9-5: Internal configuration of a dual channel instrument with IR and electrochemical channel

9-4-3 Replacing the Sensor

- Remove the weared sensor from the holder.
- Take the sealing plug off the new sensor and place this sensor into the sensor holder verifying the nameplate label is aligned to the top.
- Close the worn-out sensor's opening with the sealing plug and send it back to the EMERSON Process Management factory for waste disposal.

9-4 Replacing the Electrochemical Oxygen Sensor

9-4-4 Reassembling the Sensor Unit

Reassemble the sensor with sensor holder by performing the steps given in sections 9-4-1 resp. 9-4-2 in reverse order.

When all gas path and electrical connections have been restored perform a leak test (see 9-3) and check the sensor amplifier board settings.

9-4-5 Sensor Amplifier Adjustment

- Power on the opened instrument.
- Supply gas path with ambient air (approx. 21 % O₂)
- Connect a digital voltmeter (DVM) to Tp 1 (signal) and Tp 2 (GND) on pcb OXS (fig. 9-4 and 9-6).
- Adjust the measured signal to 3360 mV DC (± 5 mV) utilizing potentiometer R4 on pcb OXS.

Note !

Once the output signal has been adjusted for a specific sensor do never change the potentiometer settings without prior replacing the sensor!

- Power off the analyzer and close the housing.

In a next step now perform a zero and span calibration at least for the channel the replaced sensor is related to.

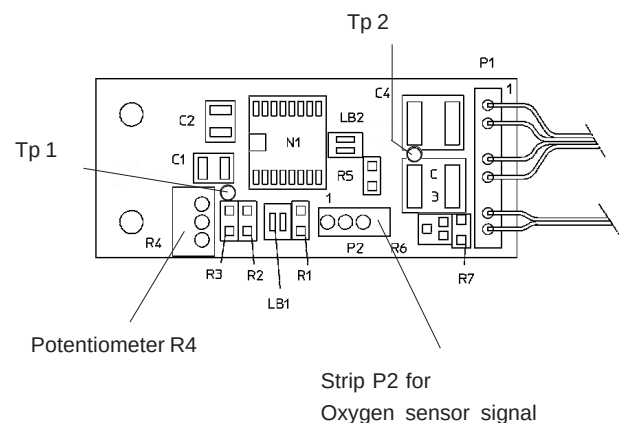


Fig. 9-6: Pcb OXS, top view

9-5 Magnetically Operated Front Panel

9-5 Magnetically Operated Front Panel

To ensure safe and faultless operation of the touch panel it is required to have the control elements (reed relays) installed within a defined distance from the inner side of the security glass.

9-5-1 Sliding the Analyzer into Position

Set the analyzer module onto the fixture. Position the dome and fix it turning it clockwise. The inner side of the front glass now should have contact to the analyzers key pad bar and push the analyzer into the CAT housing. The front of the key pad bar should now have a distance from the inner side of the dome's front glass of between 0 and max. 2 mm (fig. 10). If the distance is bigger, it has to be corrected by repeating the procedure. Next remove the CAT dome once more.

9-5-2 Fixing the Analyzer

Fix the analyzer on the fixture using the screws located at the analyzers left and right side.

9-5-3 Completing the Adjustment

Position the CAT dome and fix it. Make sure the distance is still correct.

If it is necessary to remove the CAT 100 dome for maintenance and/or inspection purposes and to loosen the internal analyzer, the following measures must be followed:

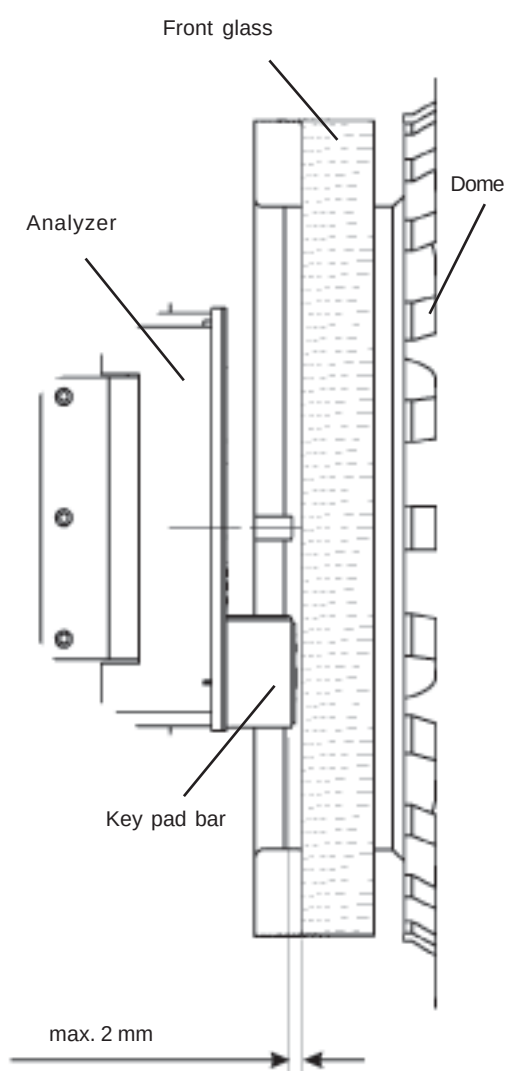


Fig. 9-7: Distance between glass and key pad bar

Appendix

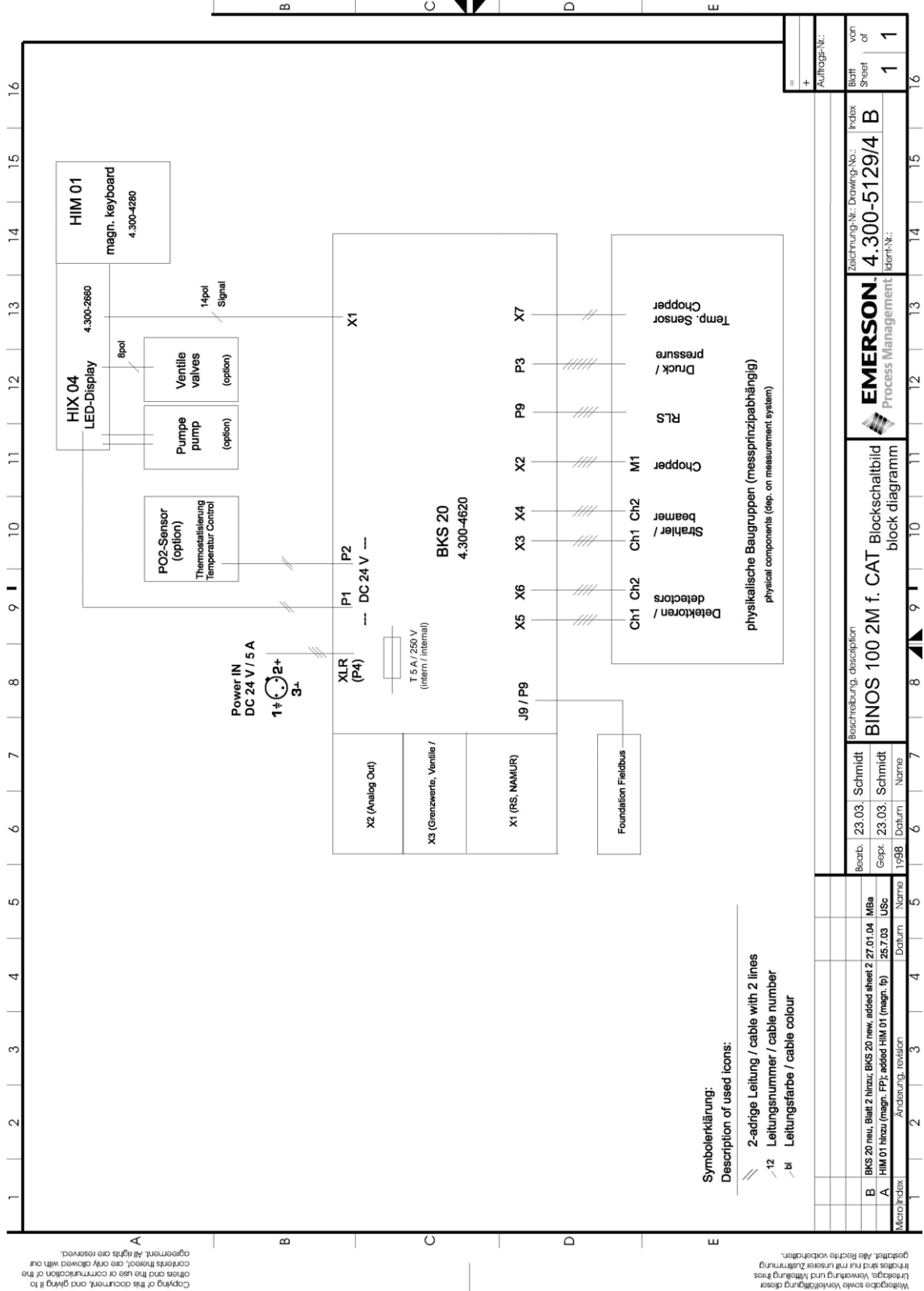
This chapter contains block diagrams for all the different analyzer variations as well as the EC declarations of conformity.

A-1 Block Diagrams

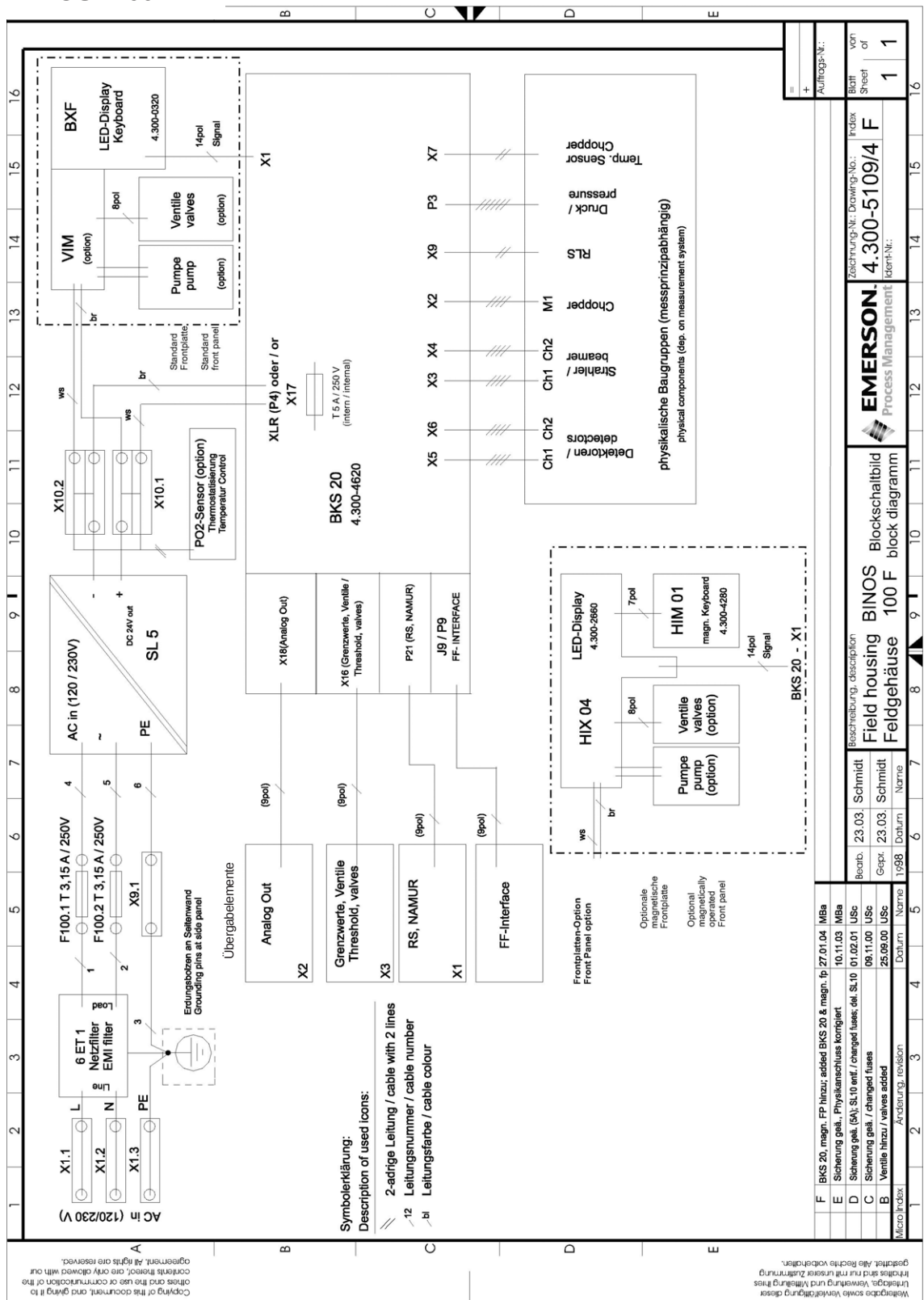
[illegible]

A-1 Block Diagrams

A-1-2 BINOS® 100 2M

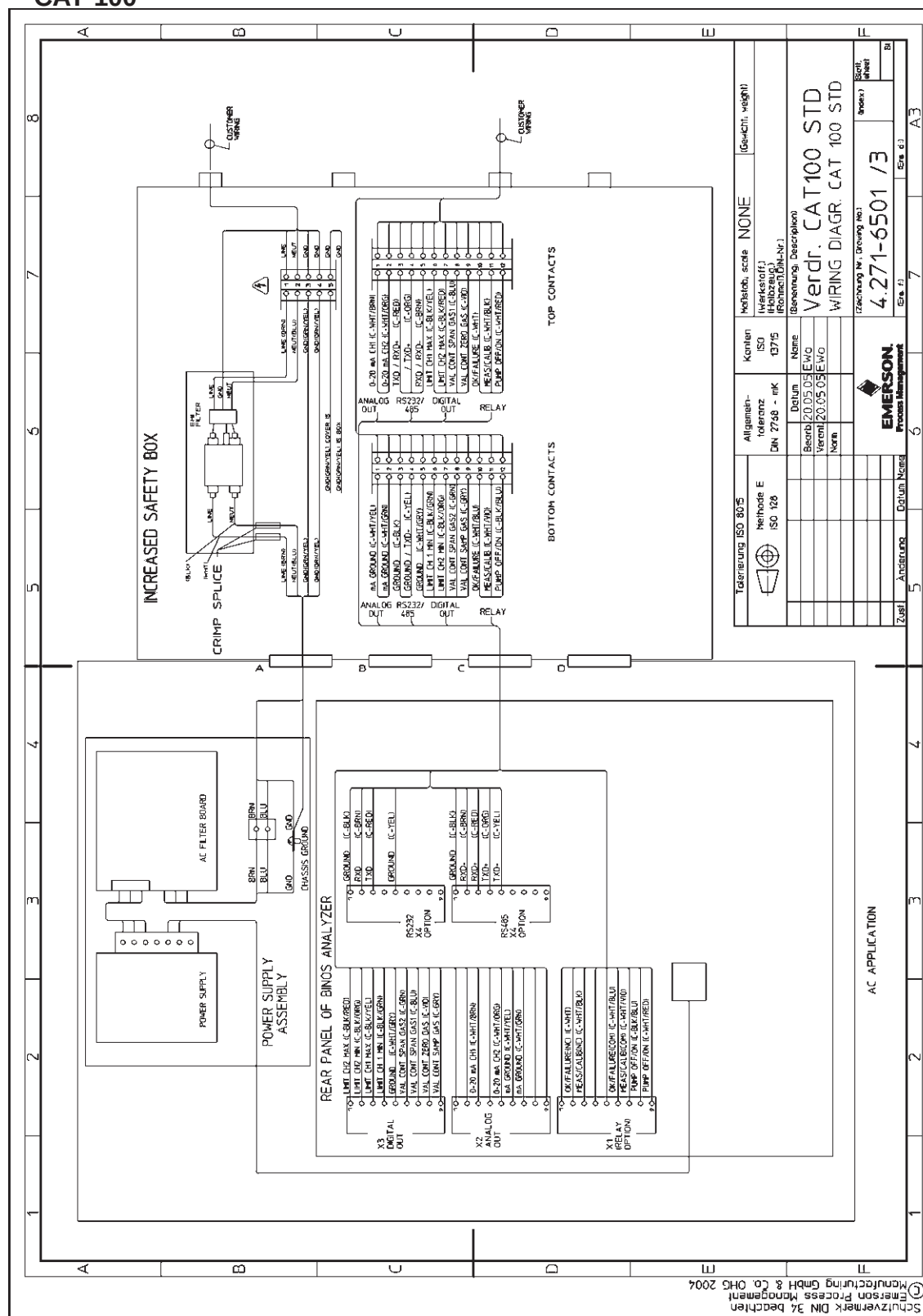


A-1-3 BINOS® 100 F

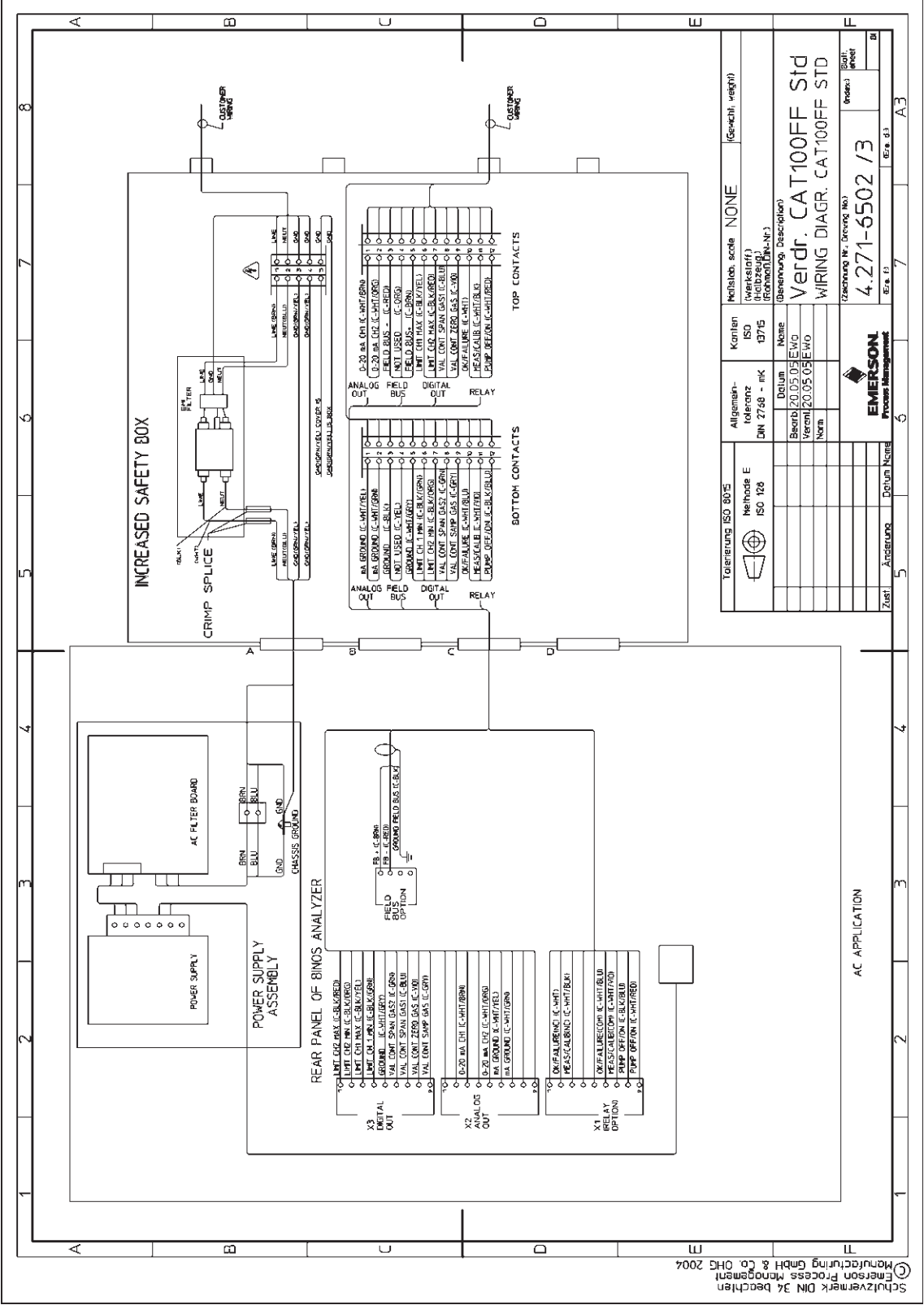


A-1 Block Diagrams

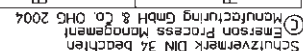
A-1-4 CAT 100



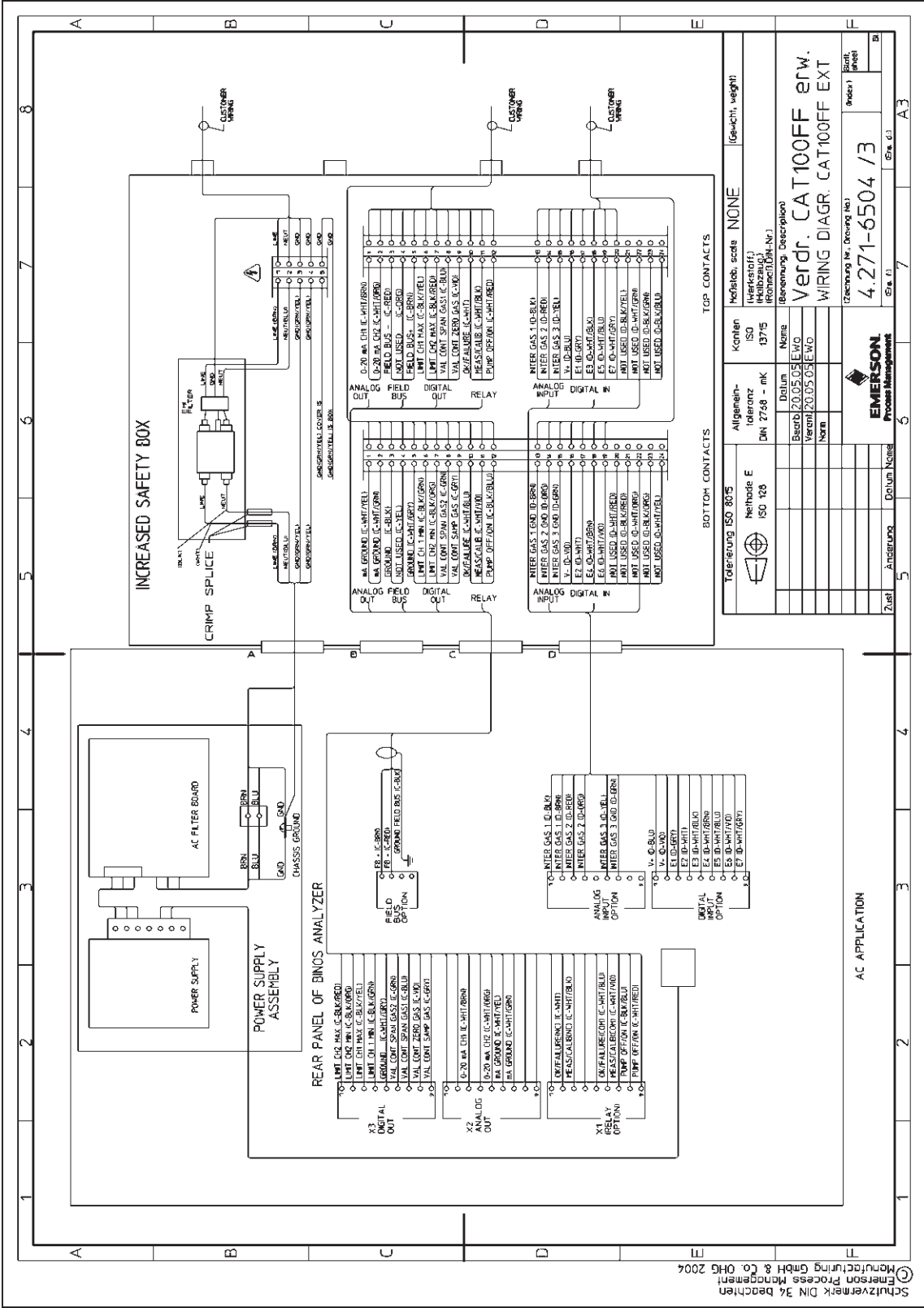
A-1 Block Diagrams



A-1 Block Diagrams



A-1 Block Diagrams



A-2 Declarations of Conformity

A-2 Declarations of Conformity

**A-2-1 BINOS® 100 (M),
OXYNOS® 100,
HYDROS® 100**

A-2-2 BINOS® 100 2M

A-2-3 BINOS® 100 F

A-2-4 CAT 100

EC DECLARATION OF CONFORMITY

Document number: RAE/BINOS100-E9

Date: February 2008

We,

Emerson Process Management GmbH & Co. OHG

located at

Industriestrasse 1, D-63594 Hasselroth, Germany

declare under our sole responsibility that our gas analyzer, type

BINOS 100

to which this declaration relates is in conformity with the provisions of:

89/336/EEC EMC Directive (changed by directive 91/263/EEC 92/31/EEC and 93/68/EEC)
with the application of the harmonized standards:

EN 61326-1:1997	Electrical equipment for measurement, control and laboratory use -
+ A1:1998 + A2:2001	EMC requirements
+ A3:2003	

97/23/EC Pressure Equipment Directive

This analyzer has been designed and manufactured considering article 3, paragraph 3 of the above mentioned directive and therefore CE marking does not refer to this directive.

BINOS 100 series does include any ¼ 19" BINOS 100(M), BINOS 100 4P, Hydros 100 or OXYNOS 100 analyzer (both rack-mountable or table-top version).

Hasselroth, February 2008


.....
(Signature)

Andy Kemish
(Name)

VP Rosemount Analytical Europe
(Function name)

ROSEMOUNT®
Analytical

This declaration confirms the compliance with announced directives but does not include the assurance of properties.
The safety and installation instructions of the documentation have to be followed.


EMERSON
Process Management

EC DECLARATION OF CONFORMITY

Document number: RAE/BINOS100 2M-E8

Date: February 2008

We,

Emerson Process Management GmbH & Co. OHG

located at

Industriestrasse 1, D-63594 Hasselroth, Germany

declare under our sole responsibility that our gas analyzer, type

BINOS 100 2M

to which this declaration relates is in conformity with the provisions of:

89/336/EEC EMC Directive (changed by directive 91/263/EEC 92/31/EEC and 93/68/EEC)
with the application of the harmonized standards:

EN 61326-1:1997

+ A1:1998 + A2:2001

+ A3:2003

Electrical equipment for measurement, control and laboratory use -
EMC requirements

73/23/EEC Low Voltage Directive (changed by directive 93/68/EEC)
with the application of the harmonized standards:

EN 61010-1:2001

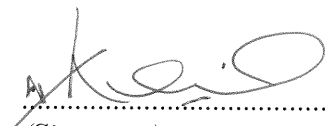
Safety requirements for electrical equipment for measurement, control and
laboratory use
Part 1: General requirements

97/23/EC Pressure Equipment Directive

This analyzer has been designed and manufactured considering article 3,
paragraph 3 of the above mentioned directive and therefore CE marking
does not refer to this directive.

BINOS 100 2M does include any ½ 19" BINOS 100 2M analyzer (both rack-mountable or
table-top version).

Hasselroth, February 2008


.....
(Signature)

Andy Kemish

(Name)

VP Rosemount Analytical Europe

(Function name)

ROSEMOUNT®
Analytical

This declaration confirms the compliance with announced directives but does not include the assurance of properties.
The safety and installation instructions of the documentation have to be followed.


EMERSON
Process Management

EC DECLARATION OF CONFORMITY

Document number: RAE/BINOS100 F-E8

Date: February 2008

We,

Emerson Process Management GmbH & Co. OHG

located at

Industriestrasse 1, D-63594 Hasselroth, Germany

declare under our sole responsibility that our gas analyzer, type

BINOS 100 F

to which this declaration relates is in conformity with the provisions of:

89/336/EEC EMC Directive (changed by directive 91/263/EEC 92/31/EEC and 93/68/EEC)
with the application of the harmonized standards:

EN 61326-1:1997

+ A1:1998 + A2:2001

+ A3:2003

Electrical equipment for measurement, control and laboratory use -
EMC requirements

73/23/EEC Low Voltage Directive (changed by directive 93/68/EEC)
with the application of the harmonized standards:

EN 61010-1:2001

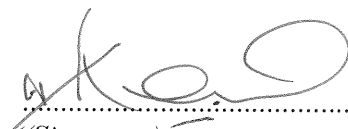
Safety requirements for electrical equipment for measurement, control and
laboratory use
Part 1: General requirements

97/23/EC Pressure Equipment Directive

This analyzer has been designed and manufactured considering article 3,
paragraph 3 of the above mentioned directive and therefore CE marking
does not refer to this directive.

BINOS 100 F does include any general purpose field housing based on BINOS 100(M),
BINOS 100 4P, Hydros 100 or OXYNOS 100 analyzers.

Hasselroth, February 2008


(Signature)

Andy Kemish

(Name)

VP Rosemount Analytical Europe

(Function name)

ROSEMOUNT
Analytical


EMERSON
Process Management

This declaration confirms the compliance with announced directives but does not include the assurance of properties.
The safety and installation instructions of the documentation have to be followed.

EC DECLARATION OF CONFORMITY

Document number: RAE/CAT-ATEX-E6

Date: February 2008

We,

Emerson Process Management GmbH & Co. OHG

located at

Industriestrasse 1, D-63594 Hasselroth, Germany

declare under our sole responsibility that our gas analyzer, type

CAT,

with flameproof enclosure, approved for service in hazardous locations and marked

CE 0035 **Ex** II 2 G
Ex demb IIB + H₂ T4
LCIE 00 ATEX 6009 X

conforms to the provisions of the following European Directives:

89/336/EEC **EMC Directive** (changed by directive 91/263/EEC 92/31/EEC and 93/68/EEC)
with the application of the harmonized standards including the latest amendments:
EN 61326-1:1997 + A1:1998 + A2:2001 + A3:2003

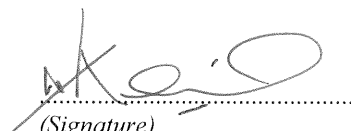
94/9/EC **Equipment and protective systems in potentially explosive atmospheres**
with the application of the harmonized standards including the latest amendments:
EN 60079-0: 2006
EN 60079-1: 2004
EN 60079-7: 2007
EN 60079-18 :2004

Production is monitored according annex VII by
TÜV RHEINLAND INDUSTRIE SERVICE GMBH
Am Grauen Stein, D-51105 Köln, Germany
Notified Body of the European Union under directive 94/9/EC, identification no. 0035

97/23/EC **Pressure Equipment Directive**
This analyzer has been designed and manufactured considering article 3, paragraph 3 of the
above mentioned directive and therefore CE marking does not refer to this directive.

The standards published in the EC's OFFICIAL JOURNAL with reference to directive 73/23/EC
(e.g. EN 61010-1) have been used to fulfill 1.2.7 of Annex II of directive 94/9/EC to eliminate electrical risks.

Hasselroth, February 2008


(Signature)

Andy Kemish
(Name)

VP Rosemount Analytical Europe
(Function name)

ROSEMOUNT
Analytical


EMERSON
Process Management

This declaration confirms the compliance with announced directives but does not include the assurance of properties.
The safety and installation instructions of the documentation have to be followed.

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